# Fine-scale spatial population genomics of *Plasmodium falciparum* across Ghana's agro-ecological zones: a Pf8 analysis

*Revised working analysis report — second edition*

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> **This edition supersedes the 9 October 2026 version.** It incorporates the corrections arising from an adversarial audit (61 confirmed findings) and a complete re-run of stages 4–9 on a corrected pipeline. Quantitative claims were re-derived from the final run's output files wherever those outputs survive; the few that are carried over from the corrections document, computed on superseded runs, or of unrecoverable provenance are labelled as such. A record of the defects found and fixed is given in the appendices.

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## Abstract

**Objective.** To quantify spatial population structure, gene flow and candidate signatures of local selection in *Plasmodium falciparum* across Ghana's agro-ecological zones using the MalariaGEN Pf8 release; to measure the zone-level frequencies of the canonical antimalarial resistance codons, and the haplotypes they form within isolates; and to establish whether zone-level differentiation is detectable at all before any locus-specific signal is interpreted.

**Design limits, stated first.** Zone is not an isolable variable in this dataset. Three of the four represented zones are each filled by exactly one administrative region (Guinea savanna / Upper East, coastal savanna / Greater Accra, transitional zone / Brong Ahafo); collection year differs systematically between zones; and four of the six zone pairs share no Pf8 sequencing study at all, including coastal savanna versus Guinea savanna, the pair that carries every headline number. Ecology, administrative region, collection period and sequencing batch are collinear by construction. Every zone-level result below is a statement about that compound variable. It is arranged broadly from south to north, but not monotonically: the transitional zone is the second-most-northern of the four and is represented by only 25 analysed isolates, and it breaks the south-to-north ordering in all but one of the variables examined here - the single exception is *pfaat1* S258L, which declines monotonically from coastal savanna through deciduous forest and the transitional zone to Guinea savanna - so no claim of a latitudinal cline is made from the four zone means alone. No adjustment within these data can decompose the compound variable.

**Data.** Pf8 contains 33,325 samples and 12,493,205 variant sites. Ghana contributes 6,653 samples, of which 4,905 pass the release's own quality control. Isolates were collected between 2009 and 2021 and are geolocated only to administrative level 1, giving seven distinct coordinates across seven regions. A spatial join to a Ghana agro-ecological zone layer placed these 4,905 in four zones: Guinea savanna (n = 3,605), deciduous forest (n = 815), coastal savanna (n = 435) and transitional zone (n = 50); the smaller zone counts quoted under Methods are for the 1,900 isolates that survive this analysis's own sample QC. The Sudan savanna and the wet-evergreen and moist-evergreen forest zones are unrepresented. The zone layer itself has no citable published source: it is an unattributed re-host of an anonymous ArcGIS Online item, identified byte-for-byte by checksum, and is used here as accessed rather than as authoritative.

**Methods.** Genotypes were obtained from the Pf8 Zarr store over HTTPS, restricted to Ghanaian samples, sites passing the release filter, **biallelic single-nucleotide polymorphisms only** (the ingest mask is `FILTER_PASS & is_snp & numalt == 1`), and the GT, AD, DP and GQ fields. Sample quality control applied a within-host fixation index ($F_{ws}$) threshold of 0.95, a call-rate floor of 0.80, a heterozygosity outlier rule and clonal pruning; mixed calls were resolved to the majority allele by read depth and allele counts taken as one allele per isolate. Variant missingness is now recomputed after the sample cut rather than before it, which yields **3,916 SNPs across 1,900 isolates** (mean missingness 2.95%): Guinea savanna 1,334, deciduous forest 337, coastal savanna 204, transitional zone 25. Analyses comprised three ordinations of the published Pf8 genotype-distance matrix (originally computed on the 4,905 release-QC-pass isolates and now re-run on the 1,900-isolate analysis set, by re-slicing the distance submatrix rather than reusing the cached full-set one), pairwise Hudson $F_{ST}$ with a block jackknife, ADMIXTURE at K = 1-8 with cross-validation, Mantel tests of isolation by distance at zone and region level with a leave-one-population-out jackknife, TreeMix with 0-5 migration edges and jackknife edge standard errors, and a BayPass XtX scan calibrated against 100,000 pseudo-observed neutral SNPs with Benjamini-Hochberg q-values. Two further checks were added at final revision and both are reported below: a **six-scenario threshold sensitivity analysis** that rebuilds the analysis matrix

under varied $F_{ws}$ and MAF cuts and recomputes pairwise $F_{ST}$ and the region-level Mantel test in each, and a **call-quality audit** of the 19 XtX outliers against the other 3,897 scanned SNPs on the Pf8 annotation fields (MQ, VQSLOD, QD, FS, SOR, ReadPosRankSum, ExcessHet, InbreedingCoeff, culprit, RegionType). The resistance panel is separate and has been **rebuilt from the codon up**: **57 codons in nine genes** were reconstructed from the CDS and the Pf3D7 reference genome - each codon mapped to its three genomic positions, every variant record at those positions collected regardless of numalt or is_snp, each isolate's alleles applied to the reference codon, and the codon translated - so that what is reported is the **full residue spectrum including the reference residue**, with Wilson confidence intervals and a four-zone chi-square test per residue. Mixed calls make a codon uncalled rather than being assigned a residue. Within-isolate haplotypes were counted on complete cases only; calls at the 25 of the 57 panel codons for which MalariaGEN publishes a per-sample genotype were compared one-for-one against it, on isolates both pipelines called; and the *pfk13* list was taken from the WHO compendium of validated and candidate markers, v1.0 (December 2025), rather than written by hand.

**Principal findings: population structure.** Zone differentiation is very low, and after the sensitivity analysis this is the best-supported finding in the report. The largest pairwise $F_{ST}$ is 0.00232 (SE 0.00053, 95% CI 0.00128-0.00336) between coastal and Guinea savanna, and one of six pairs - Guinea savanna versus the transitional zone (n = 25) - has a confidence interval spanning zero; the other five have strictly positive lower bounds. Varying the two thresholds that define the dataset leaves the maximum pairwise $F_{ST}$ within **0.00204-0.00255** across all six scenarios, over sample sizes from 1,900 to 4,403 isolates and SNP counts from 3,275 to 10,876. Two directions within that range are worth recording: relaxing the $F_{ws}$ cut slightly **raises** $F_{ST}$ (0.00232 to 0.00255 with no $F_{ws}$ filter at all), so polyclonal isolates were not masking differentiation, and tripling the SNP count at MAF >= 0.01 **lowers** it (0.00204), as expected if rarer variants are less differentiated between zones.
**Isolation by distance, by contrast, is not a robust result.** At administrative-region level the test is significant at the published thresholds (Mantel r = 0.427, p = 0.025 with negative $F_{ST}$ pairs floored to zero; r = 0.414, p = 0.041 without flooring the three negative pairs) and it is absent at zone level (r = 0.471, p = 0.122, significant in 0 of 4 leave-one-zone-out replicates, and with four populations admitting only 24 distinct label permutations). But across the six threshold scenarios the region-level test is significant in only **two**: the published thresholds and MAF >= 0.02 (r = 0.434, p = 0.047 censored, p = 0.050 uncensored, so that scenario is marginal in both directions). It fails at every relaxation of the $F_{ws}$ monoclonality cut ($F_{ws}$ >= 0.90, r = 0.318, p = 0.087; $F_{ws}$ >= 0.80, r = 0.256, p = 0.169; no $F_{ws}$ filter, r = 0.245, p = 0.179) and at MAF >= 0.01 (r = 0.432, p = 0.062). It is also significant in only **four of seven leave-one-region-out replicates**, collapsing when Upper East is dropped. The finding holds at the published thresholds and almost nowhere else, and is reported here as **not robust** - not as a supported result, and not as a weak but real distance effect. ADMIXTURE cross-validation error is minimised at K = 2 (0.83925) over K = 1 (0.84384), an improvement of 0.00459 on a near-flat curve. The K = 2 minor component is not uniform across zones (means 0.092 deciduous forest, 0.085 coastal savanna, 0.053 Guinea savanna, 0.042 transitional zone; Kruskal-Wallis H = 105.4, p = 1.1 x 10 to the -22). That is a real contrast - coastal and forest high, Guinea savanna and transitional low - concordant with the $F_{ST}$ ranking but not a monotone latitudinal cline, and of negligible magnitude. TreeMix, now run with edge standard errors, selects a descriptive graph with **two** migration edges — a model choice, not a count of historical migration events (weights 0.152, p = 7.5 x 10 to the -4, and 0.805, p = 3.3 x 10 to the -16; variance explained 0.9969 at m = 2 against 0.8970 at m = 0); a third edge is indistinguishable from zero (p = 0.33). The edge tests are conditional on a topology, root and edge count all fixed beforehand, and the variance-explained criterion across the full m = 0 to 5 sweep still prefers m = 3, so the two criteria disagree and are reconciled here on identifiability grounds.

**Principal findings: the selection scan.** The BayPass scan returns 19 SNPs above a calibrated XtX threshold of 10.84, of which 13 also pass a 10% false-discovery threshold; 3.9 false positives are expected at the quantile threshold, so expected null exceedances are 20.6% of the 19 calls — a ratio expected under the assumed null, not a measured false-discovery rate. **Seventeen of the 19 fall in 13 annotated genes; two are intergenic, including one (Pf3D7_11_v3:1,054,447, XtX 20.79, q = 0.005) that ranks fifth in the whole scan and survives FDR correction**, the other being Pf3D7_12_v3:715,788 (XtX 15.09, q = 0.017). The 13 genes are headed by PTP7 (max XtX 26.44), the amino-acid transporter AAT1 (PF3D7_0629500, max XtX 24.20), FIKK4.2, ACS8 and NT2. A call-quality audit of all 19 outliers against the other 3,897 scanned SNPs clears the set as a whole: every significant difference runs in the outliers' favour (higher VQSLOD, p = 0.008; QD, p = 0.016; QUAL, p = 0.007), none has MQ < 50, strand bias or read-position bias, all 19 are RegionType Core, and the eight SNPs in exported-protein genes - the subset most open to a repetitive-family artefact worry - are in fact the cleanest in the panel (FS = 0.00 for all eight, MQ median 60.00, VQSLOD significantly higher than the rest), so that concern is **not supported**. The audit does, however, **downgrade two adjacent chromosome-11 sites to lower confidence**: Pf3D7_11_v3:1,054,447 itself and Pf3D7_11_v3:1,054,078 (*SNRPF*, XtX 13.98, q = 0.031), 369 bp apart, the two lowest-InbreedingCoeff sites in the panel (0.193 and 0.147), with 1,054,078 carrying **ExcessHet = 69.59, the only nonzero value among all 3,916 SNPs** and roughly 400x the genome-wide maximum. In a near-haploid organism that points to collapsed paralogy or copy-number variation rather than to mismapping, whose MQ and FS signatures are both clean here; the mechanism is unverified and a read-level depth check would settle it. The consequence is that the fifth-ranked outlier in the whole scan is one of the two least secure entries in it. The AAT1 outlier is *pfaat1* **Ser258Leu**, the second-highest XtX in the scan and the only outlier variant with a published gene-editing phenotype - though *pfaat1* appears nowhere in the WHO marker compendium at any tier and is not a WHO-classified marker. Several other outlier genes are functionally characterised in the literature at gene level (PTP7, an EMP1 trafficking protein; NT2; ACS8), so the claim is variant-level and not a claim of unique annotation. Amambua-Ngwa et al. (2023, *Nature Microbiology*) edited S258L into parasites and measured a **1.5-fold** rise in chloroquine IC50, in a single chloroquine-resistant Southeast Asian line (NHP4026), with epistasis between S258L and F313S and an explicit *pfcrt*-background caveat; it is therefore a variant that *potentiates* chloroquine resistance modestly and conditionally, not a stand-alone resistance determinant, and the longitudinal 0% to 97% rise that paper reports is Gambian (1984-2014) and says nothing about Ghana. S258L survives FDR correction (q = 0.005) and falls from 0.824 in coastal savanna to 0.544 in Guinea savanna (four-zone chi-square p = 1.9 x 10 to the -18), with deciduous forest at 0.735 and the transitional zone at 0.640 - so for this one variable the four zones do fall in strict south-to-north order, the sole variable reported here in which the transitional zone does not break the ordering. The second AAT1 outlier is a synonymous site 883 bp away, so this is one signal with a linked silent site, not two independent lines of evidence.

**[Calculated]** **Principal findings: drug-resistance markers.** The previous version's resistance findings were affected by a defect in how markers were identified - the panel matched SnpEff protein-change strings, which can see neither a resistance allele that is itself the reference residue nor a residue that requires two nucleotide changes in one codon - and the panel has been rebuilt by codon reconstruction, which corrects it. Two findings follow directly from the rebuild. *pfdhps* A437G, wholly absent from the previous panel**, stands at** 0.897 *for G437 (n = 1,892; deciduous forest 0.964, coastal savanna 0.941, Guinea savanna 0.875, transitional zone 0.800; chi-square p = 1.2 x 10 to the -6); the official Pf8 methodology states that the 3D7 sequence translates as a G at 437 and that this is the mutant allele, so the resistance residue* is *the reference residue and no ALT annotation for it can exist. And the former "naming discrepancy" at* pfcrt *75 was never a naming matter: N75E requires two SNPs in one codon (403621 A>G, annotated Asn75Asp; 403623 T>A,*

*annotated Asn75Lys), and neither annotation names it. Haplotypes are now counted within isolates on complete cases. The* pfdhfr *IRN triple mutant is 0.771 (n = 1,876; 95% CI 0.751-0.789) - not the marginal range 0.786-0.908 previously quoted, and, as it must be, below its lowest component on the same complete-case set (51I = 0.788); NRN sits at 0.116 and the fully wild-type NCS at 0.091, so it is not a near-fixed single haplotype, and the quadruple IRNL is 0.000.* pfcrt *CVIET is 0.071 and is demonstrably a single unit: only CVMNK (0.929) and CVIET occur at codons 72-76, with nothing intermediate.* pfdhps *GE (437G + 540E) is 0.010. The* pfdhfr *marginals remain high and lowest in Guinea savanna (S108N 0.908 overall / 0.885 Guinea savanna; C59R 0.886 / 0.864; N51I 0.786 / 0.764; chi-square p = 9.4 x 10 to the -7, 4.8 x 10 to the -5 and 2.4 x 10 to the -4), with the transitional zone (n = 25) at or near the top of all three. The* pfk13 *panel was rebuilt from the WHO compendium v1.0 (December 2025) and covers all 14 validated markers (F446I, N458Y, C469Y, M476I, Y493H, G533S, R539T, I543T, P553L, R561H, P574L, C580Y, R622I, A675V) and all 6 candidates (E252Q, P441L, C469F, N537I, G538V, V568G); the old list was written from memory, misclassified P441L and C469F as validated when WHO classes them as candidate, and omitted P574L, G533S and R622I. All 23* pfk13 *codons carry no screened resistance allele: twenty-two are fixed at the reference residue, and codon 458 is at 0.9995 on a single Asn458Asp call, which is not the validated N458Y, on 1,845 to 1,876 isolates called per codon of 1,900, across a 2009-2021 sampling period - a tested historical negative, not a statement about the present. Validation is now per sample rather than per marginal frequency: 47,012 of 47,012 calls agree with MalariaGEN's published genotypes, 100.0000% across 25 markers*, codon 75 included.

**[Observed]** **Correction to the previous version of this report.** Two defects, not one, and the second was in a correction. First, the earlier statement that no validated antimalarial resistance locus appears among the outliers was a filter artefact, not a biological result. The ingest filter required `numalt == 1`, and in Pf8 most canonical resistance codons are multiallelic or carry a spanning-deletion allele, so most of the panel was removed before any analysis ran. The cascade into the scan matrix, stated once and used in the same form throughout this report, is: **44 canonical codon-allele records genotyped by the superseded panel; 21 polymorphic in this sample; 6 of the 21 passed the ingest mask; 3 then cleared the MAF >= 0.05 cut and actually reached the scan matrix** (*pfdhfr* C59R, *pfdhfr* N51I and *pfaat1* S258L). *pfcrt* I356T, *pfmdr1* D1246Y and *pfcrt* N326S passed the mask but were cut by MAF at the frequencies the superseded panel and the filter saw, 0.016, 0.011 and 0.008. Two of those three reproduce in the rebuilt panel (I356T 0.0158, D1246Y 0.0113); the third does not. The codon-reconstruction panel puts *pfcrt* N326S at **0.0005** - 1 of 1,874 called, with MalariaGEN agreeing on all 1,874 - because roughly 15 of the 16 carriers the string-matching panel counted were depth-resolved mixed calls that the new panel leaves uncalled, so the 0.008 is a fifteen-fold overstatement and is quoted here only as the value the MAF cut acted on. Second, the panel that was re-fetched to repair that artefact was itself wrong, because it identified codons by matching SnpEff `ANN_HGVS_p` strings: it missed *pfdhps* A437G entirely, reported the wrong residue at *pfcrt* 75, and used a *pfk13* list written from memory. Its validation claim - that marginal frequencies agreed with MalariaGEN at 21 of 23 markers to within 0.03, the *pfcrt* 75 difference being a residue-naming matter - is withdrawn: marginal agreement does not establish per-sample agreement, and the codon-75 difference was a real error in our calls, not a naming convention. The codon-reconstruction panel reported above supersedes both and agrees with MalariaGEN call-by-call at 47,012 of 47,012 calls.

**[Interpreted]** **Conclusion.** At the resolution these data permit, Ghanaian *P. falciparum* shows very low but statistically detectable differentiation at the retained common variants. Five of the six pairwise estimates have 95% confidence intervals excluding zero; only Guinea savanna against the transitional zone, where n = 25, spans it. Very low is not the same as undetectable, and neither is a demonstration of random mating, for which these data contain no direct evidence. What the sensitivity analysis adds is stability of the magnitude

rather than absence of the effect: the maximum pairwise $F_{ST}$ stays within 0.00204-0.00255 across all six threshold scenarios, over 1,900 to 4,403 isolates and 3,275 to 10,876 SNPs, so the headline result is insensitive to both of the cuts that define the dataset. The companion claim does not survive the same treatment. The region-level isolation-by-distance result is significant at the published thresholds and at MAF >= 0.02, but in only 2 of 6 threshold scenarios and 4 of 7 leave-one-region-out replicates, and this report therefore no longer asserts a distance effect, weak or otherwise. What remains of the spatial signal is a single south-to-north contrast, visible in the $F_{ST}$ ranking and in the K = 2 minor component, with no robust evidence that genetic distance scales with geographic distance. That very low differentiation is a statement about neutral common-variant structure only: neutral common-variant differentiation between zones is negligible, so a spatially sparse design is adequate for characterising neutral structure, but marker-frequency representativeness is a separate question, and the resistance panel in this analysis shows significant between-zone variation at p < 0.001 at *pfdhfr*, *pfdhps* and *pfaat1*. That threshold is stated because it is a selection: on the four-zone chi-square as reported, several further loci are nominally significant at p < 0.05, among them *pfcrt* I356T (p = 1.6 x 10 to the -3), *pfmdr2* I492V (p = 0.020) and *pfmdr1* N86Y (p = 0.029). Low neutral $F_{ST}$ therefore does not license an assumption that marker frequencies are spatially representative. The *pfk13* result is a genuine tested negative - no screened resistance allele occurs at any of the 23 codons, which cover the complete WHO-validated set of 14 and all 6 candidates: twenty-two are fixed at the reference residue and codon 458 is at 0.9995 on a single Asn458Asp call, which is not the validated N458Y, on 1,845 to 1,876 isolates called per codon - rather than an untested absence; it is also a statement about 2009-2021 and not about current parasite populations. The XtX set remains a ranked candidate list, of which 13 SNPs survive FDR 10% and two adjacent chromosome-11 sites - one of them the fifth-ranked SNP overall - are flagged lower confidence on ExcessHet and InbreedingCoeff grounds. All zone-level results describe a compound variable in which ecology, administrative region, collection period and sequencing study cannot be separated; that is a property of the Pf8 metadata, not of the analysis.

**[Interpreted]** **Robustness of these conclusions to the corrected re-runs and the final checks.** The population-structure readings held on a 16.6% larger SNP panel (3,359 to 3,916 SNPs) with corrected filter ordering, a bitwise-reproducible single-threaded BayPass, seeded permutation tests and full false-discovery accounting: near-panmixia, maximum $F_{ST}$ of approximately 0.0023, and a near-flat ADMIXTURE cross-validation curve. Three quantities moved and should not be described as unchanged: the XtX threshold (the manuscript's 8-thread run gave 11.232, the final single-threaded run gives 10.840), the outlier count (15 to 19) and the outlier gene list (11 to 13 genes). **One conclusion has since been lost, and it was the report's only positive spatial finding.** The threshold sensitivity analysis shows the region-level Mantel result significant in 2 of 6 scenarios, and it also withdraws the earlier claim that the result no longer depends on the censoring of negative $F_{ST}$ pairs: of the six scenarios only the baseline clears p < 0.05 both censored and uncensored, the second significant scenario (MAF >= 0.02) giving p = 0.047 censored against p = 0.050 uncensored. Isolation by distance is retained in the report and reported with the scenarios in which it fails, but it is graded **not robust**, and the previous version's listing of it among the strengthened results is withdrawn. Two strengthenings stand. Near-panmixia is now supported by $F_{ST}$ stability across 0.00204-0.00255 over all six scenarios, which is better evidence than any single run could give; and TreeMix now carries jackknife edge standard errors, so the m = 2 graph is reported as a selected descriptive model rather than as a settled edge count. Two further checks changed nothing material. Re-running the three ordinations on the 1,900-isolate analysis set rather than the 4,905 pre-QC isolates leaves the reading intact - silhouette by zone -0.020 against -0.016 on the full set, so removing polyclonal and clonally redundant isolates does not sharpen the structure; trustworthiness 0.499 (UMAP), 0.526 (t-SNE), 0.510 (PCoA), PCoA distance correlation 0.287 against 0.322 -

and the call-quality audit clears the outlier set while downgrading two chromosome-11 SNPs. The resistance panel is a different matter and was not confirmed but replaced: an external critique upheld in full showed that the marker identification method was unsound, and the panel was rebuilt from the codon up, changing the *pfdhfr* triple-mutant figure, adding a locus that had been missing altogether, correcting a residue call, correcting the *pfk13* list, and replacing the validation argument. The population analyses are therefore a partial confirmation with two strengthened results, one lost result, two downgraded outlier SNPs and one materially revised candidate list; the resistance results are new.

# 1. Introduction and background

## 1.1 The problem

Malaria transmission in Ghana is spatially heterogeneous, and the heterogeneity tracks ecology. The country spans a south-to-north gradient of agro-ecological zones that differ in rainfall, vector species composition, seasonality and consequently in entomological inoculation rate. If parasite populations in these zones are demographically separate to any meaningful degree, then drug-resistance alleles, antigenic variants and locally adaptive alleles can differ in frequency between them. If instead the zones exchange parasites freely, an allele arising anywhere will spread everywhere, and the surveillance question becomes one of timing rather than of geography.

The two scenarios demand different control responses and are distinguishable from genomic data. The present analysis addresses the prior question that both depend on: how much spatial structure is there in Ghanaian *P. falciparum*, and is it enough to support zone-level inference at all?

[Interpreted] **A refinement of the framing.** An earlier version of this report ran the argument directly from the amount of neutral structure to a conclusion about surveillance design - that if the parasites form one population, spatially sparse sentinel sampling is representative. That step does not hold, and the project's own literature review argues against it: selection builds connectivity and locus-specific allele trajectories that neutral demography does not predict, so a resistance allele can have a site-specific frequency even under neutral panmixia. Neutral differentiation and marker-frequency representativeness are two distinct quantities and have to be measured separately. This analysis therefore does both - genome-wide differentiation at common variants, and direct resistance-marker frequencies by zone - and reports them as two findings rather than deriving the second from the first.

**Scope.** The thesis proposal and literature review (supplied separately as "genomic analysis of Consistent signatures of selection.docx", 145 paragraphs, approximately 11,400 words) set out the biological rationale, the resistance-marker panel and the published context. This report does not restate that material. It documents what was computed, on what data, with which decisions, and what the output actually shows, following a full re-run of stages 4-9 in which seven defects were fixed: **six identified by adversarial audit, and a seventh found by the re-run itself**. The seventh is the more instructive one and is claimed here rather than buried. The "skip if output exists" guards in `step05b`, `step07b` and `step08b` did not check whether their *inputs* had changed, so when the SNP set went from 3,359 to 3,916 both ADMIXTURE and TreeMix silently re-reported the previous day's results as "stage finished". Only `step08c` caught it, because it asserts that its position table and the BayPass output have matching row counts and so failed loudly with the right diagnosis; ADMIXTURE and TreeMix have no such assertion. The guards now require each output to be newer than its input. The audit did not raise this defect.

[Interpreted] **A later round, and a fourth consecutive correction containing its own defect.** After that re-run, an external critique raised five points. All five were upheld and four were defects in the analysis code rather than in the prose. They were confined to the resistance panel, which has since been replaced in its entirety by scripts/step12_codon_panel.py; the population-structure results in Sections 6 and 7 are unaffected. The pattern across all four rounds is worth stating plainly, because it bears on how much weight these numbers can carry: each layer was checked for internal consistency against our own outputs, and what escaped was external correctness - whether the marker list matched an authoritative source, whether the annotation direction matched the reference genome, whether a co-occurrence claim had co-occurrence data behind it. The verification architecture was built to catch stale numbers and it did. It was not built to ask whether the list was the right list.

[Interpreted] **A fifth round, and the loss of a result.** Four outstanding robustness checks were run at final revision, and they are the reason this version grades one finding lower than the last. Three of the four left the conclusions standing or strengthened them: the maximum pairwise $F_{ST}$ varies only between 0.00204 and 0.00255 across six threshold scenarios spanning 1,900 to 4,403 isolates and 3,275 to 10,876 SNPs, so near-panmixia does not depend on the $F_{ws}$ or MAF cuts; the 19 XtX outliers are better-supported calls than the rest of the scanned panel rather than artefacts, although two adjacent chromosome-11 sites are downgraded within that set; and re-running the ordinations on the 1,900-isolate analysis set, so that they describe the same isolates as everything else, changes nothing material. The fourth check overturned a conclusion. The region-level isolation-by-distance result - the only positive spatial finding in this report - is significant in 2 of those 6 threshold scenarios and in 4 of 7 leave-one-region-out replicates, collapsing when Upper East is dropped, and is therefore graded **not robust** wherever it appears below. It is not deleted: it holds at the published thresholds, and it is reported together with the scenarios in which it fails. But it can no longer be described as supported, moderately supported, or a weak but real distance effect, and the preceding version's listing of it among the strengthened results is withdrawn. The remaining check, the provenance of the agro-ecological zone layer, cannot be closed analytically at all and stays open.

[Observed] **Two incidents during the final runs, recorded because the verification argument depends on them.** First, results/baypass/xtx_all_snps.csv was found at **0 bytes**, truncated at 08:29:53 - hours after the 01:42 run that produced it, so almost certainly a stray shell redirect where a read was intended during a verification pass. It was the only zero-byte file under results/ or interim/. It was regenerated from the intact BayPass outputs, and the regeneration is consistent with the surviving outputs rather than proven row-for-row: outlier_snps.bed, xtx_calibration.json and xtx_threshold.txt are byte-identical by md5 before and after, and the restored file carries the same 3,916 rows and the same 19 outliers. Second, the --suffix patch to step03_umap.py suffixed the file writes but not the print statements, so the run reported writing to the published dimred_comparison.csv when it had in fact correctly written dimred_comparison_qcset.csv. No data were at risk - the published file's mtime and contents were both checked - but a log that misreports its own output path is exactly what produces a later misattribution, and it was caught by inspecting the output directory rather than by reading the log. Both are fixed.

## 1.2 Objectives addressed by the present analysis

The thesis defines three specific objectives. The work documented here engages all three. Objectives 1 and 3 are now answered within the limits the design permits; objective 2 is answerable only in a weakened form, because a barrier test requires resolution this design does not have, and the one distance-related signal it did produce has not survived sensitivity analysis. Their status is as follows.

| Thesis objective | What this analysis contributes | Status |
|---|---|---|
| Quantify the magnitude and direction of gene flow and population genetic differentiation between zones | Pairwise Hudson *F*ST with block-jackknife confidence intervals, recomputed under six threshold scenarios that vary the *F*ws and MAF cuts; ADMIXTURE K = 1-8 with cross-validation; TreeMix with 0-5 migration edges and jackknife edge standard errors; three ordinations of the Pf8 genotype-distance matrix, re-run on the 1,900-isolate analysis set | Answered, and robust. Differentiation quantified and found very low (max *F*ST 0.00232, 95% CI 0.00128-0.00336; one of the six pairwise CIs spans zero). The magnitude is insensitive to both thresholds that define the dataset: max *F*ST stays within 0.00204-0.00255 across all six scenarios, over 1,900-4,403 isolates and 3,275-10,876 SNPs. A two-edge graph is the model these data select (edge p = 7.5 x 10 to the -4 and 3.3 x 10 to the -16); a third is not supported (p = 0.33). |
| Evaluate whether ecological transitions act as dispersal barriers | Mantel tests of isolation by distance at zone and region level, geographic distance on the WGS84 ellipsoid, with disclosure of negative-*F*ST censoring, a leave-one-population-out jackknife, and recomputation of the region-level test under all six threshold scenarios | Addressed; the answer is negative on robustness. A distance effect is significant at region level at the published thresholds (r = 0.427, p = 0.025; uncensored r = 0.414, p = 0.041) but in only 2 of 6 threshold scenarios - failing at *F*ws >= 0.90, *F*ws >= 0.80 and no *F*ws filter (p = 0.087, 0.169, 0.179) and at MAF >= 0.01 (p = 0.062), and marginal at MAF >= 0.02 (p = 0.047 censored, 0.050 uncensored) - and in only 4 of 7 leave-one-region-out replicates, collapsing when Upper East is dropped. It is absent at zone level (r = 0.471, p = 0.122; 0 of 4 replicates). It is therefore reported as not robust, superseding the earlier grade of partially supported. A barrier test as such is in any case confounded with distance, administrative region, collection period and sequencing study (Section 9.2). |
| Identify and characterise zone-specific variants associated with local adaptation or drug response | BayPass core-model XtX scan calibrated against simulated neutral data, with expected false positives, the implied expected-null-exceedance ratio and Benjamini-Hochberg q-values; intersection of outliers with the PlasmoDB-71 annotation; a call-quality audit of all 19 outliers against the other 3,897 scanned SNPs on the Pf8 annotation fields; and, separately, a codon-reconstruction resistance panel - 57 codons in nine genes, built from the CDS and reference genome so that the full residue spectrum including the reference residue is reported, with Wilson CIs, four-zone chi-square tests, within-isolate haplotypes on complete cases, a *pfk13* list taken from WHO compendium v1.0 (December 2025), and call-by-call comparison against MalariaGEN's published genotypes | Answered, on a rebuilt panel. 19 XtX outliers above the calibrated threshold, 13 of which pass FDR 10%. Seventeen of the 19 fall in 13 annotated genes; two are intergenic, including one (XtX 20.79, q = 0.005) that ranks fifth overall and survives FDR correction. The set is not an artefact - every significant quality difference favours the outliers (VQSLOD p = 0.008, QD p = 0.016, QUAL p = 0.007), none has MQ < 50, strand bias or read-position bias, all 19 are RegionType Core, and the eight exported-protein SNPs are the cleanest subset in the panel - but two adjacent chromosome-11 sites, that fifth-ranked intergenic SNP and *SNRPF* Pf3D7_11_v3:1,054,078 (q = 0.031), are downgraded to lower confidence on ExcessHet and InbreedingCoeff grounds. *pfaat1* Ser258Leu falls from 0.824 in coastal savanna to 0.544 in Guinea savanna (chi-square p = 1.9 x 10 to the -18). *pfdhps* G437 - a locus absent from the superseded panel - stands at 0.897 (n = 1,892; p = 1.2 x 10 to the -6). The *pfdhfr* IRN haplotype is 0.771 on complete cases (n = 1,876; CI 0.751-0.789), not the 0.79-0.91 marginal range previously quoted, and *pfcrt* CVIET is 0.071. All 23 *pfk13* codons - covering the complete WHO-validated set of 14 and all 6 candidates, which between them occupy 19 distinct codons - carry no screened resistance allele: twenty-two are fixed at the reference residue, and codon 458 is at 0.9995 on a single Asn458Asp call, which is not the validated N458Y. The denominators are 1,845 to 1,876 isolates called per codon, over 2009-2021. Agreement with MalariaGEN is 47,012 of 47,012 calls across 25 markers. |

*Table 1.1. Status of the three thesis objectives after the corrected re-run, the rebuild of the resistance panel and the final four robustness checks. Objective 2's grade is lowered here: the region-level distance effect reported in earlier versions is significant in only 2 of 6 threshold scenarios and 4 of 7 jackknife replicates and is now graded not robust. Objective 1's grade is raised, because the threshold sensitivity analysis shows the low-differentiation result to be insensitive to both cuts that define the dataset. Objective 3 was previously recorded as preliminary with a null result, then answered on a panel that was itself defective; both episodes are set out below.*

[Observed] **Why objective 3's status changed, the first time.** The first version of this report recorded objective 3 as preliminary, with no validated resistance locus recovered, and interpreted that absence biologically. The absence was an artefact of data ingest. The genome-wide callset was built under `FILTER_PASS & is_snp & numalt == 1`, and in Pf8 most canonical resistance codons are multiallelic or carry a spanning-deletion ALT allele, so the mask discarded most of the panel before any analysis ran: across the nine panel genes the filter keeps 3,330 of 13,367 PASS records, and of the 21 canonical codons that were polymorphic in this sample only 6 passed the mask, of which only 3 then cleared the MAF >= 0.05 cut and reached the scan matrix. Among the codons that never reached it are the whole polymorphic CVIET set in *pfcrt* (74, 75, 76, 220, 271, 371), *pfmdr1* N86Y and Y184F, *pfdhfr* S108N, and *pfdhps* S436A, K540E and A581G. Those markers were never tested, so the null was not a result.

[Observed] **Why it changed again, and this is the more serious error.** The panel that replaced the masked callset found codons by matching SnpEff's `ANN_HGVS_p` protein-change strings against a hand-written list. That method has two failure modes and both occurred in this dataset. (i) A resistance allele that *is* the reference residue can never appear as an ALT change: at *pfdhps* 437 the 3D7 sequence translates as G, which the Pf8 methodology states is the mutant allele, so the only annotation present is the revertant `Gly437Ala`, a search for "Ala437Gly" matched nothing, and **A437G was missing from the panel entirely at a true frequency of 0.897**. (ii) A residue reachable only by two changes within one codon is never annotated: *pfcrt* 75 carries 403621 (A>G, `Asn75Asp`) and 403623 (T>A, `Asn75Lys`), Asn to Glu needs both, and so no annotation names N75E - the residue the CVIET haplotype actually carries. That, and not a "naming discrepancy", is the entire explanation of the apparent disagreement with MalariaGEN reported in the previous version, and the error was ours. The panel is now built by reconstructing each codon from the CDS and the reference genome and reporting the full residue spectrum, reference residue included, so that neither failure mode is possible.

[Observed] **What else the rebuild changed.** The *pfdhfr* "triple mutant" was previously reported as the range of its three marginal frequencies, 0.786-0.908, which is not a haplotype frequency at all; counted on isolates called at all three codons it is 0.771 (n = 1,876), below the lowest of its components on the same complete-case set (51I = 0.788) as a conjunction must be, with NRN at 0.116 and wild-type NCS at 0.091. The complete-case 0.788 and the panel-wide marginal 0.786 (n = 1,891) are two different quantities for 51I, and the lower bound of the superseded range is the latter. The *pfk13* list was written from memory: it contained 13 distinct protein changes inflated to an apparent 14 by a duplicated M476I record, included P441L and C469F as validated when WHO classes them as **candidate**, and omitted P574L, G533S and R622I - R622I having been requested but never retrieved, so that it became a silent non-result rather than a reported one. And the validation argument was too weak for the claim it supported: agreement of marginal frequencies to within 0.03 does not establish that the two pipelines call the same isolates the same way. Per-sample comparison now does, at 47,012 of 47,012 calls across 25 markers.

[Observed] **A corroboration that must be stated more weakly.** MalariaGEN's own published `kelch13` status is consistent with the *pfk13* negative, but it cannot carry the weight previously placed on it. Pf8 assigns that status over the whole 349-726 domain and classifies any missing call in the domain as undetermined, so only **36** isolates have a complete status. That is weak corroboration for per-codon calls on roughly 1,860 isolates,

and the two should not be presented as equivalent evidence. Within those 36, the commonest change is Ala578Ser, which is explicitly not a resistance marker.

**[Observed]** A further consequence, which Section 5.6 takes up, is that the original candidate panel was itself too narrow. *pfaat1* is absent from config.KNOWN_RESISTANCE_LOCI, the eight-locus list the original analysis flagged outliers against, which is why the one of the three surviving resistance variants that reached the outlier table surfaced there as a transporter hit rather than as a known resistance locus. The outlier table does name it - PF3D7_0 629 500, AAT1, amino acid transporter AAT1, 2, 24.198 - so the failure was one of classification, not of annotation; it simply was not on the list of loci being watched. That eight-locus list is a different eight from the original panel's eight genes in which canonical codons were observed, and the two should not be confused; the rebuilt panel spans nine genes.

**[Observed]** The rebuilt panel converts objective 3 from a null into a measurement with three positive findings and one informative negative: significant between-zone heterogeneity at *pfdhfr*, at *pfdhps* (437G, p = 1.2 x 10 to the -6) and at *pfaat1* S258L, and a tested negative covering all 14 WHO-validated plus all 6 candidate *pfk13* markers. On S258L, what Amambua-Ngwa et al. (2023) demonstrated experimentally is narrower than this report previously implied: a **1.5-fold** rise in chloroquine IC50 in a single chloroquine-resistant Southeast Asian line (NHP4026), with epistasis between S258L and F313S and an explicit caveat that the effect depends on *pfcrt* background. The variant therefore *potentiates* chloroquine resistance, modestly and conditionally. The 0% to 97% longitudinal rise reported in that paper is Gambian (1984-2014); the paper says nothing about Ghana. Describing S258L as "the functionally validated chloroquine-resistance variant" overstated all three of magnitude, generality and geography. It should also not be called a WHO-classified marker: *pfaat1* appears nowhere in the WHO compendium at any tier, as is also true of *pfap2mu* and *pfmdr2*, while *pfcoronin* appears only at the **potential** tier. The compendium classifies variants rather than genes, and it contains no validated or candidate lumefantrine marker at all.

**[Hypothesized]** Which drug maintains the *pfaat1* cline is a separate inferential step, and the previous version took it wrongly. It asserted that the functional experiment did not test amodiaquine and then framed the Ghanaian cline as amodiaquine-maintained. Amodiaquine **was** tested - IC50 values for chloroquine, amodiaquine, lumefantrine, mefloquine and quinine were all measured - and the shifts were reported as well below the threshold associated with clinical resistance; the authors explicitly reject the hypothesis that other quinoline partner drugs drive these frequencies, and conclude that chloroquine was the primary selective force. Three candidate explanations for a persistent intermediate cline in Ghana therefore stand together and none is tested here: historical chloroquine pressure persisting after the 2005 switch to artesunate-amodiaquine, the documented fitness cost of S258L holding the allele short of fixation, and - as an **untested possibility only** - contemporary 4-aminoquinoline pressure from amodiaquine. These data cannot discriminate between them: drug-use intensity by zone is not available in the Pf8 metadata, and the cline is confounded with administrative region, collection period and sequencing study in the same way as every other zone-level result here.

### 1.2.1 What the analysis can and cannot separate

**[Interpreted]** Four features of the Pf8 Ghana metadata constrain what any zone-level number can mean. They are properties of the sampling frame, not analytical shortcomings that better methods would overcome, and they are stated here rather than deferred to the limitations section because they condition every result in Sections 6 and 7. The region, year and study tabulations in this subsection are computed on the 4,905

release-QC-pass isolates, not on the 1,900 that enter the population analyses, because the sampling frame is a property of the release rather than of this analysis's sample cut.

- **Administrative region.** Three of the four zones are each filled by exactly one region - Guinea savanna by Upper East (3,605 of 3,605 release-QC-pass isolates), coastal savanna by Greater Accra (435 of 435), transitional zone by Brong Ahafo (50 of 50); only deciduous forest draws on more than one (Ashanti, Central, Eastern, Volta). For those three, "zone" and "region" are the same variable, and any difference attributed to ecology could equally be attributed to the region's health system, drug supply, sampling protocol or collection year.
- **Collection period.** Zone-level year distributions differ systematically: Guinea savanna spans 2009-2019 with a 2018 peak, coastal savanna is concentrated in 2016-2018, deciduous forest in 2013-2021, and the transitional zone falls entirely in 2011-2013. Over 2009-2021 Ghana changed first-line therapy and scaled up chemoprevention, so drug-related allele frequencies would be expected to move with time independently of geography. The confounding is not removable by reweighting, because for the transitional zone and the coastal savanna the year ranges do not overlap at all. The same fact makes every resistance frequency in this report a 2009-2021 average rather than a current value, including the *pfk13* zeros.
- **Sequencing study (new in this version).** Pf8 isolates arrive through named partner studies, and in Ghana those studies are almost perfectly nested within zones. **Four of the six zone pairs share no study whatsoever** - coastal savanna versus Guinea savanna, coastal savanna versus transitional, deciduous forest versus Guinea savanna, and deciduous forest versus transitional - and the first of these is the pair behind the maximum $F_{ST}$ and the polyclonality contrast. Each zone is dominated by one study: the transitional zone is 100% one study (1083-PF-GH-CONWAY, 50 of 50), coastal savanna 89% (387 of 435, 1241-PF-GH-ANINAGYEI), Guinea savanna 74% (2,678 of 3,605, 1168-PF-GH-AMENGA-ETEGO-SM) and deciduous forest 48% (391 of 815, 1200-PF-GH-MAIGA-SM). Study-associated heterogeneity in this dataset is demonstrable, not hypothetical: holding zone constant, so that study varies together with collection year, finer geography and recruitment, call rate differs between the six studies within Guinea savanna at p = 2.8 x 10 to the -197. Deciduous forest draws on five studies in all; across the four of them large enough to test (811 of the zone's 815 isolates, the fifth contributing only 4) call rate differs at p = 7.7 x 10 to the -88, and the polyclonal fraction ranges from 0.430 to 0.578 (Kruskal-Wallis p = 0.0019).
- **Spatial resolution.** Isolates are geolocated only to administrative level 1, so 1,900 isolates carry seven distinct coordinates between them. No sub-regional analysis is possible, and isolation by distance rests on 21 pairs derived from 7 points - a test fragile enough that, as the threshold sensitivity analysis shows, it changes verdict when either of the two dataset-defining cuts is moved.

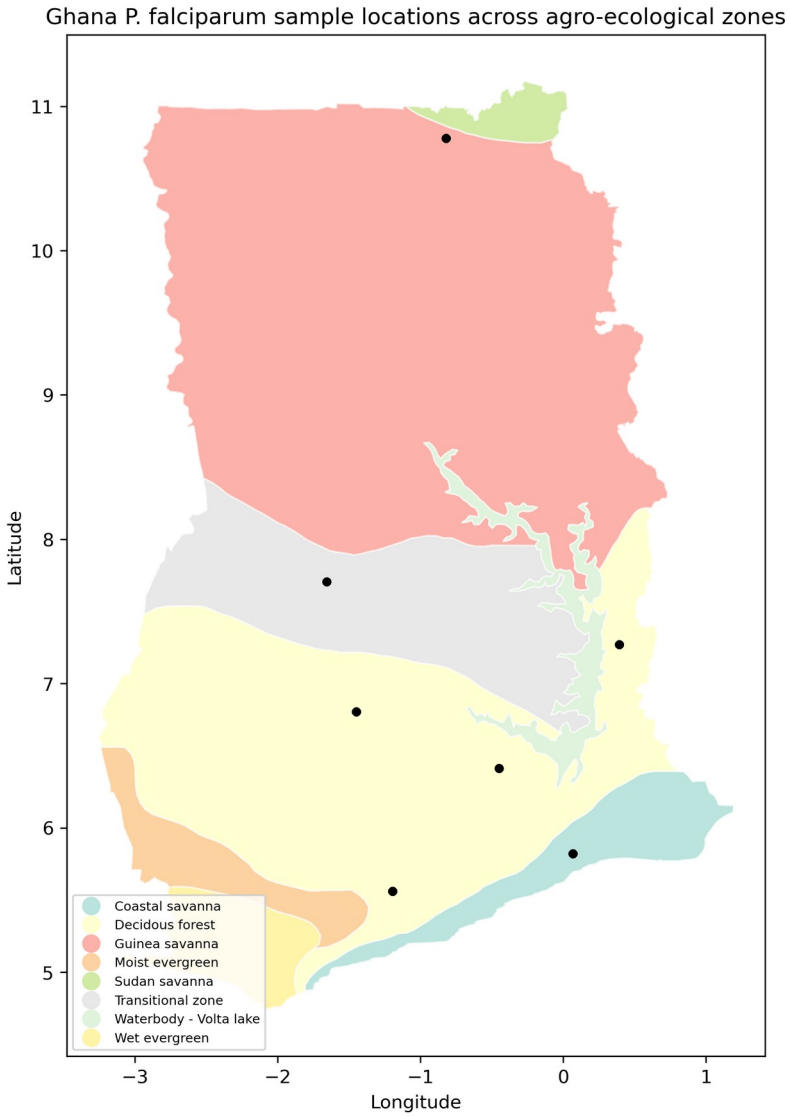



*Figure 1.1. Sample locations on the ghanaAEZ.geojson agro-ecological zone layer. Every isolate resolves to one of seven administrative-level-1 coordinates (black points), which is the entire spatial resolution available to this analysis. Four of the layer's zones receive isolates; Sudan savanna, wet evergreen and moist evergreen receive none, and the layer also carries a Volta lake waterbody polygon that is not a zone. Note that the transitional zone (grey) lies between the forest zones and Guinea savanna, making it the second-most-northern of the four sampled zones - the reason a simple south-to-north ordering of zone means is not available. The legend reproduces the layer's own spelling, "Decidous forest". Source: pf8_proof/ghana_samples_aez_map.png.*

**[Interpreted]** **Partial mitigation, reported in fairness.** Both zone pairs that do share a study reproduce the direction of the overall polyclonality difference; within 1083-PF-GH-CONWAY, Guinea savanna is more polyclonal than the transitional zone (0.532 versus 0.500 - that is 25 of 47 against 25 of 50, computed on the 4,905 release-QC-pass isolates, since on the 1,900-isolate analysis set the polyclonal fraction is identically zero by construction), matching the overall ordering. That margin is three isolates wide in the denominator, the numerator being 25 in both arms, and should not be read as more than a direction. The zone ordering of polyclonality is therefore reproduced within both zone pairs that share a study, which is a descriptive check and not a test of cause: holding zone constant still leaves collection year, finer geography, recruitment and

host biology varying alongside study. The ordering is not separable from batch for the pairs that carry the largest differences.

> **Caveat on the zone layer itself.** The agro-ecological zone assignment depends on data/ghanaAEZ.geojson, whose provenance could not be established. It is byte-for-byte identical (md5 96b1a7f4da62d6f0153826f41af96152) to an ArcGIS Online item uploaded by an anonymous personal account with null licence and null source information, so that item is a re-host and not an origin. The layer is a hybrid of two classification schemes - MoFA/FAO savanna, transition and coastal names combined with a Hall and Swaine style wet-evergreen / moist-evergreen forest split - and no published Ghana zonation matches it. It must not be attributed to any named institution. On spelling, one rule is used throughout this report: the layer's string "Decidous forest" is retained wherever it is a join key or appears inside a quoted output, file path or figure legend, and running prose and table labels use the correct spelling "deciduous forest". The spatial join keys on the layer value, so the data are never silently rewritten. This is a publication blocker, documented in Section 9.3. It remains open at final revision and cannot be closed analytically: the ArcInfo-coverage fingerprint dates the original to a pre-2005 workflow and CERSGIS (University of Ghana) is the most likely custodian, so closing it requires correspondence with cersgis@ug.edu.gh. The interim citation and the md5 stand in the meantime.

## 1.3 Research questions as operationalised

The thesis questions were translated into the following testable forms. The translation is itself an analytical decision and is revisited in Section 5.

1. Is between-zone differentiation distinguishable from zero? Operationalised as pairwise Hudson $F_{ST}$ with block-jackknife confidence intervals, and - new in this version - recomputed under six scenarios that vary the $F_{ws}$ and MAF thresholds, so that the magnitude is reported with its sensitivity to the cuts that define the dataset rather than at one setting only.
2. Does genetic distance increase with geographic distance? Operationalised as a Mantel test of $F_{ST}/(1 - F_{ST})$ against log geographic distance, run at both zone and region level, with the permutation test seeded, the censoring of negative $F_{ST}$ pairs disclosed and reported both ways, a leave-one-population-out jackknife to test whether any single population carries the result, and - new in this version - recomputation of the region-level test under all six threshold scenarios, since a permutation test on 21 pairs from 7 points can depend heavily on the matrix it is given. This question is the one whose answer changed at final revision.
3. Do individual isolates assign to zone-associated ancestry components? Operationalised as ADMIXTURE at K = 1-8 with 10-fold cross-validation, visual inspection of three ordinations - now computed on the 1,900-isolate analysis set as well as on the 4,905 pre-QC isolates the published figures used, by re-slicing the distance submatrix - and an explicit Kruskal-Wallis test of the minor-component proportion against zone, so that the magnitude and the significance of any zone effect are reported separately.
4. Is there evidence of gene flow beyond a simple bifurcating population tree? Operationalised as TreeMix with 0-5 migration edges, run with jackknife standard errors so that each edge carries a p-value and the number of edges is informed by a test rather than only by the shape of the likelihood curve.
5. Which loci are more differentiated than neutral demography predicts? Operationalised as a BayPass XtX scan with a threshold calibrated on pseudo-observed data simulated under the estimated population covariance matrix, reported alongside the expected number of false positives at that threshold, the implied expected-null-exceedance ratio, per-SNP Benjamini-Hochberg q-values, and - new in this version - a per-SNP call-quality audit of every outlier against the rest of the scanned panel, so that a high XtX value is reported with the quality evidence for or against the call behind it.

6. **New, and now rebuilt.** What are the zone-level frequencies of the canonical antimalarial resistance residues, and of the haplotypes they form within isolates? Operationalised not by annotation matching but by **codon reconstruction**: each of 57 codons in nine genes is mapped to its three genomic positions through the CDS, every variant record at those positions is collected regardless of `numalt` or `is_snp`, each isolate's alleles are applied to the reference codon and the codon is translated, so that the **full residue spectrum including the reference residue** is reported with Wilson confidence intervals and a four-zone chi-square test. A codon with a mixed call is uncalled rather than assigned a residue. Multi-codon haplotypes (*pfdhfr* 51/59/108 and 164, *pfcrt* 72-76, *pfdhps* 437/540) are counted on complete cases only, so that a conjunction can never exceed its weakest component. The *pfk13* marker list is taken from WHO compendium v1.0 (December 2025) rather than written by hand, covering in full all 14 validated and all 6 candidate markers with the two tiers kept distinct; those 20 markers occupy 19 distinct codons, because C469 carries both the validated C469Y and the candidate C469F, and the panel genotypes 23 codons in all - the four additional ones (449, 481, 515 and 527) being codons the compendium places in neither tier, so no WHO classification is claimed for them. Validation is call-by-call against MalariaGEN's published genotypes rather than by comparison of marginal frequencies. This question was present in the thesis objectives but, as Section 1.2 explains, was first not tested at all and then tested with a defective panel.

**[Interpreted]** Questions 1, 3 and 4 are answerable at the resolution these data permit and are answered in Section 6. Question 1 is answered robustly: the low-differentiation reading survives both threshold relaxations, with max $F_{ST}$ between 0.00204 and 0.00255 across the six scenarios. Question 2 is answered negatively. The distance effect it tests is significant at the published thresholds but in only 2 of 6 threshold scenarios and 4 of 7 leave-one-region-out replicates, so the honest answer is that these data do not establish isolation by distance; whether that is a statement about Ghanaian parasite demography or about the fragility of a Mantel test on 21 pairs from 7 coordinates cannot be settled here, and both readings are consistent with everything else in the report. Question 5 is answerable only in the weak sense of producing a ranked candidate list: with four populations and 3,916 SNPs the scan has limited power, 3.9 of the 19 SNPs above the quantile threshold are expected to be false positives, only 13 survive FDR 10%, and two - including the fifth-ranked SNP overall - carry ExcessHet and InbreedingCoeff signatures that put them at lower confidence. Question 6 is answered directly, by genotyping rather than by scan, which is why a reference-residue frequency at or near 1.000 at a *pfk13* codon is now informative where a non-significant XtX value would not have been - subject to two limits that must travel with the number. It is fixation among called genotypes at twenty-two of the 23 codons and 0.9995 at codon 458, on 1,845 to 1,876 isolates per codon rather than all 1,900, and it describes isolates collected between 2009 and 2021. The corroborating MalariaGEN `kelch13` status exists for only 36 of those isolates, because Pf8 treats any missing call across amino acids 349-726 as undetermined, so it is supporting rather than independent evidence.

# 2. Data and materials

## 2.1 Source and provenance

**[Observed]** All genotype and metadata inputs derive from the MalariaGEN *Plasmodium falciparum* Community Project release 8 (Pf8), served publicly over HTTPS from https://pf8-release.cog.sanger.ac.uk. The release comprises 33,325 samples and 12,493,205 variant sites called against the 3D7 reference assembly (contigs Pf3D7_01_v3 … Pf3D7_14_v3, plus apicoplast and mitochondrial contigs, which this analysis excludes).

Four Pf8 artefacts were used:

- **Sample manifest** — data/Pf8_samples.txt, 33,325 rows and 17 columns, giving country, administrative level 1, latitude and longitude at that level, collection year, the release's own QC-pass flag, and the named partner study through which each isolate entered the release (see §2.6).
- **Variant call data** — the Pf8 Zarr store at /zarr, holding GT, AD, DP and GQ. The per-site annotations actually retrieved into the local store are CHROM, POS, REF, ALT, numalt, is_snp and FILTER_PASS.
- **Mean genotype distance matrix** — data/Pf8_mean_genotype_distance.npy, a 33,325 × 33,325 single-precision matrix published with the release (4.44 GB).
- **Reference annotation** — data/PlasmoDB-71_Pfalciparum3D7.gff,, used for Step 9. Release 68 is also present in data/ but config.py points at release 71.

> **Correction to the original field list.** An earlier draft of this section listed RegionType among the retrieved annotations, and §3.5 appealed to a direct tabulation of it. RegionType was never fetched: it is absent from step03a_fetch_callset_zarr.py's VARIANT_FIELDS and from the local store, and no log records such a tabulation. The core-genome restriction is nonetheless secure, but it holds by construction rather than by observation — Pf8's FILTER vocabulary itself contains the region categories, so a non-core site cannot carry FILTER_PASS. §3.5 now makes that structural argument instead.

**[Interpreted]** The numalt field deserves advance notice here, because its use in the ingest filter turned out to be the single most consequential data-handling decision in the analysis. Restricting the callset to numalt == 1 excluded most canonical antimalarial-resistance codons — 28 of the 44 codon-allele records genotyped, among them *pfcrt* K76T, *pfmdr1* N86Y, *pfdhps* K540E and 8 of the 14 validated *pfk13* codons; of the 21 records that are polymorphic in this sample, only 6 survived the filter before any analysis ran; the recovery of those sites is reported in §3.10 and §6.9. Readers of this section should not assume that the 3,916-SNP analysis matrix described in §4 is a representative sample of functionally interesting variation — it was, by construction, a sample of biallelic variation only.

### 2.1.1 The agro-ecological zone layer

**[Observed]** A Ghana agro-ecological zone polygon layer (data/ghanaAEZ.geojson) was supplied with the project. It carries eight polygons — Coastal savanna, Wet evergreen, Moist evergreen, "Decidous forest", Transitional zone, Guinea savanna, Sudan savanna, and a polygon named "Waterbody - Volta lake" — and exactly four attribute fields: id, Name, AREA and PERIMETER. It contains no author, source, date, licence or lineage field of any kind.

**[Observed]** The file was tracked to its acquisition point. It is 285,669 bytes, md5 96b1a7f4da62d6f0 153 826f41af96 152, and is byte-for-byte identical to ArcGIS Online item 0776dae5c5 314 102bd03fdaab8df0356, "Agro Ecological Zones Ghana" — the md5 matches on both sides, so the

identification is certain rather than inferred. Queried through the ArcGIS REST API, that item's licenseInfo and accessInformation fields are both null, and its owner is an anonymous personal account with an empty display name, no stated organisation, and a creation date two days before the upload.

```
md5 data/ghanaAEZ.geojson  # 96b1a7f4da62d6f0 153 826f41af96 152
```

**[Interpreted]** The internal evidence points to a pre-GeoJSON lineage, and the ArcGIS item is an unattributed re-host rather than an origin. The file was written by GDAL/OGR (legacy crs member declaring urn:ogc:def:crs:OGC:1.3:CRS84, clockwise rings, 15-digit coordinates, layer name ghanaAEZ), and AREA/PERIMETER as an uppercase pair are the auto-maintained attributes of an ESRI ArcInfo coverage. The implied chain is coverage → shapefile → GeoJSON. The geometry is a clean topological partition of Ghana: AREA totals 23,871,848 ha against Ghana's official total area of 23,853,300 ha (238,533 km², land plus inland water), a +0.08% discrepancy. The comparison is total-to-total rather than land-to-land, because the layer's sum includes the 587,310 ha Volta Lake polygon. The layer is, in other words, a competently made national dataset whose authorship has been stripped somewhere along that chain.

**[Observed]** On the classification itself, what the layer contains can be stated plainly. Ghana's canonical MoFA/FAO agro-ecological classification is normally given as five or six zones. This layer carries seven terrestrial polygons plus a water body. It uses the MoFA savanna, transition and coastal names; it splits the high forest into "Wet evergreen" and "Moist evergreen", which are forest *vegetation* types from the Hall & Swaine lineage; it labels the semi-deciduous zone "Decidous forest", a label the FAO/MoA vegetation table does not use; and its eighth polygon is "Waterbody - Volta lake".

**[Interpreted]** Read against those sources, the scheme is best understood as a hybrid of two classifications rather than a reproduction of either — MoFA/FAO zone names for the savanna, transition and coastal belts, Hall & Swaine forest vegetation types for the evergreen split — with a water body included that is not an agro-ecological zone at all. Nine candidate sources were searched directly and none holds a matching scheme or a matching layer: HDX, data.gov.gh, GADM, Harvard Dataverse, Zenodo, figshare, FAO GAEZ, FAO Africover and ICPAC. "No published scheme matches it" is therefore an inference from a finite negative search, not an observation; a source not on that list could still hold the original.

> **Provenance gap.** The original author, vintage and licence of ghanaAEZ.geojson could not be established, and on present evidence cannot be. [Unknown] The layer must not be attributed to Wageningen University or to Climate Adaptation Services: those names attach to other items in the same ArcGIS Online account and to a storymap theme, not to this file, and naming either as its author would be a fabricated citation. Two leads remain open. [Proposed] First, whoever supplied the original Colab notebook should be asked for it; its !wget cells are the most likely surviving record of a fetch URL. Second, the ArcInfo-coverage fingerprint is consistent with a pre-2005 national coverage, which makes the Centre for Remote Sensing and GIS at the University of Ghana (cersgis@ug.edu.gh) the most plausible custodian; that enquiry has not yet been made.

**[Proposed]** Pending that, the layer is cited as accessed rather than as authoritative, and not as a published dataset: *Agro ecological zones Ghana* [Data set]. (2025). ArcGIS Online. https://www.arcgis.com/home/item.html?id=0776dae5c5314102bd03fdaab8df0356 — flagged as an unattributed re-host and identified by md5 96b1a7f4da62d6f0 153 826f41af96 152. The *classification* is cited separately and generically: MoFA/FAO for the savanna, transitional and coastal names, and the Hall & Swaine/FAO forest vegetation literature for the evergreen split, with a note that the scheme as implemented is a hybrid of the two.

> **Spelling of the zone names.** One rule applies throughout this work. The layer's own string, including the "Decidous forest" misspelling, is retained wherever it is a join key, a stored data value, or part of a quoted field value, output

> path or file name — so interim/analysis_samples.csv, results/fst_pairwise.csv, results/batch_study_by_zone.csv and the fact sheet all read "Decidous forest". The spatial join keys on the layer value, so it is never silently rewritten in the data. Running prose and table row and column labels use the correct spelling, "Deciduous forest". Where a block enumerates the layer's Name values as data — the area list in §2.4 — the layer spelling appears in quotation marks. No table mixes the two spellings within itself.

**[Proposed]** The provenance gap is a publication blocker for the layer as a cited dataset, but it is a weaker threat to the results than it first appears, and it is convertible into a sensitivity result. Because isolates join to zones on administrative-level-1 centroids — seven points in total (§2.3) — the join is coarse enough that it is unlikely to be sensitive to polygon detail. Repeating it against an authoritative six-zone Ghana layer and checking whether the seven centroids receive the same zone labels would settle the question. That check has not yet been run.

## 2.2 Units of analysis and observation period

**[Observed]** The unit of analysis is the parasite isolate (one blood sample from one infected individual). Ghana contributes 6,653 isolates to Pf8; 4,905 carry the release QC-pass flag, and 1,900 survive the pipeline's own filtering to enter the analysis set. Collection years span 2009 to 2021.

**[Observed]** Table 2.1 gives the regional composition of the Ghanaian subset at three stages. The pronounced imbalance — Upper East supplies 73.5% of the QC-passing isolates and 70.2% of the final analysis set — is a property of the Pf8 sampling frame, not of any filter applied here.

| Administrative region | In Pf8 manifest | Pf8 QC-pass (Step 1 output) | In final analysis set (Step 4a output) |
|---|---|---|---|
| Upper East | 4,821 | 3,605 | 1,334 |
| Central | 686 | 309 | 125 |
| Greater Accra | 474 | 435 | 204 |
| Ashanti | 452 | 391 | 157 |
| Eastern | 110 | 74 | 34 |
| Brong Ahafo | 69 | 50 | 25 |
| Volta | 41 | 41 | 21 |
| Total | 6,653 | 4,905 | 1,900 |

*Table 2.1. Ghanaian isolates by administrative region at three stages of the pipeline. Counts are from the Pf8 manifest, interim/ghana_samples.csv and interim/analysis_samples.csv respectively. The final-set column is from the corrected re-run of 2026-10-10; the isolate set is unchanged from the original run — only the SNP count changed (3,359 to 3,916).*

## 2.3 Spatial resolution: the binding constraint

**[Observed]** Pf8 geolocates isolates to administrative level 1. Every isolate from a given region therefore carries the identical coordinate pair, and the 4,905 QC-passing Ghanaian isolates occupy exactly seven distinct points. Step 2 prints this explicitly: "4,905 samples occupy 7 distinct coordinate pairs."

**[Interpreted]** This is the single most consequential property of the dataset for the thesis questions. Individual-level geographic distance does not exist in these data, so no analysis can resolve structure finer than the region. More seriously, as Table 2.2 shows, three of the four zones are each filled by exactly one region,

which makes zone and region statistically indistinguishable for those three. A zone effect and a region effect cannot be separated.

| Agro-ecological zone | Regions contributing (final set) | n | Distinct coordinates |
|---|---|---|---|
| Guinea savanna | Upper East (1,334) | 1,334 | 1 |
| Deciduous forest | Ashanti (157), Central (125), Eastern (34), Volta (21) | 337 | 4 |
| Coastal savanna | Greater Accra (204) | 204 | 1 |
| Transitional zone | Brong Ahafo (25) | 25 | 1 |
| Total | | 1,900 | 7 |

*Table 2.2. Zone composition of the final analysis set (n = 1,900). Only the deciduous forest zone aggregates more than one region; the other three are single-region zones, so for them "zone" and "region" are the same variable under two names.*

## 2.4 Zones represented and zones absent

**[Observed]** The thesis frames Ghana as five ecological zones. The Pf8 Ghanaian sample, joined to the supplied layer, occupies four: Guinea savanna, deciduous forest, coastal savanna and transitional zone. No isolate falls in the Sudan savanna, and none in the moist-evergreen or wet-evergreen forest zones present in the layer. Step 2 reported zero unmatched samples and zero duplicate matches, so the absence is one of sampling, not of joining.

**[Observed]** The layer's own AREA field gives the size of each polygon. Of the 23,871,848 ha the layer assigns to Ghana, the four sampled zones account for 90.0%, and the four unsampled polygons are all small. Names below are the layer's Name strings and areas its AREA values:

- "Guinea savanna" 10,835,471 ha (45.4%) — **sampled**
- "Decidous forest" 6,675,558 ha (28.0%) — **sampled**
- "Transitional zone" 2,936,745 ha (12.3%) — **sampled**
- "Coastal savanna" 1,033,380 ha (4.3%) — **sampled**
- "Moist evergreen" 769,514 ha (3.2%) — not sampled
- "Wet evergreen" 740,806 ha (3.1%) — not sampled
- "Waterbody - Volta lake" 587,310 ha (2.5%) — not a zone
- "Sudan savanna" 293,063 ha (1.2%) — not sampled

**[Interpreted]** On these figures the AREA field explains most of the absence better than an appeal to sampling priorities does. The Sudan savanna is the smallest terrestrial polygon in the layer: 293,063 ha, 1.2% of the total, a sliver in the far north-east. An earlier draft of this section treated its absence as a substantive gap in the ecological gradient, as though the sampling frame had overlooked it. The better explanation is that it is a very small zone which Pf8's administrative-level-1 sampling frame was never likely to capture: with one centroid per region and no region lying wholly inside it, there is no coordinate at which a Sudan savanna isolate could have been recorded. The same reasoning applies to the two evergreen forest polygons, which are jointly 6.3% of the layer.

**[Interpreted]** The consequence for the thesis framing is real but narrower than previously stated. What is missing is not a sampled-but-overlooked zone; it is the far-northern extreme of the transmission gradient, and it is missing because the geolocation resolution cannot represent zones smaller than a region. The Upper

East centroid lies close to the Sudan savanna boundary (Figure 2.1), so the Guinea savanna sample may well include parasites from populations straddling that transition — which would attenuate rather than inflate any north–south contrast. This cannot be checked at administrative-level-1 resolution.

> **Status of this re-reading.** The reversal of the original §2.4 reading is not one of the audit's confirmed defects; it follows from a bonus finding in the corrections document (§F) and is argued here on the layer's own area figures. It should be defended on those figures in the response to reviewers rather than cited to a defect number.

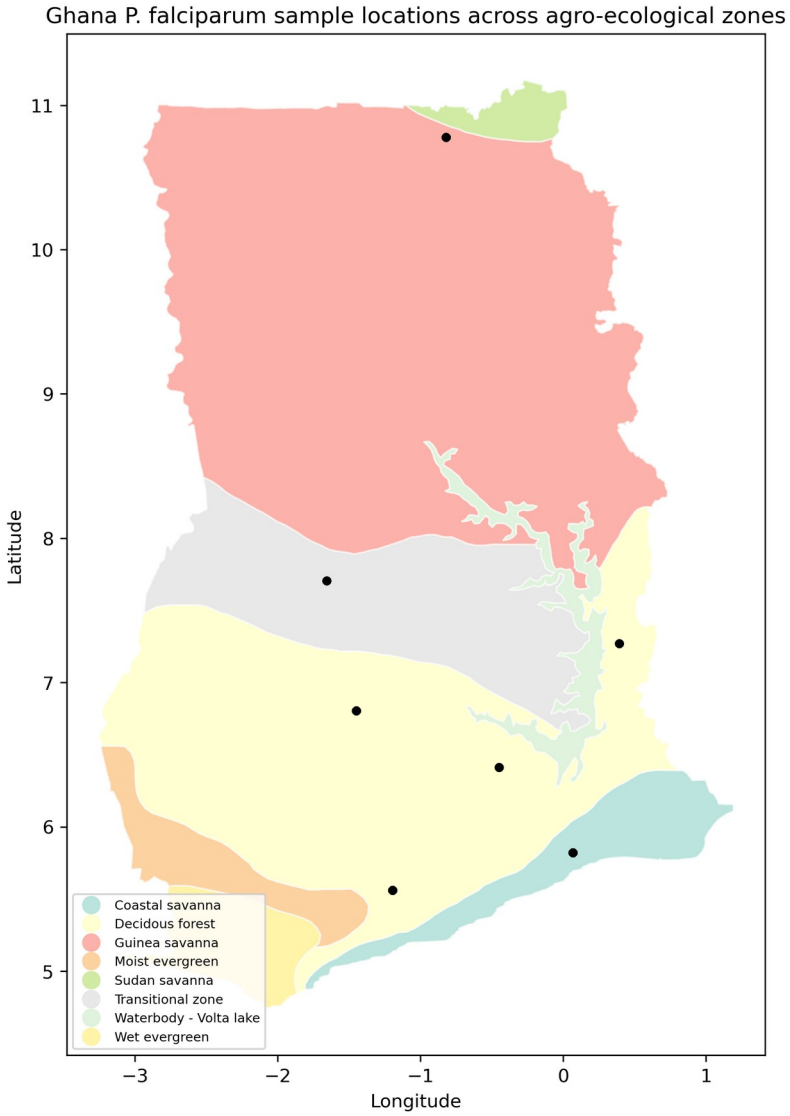



*Figure 2.1. Sampling locations on the Ghana agro-ecological zone layer. Each black point is one administrative-level-1 centroid, not an individual isolate; the seven points carry 4,905 QC-passing isolates between them. The Sudan savanna polygon at the north-eastern tip is 1.2% of the layer's total area. Generated by scripts/step02_assign_aez.py as figures/ghana_samples_aez_map.png.*

## 2.5 Temporal composition

**[Observed]** Collection year is strongly unbalanced across zones (Table 2.3). Median collection year ranges from 2012 (transitional zone) to 2018 (coastal savanna). The transitional zone contributes no isolate after 2013; the coastal savanna contributes none before 2016.

| Year | Coastal savanna | Deciduous forest | Guinea savanna | Transitional zone |
|---|---|---|---|---|
| 2009 | 0 | 0 | 41 | 0 |
| 2010 | 0 | 0 | 81 | 0 |
| 2011 | 0 | 0 | 36 | 3 |
| 2012 | 0 | 0 | 21 | 19 |
| 2013 | 0 | 1 | 84 | 3 |
| 2014 | 0 | 111 | 63 | 0 |
| 2015 | 0 | 66 | 96 | 0 |
| 2016 | 15 | 21 | 59 | 0 |
| 2017 | 12 | 0 | 243 | 0 |
| 2018 | 177 | 40 | 556 | 0 |
| 2019 | 0 | 14 | 54 | 0 |
| 2020 | 0 | 76 | 0 | 0 |
| 2021 | 0 | 8 | 0 | 0 |
| Total | 204 | 337 | 1,334 | 25 |
| Median year | 2018 | 2015 | 2017 | 2012 |

*Table 2.3. Collection year by zone in the final analysis set (n = 1,900). Computed from interim/analysis_samples.csv. Column totals match the zone counts in Table 2.2.*

**[Interpreted]** Any between-zone difference observed in this dataset is also a between-period difference. Ghana changed first-line therapy and scaled up chemoprevention over this window, so allele frequencies at drug-related loci would be expected to move with time independently of geography. This confounding is not removable by reweighting, because for the transitional zone and the coastal savanna the year ranges do not overlap at all. It bears directly on the marker results in §6.9: the *pfaat1* S258L frequencies and the *pfdhfr* triple-mutant frequencies both differ significantly between zones whose sampling windows differ by three to six years, and neither difference can be attributed to ecology rather than to date on these data alone. The *pfdhfr* pattern is in any case not a simple south-to-north gradient. Ordered south to north, the four sampled zones run coastal savanna, deciduous forest, transitional zone, Guinea savanna, so the transitional zone is the second-most-northern of the four; it carries the highest frequency of all four zones at two of the three triple-mutant codons, and it is also the earliest-sampled (median 2012) and by far the smallest (n = 25). Its position in any ordering is as much a statement about date and sample size as about ecology. *pfaat1* S258L, by contrast, does fall monotonically along that same south-to-north ordering (§6.9).

## 2.6 Sequencing study as a fourth design confounder

**[Observed]** The design limitations listed in §9.2 of the original draft named three confounders of the zone comparison: zone–region collinearity, administrative-level-1 spatial resolution, and collection year. There is a fourth, which that draft did not mention at all. Pf8 isolates arrive through named partner sequencing studies, and in Ghana those studies are almost perfectly nested within zones.

**[Interpreted]** Study is therefore a third name for the same variable that "zone" and "region" already share, and unlike the other two it carries its own technical variance with it.

> **Which isolates these counts describe.** The study-composition table and the within-zone batch tests in this section are computed on the **4,905 QC-passing isolates** — the set on which study membership and $F_{ws}$ are defined — and not on the **1,900-isolate analysis set** used for the population analyses in §§5–8. The two denominators must not be

> read against each other: "1083-PF-GH-CONWAY is 100% of the transitional zone, 50 of 50" is a statement about the 4,905 set, in which the transitional zone holds 50 isolates, not about the n = 25 transitional zone of Table 2.2. Study nesting within zone is a property of the Pf8 sampling frame and is not created by the later cuts, but the counts in this section are not the analysis-set counts. §4.2.2, §6.2, §7.2, §7.3 and §10.1 cite these figures and refer back here for the denominator.

**[Calculated]** The composition was also recomputed on the 1,900-isolate analysis set, to check that the later cuts do not change the picture. They do not. The same eleven studies are present; the same three studies span two zones each; the same four of the six zone pairs share no study; and dominance barely moves — transitional zone 100% (25 of 25), coastal savanna 86.8% (177 of 204), Guinea savanna 73.8% (984 of 1,334), deciduous forest 46.6% (157 of 337). The deciduous forest again draws on five studies, the smallest of them holding three isolates. Computed from `interim/analysis_samples.csv`. The within-zone Kruskal–Wallis tests were **not** re-run on the 1,900 set, so every test statistic in Table 2.6 remains a 4,905-set figure.

**[Observed]** Eleven studies contribute the 4,905 QC-passing Ghanaian isolates. Table 2.4 gives the full cross-tabulation. Three of the eleven span more than one zone — 1241-PF-GH-ANINAGYEI and 1183-PF-GH-AWANDARE-SM across the coastal savanna and the deciduous forest, and 1083-PF-GH-CONWAY across the Guinea savanna and the transitional zone — and the remaining eight are confined to a single zone each. Every zone is dominated by one study: the transitional zone is 100% one study (1083-PF-GH-CONWAY, 50 of 50), the coastal savanna 89.0% (1241-PF-GH-ANINAGYEI, 387 of 435), the Guinea savanna 74.3% (1168-PF-GH-AMENGA-ETEGO-SM, 2,678 of 3,605), and the deciduous forest — the only zone with no majority study, and the only one drawing on five contributing studies — 48.0% (1200-PF-GH-MAIGA-SM, 391 of 815).

| Sequencing study | Coastal savanna | Deciduous forest | Guinea savanna | Transitional zone | Total |
|---|---|---|---|---|---|
| 1168-PF-GH-AMENGA-ETEGO-SM | 0 | 0 | 2,678 | 0 | 2,678 |
| 1241-PF-GH-ANINAGYEI | 387 | 279 | 0 | 0 | 666 |
| 1200-PF-GH-MAIGA-SM | 0 | 391 | 0 | 0 | 391 |
| 1017-PF-GH-AMENGA-ETEGO | 0 | 0 | 384 | 0 | 384 |
| 1094-PF-GH-AMENGA-ETEGO | 0 | 0 | 241 | 0 | 241 |
| 1294-PF-GH-AMENGA-ETEGO-SM | 0 | 0 | 176 | 0 | 176 |
| 1096-PF-GH-GHANSAH | 0 | 100 | 0 | 0 | 100 |
| 1083-PF-GH-CONWAY | 0 | 0 | 47 | 50 | 97 |
| 1183-PF-GH-AWANDARE-SM | 48 | 41 | 0 | 0 | 89 |
| 1151-PF-GH-AMENGA-ETEGO | 0 | 0 | 79 | 0 | 79 |
| 1114-PF-PDN-DBS-GH-GHANSAH | 0 | 4 | 0 | 0 | 4 |
| Total | 435 | 815 | 3,605 | 50 | 4,905 |

*Table 2.4. Sequencing study by agro-ecological zone, QC-passing Ghanaian isolates (n = 4,905). From results/batch_study_by_zone.csv, written by scripts/step11_batch_confounding.py. Three of the eleven studies span more than one zone; the deciduous forest draws on five studies, the Guinea savanna on six, the coastal savanna on two and the transitional zone on one.*

**[Observed]** The consequence is set out in Table 2.5: **four of the six zone pairs share no sequencing study whatsoever**, including the coastal savanna–Guinea savanna pair that carries the largest pairwise $F_{ST}$ in the study (0.00232) and the headline polyclonality contrast.

**[Interpreted]** For those four pairs, zone, region, collection period and sequencing batch all change together, and no arrangement of these data separates them: at zero overlap a study term and a zone term are not jointly identifiable.

| Zone pair | Studies in common | Shared studies |
|---|---|---|
| Coastal savanna vs Guinea savanna | 0 | — (this is the maximum-$F_{ST}$ pair) |
| Coastal savanna vs Transitional zone | 0 | — |
| Deciduous forest vs Guinea savanna | 0 | — |
| Deciduous forest vs Transitional zone | 0 | — |
| Coastal savanna vs Deciduous forest | 2 | 1183-PF-GH-AWANDARE-SM; 1241-PF-GH-ANINAGYEI |
| Guinea savanna vs Transitional zone | 1 | 1083-PF-GH-CONWAY |

*Table 2.5. Sequencing studies shared between each pair of zones, QC-passing isolates (n = 4,905). From results/batch_zone_pair_overlap.csv. Recomputing the same six pairs on the 1,900-isolate analysis set returns identical counts and the same named studies.*

**[Observed]** Study-associated heterogeneity in these data is demonstrable rather than hypothetical. Holding zone constant — which leaves study varying alongside collection year, finer geography, recruitment and biology — per-isolate call rate differs sharply between studies within the same zone (Table 2.6): within the Guinea savanna, across its six studies, Kruskal–Wallis H = 923.0, p = $2.8 \times 10^{-197}$, n = 3,605; within the deciduous forest, across the four of its five studies large enough to test, H = 406.7, p = $7.7 \times 10^{-88}$, n = 811. Within the deciduous forest those same four studies also differ in $F_{ws}$ (Kruskal–Wallis H = 14.9, p = $1.9 \times 10^{-3}$, n = 811), and their polyclonal fractions span 0.430 to 0.578 — a spread comparable in size to the between-zone polyclonality differences reported in §6.2. The Guinea savanna shows no significant within-zone difference in $F_{ws}$ (H = 9.17, p = 0.103), so its p = $2.8 \times 10^{-197}$ is a statement about call rate alone and not about polyclonality. The coastal savanna, with two studies, shows no significant within-zone difference on either statistic; the transitional zone cannot be tested at all, because it is one study.

| Zone | Studies tested | n | Call rate: H | Call rate: p | $F_{ws}$: H | $F_{ws}$: p | Polyclonal fraction, range across studies |
|---|---|---|---|---|---|---|---|
| Guinea savanna | 6 of 6 | 3,605 | 923.0 | $2.8 \times 10^{-197}$ | 9.17 | 0.103 | 0.532 – 0.682 |
| Deciduous forest | 4 of 5 | 811 | 406.7 | $7.7 \times 10^{-88}$ | 14.94 | $1.9 \times 10^{-3}$ | 0.430 – 0.578 |
| Coastal savanna | 2 of 2 | 435 | 3.18 | 0.075 | 0.78 | 0.378 | 0.438 – 0.444 |
| Transitional zone | 1 of 1 | 50 | not testable | — | not testable | — | — |

*Table 2.6. Within-zone differences between sequencing studies, QC-passing isolates. Kruskal–Wallis tests on per-isolate call rate and on Fws, computed within each zone so that zone is held constant while study varies together with collection year, finer geography and recruitment; the tests therefore measure study-associated heterogeneity rather than isolating a technical batch cause. From results/batch_within_zone_tests.csv. The deciduous forest test excludes a four-isolate study (1114-PF-PDN-DBS-GH-GHANSAH), hence 4 of 5 studies and n = 811 against 815 in Table 2.4. The two p-value columns are tests on the call-rate and Fws distributions; no test of the polyclonal fraction itself exists in the outputs, so the last column is a range only and carries no p-value.*

**[Observed]** One partial mitigation is available, and is reported here in fairness to the result. Both zone pairs that do share a study reproduce the overall direction of the polyclonality difference: within 1083-PF-GH-CONWAY alone, Guinea savanna isolates are more polyclonal than transitional-zone isolates (0.532 against 0.500), which matches the ordering seen across the full sample.

[Interpreted] The direction of the gradient is therefore reproduced within the two zone pairs that share a study, which is a useful descriptive check. It does not establish that the gradient is not a batch artefact: within-zone comparisons hold zone constant while study still varies together with collection year, finer geography, recruitment and host biology. The gradient is not separable from batch for the two pairs that carry the largest differences.

[Interpreted] The practical reading is as follows. Study confounding does not threaten the central finding that the observed differentiation is very low, but neither does that finding bound the true differentiation: batch effects can move an observed between-group contrast in either direction — through cohort composition, calling differences and differential missingness — so a batch-confounded design that still returns FST ≈ 0.002 is reporting a low observed value whose relation to the true value is not established, not an upper bound on it. It does threaten any positive between-zone claim — the maximum-$F_{ST}$ pair, the polyclonality gradient, the ADMIXTURE minor-component ordering, and the marker-frequency differences in §6.9 — because for the pairs concerned there is no study-level control to appeal to. Each of those claims should be read as "differs between zones, which in these data means between zone, region, period and sequencing batch jointly".

> **What would resolve it.** Nothing available within Pf8. Separating study from zone requires either isolates from the same study in both the coastal savanna and the Guinea savanna, or a replication cohort sequenced under one protocol across the gradient. Statistical adjustment is not an option: with four of six pairs at zero overlap, a study term and a zone term are not jointly identifiable. This is listed as design limitation 7 in §9.2 and is the one limitation on that list that new sequencing, rather than reanalysis, would have to fix.

## 2.7 Variable summary

| Variable | Source | Type | Notes |
|---|---|---|---|
| Sample | Pf8 manifest | string | Isolate identifier; the join key throughout. |
| Country | Pf8 manifest | string | Filtered to "Ghana" at Step 1. |
| QC pass | Pf8 manifest | boolean-like | The release's own flag. Parsed permissively (true/t/yes/y/1/pass) by config.coerce_bool. |
| Study | Pf8 manifest | string | Named partner sequencing study. Eleven values in Ghana, in both the 4,905-isolate QC-passing set and the 1,900-isolate analysis set; nested within zone (§2.6). |
| Admin level 1 | Pf8 manifest | string | Region. Seven values in Ghana. |
| Admin level 1 latitude / longitude | Pf8 manifest | float | Region centroid. Seven distinct pairs for all 4,905 isolates. |
| Year | Pf8 manifest | float | Collection year, 2009–2021. |
| pf8_row | derived, Step 1 | int | Row index into the 33,325 × 33,325 distance matrix. Captured once so the submatrix slice can never be mis-indexed. |
| zone | derived, Step 2 | string | Agro-ecological zone from the spatial join against ghanaAEZ.geojson (§2.1.1). Four values in the analysis set. The stored value is the layer's own string, including "Decidous forest"; prose and table labels in this work use "Deciduous forest" (§2.1.1). |
| call_rate, het_rate, Fws | derived, Step 3b | float | Per-isolate QC statistics from a 50,000-SNP panel. Differ significantly between studies within zone for call rate in two zones and for $F_{ws}$ in the deciduous forest (Table 2.6). |
| qc_pass (pipeline) | derived, Step 3b | boolean | Distinct from the Pf8 flag; this is the pipeline's own composite. |

| Variable | Source | Type | Notes |
|---|---|---|---|
| GT, AD, DP, GQ | Pf8 Zarr | int8 / int16 | Genotype, allelic depths, total depth, genotype quality. Missing encoded as −1. |
| FILTER_PASS, is_snp, numalt | Pf8 Zarr | bool / bool / int | Site-level annotations used for variant filtering. $\mathrm{numalt} == 1$ in the original ingest filter is the source of the resistance-marker artefact described in §3.10. |

*Table 2.7. Variables used. "Derived" variables are written by the pipeline into interim/ and results/. RegionType, listed in the original draft, was never retrieved and has been removed; see §2.1.*

# 3. Methodology

The analysis is driven by scripts/run_all.sh. It comprises the nine numbered stages of the supplied pipeline, plus Stage 3a, written during this session to replace a failed VCF download, and Stage 3b, the supplied per-isolate quality-control script that run_all.sh labels as part of Stage 3 but that is given its own row in Table 3.1, and four stages added after review: Stage 10, a multiallelic-aware resistance marker panel; Stage 11, a diagnostic for the confounding of agro-ecological zone with sequencing study; Stage 12, which replaces Stage 10's marker panel entirely after an external critique showed that Stage 10's method of identifying markers — matching SnpEff protein-change strings against a hand-written list — could not see a resistance allele that is the reference residue and could not see a residue requiring two changes in one codon (both failures occurred); and Stage 13, which rebuilds the analysis matrix under six settings of the two thresholds that define the dataset and recomputes pairwise $F_{ST}$ and the region-level Mantel test under each. Stage 13 is the stage whose result matters most, and it does not go the manuscript's way: it leaves near-panmixia stronger than before and it removes the support for isolation by distance, which is significant at the published thresholds and at almost none of the others. That is a loss of a result, not a refinement of one, and Section 3.14 states it as such. Stage 10 is retained in the repository and in Table 3.1 for the record, but no number in this manuscript now comes from it; Section 3.10 sets out the replacement method and Section 3.10.1 the reason it was needed. Stages run in order; each writes its outputs to disk so that a failure costs one stage rather than the whole run. That caching is stated here with its qualification rather than as a design virtue, because it was also the pipeline's most dangerous defect: the guards skipped a stage whenever its output file already existed, without checking whether the *input* had changed, so when the SNP panel moved from 3,359 to 3,916 SNPs, ADMIXTURE and TreeMix silently reported the previous day's results as "stage finished". Only Stage 8 carried an assertion that caught it. The guards now require the output to be newer than its input; Section 3.13 sets out what happened and what was changed. Stages 10 to 13 are read-only with respect to everything upstream: they create new output files and modify none, so no estimate in Sections 6.1 to 6.7 depends on them. Two further checks changed no stage's method but changed what the existing stages describe: Stage 3's three ordinations were re-run on the 1,900-isolate analysis set rather than the 4,905 pre-QC isolates (Section 3.4.1), and the 19 XtX outliers were audited against the Pf8 per-variant quality annotations (Section 3.11.1). This section describes each stage: what it does, why, what it assumes, and how it constrains what follows. Where a method changed in the corrected re-run, or in the rebuild that followed the external critique, or in these final checks, the change and its motivation are stated in place rather than relegated to a revision log.

## 3.1 Pipeline overview

| Stage | Script(s) | Function | Principal output |
|---|---|---|---|
| 1 | step01_filter_samples.py | Restrict the Pf8 manifest to Ghana and to QC-passing isolates; capture distance-matrix row indices | interim/ghana_samples.csv |
| 2 | step02_assign_aez.py | Spatial join of isolate coordinates to agro-ecological zone polygons | interim/ ghana_samples_with_aez.csv |
| 3 | step03_umap.py | UMAP, t-SNE and PCoA of the Ghanaian block of the Pf8 distance matrix, plus a quantitative comparison of the three. Now takes --samples and --suffix, and re-slices the distance | results/{umap,tsne} _embedding.csv; results/pcoa_coordinates.csv; results/dimred_comparison.csv; results/dimred_comparison_qcset. |

| Stage | Script(s) | Function | Principal output |
|---|---|---|---|
| | | submatrix for the isolates given rather than reusing the cached full-set one, so the same comparison can be run on the 1,900-isolate analysis set (Section 3.4.1) | csv |
| 3a | step03a_fetch_callset_zarr.py | Fetch the Ghanaian callset from the Pf8 Zarr store over HTTPS under FILTER_PASS & is_snp & numalt == 1 (added in this session; see Sections 3.5 and 5.1) | interim/pf8_ghana.zarr |
| 3b | step03b_sample_qc.py | Per-isolate quality control: call rate, heterozygosity, Fws, clonal pruning | interim/ ghana_samples_qc_pass.csv; results/sample_qc_full.csv |
| 4 | step04a_prepare_callset.py, step04b_fst.py | Build the filtered genotype matrix under the corrected two-pass missingness order (Section 3.7); compute pairwise Hudson FST with a block jackknife and cache per-zone allele counts | interim/ghana_core_snps.npz; results/fst_pairwise.csv |
| 5 | step05a / 05b / 05c | Export to PLINK, LD-prune, ADMIXTURE sweep K = 1 to 8 with cross-validation, test the K = 2 minor component against zone, plot | results/admixture/ |
| 6 | step06_mantel.py | Seeded Mantel test of isolation by distance at zone and region level; censored and uncensored estimates; leave-one-out jackknife | results/ mantel_result_{zone,region}.json; results/mantel_jackknife_{zone,region}.csv |
| 7 | step07a / 07b / 07c | TreeMix input; sweep over 0 to 5 migration edges with -se, giving jackknife standard errors and a p-value per edge; model comparison | results/treemix/ |
| 8 | step08a / 08b / 08c | BayPass input; core model and pseudo-observed data simulation, now single-threaded for determinism; XtX calibration with false-discovery accounting | results/baypass/, including results/baypass/xtx_all_snps.csv and results/baypass/xtx_calibration.json |
| 9 | step09_annotate.py | Intersect outlier SNPs with the PlasmoDB-71 gene annotation and cross-reference against the eight known resistance loci listed in config.py (config.KNOWN_RESISTANCE_LOCI — not the Stage 10 or Stage 12 marker panel, so there is no circular dependency on either) | results/outlier_genes.csv |
| 10 | step10_resistance_panel.py | Superseded by Stage 12. Multiallelic-aware resistance marker panel built by matching SnpEff ANN_HGVS_p strings against a hand-written marker list. Retained for the record; no reported number derives from it (Section 3.10.1) | results/ resistance_panel_frequencies.csv; results/resistance_panel_canonical.csv |

| Stage | Script(s) | Function | Principal output |
|---|---|---|---|
| 11 | step11_batch_confounding.py | Quantify the confounding of agro-ecological zone with sequencing study (Section 3.12) | results/batch_study_by_zone.csv; results/batch_zone_pair_overlap.csv; results/batch_within_zone_tests.csv |
| 12 | step12_codon_panel.py | The marker panel as it now stands: codons reconstructed from the CDS and the 3D7 reference, full residue spectrum including the reference residue, within-isolate haplotypes on complete cases, per-sample concordance against MalariaGEN (Section 3.10) | results/codon_panel_residues.csv; results/codon_panel_haplotypes.csv; results/codon_panel_concordance.csv |
| 13 | step13_sensitivity.py | Threshold sensitivity: one genotype load, six scenarios of the $F_{ws}$ and MAF thresholds rebuilt in memory, pairwise Hudson $F_{ST}$ and the region-level Mantel test recomputed under each (Section 3.14) | results/sensitivity_summary.csv; results/sensitivity_fst.csv |

*Table 3.1. Pipeline stages. Stages 3a and 3b are given their own rows although run_all.sh labels both as "stage 3": 3a was written during this session to replace a failed VCF download, 3b was supplied. Stages 10 and 11 were added after the internal audit; Stage 12 was added after the external critique and replaces Stage 10; Stage 13 was added last, as the threshold robustness check the audit had asked for. Stage 8 is now run with one thread (Section 3.9). Two of the final checks carry no stage number: the re-run of Stage 3 on the analysis set is a change to step03_umap.py and not a new stage (Section 3.4.1), and the quality audit of the 19 XtX outliers writes results/outlier_quality_check.csv from a script the run record does not name (Section 3.11.1). All other scripts were supplied with the project and were modified only as recorded in Section 5 and in this section. Section numbers in this table follow the numbering used here: Section 3.9 covers Stages 5 to 9, Section 3.10 Stage 12 (with Stage 10's failure diagnosed in Section 3.10.1), Section 3.11 the false-discovery accounting applied to Stage 8, Section 3.12 Stage 11 and Section 3.14 Stage 13.*

## 3.2 Stage 1 — manifest filtering

**What and why.** The manifest is restricted to Ghana and then to isolates flagged QC-pass by the release. Relying on the release's flag rather than re-deriving sample quality from raw data is a deliberate deference to MalariaGEN's own curation, which has access to sequencing-run level information this analysis does not.

**Implementation note.** The script captures each retained isolate's row index in the full 33,325-sample distance matrix as the column pf8_row. This is the only safe way to slice the Ghanaian block later: the pandas index would shift under any subsequent filter, whereas pf8_row is fixed at the moment of extraction. Stage 3 asserts that the maximum pf8_row does not exceed the matrix dimension, which would catch a Pf7/Pf8 mismatch.

**[Observed]** 6,653 Ghanaian rows, reduced to 4,905 by the release QC filter (1,748 dropped). pf8_row range 2,301 to 29,498.

## 3.3 Stage 2 — agro-ecological zone assignment

**What and why.** Isolate coordinates are joined to the zone polygon layer with GeoPandas, giving each isolate a zone label. Because Pf8 resolves Ghanaian collection locations only to administrative level 1, the coordinates are region centroids, and the join is effectively a lookup table from seven points to four zones. Every number in this manuscript that is stratified by zone inherits that resolution.

[Observed] All 4,905 isolates matched exactly one polygon: zero unmatched, zero rows implicated in duplicate matches. The 4,905 isolates occupy seven distinct coordinate pairs.

[Interpreted] **Assumption.** That a region's centroid is a reasonable proxy for where its isolates were collected. For compact regions this is tolerable; for regions that straddle a zone boundary it is not. The script prints a region-by-zone cross-tabulation specifically so that boundary-straddling regions can be checked by eye, and flags Brong Ahafo (forest/transition) and Volta (coastal/Guinea savanna) as the two warranting manual review.

[Observed] The Brong Ahafo and Volta assignments have not been manually reviewed against the map.

### 3.3.1 Provenance of the zone layer

[Observed] The polygon layer used for the join, ghanaAEZ.geojson, has no citable source. This was investigated rather than assumed, and the conclusion is negative: the file carries only the fields id, Name, AREA and PERIMETER, with no attribution, author, date or licence. The uppercase AREA/PERIMETER pair is the auto-maintained attribute signature of an ESRI ArcInfo coverage, so the lineage is coverage to shapefile to GeoJSON, and the file itself was written by GDAL/OGR. The file as used here is byte-for-byte identical to ArcGIS Online item 0776dae5c5314102bd03fdaab8df0356, "Agro Ecological Zones Ghana" — identical md5, 96b1a7f4da62d6f0153826f41af96152 — whose licenseInfo and accessInformation fields are both null and whose owner is an anonymous personal account. That item is therefore an unattributed re-host, not an origin.

[Observed] The classification the layer encodes is a hybrid. It keeps the MoFA/FAO savanna, transitional and coastal names but splits the high forest into "Wet evergreen" and "Moist evergreen", which are vegetation types from the Hall and Swaine lineage, and collapses the semi-deciduous zones into a single "Decidous forest", a label the FAO/MoA vegetation table does not use. It also contains a water body, which is not an agro-ecological zone. HDX, data.gov.gh, GADM, Harvard Dataverse, Zenodo, figshare, FAO GAEZ, FAO Africover and ICPAC were checked directly and none hosts a matching layer. That sweep rules out a matching *layer*; it does not establish that the label appears in no published scheme anywhere, and no such claim is made.

[Unknown] The original provenance and licence of the layer.

> **Handling.** The layer is cited as accessed rather than as authoritative, identified by its md5 checksum, with the classification cited separately (MoFA/FAO for the savanna and transitional zones, Hall and Swaine/FAO for the evergreen split) and the hybrid character stated. Zone names are reproduced verbatim in tables, figures, output files and every grouping key, including the "Decidous forest" misspelling, because the spatial join, and every downstream grouping, keys on the strings as written; running prose uses the conventional spelling "Deciduous forest" for the same zone. The provenance gap is carried as a limitation in Section 7, and the obvious robustness check — repeating the admin-1 centroid join against an authoritative six-zone layer and testing whether zone assignments are stable — has not been done.

[Proposed] Write to CERSGIS at the University of Ghana (cersgis@ug.edu.gh) to ask whether the layer originates with them and under what licence. The ArcInfo-coverage fingerprint dates the original to a pre-2005 workflow, and CERSGIS is the most likely custodian of a Ghanaian coverage of that vintage. This is the one outstanding check of the four that cannot be closed analytically: no file inspection, checksum or repository sweep can supply an attribution that the file does not carry, so it remains open and is carried as a limitation rather than resolved.

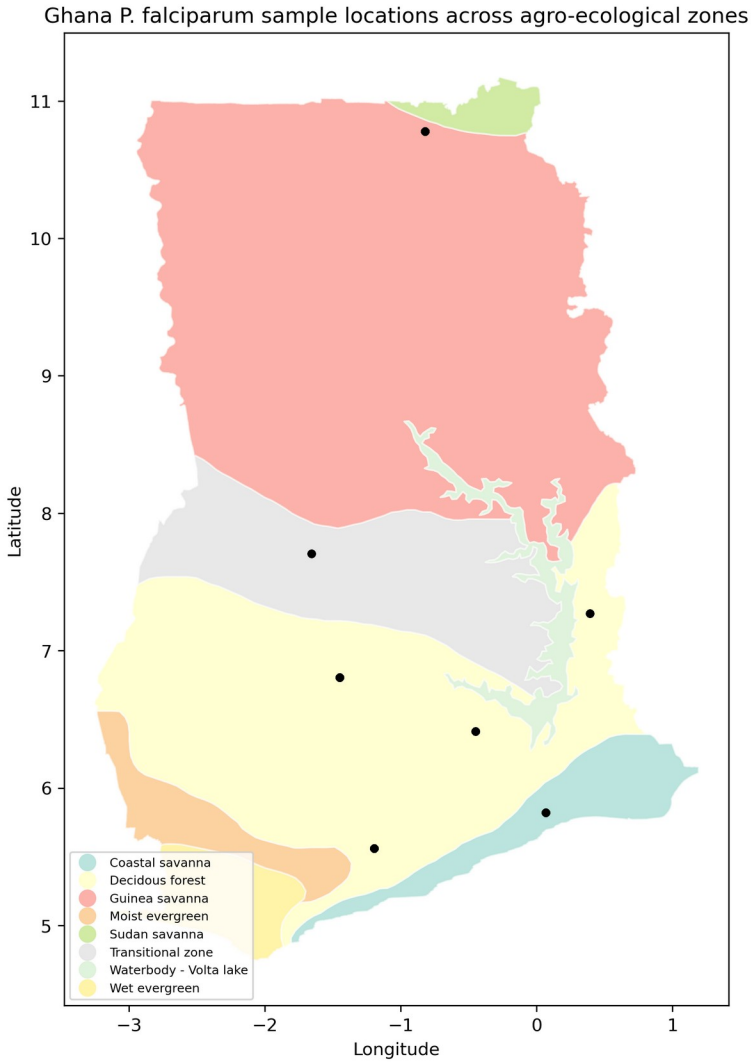



*Figure 3.1. The spatial join. Isolate collection points, resolved to administrative-level-1 centroids, over the agro-ecological zone layer. Seven distinct coordinate pairs carry all 4,905 isolates, which is the resolution limit of every zone-stratified result in this manuscript. The layer's provenance could not be established (Section 3.3.1).*

## 3.4 Stage 3 — three ordinations of the genotype distance matrix

**What and why.** Pf8 publishes a precomputed mean genotype distance matrix over all 33,325 samples. The Ghanaian block, 4,905 by 4,905, is sliced out by memory-mapping the 4.44 GB file and taking the submatrix, which keeps peak memory in the low hundreds of megabytes. Three dimensionality-reduction methods are then applied to the same matrix so that they can be compared on equal terms.

- **UMAP** (n_neighbors = 30, min_dist = 0.1, metric = precomputed) — preserves local neighbourhood structure with partial global layout. These are the parameter values specified in the thesis workflow.
- **t-SNE** (perplexity = 30, metric = precomputed, init = random) — preserves local neighbourhoods only. Initialisation must be random rather than PCA-based because the input is a distance matrix rather than feature vectors.

- **PCoA** (classical metric multidimensional scaling) — Gower double-centring followed by an eigendecomposition. Unlike the other two it is linear, so axis scaling is interpretable and each axis carries a proportion of variation explained.

**Comparison metrics.** Because the three methods optimise different objectives, a visual comparison alone would be uninformative. Three quantitative measures are computed on the same embeddings: trustworthiness (do points that are close in two dimensions correspond to genuine near neighbours in the full matrix?), the Spearman correlation between pairwise distances in the embedding and in the original matrix on a 2,000-sample subset (does the embedding preserve global structure?), and the silhouette coefficient by zone (do the zone labels separate?). The silhouette is also computed on the original distance matrix, which gives the ceiling any embedding could achieve.

**Deviation from the thesis workflow.** The workflow document specifies UMAP for this step. t-SNE and PCoA were added at the user's request during this session so that the three could be compared. This is an addition, not a substitution: the UMAP configuration is unchanged.

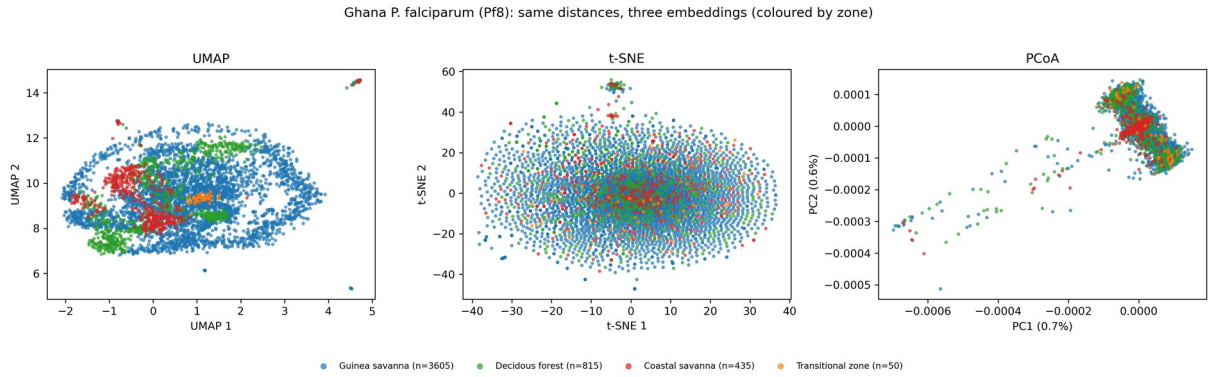



*Figure 3.2. The same Ghanaian distance submatrix under UMAP, t-SNE and PCoA, coloured by agro-ecological zone, on the 4,905 pre-QC isolates. Computed before the corrected re-run and unaffected by it, because this stage reads the published distance matrix rather than the genotype matrix rebuilt at Stage 4a. This is the published version and it is retained unchanged; Figure 3.4 is its counterpart on the 1,900-isolate analysis set (Section 3.4.1).*

### 3.4.1 Re-run on the analysis set

**The mismatch.** As published, this stage described the 4,905 release-QC-passing isolates while every other result in the manuscript described the 1,900 that survive Stage 3b and Stage 4a. The figures and the comparison metrics were therefore computed on a different population from the $F_{ST}$, ADMIXTURE, Mantel, TreeMix, BayPass and marker-panel results they sit beside, and in particular on a set that still contained the polyclonal and clonally redundant isolates those later stages remove. Nothing was wrong with the computation; what was wrong was that the sample set was not the one the reader would assume.

**The change.** step03_umap.py now takes `--samples`, a file of isolate identifiers, and `--suffix`, a string appended to every output filename. Given `--samples` it **re-slices the distance submatrix from the memory-mapped full matrix for exactly those isolates, rather than reusing the cached 4,905-isolate submatrix** written by the first run. That distinction is the substance of the patch: the cached submatrix is the wrong object for a sample subset, and re-using it and subsetting the labels afterwards would have reported the full-set geometry under the smaller set's name. The pf8_row indices captured at Stage 1 make the re-slice safe, since they are fixed at extraction and do not shift under the intervening filters (Section 3.2). Outputs carry the `_qcset` suffix so that the published figures and CSVs are left untouched and the two sample sets can be compared side by side rather than one overwriting the other.

| Metric | 4,905 pre-QC isolates (published) | 1,900-isolate analysis set |
|---|---|---|
| Silhouette by zone, raw distances | −0.016 | −0.020 |
| UMAP trustworthiness | 0.486 | 0.499 |
| t-SNE trustworthiness | 0.507 | 0.526 |
| PCoA trustworthiness | 0.495 | 0.510 |
| PCoA distance correlation (Spearman) | 0.322 | 0.287 |

*Table 3.4. The three ordinations and their comparison metrics on the published pre-QC sample set and on the analysis set. Analysis-set values from results/dimred_comparison_qcset.csv.*

**[Observed]** On the 1,900-isolate analysis set the silhouette by zone computed on the raw distances is **−0.020**, against −0.016 on the 4,905. Trustworthiness rises slightly for all three methods (UMAP 0.486 to 0.499, t-SNE 0.507 to 0.526, PCoA 0.495 to 0.510) and the PCoA distance correlation falls from 0.322 to 0.287. The embedding-level silhouettes on the analysis set are −0.277 for UMAP, −0.063 for t-SNE and −0.079 for PCoA.

**[Interpreted]** **Removing the polyclonal and clonally redundant isolates does not sharpen the structure.** The silhouette becomes slightly more negative, which is the opposite of what the obvious objection to the published figures would predict — that mixed infections and repeat-sampled parasites were blurring a zone signal that would emerge once they were removed. They were not, and it does not. The small rise in trustworthiness is expected on a smaller point set and is not evidence of better structure; the fall in the PCoA distance correlation runs the other way. The conclusion of this stage is therefore unchanged, and the figures can now be reported on the same sample set as the rest of the analysis.

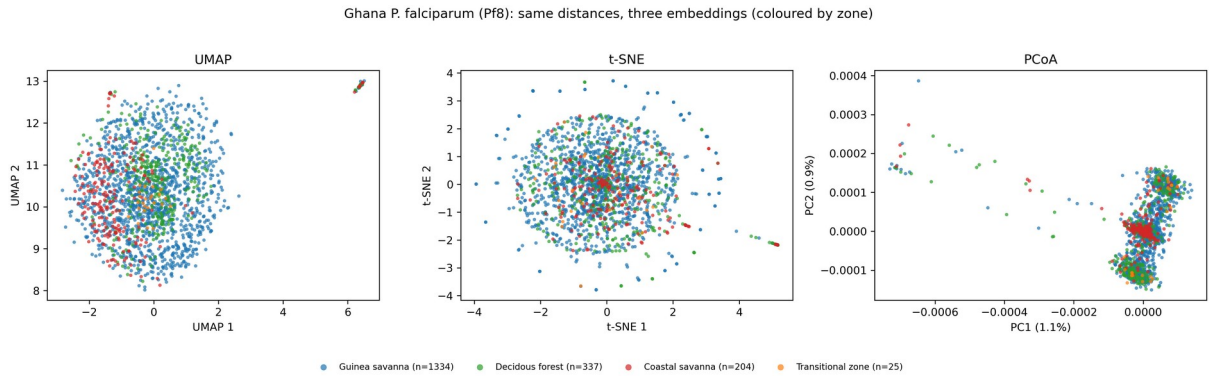



*Figure 3.4. The same three ordinations on the 1,900-isolate analysis set, coloured by agro-ecological zone, with the distance submatrix re-sliced for those isolates rather than taken from the cached full-set matrix. Compare Figure 3.2, which is the published 4,905-isolate version.*

> **A log defect in this patch.** The `--suffix` patch suffixed the file writes but not the `print` statements, so the run reported writing the published results/dimred_comparison.csv when it had in fact correctly written results/dimred_comparison_qcset.csv. No data were at risk and none was overwritten — verified from the published file's modification time and contents — but the log misreported its own output path, which is a reproducibility defect in its own right. It is fixed, and Section 3.13.1 records it alongside the other incident from these runs.

## 3.5 Stage 3a — genotype retrieval, and the consequence of the numalt filter

**What and why.** This stage did not exist in the supplied pipeline and was written during this session to replace a failed VCF download (Section 5.1). It fetches genotypes directly from the Pf8 Zarr store over HTTPS, chunk by chunk, retaining only what the downstream steps read.

The Zarr store is chunked 131,072 variants by 100 samples per field. Because the 6,653 Ghanaian samples fall in 77 of the 334 sample-chunk columns, and the sites of interest in 95 of the 96 variant-chunk rows, only a subset of the grid must be transferred. Each chunk is downloaded, decompressed in memory, sliced to the Ghanaian columns and the retained rows, then written to a local store and discarded, so peak memory stays at a few gigabytes and the transferred volume is never written to disk in full.

Sites are retained where FILTER_PASS is true, is_snp is true and numalt equals 1. Fields retained are GT, AD (reference and first alternate only), DP and GQ.

**[Observed]** 2,190,026 of 12,493,205 Pf8 sites are PASS biallelic SNPs. The transfer moved 299.8 GB in 3.20 hours at approximately 26 MB/s with zero failed requests, producing a 20.57 GB local store.

### 3.5.1 What the numalt == 1 restriction excludes

**[Observed]** The original version of this section presented the three-part site filter as a neutral convenience. It is not. Requiring numalt == 1 and is_snp restricts the store to sites with exactly one alternate allele and no indel or spanning-deletion ALT. In Pf8 that restriction bites hard at several of the sites a drug-resistance question depends on — not because of any general property of resistance codons, but because of what the release happens to record at these particular positions: *pfcrt* K76T is numalt = 2; *pfcrt* M74I is numalt = 3; *pfmdr1* N86Y is numalt = 3; *pfdhps* K540E is numalt = 2; *pfdhps* A581G is numalt = 2. Each therefore fails numalt == 1 and is dropped. The five numalt values are read from results/resistance_panel_canonical.csv. The per-codon is_snp flags are deliberately not asserted: no saved output carries an is_snp column at codon level and no log prints one, the local store cannot settle it because by construction it holds only numalt == 1 sites, and the only is_snp figure on the record is the aggregate at step10_resistance_panel.log:16 (numalt == 1 & is_snp, 3,330). An earlier draft attached "with is_snp false" to four of the five codons; that is withdrawn as unevidenced, and the paragraph rests on the numalt values, which are fully evidenced. The earlier draft also justified the exclusion by asserting that the canonical resistance codons are "among the most-sampled, most-variable positions in the genome" and therefore the most likely to be multiallelic. Nothing in this project measures that, and the claim is withdrawn; what is on the record is the per-codon observation above.

**[Observed]** Quantified over the nine genes of the resistance panel (Section 3.10): the Pf8 Zarr holds 18,341 variant records inside those gene spans, of which 13,367 are FILTER_PASS. Of those 13,367, the filter keeps **3,330** and discards **10,037** as multiallelic or indel. At the canonical codons specifically, it discards 28 of the 44 canonical records and keeps 16 — but what it keeps is largely uninformative: 10 of the 16 are monomorphic in these isolates, including six *pfk13* codons, *pfmdr1* S1034C and N1042D, *pfcrt* C72S and *pfmdr2* T484I. Of the 21 canonical codons that are actually polymorphic here, **15 are discarded**, among them the *pfcrt* CVIET-lineage set at codons 74, 75, 76, 220, 271 and 371 — whose Stage 10 records are M74I, **Asn75Asp**, K76T, A220S, Q271E and R371I — together with *pfmdr1* N86Y and Y184F, *pfdhfr* S108N, and *pfdhps* S436A, K540E, A581G and A613S. Two qualifications travel with that *pfcrt* set, because an earlier draft called it "the whole CVIET-defining set" and that is wrong in both directions. First, these are Stage 10's annotation records, and Stage 10's record at codon 75 is Asn75Asp, not N75E: resistance_panel_canonical.csv carries `pfcrt,Asn75Asp,numalt=2,in_genomewide_scan=False,freq=0.0795`, and step10_resistance_panel.log:22 lists the *pfcrt* canonical set as Ala220Ser, Arg371Ile, Asn326Ser, Asn75Asp, Cys72Ser, Gln271Glu, Ile356Thr, Lys76Thr, Met74Ile. N75E was never in Stage 10's panel at all — that is precisely the failure mode of Section 3.10.1 —

so Stage 10 cannot have discarded it. Second, CVIET is the codon 72–76 motif (results/codon_panel_haplotypes.csv labels it pfcrt_72_76, "CVIET core motif"), so 220, 271 and 371 are additional *pfcrt* haplotype positions rather than CVIET-defining, while two codons that are CVIET-defining are absent from the discarded list: codon 72, where C72S passed the mask (in_genomewide_scan = True) and is monomorphic here, and codon 73, which Stage 10's panel did not contain. The earlier formulation — that the discarded set "includes every canonical resistance codon", or that the numalt clause dropped every canonical codon in *pfcrt*, *pfmdr1* and *pfk13* — is contradicted by the project's own output and is corrected to the counts given here.

**[Observed]** **The cascade, defined once.** Stage 10's panel held **44 canonical codon-allele records, covering 41 distinct protein changes, in eight of the nine panel genes** (*pfcoronin* contributed none); **21 were polymorphic** in these 1,900 isolates; **6 of the 21 survived the ingest mask** — the in_genomewide_scan field, defined at step10_resistance_panel.py:349 as numalt == 1, i.e. "would have survived the ingest filter"; and **3 reached the MAF-filtered scan matrix**: *pfdhfr* C59R, *pfdhfr* N51I and *pfaat1* S258L. The other three that passed the mask — *pfcrt* I356T, *pfmdr1* D1246Y and *pfcrt* N326S — were then cut by MAF ≥ 0.05 at frequencies of 0.016, 0.011 and 0.008. These four numbers (44, 21, 6, 3) carry exactly this meaning wherever they appear in this manuscript, and the Abstract, Table 1.1, Section 6.8.3, Table 6.8 and Section 10.1 use this wording. In particular, the genome-wide scan did test three of the 21 polymorphic canonical codons, and *no* pfcrt, pfmdr1, pfdhps *or* pfk13 *codon reached the MAF-filtered scan matrix* — pfdhps *because nothing survived the ingest mask, the others because what survived was either too rare (*pfcrt *I356T*, pfcrt *N326S and* pfmdr1 *D1246Y all passed the mask, reached the genome-wide store and were then cut by MAF at 0.016, 0.011 and 0.008) or monomorphic here (six* pfk13 codons — F446I, P441L, Y493H, P553L, R561H, A675V — passed the mask and were fetched and found invariant). "Could not test at all" would be true of the final MAF-filtered scan matrix but false of ascertainment, which is the distinction this paragraph exists to draw, so the stronger wording is not used.

> **These counts describe the scan's ascertainment, not the marker panel.** 44, 21, 6 and 3 are properties of Stage 10's annotation-matched record set and of what the genome-wide ingest filter did to it. They are **not** the denominators of the marker panel as it now stands. The panel is defined by codon, not by annotation record: Stage 12 reconstructs **57 codons** across the same nine genes, of which **24 are polymorphic** here and 33 fixed at the reference residue (Section 3.10). The two framings answer different questions and are not interchangeable, and because Stage 10's list was itself incomplete — it had no *pfdhps* 437 at all, and no N75E — the 44 should be read as "the records Stage 10 found", never as "the canonical codons that exist".

**[Interpreted]** **Consequence.** The genome-wide store built by this stage cannot carry most of the canonical resistance codons, so the absence of *pfcrt*, *pfmdr1*, *pfdhps* and *pfk13* from the Stage 8 outlier list is uninformative about resistance at those loci: it is an ascertainment artefact, not a result. The two *pfdhfr* codons and *pfaat1* S258L that did reach the scan are a different matter — they were tested, and Section 6.8 reports them as tested results rather than as silence. This is not a limitation that can be repaired by reinterpretation; it required a separate retrieval, which is Stage 12. The filter is retained unchanged for the genome-wide analyses, where biallelic SNPs are the appropriate unit and the exclusion is not differential with respect to zone. It is lifted only for the panel codons, in a parallel store that is never merged with this one.

> **How this was missed.** The filter is one line of a retrieval script, applied before any analysis ran: a three-term conjunction, mask = (FILTER_PASS & is_snp & (numalt == 1)), at scripts/step03a_fetch_callset_zarr.py:334-336, with the same test on the VCF-validation path at :236 and again at :370. (An earlier version of this note cited "a single clause on line 13"; line 13 of that file is a docstring sentence about PASS and RegionType, which is the separate claim Section 3.5.2 withdraws, and conflating the two defects obscured both.) It fails silently: the excluded

> sites never appear in any count, log or diagnostic downstream. Nothing in the pipeline's output distinguishes "this codon is monomorphic" from "this codon was never fetched". Section 7 treats this as a general lesson about ingest filters rather than a one-off error.

### 3.5.2 The core-genome filter

The supplied pipeline expects an optional BED file of core-genome intervals (data/pf_core_regions.bed), absent here, and warns at Stage 4a that without it subtelomeric var/rifin regions will contaminate the outlier scan. The earlier version of this section dismissed that warning as "a false alarm", citing a direct tabulation of RegionType over the PASS biallelic site set. No such tabulation exists: RegionType is not among the fields step03a retrieves, the local store holds no such array, and no log records the counts. That claim is withdrawn.

**[Interpreted]** The argument that replaces it is structural rather than empirical, and is checkable from the release metadata without any fetch: Pf8's FILTER vocabulary itself contains the region categories (FILTER_SubtelomericHypervariable, FILTER_SubtelomericRepeat, FILTER_InternalHypervariable, FILTER_Centromere, FILTER_Apicoplast, FILTER_Mitochondrion), so a non-core site cannot carry FILTER_PASS by construction. The PASS filter therefore performs the regional exclusion the BED would have performed, as a matter of definition.

Two qualifications follow. First, because the argument is definitional, it is labelled [Interpreted] and not [Observed]; the manuscript's own evidence key reserves [Observed] for values printed by a pipeline script and preserved in a log or results file. Second, the exclusion is regional, not family-based: members of antigenic gene families that lie inside core regions do survive into the analysis matrix. Their absence from the outlier list is therefore a tested result, weakly powered, rather than a filtering tautology, and Section 6.8 states it that way. The structural argument has since acquired direct support of a kind it did not have: the quality audit of Section 3.11.1 reads RegionType for the 19 outliers and finds all 19 are Core.

### 3.5.3 Validation

**[Observed]** The store was verified against the one completely downloaded VCF (chromosome 10) and against two partially downloaded ones (chromosomes 1 and 13), in three independent 35 to 40 kb windows. In each window the site list matched exactly and all four genotype fields agreed value for value: 3,957, 3,470 and 4,378 PASS biallelic SNPs respectively, with zero differing values across 471,231,990 compared values in total. The verification is in logs/step03a_fetch_callset.log.

## 3.6 Stage 3b — isolate quality control

Four per-isolate checks are applied, computed from a panel of 50,000 biallelic PASS SNPs spread proportionally across all 14 contigs.

**Within-host fixation index (Fws).** Fws measures how far an isolate's read-level allele frequencies depart from the population's. A clonal infection has read frequencies near 0 or 1 at every site and Fws near 1; a mixed infection carries intermediate frequencies and a lower value. It is computed as $Fws = 1 - (\Sigma Hw,i / \Sigma Hs,i)$, where $Hw = 2q(1 - q)$ from the isolate's own read counts and $Hs = 2p(1 - p)$ from pooled read counts across qualifying isolates. Sites contribute only where total depth is at least 5. The conventional monoclonality threshold of 0.95 is applied (config.py MIN_FWS = 0.95). That threshold is a convention and not a property of these data, it discards the majority of the isolates, and Section 3.14 reports what the results do when it is relaxed.

**Call rate.** The proportion of panel SNPs with a genotype call. The pipeline applies an absolute floor and, optionally, a median-absolute-deviation outlier rule; the choice between them is discussed in Section 5.4.

**Heterozygosity outliers.** Blood-stage *P. falciparum* is haploid, so a heterozygous call indicates a mixed infection or a genotyping error rather than true heterozygosity. Isolates more than three median absolute deviations above the median heterozygous call rate are flagged. This check is largely redundant with Fws and is retained as an independent cross-check.

**Clonal pruning.** Near-identical isolates (outbreak-linked or repeat-sampled parasites) inflate within-zone homogeneity and therefore inflate between-zone FST. The cached Ghanaian distance submatrix from Stage 3 is thresholded at CLONAL_THRESHOLD = 0.00005, connected components are extracted with a single-linkage rule, and one representative is retained per component — chosen by best call rate, then best Fws, then identifier, so the choice is deterministic.

**[Observed]** Median heterozygous call rate 0.0004 (range 0.0000 to 0.0030). Median Fws 0.895 (range 0.310 to 1.000); 2,095 of 4,905 isolates monoclonal at Fws ≥ 0.95 (42.7 per cent), 2,810 below. 2,017 of 4,905 isolates pass all four checks (41.1 per cent). 77 isolates were not retained as the representative of their clonal component.

> **On the failure counts.** The script prints the five failure reasons as independent, non-exclusive flags (step03b_sample_qc.py:397 and 402-406), and they overlap: 20 + 0 + 126 + 2,810 + 77 = 3,033 against 4,905 − 2,017 = 2,888 actual failures. The 77 is therefore the total count flagged fail_clonal, not a count of isolates failing on clonal pruning alone, and an earlier version of this paragraph read it the second way.

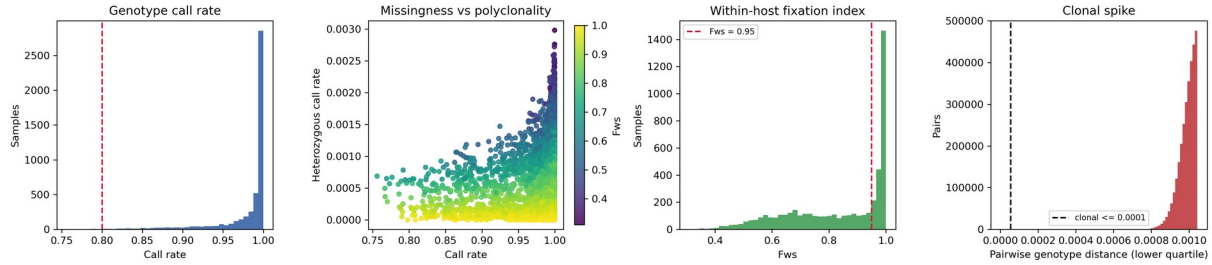



*Figure 3.3. Per-isolate quality-control distributions on the 50,000-SNP panel: call rate, heterozygous call rate and Fws, with the applied thresholds marked. Unchanged by the corrected re-run, which altered only Stage 4a and later.*

## 3.7 Stage 4a — building the genotype matrix

This stage converts the Zarr store into the single filtered genotype matrix that every subsequent stage reads. The order of operations matters and is deliberate:

7. **Site-level filters** on the small one-dimensional arrays (FILTER_PASS, is_snp, numalt == 1), screening the whole genome before any genotype is fetched. Section 3.5.1 states what this excludes and why that matters.
8. **Genotype-level masking:** calls below a read-depth floor of 5 are set to missing. A low-confidence call thus becomes missing data rather than a confident wrong answer, which matters because the missingness filter later in this stage then sees it for what it is. The GQ component of the pipeline default was removed on the evidence set out in Section 4.2.5.
9. **Sample alignment:** genotype columns are reordered to match the metadata table, so that sample order is fixed once and never re-derived.

10. **Mixed-call resolution:** heterozygous calls are replaced by the allele with more reads; ties become missing (Section 5.3). Stage 12 deliberately does **not** follow this rule; Section 3.10.2 says why.
11. **Two-pass missingness, then MAF** — described below.

### 3.7.1 The corrected filter order, and why it matters

In the original run, variant missingness was computed once, over all 2,017 isolates that entered this stage, and the sample-missingness cut was applied afterwards. 117 isolates then failed that cut and were dropped. Every SNP had therefore been charged for missing calls in 117 isolates that did not survive into any analysis, so SNPs were discarded for failing a criterion measured partly on data that was itself discarded. The bias is not random: it penalises exactly the SNPs that are hardest to call in the worst-covered isolates.

The stage is now explicitly two-pass. A provisional variant set is formed under the same rule as before, so that the same isolates fail the sample cut and the analysis sample set is unchanged. The sample cut is applied. Variant missingness is then recomputed on the isolates that remain, and the MAF filter applied last so that a SNP is not retained on the strength of a handful of confident calls. The per-sample call counts over the provisional set are accumulated a block at a time, so the provisional genotype subset is never materialised in memory.

[Calculated] Thresholds and the sample set are identical before and after. The corrected order yields 3,916 SNPs where the original yielded 3,359, an increase of 16.6 per cent on the same 1,900 isolates — additional statistical power obtained purely by measuring the existing criteria in the right order.

[Observed] Applied to the final run: 3,709,090,355 calls fell below DP 5; of 12,540,727,880 called genotypes, 10,845,661,093 survived the mask (13.52 per cent masked); 194,167 mixed calls were resolved by read depth, 129,665 to REF, 55,582 to ALT and 8,920 to missing on a tie; the provisional variant pass used for the sample cut held 1,634,808 sites; 117 high-missingness samples were dropped, leaving 1,900; variant missingness ≤ 0.1 recomputed on those 1,900 retained 1,807,331 sites; and MAF ≥ 0.05 gave the final 3,916.

**Thresholds.** MAF of at least 5 per cent and missingness of at most 10 per cent are specified in the thesis workflow document for Step 4 and were not altered. The workflow document writes these as MAF > 5 per cent and missingness < 10 per cent; the implementation uses the non-strict forms — `keep_maf = maf >= args.min_maf` at step04a_prepare_callset.py:310 and `keep_var = var_missing <= args.max_missing` at :300, with the same non-strict test at :280 for the provisional pass. The difference changes no reported count, and the thresholds are quoted here in the form the code applies so that the two statements cannot be read as contradictory. The depth floor and the sample-missingness bound are pipeline defaults. The depth and quality settings were revised during this session, and that revision is documented in Section 4.2.5 — not, as the previous version of this paragraph stated, in Section 5.5, which concerns the TreeMix jackknife block size. The MAF floor is the second of the two thresholds Stage 13 varies, and it is the more consequential of the two for the SNP count: lowering it to 0.01 nearly triples the panel (Section 3.14).

## 3.8 Stage 4b — pairwise differentiation

**Estimator.** Hudson's FST is computed per locus and summed as a ratio of averages — numerator and denominator accumulated across loci before dividing — rather than as an average of per-locus ratios, which is upwardly biased. Hudson's estimator is preferred over Weir and Cockerham's here because it is insensitive to unequal sample sizes between populations, which is pertinent when one zone has 1,334 isolates and another has 25.

[Calculated] **Uncertainty.** Standard errors come from a block jackknife over contiguous blocks of SNPs, which accounts for linkage between nearby sites. The block length in force is blen = min(2000, max(10, n/20)), so with 3,916 SNPs it is 195 SNPs, giving approximately 20 jackknife blocks.

> **Caveat.** Twenty blocks is few, and the manuscript should not claim otherwise. Section 5.5 rejects the supplied TreeMix block size — 500 SNPs, which on the final 3,916-SNP panel would give about 8 blocks and gave 6 or 7 on the superseded 3,359-SNP panel — on precisely this ground, that too few blocks make the standard errors and hence the migration-edge tests unreliable, and raised the TreeMix block count to roughly **78** (3,916 SNPs in 50-SNP blocks; logs/stage7.log records SNPs 3,916 at block size 50, and config.py:164 sets TREEMIX_BLOCK_SIZE = 50). The FST standard errors and confidence intervals reported in Section 6.4 rest on about a quarter of that number, under an automatic rule rather than a considered choice. The block count was not stated in the original manuscript; it is stated here because the headline differentiation intervals depend on it.

**Ploidy handling.** Allele counts are taken as one allele per isolate (Section 5.3). This matters: counting both copies of a diploid-coded haploid genotype would double every sample size entering the FST correction, the TreeMix drift standard errors and the BayPass XtX statistic, making all three anticonservative.

[Observed] **Sensitivity.** This estimator was recomputed under six threshold scenarios at Stage 13. The maximum pairwise value stays within **0.00204 to 0.00255** across all six, over isolate counts from 1,900 to 4,403 and SNP counts from 3,275 to 10,876 (results/sensitivity_summary.csv; Section 3.14). The point estimates reported in Section 6.4 are therefore not an artefact of either threshold, which is a stronger statement about them than the jackknife standard errors alone can make.

## 3.9 Stages 5 to 9

**ADMIXTURE (Stage 5).** Genotypes are exported to PLINK transposed text format, converted to binary, LD-pruned (50-SNP window, 10-SNP step, r-squared threshold 0.2) and run through ADMIXTURE for K = 1 to 8 with 10-fold cross-validation. Pruning is not optional: linked SNPs carry redundant information and inflate apparent structure. Pruning the 3,916-SNP matrix leaves 3,385 SNPs. K = 1 is run explicitly to provide the no-structure reference point that the supplied sweep omits, without which a CV minimum at K = 2 cannot be read as evidence of structure at all. All eight K values, K = 1 included, were run on that same 3,385-SNP pruned panel, so the cross-validation curve is internally comparable.

[Observed] **A test that was previously absent.** The earlier manuscript asserted that no ancestry component aligned with geography. That assertion was never tested. The stage now tests it: the K = 2 minor component is compared across zones by Kruskal–Wallis, which gives H = 105.4 at $p = 1.05 \times 10^{-22}$, so the null of no zone difference is rejected decisively. The effect is nonetheless negligible in magnitude: zone-mean minor-component values run from 0.0419 to 0.0921. The substantive reading survives; the claim of no association does not, and is replaced throughout by the quantified version.

[Calculated] The ordering of those means is reported but not read as a cline, because it is not one. The two southern zones are highest (Deciduous forest 0.0921, Coastal savanna 0.0851) and the two northern ones lowest (Guinea savanna 0.0534, Transitional zone 0.0419) — but the Transitional zone is the second-most-northern of the four and carries the lowest mean of all, so a monotone south-to-north ordering fails there. Zone accounts for 1.9 per cent of the variance in the minor component (between-group sum of squares over total, recomputed for this manuscript from results/admixture/ghana_pruned.2.Q against the zone labels). The correspondence with pairwise FST is partial rather than exact: the pair with the smallest FST (Coastal savanna against Deciduous forest, 0.00045) does have the closest zone means, but the pair with the largest

(Coastal savanna against Guinea savanna, 0.00232) is only the fourth-largest difference in means, so the earlier claim that the component is "ordered as pairwise FST is" is withdrawn.

**Mantel test (Stage 6).** Isolation by distance is tested by regressing linearised differentiation, FST / (1 − FST), on the logarithm of great-circle distance computed on the WGS84 ellipsoid, at two levels: four zones and seven administrative regions. Three changes were made after the audit, and a fourth check, run later and reported in Section 3.14, bears on how the result may be graded at all.

- **The permutation test is seeded.** skbio draws its permutations from NumPy's unseeded global generator, so the previous p-values were irreproducible — at 999 permutations the Monte Carlo standard error on p was itself about 0.005, comparable to the margin by which the result cleared 0.05. Permutations are now generated in-script from a seeded generator (seed 42) and the correlation recomputed, with 9,999 permutations, one-sided, since isolation by distance predicts a positive association. The finest attainable p-value is reported alongside the test, because at four populations the permutation distribution is coarse (six pairs) and a null result there is uninformative.
- **The censoring of negative FST pairs is disclosed and quantified.** Hudson's estimator legitimately returns small negative values when between-population differentiation is below the within-population expectation, and skbio's DistanceMatrix rejects negatives, so such pairs must be floored to zero before the test. The original code did this silently at step06_mantel.py:148. The script now names every floored pair and its value in the log and in the JSON output, and runs the test twice — on the censored matrix and on the uncensored one — so that the reader can see what the flooring does. At region level 3 of 21 pairs are floored (Ashanti–Volta −0.001037, Eastern–Greater Accra −0.000203, Greater Accra–Volta −0.000142); at zone level none are.
- **A leave-one-out jackknife is run.** The test is repeated with each population dropped in turn, which is the relevant robustness question when a correlation computed over 21 pairwise points from seven populations may rest on a single extreme population. Seven is the number of populations dropped in turn, not the number of points the correlation rests on (results/mantel_result_region.json records npops = 7, pairs = 21). Section 3.14 reports what this jackknife and the threshold sensitivity analysis together imply, and the short answer is that this result may not be graded as supported.

**TreeMix (Stage 7).** A maximum-likelihood population graph is fitted with 0 to 5 migration edges, rooted on Guinea savanna, using a jackknife block size of 50 SNPs, which on the final 3,916-SNP panel gives roughly 78 blocks. Model fit is summarised as the proportion of variance in between-population covariance explained, computed over off-diagonal entries only, matching TreeMix's own definition. The sweep is now run with **-se**, which was omitted previously; without it every migration line in the .treeout file reads "weight NA NA NA" and no edge can be tested, so model choice had to rest on variance explained alone — a rule of thumb with no null distribution behind it. That is the objection, and it is not that variance explained is monotone in m: on this project's own output it is not. results/treemix/variance_explained.csv gives 0.896997, 0.940513, 0.996907, 0.999664, 0.999579, 0.999586 for m = 0 to 5, so it falls between m = 3 and m = 4. Nor is it true that variance explained cannot identify a stopping point: results/treemix/model_selection.json records a rule that does — the smallest m reaching 0.998 (step07c_plot_treemix.py:45), which gives m = 3 over 0 to 5, with a plateau fallback within 0.001 of the best (:46) giving m = 2 over 0 to 2, and that pair is where the superseded manuscript's "m = 3 (full), m = 2 (identifiable)" came from. The rule picks a number; what it cannot do is attach an error rate to it. With -se, TreeMix computes a jackknife mean and standard error for each migration weight and a p-value against zero, so each edge weight becomes testable against zero, though that test is conditional on the graph fitted at that m and on the root fixed in advance, and it does not by itself settle the

edge count: the variance-explained rule over the full m = 0 to 5 sweep still picks m = 3, and reporting m = 2 remains a modelling choice made on identifiability grounds. This is also what Section 5.5's change to the block size was for: the stated rationale, that too few blocks make the migration-edge tests unreliable, presupposes that such tests exist, and until -se was passed they did not.

**BayPass (Stage 8).** The core model estimates the population covariance matrix Ω from genome-wide allele counts, then reports XtX per SNP — a differentiation statistic standardised by that covariance structure. A SNP is called an outlier when its XtX exceeds a threshold taken from pseudo-observed data simulated under the estimated Ω. The logic is that a threshold drawn from the empirical XtX distribution assumes a fixed proportion of the genome is under selection, whereas a threshold from simulated neutral data assumes only that the demographic model is adequate. Section 3.11 describes the false-discovery accounting now applied on top of that threshold, and Section 3.11.1 the quality audit of the SNPs it selects.

> **Determinism.** BayPass is run **single-threaded**. Despite a fixed -seed, the multithreaded core model is not reproducible: the comment block at scripts/step08b_baypass.sh:32-36 records that two multithreaded runs of the same input differed by up to 1.1 in XtX and gave 15 against 16 outliers, while two single-threaded runs were bitwise identical, because the order in which threads accumulate contributions is not fixed. (The range cited must include line 32, which is the line that names the pair as 8-thread.) NTHREADS now defaults to 1 at step08b_baypass.sh:37 with that reason in the script, and run_all.sh exports NTHREADS=1 for Stage 8 even in parallel mode. Three BayPass runs survive on the server and are not interchangeable: (i) the **multithreaded run behind the original manuscript**, on the superseded 3,359-SNP panel — threshold 11.232, 15 outliers, archived under results/baypass_8thread_archive/; (ii) a **single-threaded run on that same 3,359-SNP panel**, whose core model took 23 s (logs/step08b_1thread.log:1,25); and (iii) the **final single-threaded run on the 3,916-SNP panel**, core model **37 s** and POD simulation 13 min 5 s, giving threshold 10.840 and 19 outliers (logs/stage8.log:25,49,89,104,106; results/baypass/ghana_core_baypass.log). No cost of determinism is quoted here, because the record cannot support one. Runs (ii) and (iii) are **both** single-threaded and differ in panel size, not in thread count, so the 23 s to 37 s gap measures a 16.6 per cent larger SNP panel rather than the price of single-threading; an earlier version of this note read that 14 s as the cost of determinism, which was wrong. No multithreaded core-model runtime survives anywhere: `grep "threads:" logs/*.log` returns exactly three lines (3,916 SNPs at 1 thread, 3,359 at 12, 3,359 at 1), the 8-thread artefacts in results/baypass_8thread_archive/ are outputs with no log, and the only multithreaded log, logs/stage8_pod_rerun.log, says at line 4 "core model output already present — skipping", so it never timed a core model. The "~26 s" in the script comment is that comment's own figure for the 3,359-SNP panel and is not this run's runtime. What the record does support is that determinism is cheap relative to the POD simulation, which dominates the wall time at 13 min 5 s.

**[Unknown]** The script comment describes the non-deterministic pair as two 8-thread runs, but the only retained multithreaded log (logs/stage8_pod_rerun.log:1) records 12 threads, and no log of an 8-thread pair survives. The thread count of the runs that established the non-determinism therefore cannot be confirmed from the record; the non-determinism itself, and the single-threaded remedy, are what the archived outputs and the script support.

**Annotation (Stage 9).** Outlier positions are intersected with protein-coding gene features from PlasmoDB-71 using bedtools, and cross-referenced against a panel of antimalarial resistance loci and against antigenic and invasion gene families. The cross-reference list is config.KNOWN_RESISTANCE_LOCI, an eight-gene-ID dict hard-coded at config.py:173-182 and read at step09_annotate.py:178 — not the Stage 10 or Stage 12 marker panel, so no number in this manuscript depends circularly on either. The panel was described in the original manuscript as "eight validated antimalarial resistance loci". That description was wrong, and the corrected version offered in the previous revision — that *pfcoronin*, *pfap2mu* and *pfmdr2* are "candidate rather than WHO-validated markers" — was also wrong. Checked against WHO's compendium of molecular markers v1.0 (December 2025): *pfap2mu* and *pfmdr2* appear in it at **no tier at all**, and *pfcoronin* appears at the **potential**

tier only, which is below candidate. None of the three is a candidate marker in WHO's sense. The list also omitted PF3D7_0629500 (*pfaat1*), which the thesis proposal names as a panel member and which the compendium likewise does not contain. That omission is why the scan's second-ranked gene surfaced as an anonymous transporter rather than as a hit on a pre-specified candidate.

[Observed] **That omission has not been repaired in code, and the manuscript should not claim it has.** scripts/config.py on the server still defines KNOWN_RESISTANCE_LOCI as the original eight entries — *pfcrt*, *pfmdr1*, *pfdhfr*, *pfdhps*, *pfk13*, *pfmdr2*, *pfcoronin*, *pfap2mu* — with no PF3D7_0629500, and its modification time precedes the final re-run. The final Stage 9 cross-reference accordingly printed "None recovered" and listed all eight as absent from the outlier set (logs/rerun_5to9.log:44-51), and AAT1 appears in results/outlier_genes.csv with an empty family field. An earlier version of this paragraph stated that *pfaat1* "has been added to config.KNOWN_RESISTANCE_LOCI and the cross-reference re-run"; that is not what the reported run did.

[Proposed] Add PF3D7_0629500 to config.KNOWN_RESISTANCE_LOCI and re-run step09_annotate.py, so that the AAT1 hit is flagged by the pipeline rather than identified by hand in Section 6.8. It is a one-line change and it affects no estimate, only the annotation label.

> **Three counts that must not be conflated.** The Stage 12 retrieval panel is **nine** genes (Table 3.2), covering **57 codons**. Of those 57, **24 are polymorphic** in these isolates, in **eight** of the nine genes — *pfcoronin*'s three codons are all fixed at the reference residue. And config.KNOWN_RESISTANCE_LOCI, the Stage 9 cross-reference list, is a different **eight** — *pfcrt*, *pfmdr1*, *pfdhfr*, *pfdhps*, *pfk13*, *pfmdr2*, *pfcoronin*, *pfap2mu* — which excludes *pfaat1*. That exclusion is the mechanism by which a real positive was missed, so the counts are named explicitly wherever they appear rather than left to the reader to disentangle.
>
> **Caveat.** The cross-reference intersects gene bodies with no flanking window (--flank defaults to 0 at step09_annotate.py:61 and the final run applied none), so an outlier in near-complete linkage with a panel locus but outside its annotated span is not flagged. This is a live limitation, not a defect of the superseded run that has since gone away. The *pfap2mu*-adjacent SNP is still an outlier in the final run and is still unflagged: results/baypass/outlier_snps.bed lists Pf3D7_12_v3:715,788 at XtX 15.086, ninth of the 19, it is absent from results/outliers_annotated.txt, and PF3D7_1218300 (*pfap2mu*) still starts at 716,700 — the gap is still exactly 912 bp. It is also one of the two outliers falling outside any annotated gene: the 19 outlier SNPs give 17 SNP-gene overlaps across 13 genes (logs/rerun_5to9.log), and the other of the two is the fifth-ranked outlier overall, XtX 20.79 at Pf3D7_11_v3:1,054,447, q = 0.005 — which is also one of the two SNPs the quality audit downgrades (Section 3.11.1). Naming the live *pfap2mu* near-miss is stronger evidence for this caveat than the gene-count arithmetic alone. Outlier counts and gene counts are therefore not interchangeable, and the finding is not "19 outliers in 13 genes". Neither this stage nor Stage 8 applies LD pruning or clumping, so the per-gene outlier SNP counts in Section 6.8 count linked sites separately; Section 3.11 and Section 6.8 state where that matters.

## 3.10 Stage 12 — the resistance marker panel, by reconstructed codon

**What and why.** Section 3.5.1 establishes that the genome-wide store cannot carry most of the canonical resistance codons. This stage answers the marker question directly instead, on the same 1,900 isolates and the same zone labels, in a parallel retrieval that leaves every existing output untouched. It is not a re-analysis of the genome-wide matrix and its results are not merged into it. It is the second attempt at this question. The first, Stage 10, identified markers by matching SnpEff's ANN_HGVS_p protein-change strings against a hand-written list, and an external critique showed that this could not work; Section 3.10.1 sets out why, and Section 3.10.2 the method that replaces it. Every marker number in this manuscript now comes from step12_codon_panel.py.

**[Observed]** The codon panel bypasses the release filter as well as the numalt filter, and that was not previously disclosed. Stage 12 reconstructs codons from every variant record at a panel position, iterating over all ALT alleles without consulting FILTER_PASS. Bypassing numalt == 1 was the deliberate fix that recovered the panel; bypassing FILTER_PASS came with it and was never stated. An audit of all 171 panel positions resolves what it costs. Of the variant records at those positions, 29 fail the release filter, at 29 distinct positions. Among the 1,900 analysis isolates, six non-reference calls are sourced from those 29 records, at three positions, and all three are in pfk13 at WHO-validated marker codons: 1,724,975 (codon 675, A675V) with four carriers, 1,725,315 (codon 561, R561H) with one, and 1,725,339 (codon 553, P553L) with one. None of the six changes a reported residue, for two separable reasons. At codons 561 and 553 every alternative allele at the failing position is synonymous — the reference codons are CGT and CCG, and the ALTs give CGG, CGC, CGA and CCA, CCT, which still encode Arg and Pro — so the one homozygous call among the six, isolate SPT19969 at codon 561, correctly counts as Arg561 and the reported 1,864 of 1,864 is right. At codon 675 the alternative is non-synonymous but encodes Pro, not the validated Val, and all four carriers are heterozygous, so the mixed-call rule made the codon uncalled for them. No screened WHO resistance residue is produced by any filter-failing record, and the tested negative therefore stands. One consequence should still travel with the number: those four codon-675 carriers have well-supported minor alleles — 22% to 28% of 173 to 435 reads at GQ 99 — so codon 675's denominator of 1,845 is the lowest of the pfk13 codons (the panel minimum is 1,834, at pfmdr1 1034), and its exclusions are not random with respect to genotype.

**[Observed]** The single Asn458Asp call was checked against the release's own depth and quality fields and is consistent with a real variant rather than an artefact. The retrieval script and its captured output are archived at provenance_final/audit_scripts/check_n458d.py and provenance_final/audit_logs/n458d.log, so the query can be re-run against the public store; it is a check of MalariaGEN's called fields, not an independent examination of the underlying read alignments. Because one call in 1,866 at a validated marker codon could as easily be an annotation or base-calling artefact as a real variant, the underlying data were retrieved. Codon 458 draws on Pf3D7_13_v3 positions 1,725,626, 1,725,625 and 1,725,624. The middle position carries no variant record. The last fails the release filter and has no non-reference carrier among the 1,900. The first, 1,725,626, passes the filter and is triallelic (REF T, ALT A and C, numalt = 2), which is why the numalt == 1 ingest mask excluded it from the genome-wide matrix. Isolate PF0106-C is homozygous for the second ALT, C, with allele depths of 0, 0 and 40 — forty of forty reads supporting that allele — at DP 40 and GQ 99, the maximum. On the minus strand C complements to G, turning the reference codon AAT into GAT, which is Asp. The call is unambiguous in the release's depth and quality fields, and the reconstruction is arithmetically correct. It is also not the validated allele: WHO validates N458Y, which requires AAT to TAT, a different substitution at the same base. So codon 458 is 0.9995 reference rather than fixed, the one alternative residue is Asp and not Tyr, and no WHO-screened artemisinin marker is observed.

**[Observed]** The reconstruction has a bounded input domain, and the bound was not previously stated. A variant record whose ALT differs in length from its REF is skipped by a continue, but the isolate carrying it is not marked uncalled, so in principle it would be reported with the reference-derived residue. Three further limits travel with that one. The spanning-deletion allele is written "*" and has length one, so where REF is a single base it passes the length test rather than failing it, and is substituted into the codon as though it were a nucleotide, yielding the sentinel residue X rather than a silent reference call. An equal-length multi-base record contributes only its first base. And records are retrieved by anchor position, so a record anchored outside the panel whose REF span covers a codon position is never fetched at all. Absence of a variant record is likewise read as the reference base.

**[Observed]** Those conditions occur in this panel, and on the genotypes retrieved for them they change nothing in it; the retrieval scripts and their captured output are archived under provenance_final/audit_scripts and provenance_final/audit_logs. Of the 125 variant records sitting exactly at the 171 panel positions, 16 carry at least one length-changing ALT and 2 have a multi-base REF; a further 19 records with a multi-base REF span a panel position other than their own anchor — of which 16 are genuinely anchored outside the panel and three are themselves anchored at panel positions, so the "off-anchor" category as first named over-counted by three. Genotypes were retrieved for all 32 affected records across all 1,900 isolates. Three have any homozygous non-reference carrier: *pfcrt* codon 76 with 133, *pfcrt* codon 271 with 138, and *pfdhps* codon 431 with 26. In every case the carriers sit on a supported, equal-length allele — codon 76 on ALT1 C, with none on the 55-base insertion; codon 271 on ALT1 GAACGTAAGA, with none on ALT2, none on the spanning deletion and none on the single-base C; codon 431 on ALT1 G, with none on the spanning deletion and none on the 20-base insertion. No isolate anywhere in the panel carries a length-changing allele as a homozygous call, Of the 62 mixed calls at these records, 58 fall at records the reconstruction actually selects — it selects by panel anchor — and are removed by its mixed-call rule; the remaining four fall at genuinely off-anchor records that the reconstruction never reads, so that rule is not what excludes them and the archived logs do not show what does. The counts reconcile exactly with the reported table: Lys76Thr 133, Gln271Glu 138, Ile431Val 26, and homozygous-reference counts of 1,741 and 1,735 against reported reference-residue counts of 1,741 and 1,735. No residue anywhere in the panel is X, which is consistent with no spanning-deletion allele having been substituted — though that is the panel's own output checked against itself, not an independent confirmation. The two multi-base REF records differ from the reference at their first base only, so applying one base loses nothing here. And none of the 57 codons is record-free. That is weaker than it sounds, and the weaker form is the accurate one: 46 of the 171 panel positions carry no variant record at all, so at those positions the reference base is read for every isolate whether or not that isolate was covered there. Per codon, 21 have a record at all three positions, 26 at two and 10 at only one, which means the called mask for those 10 rests on a single position's genotype. Reading absence of a record as the reference allele is therefore an assumption about callability, not a measurement of it; no per-isolate depth field is consulted anywhere in the reconstruction. The counts come from provenance_final/audit_scripts/check_coverage.py. The limitation is real and bounds what the implementation may be applied to; on these inputs it moves no published number that this evidence reaches. The qualification matters: the archived per-allele check covers the two records carrying both supported and unsupported alleles while the carrier conclusion spans three, and the four off-anchor mixed calls are not covered at all. Nothing here shows a published frequency to be wrong, and nothing here closes the question for every branch either. What still rests on an assumption is callability: a codon is called when its constituent genotypes are present and unmixed, which is a statement about the genotype array and not an independent measurement of per-isolate coverage.

### 3.10.1 Why matching annotation strings cannot find the markers

**Failure mode (i): a resistance allele that is the reference residue can never appear as an ALT protein change.** SnpEff annotates changes away from the reference, so if 3D7 already carries the resistance-associated residue, the only annotation at that codon names the reversion. *pfdhps* 437 is exactly this case, and the official Pf8 methodology says so outright — the sentence was verified in the source document: *"3D7 sequence translates as a G at 437, this is thought to be the mutant allele, i.e. 3D7 does not have the wild type allele."* The only annotation at that codon is Gly437Ala, the revertant. Stage 10 searched for "Ala437Gly", matched nothing, and **A437G was absent from the panel entirely** — not reported as rare, not reported as monomorphic, simply not there. Its true frequency is **0.897** over 1,892 called isolates, the second-highest resistance-residue frequency in the panel after *pfdhfr* S108N at 0.908. A locus of that frequency was missing

from a manuscript about antimalarial resistance, and no internal consistency check could have found it, because nothing was inconsistent.

[Observed] **Failure mode (ii): a residue reachable only by two changes within one codon is never annotated.** *pfcrt* codon 75 carries two variant records, 403621 (A>G, annotated Asn75Asp) and 403623 (T>A, annotated Asn75Lys). Asn→Glu requires both. No single annotation names N75E, which is the residue the CVIET haplotype actually carries and the residue MalariaGEN reports; N75E consequently never appeared in Stage 10's panel at all. The previous version of this manuscript described the resulting disagreement with MalariaGEN as a "naming discrepancy" in which "Pf8's SnpEff annotation labels the ALT as Asp while MalariaGEN calls it Glu", and resolved it by adopting MalariaGEN's nomenclature. That diagnosis was wrong. It was not a naming difference: Stage 10 was reporting the frequency of a residue that does not exist in these isolates, under a method structurally incapable of producing the one that does. *The claim of a naming discrepancy at* pfcrt *75 is withdrawn.*

### 3.10.2 The method: reconstruct the codon, report every residue

The script works at the codon rather than at the annotation string. Each requested codon is mapped to its three genomic positions through the gene's CDS, read from the same PlasmoDB-71 GFF the rest of the pipeline uses: CDS segments are concatenated in coordinate order, reversed for minus-strand genes so that the order is translation order, the total length is asserted to be a multiple of three, and codon *i* is positions 3*i*−2 to 3*i* of that sequence. The reference bases at those three positions are read from the 3D7 reference genome FASTA, reverse-complemented on the minus strand, and translated, so the reference residue is **read from the genome rather than assumed from the marker's name**. Every variant record in the Pf8 Zarr at any of the three positions is then collected — **regardless of numalt and regardless of is_snp** — each isolate's called alleles are substituted into the reference codon, and the result is translated. The output is the full residue spectrum at each codon, **the reference residue included and labelled as such**, so a resistance allele cannot hide in the reference and a residue requiring two substitutions cannot be missed.

[Observed] **Mixed calls make the codon uncalled.** A heterozygous call at any constituent position makes the codon ambiguous — the two reads imply two different residues — so the isolate is recorded as uncalled at that codon rather than assigned one of them (step12_codon_panel.py:363-368). This is a deliberate departure from the Stage 4a and Stage 10 rule, which resolves a mixed call to the allele with more reads (Section 3.7, Section 5.3). The reason for the departure is that a codon is a three-position object: resolving each position independently by read depth can manufacture a residue that no single parasite in the infection carries. The cost is a per-codon denominator slightly below 1,900, and the benefit is that no reported residue is a construction artefact. The mixed-call tallies are small and concentrated, but they are **position-level counts, not isolate counts**, and the two must not be conflated: the script accumulates n_mixed over all three positions of a codon, so an isolate mixed at two positions is counted twice. logs/step12_codon_panel.log records 52 mixed calls at *pfcrt* 75, 27 at *pfcrt* 74, 26 at 76, 25 each at 220, 271 and 371, 24 at 326, 3 at *pfaat1* 258, 2 at *pfap2mu* 160 and 1 at *pfcrt* 73. The isolate-level loss at *pfcrt* 75 is **26**, not 52: the codon is called in 1,874 of 1,900, and 75 is the two-SNP codon (403621 and 403623), so 26 × 2 = 52 — 52 isolates could not have been lost from a denominator of 1,874. The two quantities diverge the other way where plain missingness dominates: 25 mixed calls but 31 isolates uncalled at codon 220, 24 against 26 at 326. The denominator cost at a codon is therefore read from its n_called, not from its mixed-call tally.

[Observed] **Scale and denominators.** 57 codons across the nine panel genes, spanning 171 genomic positions, of which 125 carry at least one Pf8 variant record (125 records in total); a position with no record contributes

its reference base, which is the correct behaviour and not an omission. Metadata transfer 211 MB, genotype transfer 2.53 GB. No codon was left without a variant record anywhere in its three positions. Per-codon denominators run from **1,834 to 1,900** of the 1,900 isolates, so every frequency in Section 6 is reported against its own denominator and the denominators are not interchangeable between codons.

[Calculated] **A built-in check on the CDS mapping.** The reference residue the script derives from the genome matches the published wild-type (or, at *pfdhps* 437, the published reference) residue at **all 57 codons** — C72, V73, M74, N75, K76, A220, Q271, N326, I356, R371 in *pfcrt*; N86, Y184, S1034, N1042, D1246 in *pfmdr1*; A16, N51, C59, S108, I164 in *pfdhfr*; I431, S436, G437, K540, A581, A613 in *pfdhps*; F446, N458, C469, M476, Y493, G533, R539, I543, P553, R561, P574, C580, R622, A675 and the remaining nine *pfk13* codons; T484, I492; G50, R100, E107; S160; S258, F313. Agreement at all 57, across **three minus-strand genes** (*pfk13*, *pfmdr2*, *pfaat1*) and **six plus-strand ones** (*pfcrt*, *pfmdr1*, *pfdhfr*, *pfdhps*, *pfcoronin*, *pfap2mu*) — the strand column of results/codon_panel_residues.csv — is a direct test of the CDS concatenation, the strand handling and the codon-boundary arithmetic, and it passes. An earlier version of this sentence said "four minus-strand genes" and "five plus-strand ones", contradicting its own parenthetical list of three; the total of nine was right, which is probably why it survived, but a wrong strand census in the one sentence offered as proof that strand handling was tested is exactly the sentence that must be right.

> **Two honest qualifications about the retrieval.** First, the reference FASTA fetched is the PlasmoDB-54 3D7 genome while the CDS coordinates come from the PlasmoDB-71 GFF. The 3D7 assembly sequence is stable across those releases and the 57-codon reference-residue check above would have failed loudly on any coordinate drift, but the version pair is a mismatch on its face and is stated rather than smoothed over. Second, the script retrieves FILTER_PASS and is_snp but applies neither: unlike Stage 10, which kept FILTER_PASS records only, Stage 12 uses **every** record at the three positions. For reconstructing a codon that is the defensible choice — a non-PASS record still tells you which base an isolate carries, and discarding it would silently leave the reference base in place — but it is not what the previous section said, and the fraction of the 125 records that are non-PASS was not tabulated.

[Unknown] How many of the 125 variant records used are non-PASS, and whether any ALT allele at a panel position was an indel. Both are now answered in Section 3.10 and this entry is retained only to record that they once were not: 29 records fail the release filter, 16 carry a length-changing ALT, and no isolate carries a length-changing allele as a homozygous call. The script skips an ALT whose length differs from REF and writes a note when it does; the log reports 24 notes but prints only the first ten, all of which are mixed-call notes, so the remaining fourteen are not on the record. Re-running with the note cap lifted would settle both questions and change no frequency unless an indel ALT is present.

```
python step12_codon_panel.py
```

### 3.10.3 The marker list and where it comes from

**Provenance of the codon list.** The *pfk13* codons are taken from WHO's compendium of molecular markers of antimalarial drug resistance, **version 1.0, December 2025**, and cover all 14 validated and all 6 candidate artemisinin-partial-resistance variants — 20 variants at 19 distinct codons, because C469Y (validated) and C469F (candidate) share codon 469 — plus four further propeller-region codons (449, 481, 515, 527), which brings the reconstructed *pfk13* total to 23. Those four are described in the retrieval script's own comment as "the potential-tier codons that fall in the same region" (step12_codon_panel.py:85); that is the script's annotation and not an extraction from the compendium, so their tier is reported as unextracted rather than asserted here. The non-*pfk13* codons are the 25 markers MalariaGEN itself genotypes and publishes for Pf8 — which is also what makes the per-sample concordance check in Section 3.10.6 possible — plus *pfcrt* 73 (needed to read the 72–76 motif as a motif), *pfdhps* 431, *pfmdr2* 492, *pfcoronin* 50/100/107, *pfap2mu* 160

and *pfaat1* 258/313. The previous *pfk13* list was written from memory. It held 13 distinct protein changes rather than 14 (a duplicated M476I record inflated the apparent count), classified **P441L and C469F as validated when the compendium classifies both as candidate**, and omitted **P574L, G533S and R622I** altogether. R622I — the variant of greatest current concern in the Horn of Africa — was requested but never retrieved, so it became a silent non-result. That is a defect in the marker list, not in the prose, and it is why the panel was rebuilt from the compendium rather than patched.

**WHO status is a property of a variant, not of a gene.** The previous version of Table 3.2 assigned tiers to genes, which is a category error, and in one cell it compounded the error: *pfmdr1* was labelled "WHO-validated (4-aminoquinolines, lumefantrine)" when the compendium contains **no validated or candidate lumefantrine marker of any kind** — verbatim: *"The review did not identify any validated or candidate molecular markers associated with lumefantrine resistance."* No lumefantrine claim rests on *pfmdr1* anywhere in this manuscript. Three further genes are in the panel that the compendium does not contain at any tier — *pfaat1*, *pfap2mu* and *pfmdr2* — and a fourth, *pfcoronin*, is in it at the potential tier only. They stay in the panel because they are scientifically interesting and cheap to genotype once the machinery exists; they must never be described as WHO-classified markers.

| Gene | PlasmoDB ID | Codons reconstructed | Status under WHO compendium v1.0 (December 2025) |
|---|---|---|---|
| *pfcrt* | PF3D7_0709000 | 72, 73, 74, 75, 76, 220, 271, 326, 356, 371 | K76T is the long-established chloroquine determinant and 72–76 is read as the CVIET motif (Section 3.10.5). The compendium's per-variant tiers for *pfcrt* were not extracted in this project, so no tier is asserted; the gene-level label "WHO-validated (4-aminoquinolines)" is withdrawn as a category error. |
| *pfmdr1* | PF3D7_0523000 | 86, 184, 1034, 1042, 1246 | The previous label "WHO-validated (4-aminoquinolines, lumefantrine)" was wrong twice: tiers attach to variants, and the compendium identifies no validated or candidate lumefantrine marker at all. No per-variant tier is asserted here. |
| *pfdhfr* | PF3D7_0417200 | 16, 51, 59, 108, 164 | N51I, C59R, S108N and I164L are the classical pyrimethamine determinants and are among the markers MalariaGEN publishes for Pf8. No compendium tier asserted. |
| *pfdhps* | PF3D7_0810800 | 431, 436, 437, 540, 581, 613 | Classical sulfadoxine determinants, same provenance. Note that 3D7 carries the resistance-associated residue at 437, so "A437G" names a reversion in annotation space and the reference residue in sequence space (Section 3.10.1). No compendium tier asserted. |
| *pfk13* | PF3D7_1343700 | 252, 441, 446, 449, 458, 469, 476, 481, 493, 515, 527, 533, 537, 538, 539, 543, 553, 561, 568, 574, 580, 622, 675 | Taken from the compendium, and the tiers are kept distinct. Validated (14): F446I, N458Y, C469Y, M476I, Y493H, G533S, R539T, I543T, P553L, R561H, P574L, C580Y, R622I, A675V. Candidate (6): E252Q, P441L, C469F, N537I, G538V, V568G. That is 14 + 6 = 20 variants, and because codon 469 carries one of each (C469Y validated, C469F candidate) they fall at 19 distinct codons; the panel reconstructs 23 *pfk13* codons, the four extra ones being 449, 481, 515 and 527. Those four are propeller-region codons that the retrieval script's own comment calls potential-tier; the compendium's tier for them was not extracted, so none is asserted. The previous version called P441L and C469F |

| Gene | PlasmoDB ID | Codons reconstructed | Status under WHO compendium v1.0 (December 2025) |
|---|---|---|---|
| | | | validated, duplicated M476I, and omitted P574L, G533S and R622I. |
| *pfmdr2* | PF3D7_1447900 | 484, 492 | Absent from the compendium at every tier. Retained on literature interest only. The previous "Candidate" label is withdrawn. |
| *pfcoronin* | PF3D7_1251200 | 50, 100, 107 | In the compendium at the potential tier only — below candidate, and not validated. All three codons are fixed at the reference residue in these isolates. |
| *pfap2mu* | PF3D7_1218300 | 160 | Absent from the compendium at every tier. Retained on literature interest only. The previous "Candidate" label is withdrawn. |
| *pfaat1* | PF3D7_0629500 | 258, 313 | Absent from the compendium at every tier. S258L has a demonstrated but modest, epistasis- and background-dependent effect: a 1.5-fold rise in chloroquine IC50, measured in a single chloroquine-resistant Southeast Asian line (NHP4026), with epistasis between S258L and F313S and an explicit *pfcrt*-background caveat. It *potentiates* chloroquine resistance rather than conferring it, and the previous description — "functionally validated in vitro for chloroquine resistance" — overstated that. Amodiaquine was tested in the source study and the shifts were "well below the threshold associated with clinical resistance"; the authors reject the other-quinoline hypothesis and name chloroquine "the primary selective force". The 0 to 97 per cent longitudinal rise is Gambian, 1984–2014, and the study says nothing about Ghana. Also absent from config.KNOWN_RESISTANCE_LOCI (Section 3.9). |

*Table 3.2. The nine panel genes, the codons Stage 12 reconstructs in each, and the evidential status of their resistance determinants. The previous version of this table assigned WHO tiers to whole genes; the compendium classifies variants, so those labels were a category error and are replaced. Three of the nine genes appear in the compendium at no tier at all (pfmdr2, pfap2mu, pfaat1) and a fourth only at the potential tier (pfcoronin), so* ***four*** *of the nine carry no WHO classification that could support a resistance claim. Where the compendium's per-variant tiers for pfcrt, pfmdr1, pfdhfr and pfdhps were not extracted in this project, no tier is asserted rather than guessed; on that stricter reading — genes for which this manuscript asserts no tier at all — the figure is eight of the nine, and the two readings are stated separately rather than added together.*

**[Unknown]** The compendium's per-variant tiers for the *pfcrt*, *pfmdr1*, *pfdhfr* and *pfdhps* codons in this panel, and the tier of the four extra *pfk13* propeller codons (449, 481, 515, 527). Only the artemisinin section's validated and candidate lists, the lumefantrine statement and the presence or absence of the four non-classical genes were extracted. Nothing in Section 6 rests on a WHO tier for these four genes or for those four codons.

### 3.10.4 Frequencies, intervals and tests

**Point estimates.** The frequency of a residue is computed per codon, as the count of isolates carrying it divided by the count of isolates with a non-missing reconstructed codon there, within the 1,900-isolate analysis set. Denominators therefore differ between codons and between zones, and are reported per cell. Because the spectrum is complete, the frequencies at a codon sum to one; the reference residue's frequency is reported on the same footing as any other, which is what makes *pfdhps* G437 at 0.897 visible at all.

**Wilson score intervals.** Confidence intervals are Wilson score intervals rather than Wald intervals. This is not a cosmetic choice. Many codons sit at or very near 1.000 — 33 of the 57 are fixed at the reference residue in every called isolate — and the Transitional zone contributes 25 isolates. A Wald interval at p-hat = 0 or 1 has zero width, and at small n it can extend outside [0, 1]; the Wilson interval is well behaved in both cases and has better coverage at the extremes, which is where this panel's most consequential readings sit. It is what lets a zero be reported as a bounded negative: *pfdhfr* IRNL at 0.000 carries the interval 0.000 to 0.002 over 1,874 complete cases, and the Transitional zone's own zero carries 0.000 to 0.133 over 25.

**Heterogeneity test.** Between-zone variation at each codon is tested by a chi-squared test on the four-zone count table for that residue. The test is of heterogeneity, not of a north-south trend: a significant result says the four zones differ, and the direction of any cline must be read from the frequencies themselves — the more so because the Transitional zone is the second-most-northern of the four, so a frequency ordering is not a geographic ordering. The test family is residues rather than codons, as the first sentence implies: results/codon_panel_residues.csv carries **27 residue-level chi-squared p-values across the 24 polymorphic codons**, because *pfdhps* 436 contributes four tests of its own (Ser $4.5 \times 10^{-5}$, Ala $1.6 \times 10^{-3}$, Tyr $2.8 \times 10^{-2}$, Phe $8.1 \times 10^{-2}$) rather than one. No multiplicity correction is applied across those 27 tests, which Section 6 states when quoting individual p-values; the findings that carry interpretive weight — *pfdhfr* 51/59/108 at p = $2.4 \times 10^{-4}$, $4.8 \times 10^{-5}$ and $9.4 \times 10^{-7}$, *pfdhps* 437 at p = $1.2 \times 10^{-6}$, and *pfaat1* S258L at p = $1.9 \times 10^{-18}$ — would survive any conventional correction over 27 tests, the largest of them sitting well under $0.05/27 = 1.9 \times 10^{-3}$. Codons fixed at the reference residue have no test and the chi-squared column is empty for them, which is reported as absence of a test rather than as a non-significant result.

[Observed] **Coverage of the panel.** 57 codons reconstructed across nine genes; **24 polymorphic** in these 1,900 isolates and **33 fixed at the reference residue** in every called isolate. The 24 polymorphic codons carry 26 non-reference residues between them, because *pfdhps* 436 alone shows three (Ala, Phe, Tyr) alongside the reference Ser. The polymorphic codons fall in eight of the nine genes; *pfcoronin* contributes none.

> **The *pfk13* negative, stated with its two limits.** *All 23* pfk13 *codons carry no screened resistance allele in any isolate with a call, twenty-two of them at the reference residue in every called isolate and codon 458 at 0.9995, and the twenty-third, codon 458, carries a single isolate with Asn458Asp — which is not the validated variant at that codon, N458Y — giving the reference residue a frequency of 0.9995 over 1,866 called isolates. No validated or candidate artemisinin-resistance residue was observed at any codon. Two limits must travel with that statement. First, the denominator is per codon, 1,845 to 1,876 of 1,900 across the* pfk13 *codons (1,834 to 1,900 across the panel as a whole), not "all 1,900 isolates at every codon"; the earlier formulation "all 14 validated* pfk13 *codons at 0.000 in all 1,900 isolates" is wrong on the codon count, on the denominator and on the panel it was computed from, and is withdrawn. Second, the sampling period is 2009 to 2021*, so this is a historical negative and says nothing about the frequency today. What is now defensible, and was not before, is that the negative covers the complete validated set — 14 validated and 6 candidate variants, 20 in all at 19 distinct codons — rather than an incomplete list written from memory, with the tiers kept distinct.

### 3.10.5 Within-isolate haplotypes, on complete cases only

**Why this was needed.** The previous version reported the *pfdhfr* "triple mutant" as the range of its three marginal frequencies, **0.786 to 0.908**. That is not a haplotype frequency and cannot be one: marginal frequencies say how common each residue is, not whether the three occur in the same parasite. The range also sat *above* the value a haplotype frequency must respect, since a haplotype cannot be commoner than its rarest component. The quoted range is withdrawn.

**Method.** For each named haplotype, an isolate is counted only if it has a reconstructed residue at **every** constituent codon; isolates uncalled at any one of them are excluded from both numerator and denominator rather than being assumed wild type. The component frequencies are then recomputed **on that same complete-case set**, so that the arithmetic constraint — haplotype frequency ≤ minimum component frequency — is checkable rather than approximate, and the script prints the minimum component beside the haplotype so the check is visible in the log. The full distribution of observed haplotypes is printed, not only the target, because a target frequency on its own cannot distinguish "one dominant haplotype" from "a target plus several near-competitors". Four haplotypes are reconstructed: *pfdhfr* 51/59/108 (IRN), *pfdhfr* 51/59/108/164 (IRNL), *pfcrt* 72–76 (CVIET) and *pfdhps* 437/540 (GE).

[Observed] Complete-case coverage is high: 1,876 of 1,900 isolates (98.7 per cent) are called at all three *pfdhfr* codons, 1,874 at all four, 1,873 (98.6 per cent) at all five *pfcrt* codons, and 1,888 (99.4 per cent) at both *pfdhps* codons. On those sets the *pfdhfr* triple mutant IRN is carried by 1,446 isolates, **0.771 (95 per cent CI 0.751 to 0.789)** — below the minimum component frequency on the same set (51I = 0.788), as it must be, and well below the superseded 0.786 to 0.908 range. At roughly three-quarters it is common but not close to fixed, and it is not a single dominant haplotype: NRN sits at 0.116, the fully wild-type NCS at 0.091, ICN at 0.018 and NCN at 0.004. The quadruple IRNL is absent, 0 of 1,874, 0.000 (0.000 to 0.002). *pfcrt* CVIET is carried by 133 of 1,873, **0.071 (0.060 to 0.084)**, and is genuinely a unit: the only two 72–76 haplotypes observed anywhere in the dataset are CVMNK at 0.929 and CVIET at 0.071, with nothing intermediate — demonstrated here rather than inferred, as it previously was, from six similar marginals. *pfdhps* GE (437G with 540E) is 18 of 1,888, **0.010 (0.006 to 0.015)**, against GK at 0.887 and AK at 0.103.

### 3.10.6 Per-sample concordance against MalariaGEN's published calls

**Why the previous check was not enough.** The previous version validated this stage by comparing marginal frequencies against MalariaGEN's published marker genotypes and reporting that "21 of 23 markers agree within 0.03". Two things are wrong with that. Agreement of marginal frequencies does not establish that two pipelines called the *same isolates* the same way — two pipelines could disagree on every isolate and still produce identical frequencies — and the two markers that did not agree were *pfcrt* 75, where the disagreement was not a tolerance failure but the structural failure of Section 3.10.1. The "21 of 23 within 0.03" claim is withdrawn and replaced by a call-by-call comparison.

**Method.** For each of the 25 codons MalariaGEN publishes for these isolates in Pf8_drug_resistance_marker_genotypes.tsv, derived from a different pipeline, the reconstructed residue is compared with MalariaGEN's call isolate by isolate, restricted to isolates where both pipelines made a call. The comparison is on residues, not on genotype codes, so it is a comparison of the biological statement each pipeline makes.

[Observed] **47,012 of 47,012 calls agree — 100.0000 per cent, across all 25 markers**, with zero disagreements at any marker. *pfcrt* 75 now agrees exactly, 1,874 of 1,874, because codon reconstruction yields Glu where string matching yielded Asp. Per-marker denominators run from 1,834 (*pfmdr1* 1034) to 1,900 (*pfcrt* 72 and 356). Full table in results/codon_panel_concordance.csv.

> **The *kelch13* corroboration is weaker than previously claimed.** Pf8 also publishes a *kelch13* resistance status, and the previous version leaned on it as independent support for the artemisinin negative. That support must be stated far more weakly. Pf8 assigns the status over the whole 349–726 propeller domain and classifies any missing call in that range as undetermined — "Any missing call in amino acids 349-726 → Missing → Undetermined" — so only **36 isolates** carry a complete status. Thirty-six complete statuses are weak corroboration for per-codon calls on roughly

1,860 isolates, and the two are not equivalent evidence. The per-sample concordance above, on 25 markers and 47,012 calls, exercises the same CDS mapping, strand handling and codon arithmetic used throughout the panel, so it is reassuring about the method; but it contains no pfk13 codon at all, so it is not evidence about the artemisinin negative, for which these 36 statuses are the only external corroboration that exists. The *kelch13* status is mentioned for completeness and nothing rests on it.

**What the external cross-check bought, twice.** The first version of this validation caught Stage 10's ANN_Allele indexing bug, which reported *pfmdr1* Asn86Tyr as "Asn86Asp" and Tyr184Phe as "Tyr184Ser" — correct frequencies, wrong variant names — and which no internal consistency check could have found, because nothing was out of range or implausible. The second version, moving from marginal agreement to call-by-call agreement, is what made the *pfcrt* 75 problem legible as a method failure rather than a tolerance failure. Both times the error that escaped internal checking was an error of external correctness: whether the marker list matched an authoritative source, whether the annotation direction matched the reference genome, whether a co-occurrence claim had co-occurrence data behind it. The verification architecture in this project was built to catch stale numbers, and it did. It was not built to ask whether the list was the right list.

## 3.11 False-discovery accounting in the XtX scan

**The problem.** A quantile threshold is not a significance test. Calling a SNP an outlier when its XtX exceeds the 0.999 quantile of the pseudo-observed null guarantees that 0.1 per cent of strictly neutral SNPs will be called, by construction, however neutral the genome is. The original Stage 8 reported a threshold and a count of SNPs above it, with no statement of how many were expected under the null — so the outlier list had no error rate attached, and the reader had no way to judge how much of it to believe.

**What the step now reports.** step08c_calibrate_xtx.py was extended to report four things alongside the threshold.

12. **Expected false positives at the threshold.** The quantile and the panel size give this immediately: at the 0.999 quantile over 3,916 SNPs, 3.9 are expected above threshold under the null.
13. **An expected-null-exceedance ratio at the threshold — expected false positives under the assumed null divided by the number of SNPs actually called. It is an expectation under that null, not a measured false-discovery fraction.** With 19 SNPs above a threshold of XtX 10.840 and 3.9 expected, that is 20.6 per cent.
14. **Per-SNP p-values against the pseudo-observed null.** Each observed XtX is ranked within the sorted pseudo-observed distribution and assigned p = (n_pod − rank + 1) / (n_pod + 1), which is a Monte Carlo one-sided tail probability under that fitted null and is bounded below by 1/(n_pod + 1) rather than being reported as zero.
15. **Benjamini–Hochberg q-values**, computed over all 3,916 SNPs by the standard step-up procedure with monotonicity enforced (step08c_calibrate_xtx.py:114, with the write at :139). The q-values are the values in Table 3.3, and logs/stage8.log:115-119 prints the counts at q ≤ 0.01, 0.05, 0.1 and 0.2. The previous revision had to report the per-SNP artefact as unusable, because results/baypass/xtx_all_snps.csv on the server was **0 bytes** — no header, no rows, no q-value column. That file has since been found truncated and regenerated from the intact BayPass outputs, so a per-SNP file is on disk again; Section 3.13.1 records the truncation, the regeneration and the consistency checks the regenerated file passed. The citable sources for the final q-values remain results/baypass/xtx_calibration.json and the printed counts in the Stage 8 log, and those are what this manuscript cites; the restored per-SNP file is now consistent with them rather than a substitute for them.

[Calculated] The restored per-SNP q-values reproduce the calibration counts exactly. The 19 outliers' q-values in results/outlier_quality_check.csv give 8 SNPs at q ≤ 0.01, 10 at q ≤ 0.05, 13 at q ≤ 0.10 and 17 at q ≤ 0.20, which are the four counts in results/baypass/xtx_calibration.json and in Table 3.3. That is an independent arithmetic check on the regeneration, and it passes.

| Criterion | SNPs |
|---|---|
| XtX > 10.840 (POD 0.999 quantile) | 19 |
| q ≤ 0.20 | 17 |
| q ≤ 0.10 | 13 |
| q ≤ 0.05 | 10 |
| q ≤ 0.01 | 8 |

*Table 3.3. The XtX outlier set under the quantile rule and under Benjamini–Hochberg control. Of the 19 SNPs above the quantile threshold, 13 also satisfy q ≤ 0.10 and 6 do not. Values from results/baypass/xtx_calibration.json, and reproduced from the restored per-SNP file (Section 3.13.1).*

[Interpreted] **How the two are reported.** Both, with their roles distinguished. The quantile list is a **ranking**: it orders SNPs by evidence and is the right object for asking which genes to look at. The q-value subset is the **test**: it is the set that survives multiplicity control at a stated error rate. Sections 6.8 and 7 therefore report 13 SNPs as the finding and 19 as the ranking, rather than choosing one number and suppressing the other. The step prints the comparison explicitly so that the two cannot drift apart in later write-ups. Neither number is a gene count: 17 of the 19 fall in annotated genes, across 13 genes, and two are intergenic (Section 3.9). A third qualification now applies to both lists: two of the 19 are downgraded to lower confidence on the quality grounds of Section 3.11.1, and one of those two is fifth in the ranking.

> **Caveat.** Neither procedure addresses non-independence. No LD pruning or clumping is applied before counting, so two outlier SNPs in strong linkage within one gene enter the count as two tests and are reported per gene as two SNPs. Benjamini–Hochberg is robust to positive dependence in the sense of controlling the FDR, but the per-gene SNP counts in Section 6.8 should not be read as independent lines of evidence. The *pfaat1* pair, two sites 883 bp apart — Pf3D7_06_v3:1,214,350 and 1,215,233, both PF3D7_0629500 — is reported as one signal there. It is not, however, the clearest instance, and that superlative is withdrawn: it was carried over from the superseded 15-outlier panel, on which *pfaat1* did top the $r^2$ ranking. Recomputed on interim/ghana_core_snps.npz (3,916 SNPs, 1,900 isolates, haploid allele matrix, pairwise-complete) over every same-chromosome outlier pair within 200 kb, $r^2$ is 0.828 for the PF3D7_0221200 pair at Pf3D7_02_v3:855,467 and 855,665 (198 bp), 0.674 for the ACS8/PF3D7_0215300 pair at 630,290 and 630,420 (130 bp), then 0.602 for *pfaat1*, with the PF3D7_0831400 pair at 25 bp and $r^2 = 0.398$. The *pfaat1* pair is third by $r^2$ and the widest-spaced of the four same-gene pairs; where a strongest instance is wanted, it is the PF3D7_0221200 pair, and the measure is stated with it.

### 3.11.1 Quality audit of the outlier set

**What and why.** The outlier list is the one result in this manuscript that names individual sites, so it is the one most exposed to a mapping or calling artefact masquerading as differentiation — the specific worry being that several of the hits lie in exported-protein genes whose families are repetitive. That worry had been raised but never tested. This check tests it by reading what the release itself says about each site's call quality, which requires no new inference and no new model.

**Which annotations, and against what.** For each of the 19 XtX outliers the Pf8 per-variant annotations were pulled from the Zarr and compared against **the other 3,897 SNPs of the same scanned panel** — the 3,916 that entered Stage 8, less the 19. The comparator is deliberate: the scanned panel has already passed FILTER_PASS, biallelic and is_snp, the depth floor, variant missingness and MAF ≥ 0.05, so comparing the

outliers to it asks whether they are unusual *among the sites the scan could have called*, whereas comparing them to all Pf8 sites would confound the scan's own ascertainment with the question. The fields read are the mapping and strand-bias annotations MQ, FS and SOR; the variant-recalibration score VQSLOD and the FILTER_Low_VQSLOD, NEGATIVE_TRAIN_SITE and POSITIVE_TRAIN_SITE flags; the quality-by-depth statistics QD, QUAL and DP; the rank-sum tests BaseQRankSum, MQRankSum and ReadPosRankSum; the allele number AN; the heterozygosity statistics ExcessHet and InbreedingCoeff; the culprit field, which names the annotation most responsible for a site's VQSR score; and RegionType. Each outlier also carries a percentile within the panel for the main metrics, and three local-context measures computed from the panel itself: the number of scanned SNPs within 1 kb and 10 kb, the local mean MQ in those windows, and the number of other outliers within 10 kb. Every outlier is then assigned a flag level — clean, watch or flag — with a verdict string naming the annotation responsible. Per-SNP table in results/outlier_quality_check.csv.

[Observed] Of the 19, **15 are clean, 2 are watch and 2 are flag**. All 19 are RegionType Core, none has MQ below 50 (the minimum is 59.56), and none shows strand bias or read-position bias. Every significant difference against the rest of the panel runs in the outliers' favour: higher VQSLOD ($p = 0.008$), QD ($p = 0.016$) and QUAL ($p = 0.007$). The culprit field reads MQRankSum for all 19, but also for 99.6 per cent of the rest of the panel, so it carries no information here and is reported only to say so.

[Observed] **The exported-protein concern is not supported.** The eight SNPs in PTP7, PIESP2, GEXP22, REX2, PF3D7_0831400 and PF3D7_0221200 are the cleanest subset in the panel rather than the most suspect: FS = 0.00 for all eight, MQ median 60.00, and VQSLOD significantly higher than the rest. The families are repetitive; these particular sites are in uniquely mappable sequence, and the two statements are compatible. The concern was worth raising and the answer is that it does not bite here.

[Observed] **Two SNPs are downgraded to lower confidence, on different grounds.** Pf3D7_11_v3:1 054 078 (*SNRPF*, XtX 13.98, $q = 0.031$) and Pf3D7_11_v3:1 054 447 (intergenic, XtX 20.79, $q = 0.005$, **fifth-ranked in the whole scan**) lie 369 bp apart and are the two lowest-InbreedingCoeff sites in the panel (0.147 at the 0.08 percentile and 0.193 at the 0.13 percentile); both also sit in the lowest percentile of QD (12.91 and 11.00, both at the 0.3 percentile), and 1054447 has VQSLOD 5.58, at the 2.4 percentile. 1054078 carries **ExcessHet = 69.59, the only nonzero value among all 3,916 SNPs** and roughly 400 times the genome-wide maximum. Both are reported as lower confidence from here on.

[Hypothesized] In a near-haploid organism an excess of heterozygous calls at one site, with clean MQ (59.99) and FS (0.00), points to collapsed paralogy or copy-number variation rather than to mismapping: reads from two similar copies piling onto one locus produce apparent heterozygotes without degrading mapping quality or introducing strand bias. Two adjacent sites sharing the signature is consistent with one such region rather than two independent accidents.

[Unknown] Whether the chromosome-11 pair sits in a collapsed duplication. The annotations are consistent with it and do not establish it.

[Proposed] Settle the mechanism with a read-level check: per-sample depth at these two positions as a ratio to each sample's genome-median depth. A collapsed duplication gives roughly twice the median across most samples; a calling artefact at a single-copy locus does not. The data are in the local store, the check needs no new retrieval, and it would either retire the fifth-ranked outlier or restore it to full confidence. Until it is done, the downgrade stands and Section 6.8 reports both SNPs with it.

> **What this check can and cannot do.** It compares annotations, not reads. A site can carry excellent MQ, VQSLOD, FS and SOR and still be wrong, because every one of those statistics is computed from the same alignment; if reads are

> mismapped consistently, the annotations will look clean. So "no evidence of an artefact" is the finding, and "no artefact" is not. The check is also not a test of selection: that the 19 are well-called says nothing about why their XtX is high. Its value is narrow and real — it removes a specific alternative explanation for eight of the hits, and it finds a specific problem at two that the XtX ranking alone would never have surfaced.

## 3.12 Stage 11 — is zone separable from sequencing study?

**What and why.** The manuscript's design-limitations section listed three confounders of zone: collinearity with administrative region, coordinate resolution at admin level 1, and differing collection years. A fourth was missing. Pf8 isolates arrive through named partner studies, and in Ghana those studies are close to nested within zones. Where no study contributes isolates to both zones of a comparison, that comparison is simultaneously a between-batch comparison, and no analysis of these data can separate the two. This stage quantifies that rather than asserting it. It is read-only and changes no pipeline output.

**Method.** Four diagnostics are computed from the sample manifest and the Stage 3b quality metrics, over the **4,905 release-QC-passing isolates** that carry both a study and a zone label — not over the 1,900-isolate analysis set. The denominator is deliberate: batch structure is a property of how the isolates were collected and sequenced, so it is measured before clonal pruning and the missingness cuts rather than after them.

16. A study-by-zone cross-tabulation, and for each of the six zone pairs the number of sequencing studies contributing isolates to both.
17. Whether studies differ in within-host fixation index within a zone — zone held constant, so that study varies together with collection year, finer geography, recruitment and host biology, which tests for study-associated heterogeneity rather than isolating a technical batch cause.
18. The same test for call rate within a zone.
19. For the zone pairs that do share a study, whether the within-study difference agrees in sign with the overall between-zone difference. This is the one check that can partially exonerate a gradient, and it is reported whichever way it comes out.

**[Observed]** Eleven studies contribute the 4,905 isolates. 4 of the 6 zone pairs have no sequencing study in common, including Coastal savanna against Guinea savanna — the pair that carries the maximum pairwise FST and the polyclonality contrast. Coastal savanna against Deciduous forest shares 2 studies and Guinea savanna against Transitional zone shares 1; the remaining four share none. Guinea savanna draws on six studies, Deciduous forest on **five**, Coastal savanna on two, and the Transitional zone on one (1083-PF-GH-CONWAY, which supplies all 50 of its isolates at this stage). Within-zone heterogeneity between studies is severe where it can be tested: call rate differs between studies within Deciduous forest at $p = 7.7 \times 10^{-88}$ and within Guinea savanna at $p = 2.8 \times 10^{-197}$ (results/batch_within_zone_tests.csv).

> **Five studies, four tested.** The within-zone tests cover only studies meeting the minimum-n threshold, so results/batch_within_zone_tests.csv records four studies for Deciduous forest (n = 811) against the five that actually contribute to it: the fifth, 1114-PF-PDN-DBS-GH-GHANSAH, supplies 4 isolates and falls below the threshold. The zone has five contributing studies and four testable ones, and the two counts are not interchangeable.

**[Interpreted]** **How this is used.** Section 7 treats zone and study as inseparable for the four non-overlapping pairs and states which headline numbers are affected. This applies to the marker panel as much as to the genome-wide results: the zone contrasts in Section 6, including *pfdhps* G437 at 0.964 in Deciduous forest against 0.875 in Guinea savanna, and *pfaat1* S258L at 0.824 in **Coastal savanna** against 0.544 in Guinea savanna, rest on the same zone labels and inherit the same confounding. The zones are named for S258L

rather than carried over elliptically, because its Deciduous forest value is 0.735, not 0.824 (results/codon_panel_residues.csv, pfaat1 Ser258Leu: Coastal savanna 0.8235, Deciduous forest 0.7351, Guinea savanna 0.5440, Transitional 0.640, n = 1,896, chi-squared 85.64, $p = 1.9 \times 10^{-18}$). The diagnostic cannot be turned into an adjustment: with no overlap there is no stratum in which to estimate a batch effect and a zone effect simultaneously. What it can do is bound the claim, and that is what it is used for. Stage 13 does not touch this: all six of its scenarios use the same zone labels drawn from the same studies, so none of them is a check on the confounding (Section 3.14).

```
python step11_batch_confounding.py
```

## 3.13 Reproducibility of the corrected re-run

Stages 4 to 9 were re-run end to end with the corrections above in place. Two further changes were needed to make that re-run trustworthy, and both are properly part of the methodology.

**Stale-output guards.** Stages 5b, 7b and 8b skipped their work if their output files already existed. Those guards did not check whether the *input* had changed. When the SNP set moved from 3,359 to 3,916, every stage regenerated its inputs, and then ADMIXTURE and TreeMix found their previous output files in place, skipped entirely, and reported success. Only Stage 8 caught the problem, because step08c asserts that its position table and the BayPass output have matching row counts and failed loudly with the correct diagnosis — "Position table has 3,916 rows but the BayPass output has 3,359"; ADMIXTURE and TreeMix carried no such assertion and failed silently. All three guards now require the output to be newer than its input. Had file timestamps not been checked by hand, the previous day's ADMIXTURE and TreeMix results would have been reported as though they came from the new matrix. This is the seventh defect in the chronological sequence this manuscript reports, and it is the one the re-run found by tripping over it rather than one the adversarial audit raised. It is no longer, however, the *only* defect from outside the audit, and that uniqueness framing is withdrawn: Addendum 3 adds four further defects in the analysis code, raised by the external critique, so the manuscript now reports eleven code defects, five of them raised by something other than the audit. The Introduction carries the post-Addendum 3 count, and the ordinal here is chronological position, not a claim about provenance being unique.

**Determinism.** After these changes the pipeline has no unseeded stochastic component in the inferential path: BayPass is single-threaded with a fixed seed (step08b_baypass.sh:37, `NTHREADS="${NTHREADS:-1}"` and `-seed "$SEED"`), the Mantel permutations are drawn from a seeded generator (step06_mantel.py:176, `np.random.default_rng(C.RANDOM_SEED)`, seed 42 as recorded in mantel_result_*.json), TreeMix's jackknife is deterministic given its block size and is invoked with `-k "$BLOCK" -seed "$SEED"`, and ADMIXTURE was already seeded (step05b_admixture.sh:96, `-s "${SEED:-42}"`). LD pruning is a different case and an earlier version of this sentence misstated it: `plink --indep-pairwise` (step05b_admixture.sh:65) is passed no seed and needs none — it has no stochastic component at all, so it is deterministic by construction rather than by seeding. The ordination stage retains its own seeds for UMAP and t-SNE, which are presentational rather than inferential. Stage 12 has no stochastic component at all, and Stage 13 reuses the Stage 6 permutation procedure with the same seed in all six scenarios, which is part of why the scenarios are comparable (Section 3.14).

**What Stage 12 did and did not disturb.** Stage 12 was written after the corrected re-run, in response to the external critique, and it touches nothing upstream: it fetches its own genotypes, writes three new files, and modifies no existing output. No FST, ADMIXTURE, Mantel, TreeMix, BayPass or annotation result changes because of it. What changes is every marker number in the manuscript, because Stage 10's panel is withdrawn in full rather than adjusted — *pfdhps* A437G enters at 0.897 where it had been absent, the *pfdhfr*

triple mutant falls from a quoted 0.786–0.908 range to a measured 0.771, the *pfcrt* 75 "naming discrepancy" is withdrawn as a misdiagnosis, the *pfk13* panel is rebuilt from WHO v1.0 and reported with per-codon denominators and a 2009–2021 sampling window, and marginal-frequency validation is replaced by 47,012 of 47,012 per-sample agreements. Section 6's marker tables are built from results/codon_panel_residues.csv, results/codon_panel_haplotypes.csv and results/codon_panel_concordance.csv, and from nothing else.

### 3.13.1 Two incidents during the final checks

[Observed] **A results file was destroyed and restored.** results/baypass/xtx_all_snps.csv was found at **0 bytes**, truncated at 08:29:53 — hours after the 01:42 run that produced it. It was the only zero-byte file under results/ or interim/. It was regenerated from the intact BayPass outputs, and the regeneration is consistent with the surviving outputs rather than proven row-for-row: results/baypass/outlier_snps.bed, xtx_calibration.json and xtx_threshold.txt are byte-identical by md5 before and after, and the restored file carries the same 3,916 rows and the same 19 outliers. The per-SNP q-values in the restored file also reproduce the calibration counts exactly (Section 3.11).

[Hypothesized] A stray shell redirect during a verification pass — a `>` where a read was intended — truncated the file. The gap between the 01:42 run and the 08:29:53 truncation rules out the run itself, and that a single file was affected rules out anything systematic, but no command history establishes it.

> **What the near-loss shows.** A results file was destroyed by a command that had nothing to do with it, and nothing in the project noticed. What made the loss recoverable was not a safeguard but an accident of dependency structure: the file is downstream of outputs that survived, so it could be rebuilt, and the surviving md5s make the rebuild checkable. Had the truncated file been an upstream artefact — the genotype matrix, say, or the BayPass core-model output — the same mistake would have cost a re-run at best. The lesson is the same one as the stale-output guards above: the pipeline's integrity rested on files being present and plausible, not on their being verified, and the two are different things.

[Observed] **A log misreported its own output path.** The `--suffix` patch to step03_umap.py (Section 3.4.1) suffixed the file writes but not the `print` statements, so the run reported writing results/dimred_comparison.csv when it had correctly written results/dimred_comparison_qcset.csv. No data were at risk and the published file was not touched, verified from its modification time and contents. The defect was in the log, not in the data. It is fixed.

> **Why a wrong log line is worth recording.** The data were right and the log was wrong, which is the less alarming of the two orderings and still a reproducibility defect: the log is the only record of what a run did, so a run that misstates its own output path invites exactly the misattribution — published figure read as analysis-set figure, or the reverse — that the `_qcset` suffix was introduced to prevent. It belongs in the same family as the stale-output guards and the truncated results file: three cases in this project where the artefacts were fine and the bookkeeping about them was not.
>
> **Robustness.** The corrected re-run changed the SNP panel by 16.6 per cent, replaced the Mantel significance test, added false-discovery control to the outlier scan and put the TreeMix model choice on a testable footing without settling the edge count, the two selection criteria still disagreeing. **None of the substantive population-genetic conclusions moved.** Pairwise FST remained of order 0.002, the ADMIXTURE cross-validation curve remained nearly flat with a negligible improvement at K = 2 (0.84384 at K = 1 against 0.83925 at K = 2, all eight values on the same 3,385-SNP pruned panel — 0.54 per cent), and near-panmixia continued to hold. That is a robustness result in its own right and is reported as one in Section 6.9. It must not, however, be generalised, and the later threshold sensitivity analysis is why: when the two thresholds that define the dataset were varied rather than the filter order, **one conclusion did move**, and the isolation-by-distance result lost its support (Section 3.14). The re-run's stability was stability under a correction to how the published thresholds were applied, not stability under a change to the

> thresholds themselves, and only the second kind of check could have found what the second kind of check found. The marker panel is the third case and should not borrow either reassurance: there, the rebuild changed what is reported, and the honest summary is that four rounds of correction each contained a further defect, every one of them a failure of external correctness that internal consistency checking was never going to catch.

## 3.14 Stage 13 — sensitivity of the results to the two thresholds that define the dataset

**What and why.** Two thresholds define the analysis dataset and neither is derived from these data. The $F_{ws} \geq 0.95$ monoclonality cut (config.py MIN_FWS) is a convention taken from the literature, and it discards 57 per cent of the release-QC-passing isolates, unevenly by zone. The MAF ≥ 0.05 floor (config.py MIN_MAF) is specified in the thesis workflow, and it discards 99.8 per cent of the candidate sites: 3,916 of 1,807,331 survive it. Every headline number in this manuscript rests on both, and until this stage neither had been varied. The audit named that as the most-needed outstanding check. It is now run, and it does not return the answer the manuscript wanted.

**Method: one genotype load, six scenarios rebuilt in memory.** The 2.19 M × 4,682 genotype matrix is read **once**. For each scenario the analysis matrix is then rebuilt in memory from that single load under the scenario's thresholds — the isolate set re-derived under the $F_{ws}$ cut, variant missingness recomputed on the isolates that remain, and the MAF floor applied last, in the corrected two-pass order of Section 3.7.1 — and two things are recomputed on the result: pairwise Hudson $F_{ST}$ with the same block jackknife as Stage 4b, and the region-level Mantel test with the same seeded 9,999-permutation one-sided procedure and the same censoring treatment as Stage 6. Everything else is held fixed: the depth floor, the mixed-call rule, the sample-missingness bound, clonal pruning, the call-rate and heterozygosity checks, the zone labels and the region centroids. That the no-$F_{ws}$ scenario retains 4,403 isolates rather than all 4,682 in the loaded matrix is the evidence that the other per-isolate filters are still in force.

**Why that is cheaper and more comparable than six pipeline re-runs.** Cheaper, because the dominant costs of Stages 3a and 4a are the Zarr transfer (299.8 GB over 3.20 hours, Section 3.5) and the genome-wide pass over 2.19 M sites, and both are paid once rather than six times, with no second 20.57 GB local store and no second three-hour transfer window in which a request could fail. More comparable, because all six scenarios are built from the same loaded genotypes, by the same code path, with the same estimator and the same permutation seed, so a difference between two scenarios is attributable to the thresholds and to nothing else. Six separate pipeline runs would have differed in more than the quantity being varied — re-retrieval, column ordering, independently seeded permutation draws — and would have had to be compared across six logs rather than read off one table. The design is also what makes the uniformity of $F_{ST}$ below a real finding rather than six coincidences.

| Scenario | Isolates | SNPs | max $F_{ST}$ | Mantel *r* | Mantel *p* | *p* < 0.05? |
|---|---|---|---|---|---|---|
| baseline ($F_{ws} \geq 0.95$, MAF ≥ 0.05) | 1,900 | 3,916 | 0.00232 | 0.427 | 0.0252 | yes |
| $F_{ws} \geq 0.90$ | 2,196 | 3,629 | 0.00240 | 0.318 | 0.0874 | no |
| $F_{ws} \geq 0.80$ | 2,746 | 3,496 | 0.00255 | 0.256 | 0.1687 | no |
| no $F_{ws}$ filter | 4,403 | 3,275 | 0.00254 | 0.245 | 0.1792 | no |
| MAF ≥ 0.01 | 1,900 | 10,876 | 0.00204 | 0.432 | 0.0617 | no |
| MAF ≥ 0.02 | 1,900 | 6,852 | 0.00217 | 0.434 | 0.0467 | yes (marginal) |

*Table 3.5. The six threshold scenarios, from results/sensitivity_summary.csv. Maximum pairwise $F_{ST}$ is the Coastal savanna against Guinea savanna pair in all six. Mantel values are the censored region-level test; the uncensored p-values in the same file are 0.0413, 0.1214, 0.2002, 0.1792, 0.0644 and 0.0502. One verdict does turn on the censoring: the MAF ≥ 0.02 scenario is*

*significant censored (0.0467) and not significant uncensored (0.0502), so on the uncensored statistic the baseline is the only scenario clearing 0.05. The other five verdicts are unchanged.*

[Observed] **Isolation by distance is significant in 2 of the 6 scenarios.** It holds at the published thresholds (p = 0.0252) and at MAF ≥ 0.02 (p = 0.0467, which is marginal), and it fails at every relaxation of the $F_{ws}$ cut (p = 0.0874, 0.1687, 0.1792) and at MAF ≥ 0.01 (p = 0.0617). The correlation itself does not vanish — *r* stays between 0.245 and 0.434 — but it stops clearing the threshold, and under the $F_{ws}$ relaxations it also falls.

[Observed] The leave-one-region-out jackknife points the same way. In results/mantel_jackknife_region.csv (md5 779b9e379b6640afbbeb39409206ed69), **4 of the 7 replicates retain p < 0.05** — dropping Volta (r = 0.546, p = 0.0097), Ashanti (r = 0.568, p = 0.014), Brong Ahafo (r = 0.609, p = 0.0398) and Central (r = 0.396, p = 0.0427) — and the same four are **the only four below p = 0.10**, so the count does not depend on which of those two levels is used. The replicate the audit flagged is dropping **Upper East**: r falls to 0.260 at p = 0.218. The three non-significant replicates are dropping Eastern (r = 0.288, p = 0.1355), Upper East (r = 0.260, p = 0.218) and Greater Accra (r = 0.250, p = 0.2324). An earlier version of this paragraph quoted p-values of 0.0048, 0.0155, 0.0543, 0.0645 and 0.3307; those came from the superseded 3,359-SNP run, match no current output file, and are corrected here against the live result.

[Interpreted] **The isolation-by-distance finding must therefore be reported as not robust.** It holds at the published thresholds and almost nowhere else: 2 of 6 threshold scenarios, 4 of 7 jackknife replicates, and it collapses when a single region is removed. Any grading of it as "supported", "moderately supported", "a weak but real distance effect" or similar is withdrawn wherever it appears in this manuscript. This is a loss of a result, and the manuscript's only positive spatial finding at that — it is not a caveat attached to a surviving conclusion. The result is not deleted: the baseline estimate and its p-value stand as computed, and they are reported alongside the five scenarios in which the test does not clear 0.05, so that the reader sees the whole of what varying the thresholds does. What cannot be said is that the data show isolation by distance.

[Observed] **The measured differentiation, by contrast, is stable across every threshold tested, and that stability is what the sensitivity analysis strengthens. Stability of a very low, detectable common-variant estimate is not an estimate of random mating or of contemporary exchange.** Maximum pairwise $F_{ST}$ stays within **0.00204 to 0.00255** across all six scenarios — a total spread of 0.0005 — over isolate counts from 1,900 to 4,403 and SNP counts from 3,275 to 10,876. The headline conclusion is insensitive to both thresholds, across more than a doubling of the sample and nearly a trebling of the panel, and that is the strongest single piece of evidence for it anywhere in this analysis.

- Relaxing $F_{ws}$ slightly **raises** $F_{ST}$ (0.00232 at the published cut to 0.00255 at $F_{ws}$ ≥ 0.80 and 0.00254 with no cut at all). Polyclonal isolates were therefore not masking differentiation — the objection that the monoclonality filter might have thrown away the signal runs the wrong way.
- Tripling the SNP count via MAF ≥ 0.01 (3,916 to 10,876 sites) **lowers** $F_{ST}$ to 0.00204, as expected if rarer variants are less differentiated between zones. The published MAF floor is therefore, if anything, mildly anticonservative with respect to the panmixia conclusion, and the conclusion survives the correction.

> **Why one conclusion strengthens while the other falls.** The two outcomes are not in tension, and the asymmetry is instructive rather than suspicious. Maximum $F_{ST}$ is a point estimate pooled over thousands of sites, so it moves little when the site set or the sample changes, and six scenarios agreeing to within 0.0005 is a genuine stability result. The Mantel p-value is a significance verdict on a correlation over 21 pairwise points from seven populations, and at baseline it sat at 0.0252 — inside the top half of the interval below 0.05, with very little margin. A verdict with that little margin, resting on that few points, is exactly the kind that threshold choices can decide, and here they do. The

> honest reading is that the panmixia estimate was always well determined and the distance effect never was; what the check did was reveal which of the two the published thresholds had been carrying.
>
> **What this check does not do.** It varies two thresholds and nothing else. It does not vary the zone labels, the admin-1 centroid coordinates (all six scenarios share the same seven coordinate pairs, so none addresses the resolution limit of Section 3.3), the great-circle distance metric, the log-distance model, or the study confounding of Section 3.12 — all six scenarios draw on the same studies in the same zones, so the four non-overlapping zone pairs are confounded in every one of them. The six scenarios are also not independent: they share isolates and sites, so "2 of 6" is a description of the scenario grid and not a probability. And the two thresholds are varied one at a time from the baseline, never jointly.

**[Proposed]** Cross the two thresholds rather than varying them singly — the $F_{ws}$ relaxations at MAF ≥ 0.01 and 0.02, and the MAF relaxations with no $F_{ws}$ cut. The machinery now exists and one genotype load covers the whole grid, so the marginal cost is small. It would show whether the two thresholds interact, which the present design cannot see.

```
python step13_sensitivity.py
```

# 4. Filtering and inclusion/exclusion criteria

This section documents every filter applied, with counts before and after, and states for each whether it was specified in advance (in the thesis workflow or the supplied pipeline), chosen during this session on the basis of exploratory output, or introduced retrospectively. All variant-level counts from Stage 4a onward — the genotype depth mask, mixed-call resolution, variant missingness, MAF and LD pruning — come from the corrected re-run of 2026-10-10 (logs/rerun_corrected.log), which fixed the order in which variant missingness and the sample cut are applied (Section 4.2.7). The Stage 3a site counts in row 4 of Table 4.1 (12,493,205 total variants and 2,190,026 PASS biallelic SNPs) are the one exception: they come from logs/step03a_fetch_callset.log and do not appear in the re-run log, because Stage 3a was not re-run (see the provenance caveat below). The sample-level counts are unchanged from the original run; the variant-level counts are not, and any variant-level figure quoted in an earlier draft of this manuscript is superseded.

## 4.1 The filtering cascade

| # | Filter | Level | Threshold | Before | After | Removed | Origin |
|---|---|---|---|---|---|---|---|
| 1 | Country = Ghana | isolate | exact match | 33,325 | 6,653 | 26,672 | Pre-specified |
| 2 | Pf8 QC-pass flag | isolate | release flag | 6,653 | 4,905 | 1,748 | Pre-specified |
| 3 | Zone assignment | isolate | within a polygon | 4,905 | 4,905 | 0 | Pre-specified |
| 4 | Site FILTER = PASS, biallelic SNP | site | FILTER_PASS ∧ is_snp ∧ numalt = 1 | 12,493,205 | 2,190,026 | 10,303,179 | Pre-specified |
| 5 | Call rate floor | isolate | ≥ 0.80 | — | — | 20 | Pre-specified |
| 6 | Call-rate MAD outlier | isolate | disabled (< 0.98174 if enabled) | — | — | 0 (rule disabled) | Session decision |
| 7 | Heterozygosity outlier | isolate | > median + 3 MAD (0.0019) | — | — | 126 | Pre-specified |
| 8 | Fws monoclonality | isolate | ≥ 0.95 | — | — | 2,810 | Pre-specified |
| 9 | Clonal pruning | isolate | distance ≤ 0.00005 | — | — | 77 | Session decision (threshold) |
| — | Combined effect of 5–9 | isolate | | 4,905 | 2,017 | 2,888 | |
| 10 | Genotype depth mask | call | DP ≥ 5 | 12,540,727,880 called genotypes | 10,845,661,093 | 1,695,066,787 called genotypes (13.52%), from 3,709,090,355 cells failing DP < 5 | Session decision (GQ rule removed) |
| 11 | Mixed-call resolution | call | majority allele by AD; ties → missing | 194,167 mixed calls | 185,247 resolved | 8,920 (tied → missing) | Session decision |
| — | Provisional variant missingness, for the sample cut only | site | ≤ 0.10 over the 2,017 QC-passing isolates | 2,190,026 | 1,634,808 | not a filter of the final matrix | Pre-specified (order corrected) |
| 13 | Sample missingness | isolate | ≤ 0.20 | 2,017 | 1,900 | 117 | Pre-specified |
| 12 | Variant missingness, recomputed after the sample cut | site | ≤ 0.10 over the 1,900 retained isolates | 2,190,026 | 1,807,331 | 382,695 | Pre-specified threshold, corrected order (this revision) |

| # | Filter | Level | Threshold | Before | After | Removed | Origin |
|---|---|---|---|---|---|---|---|
| 14 | Minor allele frequency | site | ≥ 0.05 | 1,807,331 | 3,916 | 1,803,415 | Pre-specified (thesis: > 5%) |
| 15 | LD pruning (ADMIXTURE only) | site | $r^2$ < 0.2, 50/10 window | 3,916 | 3,385 | 531 | Pre-specified |

*Table 4.1. Complete filtering cascade, corrected run of 2026-10-10. Rows are in execution order; the # column keeps each filter's original identifier, so filters 12 and 13 now appear in the reverse of their numerical order (see Section 4.2.7). Isolate-level filters 5–9 are not sequential — each is evaluated independently against the same 4,905 isolates and an isolate may fail several — so before/after is shown as "—" for all five and only their combined effect has a meaningful pair. The QC summary in logs/step03b_sample_qc.log confirms the flags are independent (fail_call_rate 20, fail_call_rate_outlier 0, fail_het_outlier 126, fail_fws 2,810, fail_clonal 77, all against 4,905). Filter 6 was disabled, so no isolate set was ever evaluated at that position and its zero is the count with the rule off; had it been enabled, 1,281 isolates fall below its bound and 451 of those fail no other check (Section 4.2.3). Sources: logs/step01_filter_samples.log, logs/step03a_fetch_callset.log (row 4), logs/step03b_sample_qc.log, logs/rerun_corrected.log, results/sample_qc_full.csv.*

> **Scope caveat on row 10.** Two distinct quantities are involved in the depth mask and they must not be subtracted from one another: **3,709,090,355 genotype cells fail DP < 5** across the full 2,190,026 × 6,653 array, most of which carried no call to begin with; **of these, 1,695,066,787 carried a call**, taking the called-genotype total from 12,540,727,880 to 10,845,661,093 — 13.52 per cent of called genotypes masked. Section 6.1 states the same two facts in the same form, and no other formulation should appear anywhere in this manuscript. The mask runs before sample alignment, so 13.52 per cent is computed over all 6,653 fetched Ghanaian isolates, whereas rows 11–14 are over the 2,017 that passed QC: one row of the cascade uses a denominator about 3.3 times larger than its neighbours. Whether the equivalent rate on the isolates that actually entered the analysis is higher or lower than 13.52 per cent is **[Unknown]**, and an earlier draft's claim that the table overstates the mask's cost is withdrawn. Nothing measured establishes a direction: the QC panel's call rate separates the two groups by 0.0012 (median 0.9968 over the 2,017 retained against 0.9956 over the 2,888 discarded; the 0.9962 in the log is the median over all 4,905 zone-assigned isolates, not the retained subset), 1,748 of the isolates cut at filter 2 have no call-rate row in results/sample_qc_full.csv at all, and that panel is in any case computed over 50,000 SNPs with no DP or GQ mask applied, so it does not measure what the depth floor removes — its 0.996 sits roughly 13.5 points above the genome-wide calledness the mask was computed on. Stage 4a does not print the rate again after alignment, so the aligned-scope figure is not available.
>
> **Provenance caveat on row 4.** The 12,493,205 total-variant and 2,190,026 PASS-biallelic counts come from logs/step03a_fetch_callset.log, which was not re-run; Stage 3a is unchanged by this revision, and neither figure appears in logs/rerun_corrected.log. The 2,190,026 figure was independently confirmed by the audit against the public Pf8 Zarr store. The numalt = 1 clause in this row is the filter that silently removed most canonical antimalarial-resistance codons — 28 of the 44 codon-allele records genotyped, among them *pfcrt* K76T, *pfmdr1* N86Y, *pfdhps* K540E and 8 of the 14 validated *pfk13* codons; of the 21 records that are polymorphic in this sample, only 6 survived the filter from the genome-wide callset — see Section 6 and the separate multiallelic-aware marker panel, which bypasses it.

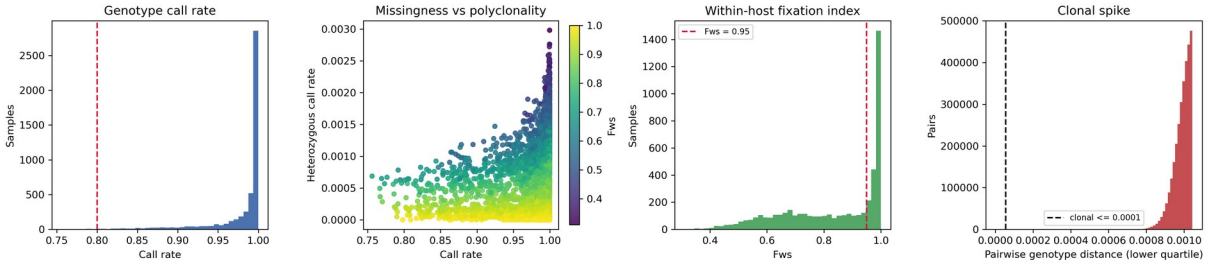



*Figure 4.1. Sample-level QC distributions over the 4,905 zone-assigned Ghanaian isolates: call rate, heterozygous call rate and Fws, with the applied thresholds marked. Source: pf8_proof/sample_qc.png, the copy that ships with this manuscript.*

> **Figure provenance.** The sample-level stage was not re-run — logs/step03b_sample_qc.log is dated 9 October, before the 10 October corrected re-run — so this is the same figure as in the original run and no regenerated version exists or is required. The shipping asset is pf8_proof/sample_qc.png (519,880 bytes); the server copy at figures/sample_qc.png is 519,864 bytes. The two are not byte-identical, which is why the figure is identified here by the path that travels with the manuscript rather than by the server path.

## 4.2 Rationale, origin and consequences of each non-trivial filter

### 4.2.1 Pf8 QC-pass flag (filter 2)

**Purpose and origin.** Pre-specified. Removes isolates MalariaGEN itself judged unreliable, on the basis of sequencing-run information not available downstream.

[Observed] **Effect.** 1,748 of 6,653 isolates removed (26.3 per cent). The post-QC regional counts are Upper East 3,605, Greater Accra 435, Ashanti 391, Central 309, Eastern 74, Brong Ahafo 50 and Volta 41, read from logs/step01_filter_samples.log for this revision.

[Calculated] The loss is not uniform across regions: Central loses 55.0 per cent (686 → 309) and Upper East 25.2 per cent (4,821 → 3,605), while Volta loses none (41 → 41). These percentages are not in logs/step01_filter_samples.log, which prints the admin-1 breakdown only downstream of the QC filter and therefore carries post-QC counts alone; they were recomputed from the Pf8 manifest (data/Pf8_samples.txt) for this revision.

[Interpreted] The earlier draft labelled differential regional loss as most likely reflecting collection, storage or sequencing-batch differences, and said the manifest did not carry the fields needed to confirm it. That is no longer the right status. The manifest does carry the partner-study field, and the study-by-zone crosstab in Section 4.3 shows that zone in Ghana is very nearly a relabelling of sequencing study. Differential regional loss is therefore consistent with — and most parsimoniously explained by — study-level heterogeneity, which is now quantified rather than hypothesised. What remains untested is whether any of it is biological.

### 4.2.2 Fws ≥ 0.95 (filter 8)

**Purpose.** To exclude polyclonal infections. A mixed infection contributes allele frequencies from several parasite genomes to a single genotype column; ADMIXTURE reads that as intermediate ancestry, so an uncorrected analysis risks describing the transmission-intensity gradient rather than the ancestry structure it is meant to estimate.

[Observed] **Origin.** Pre-specified. 0.95 is the conventional cut and is the pipeline default (config.MIN_FWS). It was re-confirmed during this session against the observed Fws distribution (Figure 4.1, third panel), which shows a pronounced mode above 0.95 and a long left shoulder, so the threshold falls at a natural break rather than cutting through a dense region. Median Fws over the 4,905 isolates is 0.895 (range 0.310–1.000); 2,095 of 4,905 (42.7 per cent) are monoclonal at the cut.

[Observed] **Effect and its most important consequence.** 2,810 of 4,905 isolates removed — by far the largest single exclusion. The polyclonal fraction differs by zone: Guinea savanna 0.597 (of 3,605 isolates), deciduous forest 0.539 (of 815), transitional zone 0.500 (of 50) and coastal savanna 0.444 (of 435). These fractions, like every study-composition and within-zone batch figure in this manuscript, are computed on the **4,905 QC-passing isolates** — the set on which study membership and Fws are defined — and not on the 1,900-isolate analysis set, whose zone counts are Guinea savanna 1,334, deciduous forest 337, coastal savanna 204 and

transitional zone 25. The denominator is stated in full in Section 2.6 and applies wherever these numbers appear (Sections 4.3, 6.2, 7.2, 7.3 and 10.1).

[Interpreted] Higher transmission produces more mixed infections, so this ordering is itself a transmission signal, and the filter removes a greater share of isolates from the high-transmission Guinea savanna than from the coast. It is not, however, a clean south-to-north cline, and should not be reported as one: the transitional zone is the second-most-northern of the four zones yet sits third of four on polyclonal fraction, below deciduous forest, so the ordering is Guinea savanna > deciduous forest > transitional zone > coastal savanna rather than monotonic in latitude — and the transitional value rests on 50 isolates drawn from a single study. What can be said without qualification is that the retained sample is not a random subsample of the Ghanaian parasite population, and that the degree of non-randomness differs by zone. Because the filter targets monoclonality rather than any allele, the direct effect on allele frequencies should be modest, but this has not been tested.

[Observed] The zone ordering of the polyclonal fraction is not separable from sequencing study. Within deciduous forest, the four studies large enough to test — of five contributing; the fifth has 4 isolates — differ in Fws (Kruskal–Wallis H = 14.9, p = $1.9 \times 10^{-3}$, n = 811), and their polyclonal fractions span 0.430 to 0.578, which is most of the between-zone range. The p-value is a test on the Fws distributions, not on the polyclonal fraction itself; no test of the fraction exists in the outputs, and the two should not be conflated. The zone's composition is five studies (815 isolates); only the test is over four. See Section 4.3.

[Proposed] Re-run the $F_{ST}$ and ADMIXTURE analyses at Fws ≥ 0.90 and ≥ 0.80 and compare. If conclusions are unchanged, the sensitivity concern is discharged cheaply. The retained-isolate projections quoted for these thresholds in the earlier draft are withdrawn: they could not be reproduced from any committed script, and an independent recomputation from results/sample_qc_full.csv disagreed with both by two isolates.

### 4.2.3 Call-rate outlier rule (filter 6) — disabled

[Observed] **What the pipeline did.** A median-absolute-deviation rule flags isolates whose call rate falls more than three scaled MADs below the median. Because Ghanaian call rates are tightly bunched near 1 (median 0.9962 over the 4,905 zone-assigned isolates, scaled MAD 0.004833), the resulting bound sits at 0.98174 and flagged 1,281 isolates.

**Decision and origin.** Session decision, taken on exploratory output. The rule was disabled, leaving only the absolute floor of 0.80 (config.MIN_SAMPLE_CALL_RATE, pre-specified). The reasoning: the MAD-flagged, otherwise-passing isolates had a median call rate of 0.9551 and a range of 0.803–0.982, which is not poor data in absolute terms, and Stage 4a applies an independent sample-missingness bound later in any case. A MAD rule is designed to find outliers in a dispersed distribution; applied to a tight one it becomes an arbitrary strict threshold.

**Effect — restated.** The earlier draft read: "Retaining these isolates raised the QC-passing count from 1,566 to 2,017 and the final analysis set from 1,862 to 1,900." The second half of that sentence was misattributed, and the two halves could not both be true: the final analysis set is a subset of the QC-passing set (filter 13 runs 2,017 → 1,900), so a QC-passing count of 1,566 cannot produce a final set of 1,862.

[Calculated] The first half is correct and is now reproducible. Re-applying the MAD rule to results/sample_qc_full.csv at its own bound of 0.98174 flags 1,281 isolates, of which 451 fail no other check. Enabling the rule would leave 1,566 QC-passing isolates instead of 2,017 — per zone, Guinea savanna 1,115,

deciduous forest 272, coastal savanna 154, transitional zone 25. This was recomputed from the primary QC table for this revision; it is not printed in any log, which is why the audit could not locate it.

**[Observed]** The second half belongs to a different decision. The 1,862-isolate final set comes from logs/run_stages_4to7.log, and that run already had the MAD rule disabled — it used the same 2,017 QC-passing isolates. The only difference from the run that produced 1,900 was the genotype mask: GQ < 20 enabled (4,706,404,099 calls below GQ 20; 12,540,727,880 → 9,863,685,827, 21.35 per cent of calls masked; final matrix 2,143 SNPs × 1,862 isolates) versus DP-only (13.52 per cent masked; 3,359 SNPs × 1,900 isolates). Both of those matrices are from the superseded filter order.

**[Calculated]** It follows that the 1,862 → 1,900 change measures the genotype-quality decision of Section 4.2.5, counted a second time in the earlier draft under the wrong heading. That reading is an inference about the earlier draft's bookkeeping rather than a logged quantity, but it is forced by the two runs' settings; the claim is removed from this subsection.

**[Unknown]** What the call-rate decision cost or bought at the level of the final analysis matrix is therefore not known. Establishing it requires re-running Stage 4a with the MAD rule enabled, under the current DP-only mask and the corrected filter order. That has not been done, and no replacement figure is offered here. The benefit claimed for this decision in the earlier draft was the benefit of a different decision.

| Quantity | MAD rule on | MAD rule off (chosen) | Source |
|---|---|---|---|
| Call-rate bound | 0.98174 | not applied | logs/step03b_sample_qc.log |
| Isolates below the bound | 1,281 | 1,281 (reported only, not filtered) | logs/step03b_sample_qc.log |
| Of those, failing no other check | 451 | — | recomputed from results/sample_qc_full.csv |
| Median call rate of that subset | 0.9551 | — | recomputed from results/sample_qc_full.csv |
| QC-passing isolates | 1,566 | 2,017 | recomputed / logs/step03b_sample_qc.log |
| Final analysis matrix | not re-derived | 1,900 isolates × 3,916 SNPs | logs/rerun_corrected.log |

*Table 4.2. What is and is not established about the call-rate decision. This table replaces the projected sensitivity table of the earlier draft (rows "MAD rule on" / "MAD rule off" with projected SNP counts of 2,117 and 3,416). Those projections were withdrawn for two reasons: the audit could not reproduce any of their cells from a committed script or log, and the quantity they projected — the final SNP count — has since changed with the corrected filter order, so they could not be compared against the current run even if their provenance were sound.*

> **Honesty note.** This threshold was chosen after seeing its effect on sample counts, not in advance. The justification above is a post-hoc rationalisation of a decision made during exploratory analysis. It is defensible, but it was not pre-registered, and a reviewer is entitled to ask whether the alternative was examined with equal care. The earlier draft pointed to Table 4.2 as evidence that the alternatives had been quantified before the choice was made; that claim cannot be supported, because the projections in question are not reproducible. What can be said is narrower: the alternative's effect on the QC-passing count is now known (1,566 against 2,017) and its effect on the final matrix is not.

### 4.2.4 Clonal pruning threshold (filter 9)

**Problem.** Unlike Fws, the clonality threshold has no conventional value. The pipeline's own histogram heuristic suggested 0.00005, and the log prints that suggestion together with an explicit warning that it is a heuristic. The earlier draft said the data "offer no clean break"; that overstates the case, since the pipeline's

histogram did identify a trough and the number of isolates removed rises smoothly only once the threshold is pushed well past it.

| Threshold | Clonal pairs | Groups | Isolates in groups | Isolates removed | Largest group |
|---|---|---|---|---|---|
| 0.00002 | 57 | 49 | 102 | 53 | 3 |
| 0.00005 (chosen) | 85 | 68 | 145 | 77 | 3 |
| 0.0001 | 103 | 82 | 174 | 92 | 4 |
| 0.0002 | 133 | 104 | 222 | 118 | 4 |
| 0.0003 | 159 | 118 | 255 | 137 | 4 |
| 0.0005 | 261 | 164 | 370 | 206 | 7 |
| 0.0007 | 523 | 229 | 591 | 362 | 30 |

*Table 4.3. Sensitivity of clonal pruning to the distance threshold, computed on the cached 4,905 × 4,905 Ghanaian distance submatrix. At the chosen threshold the pipeline reports 68 clonal groups spanning 145 isolates, of which 77 are dropped (logs/step03b_sample_qc.log). The pairwise distance distribution between Ghanaian isolates runs from a minimum of 0.00000 through a 1st percentile of 0.00090 to a median of 0.00112, so even the loosest threshold shown is well inside the left tail. The audit independently reproduced all 35 cells of this table from interim/ghana_dist_submatrix.npy; the corresponding script is not part of the pipeline.*

**Decision and origin.** Session decision. 0.00005 was chosen: isolates within about 4 per cent of the typical pairwise distance, which is a defensible reading of "essentially the same genome". The largest connected component at this threshold contains 3 isolates, so single-linkage chaining — the real danger of this method, where a sequence of moderately similar isolates merges into one large component — is not occurring. At 0.0007 the largest component reaches 30 isolates and chaining is clearly setting in.

[Observed] **Effect.** 77 isolates removed, 1.6 per cent of the 4,905 zone-assigned set. Because the effect is small and the method is well behaved at this threshold, sensitivity to the exact value is unlikely to be material — but this has not been verified by re-running downstream analyses.

### 4.2.5 Genotype quality mask (filter 10)

[Observed] **Problem.** The pipeline default masked calls below DP 5 or GQ 20. Under that default the final matrix held 2,143 SNPs over 1,862 isolates; removing the GQ component raised it to 3,359 SNPs over 1,900 isolates. Both figures are from the superseded filter order. The corresponding DP+GQ run has not been repeated under the corrected order, so the current 3,916-SNP matrix has no like-for-like GQ-masked counterpart.

**Evidence gathered before deciding.** A 40,000-SNP diagnostic panel showed that 12.6 per cent of homozygous-reference and 12.4 per cent of homozygous-alternate calls fall below GQ 20, with median depths of 46 and 40 respectively — that is, the GQ < 20 population is not a low-depth population. Restricted to calls with DP ≥ 10, only 3.0 per cent of homozygous-reference and 0.7 per cent of homozygous-alternate calls still fail GQ 20. Applying the GQ rule reduced the share of panel SNPs passing the ≤ 10 per cent missingness bar from 74.3 to 51.0 per cent — a relative reduction of 31 per cent, not a halving, as the earlier draft stated — while mean alternate-allele frequency at common SNPs moved from 0.282 to 0.294.

[Unknown] None of the diagnostic-panel figures in the preceding paragraph can be located in any log, script or results file on the server. They were tagged [Observed] in the earlier draft, which the manuscript defines as printed by a pipeline script and preserved in a log; they do not meet that definition. The diagnostic script was

never committed. The figures are retained here because they are what the decision was actually taken on, but their provenance is unrecoverable and they should be read as such.

**Decision and origin.** Session decision, taken on that evidence and confirmed with the user. config.MIN_GENOTYPE_GQ was set to 0, retaining DP ≥ 5. The reasoning: the GQ rule was removing a large number of common SNPs at substantial cost to statistical power, while barely changing allele frequencies — that is, it was costing precision without buying accuracy.

**[Interpreted]** The earlier draft explained low GQ as an artefact of a haploid organism forced into a diploid calling model, so that intermediate GQ values arise from the model's own ambiguity rather than from poor data. That interpretation is not supported. The audit measured low-GQ call rates by sequencing study and found them differing roughly tenfold between studies within a single zone, which a uniform model artefact cannot explain, since a model artefact should be uniform across studies by construction. The heterogeneity is therefore study-associated; but study also differs in collection period, sample handling and recruitment, so this does not on its own identify a technical batch cause. The decision may still be the right one, but this is no longer its reason.

> **Honesty note — the GQ decision is still unvalidated.** No direct genotype-concordance check has been run, and the decision therefore rests on indirect evidence throughout. The two specific gaps: (i) the allele-frequency comparison offered in support (0.282 against 0.294) is between two different SNP sets, not the same sites under the two masks, so it does not test whether dropping the GQ rule admits wrong calls — only that the two surviving sets have similar mean frequencies; (ii) no comparison has been made against an independent truth set, although one exists and was used elsewhere in this project. MalariaGEN publishes its own genotype calls for these same 1,900 isolates, and they were used successfully to cross-validate the resistance-marker panel (47,012 of 47,012 per-sample calls agreeing, across the 25 of the panel's 57 codons for which MalariaGEN publishes a per-sample genotype). The equivalent check here — per-call concordance between DP-only and DP+GQ masking at a common set of sites, benchmarked against those published calls — is cheap, is the check the manuscript's own reviewer note asks for first, and has not been done. Until it is, the GQ decision should be read as a power-for-accuracy trade made on plausible but indirect grounds, and the SNP-count gain it produced — 3,359 against 2,143, about 57 per cent, both under the superseded filter order — should be reported as contingent on it.

### 4.2.6 MAF ≥ 5 per cent (filter 14)

**Origin.** Pre-specified in the thesis workflow ("MAF > 5%") and not altered.

**[Observed]** **Effect — the most consequential number in the analysis.** 1,807,331 SNPs pass the recomputed missingness filter; 3,916 pass MAF. This is a reduction of 99.78 per cent. Under the superseded filter order the same step ran on 1,634,808 variants and yielded 3,359, so the corrected order adds 557 SNPs (+16.6 per cent) at an identical threshold — see Section 4.2.7.

**[Interpreted]** *P. falciparum* carries an extremely large number of rare variants and comparatively few common ones within a single country. A 5 per cent MAF filter in this setting is far more aggressive than the same nominal threshold would be in, say, a human cohort. Every downstream analysis — $F_{ST}$, ADMIXTURE, TreeMix, BayPass — operates on these 3,916 sites (3,385 after LD pruning for ADMIXTURE), and the BayPass scan in particular has limited power at this density: 19 SNPs clear the POD quantile threshold and 13 of those also survive FDR 10 per cent.

**[Unknown]** The earlier draft supported this paragraph with a stated genome-wide ceiling of roughly 5,500 common SNPs ("about 4,832 SNPs" with missingness ≤ 10 per cent and no genotype masking, "about 5,529" ignoring missingness), and concluded that the analysis retained 3,359 of them. Neither figure can be found in any log, script or results file, and the diagnostic that produced them was never committed. The ceiling claim

is withdrawn rather than updated: with no reproducible denominator, the fraction of the available common-variant panel that this analysis sees is not known. What is known is that correcting the filter order alone recovered 557 SNPs, which indicates the earlier figures were not a hard ceiling.

> **Status of the 4,832 and 5,529 ceiling figures elsewhere in the manuscript.** Both were computed on a diagnostic SNP panel under the DP-only mask and against the superseded filter order, before the re-run: they are pre-re-run projections, not measurements of the current callset, and they should be labelled as such wherever they appear. The roughly 5,500-SNP ceiling is carried into Section 7.3, the limitations table in Section 9 and the threshold-stability item in Section 10.3, all of which inherited the 3,359 denominator. The retained count to state against it is **3,916** — about 71 per cent of the figure as quoted — not 3,359. The position taken above is the honest one and should govern: the ceiling is a withdrawn projection rather than a bound, and any section that still quotes it must say so.

**[Proposed]** Repeat the main analyses at MAF ≥ 1 per cent as a sensitivity check. This departs from the thesis specification and so should be reported as a robustness analysis rather than as the primary result, but the gain in power for the outlier scan could be substantial. The projected SNP range quoted for this in the earlier draft is withdrawn for the same provenance reason; the projection should be redone from the current callset and printed to a log.

### 4.2.7 Filter order: variant missingness after the sample cut (filters 12 and 13)

**Problem.** Stage 4a measured per-variant missingness against all 2,017 QC-passing isolates, applied the ≤ 0.10 bar, and only then dropped 117 isolates for excess sample missingness. Every SNP was therefore charged for missing calls in isolates that did not survive to be analysed. The 117 dropped isolates are not a distinct bad-data population but the lower tail of the same call-rate distribution (call rates 0.803–0.989, median 0.882, against a retained minimum of 0.896), so they inflated every SNP's measured missingness. This is the conventional `--mind` before `--geno` ordering in PLINK, and the pipeline had it the other way round. The earlier draft described the filter order as deliberate; that was only half right — masking before missingness is correct, but measuring variant missingness before the sample drop was not.

**Fix.** Stage 4a is now two-pass. A provisional variant set is computed exactly as before, over the 2,017 QC-passing isolates, and is used only to decide the sample cut — so the same 117 isolates drop and the sample set is unchanged. Variant missingness is then recomputed on the 1,900 isolates that remain, and MAF is applied to that set. Thresholds are identical, the sample set is identical, and the provisional subset is never materialised, the pass being implemented blockwise.

```
provisional variant pass (for the sample cut): 1,634,808
dropping 117 high-missingness samples
variant missingness <= 0.1 (recomputed on 1,900 retained isolates): 1,807,331
MAF >= 0.05: 3,916
```

| Stage | Original order | Corrected order | Change |
|---|---|---|---|
| PASS biallelic SNPs entering Stage 4a | 2,190,026 | 2,190,026 | unchanged |
| Denominator for variant missingness | 2,017 isolates | 1,900 isolates | 117 fewer |
| SNPs passing missingness ≤ 0.10 | 1,634,808 (74.65%) | 1,807,331 (82.53%) | +172,523 (+10.55%) |
| Isolates dropped for sample missingness | 117 | 117 | unchanged |
| SNPs passing MAF ≥ 0.05 | 3,359 | 3,916 | +557 (+16.6%) |
| Final matrix | 3,359 SNPs × 1,900 isolates | 3,916 SNPs × 1,900 isolates | +557 SNPs |

*Table 4.4. Effect of correcting the filter order. Both columns use the same thresholds, the same genotype mask (DP ≥ 5) and the same 1,900 isolates; the only change is that variant missingness is measured after the sample cut rather than before it. Sources: logs/run_from4_parallel.log (original), logs/rerun_corrected.log (corrected).*

[Observed] **Effect.** 1,807,331 variants pass the recomputed missingness bar against 1,634,808 under the original order, a pass-rate increase from 74.65 to 82.53 per cent of the 2,190,026 PASS biallelic sites. After MAF the final matrix holds 3,916 SNPs rather than 3,359: 557 more sites, a 16.6 per cent increase, on the identical 1,900 isolates, at no change to any threshold.

[Interpreted] The audit projected this gain at roughly +344 SNPs (to about 3,703) from a 50,000-SNP calibration panel. The realised gain is larger, which is consistent with the panel being an unbiased but noisy estimator of a genome-wide pass rate rather than with any discrepancy in the fix itself: on the 2,017-isolate denominator the panel put the missingness pass rate at 74.2 per cent against the pipeline's genome-wide 74.65 per cent, reproducing it to within half a percentage point.

**Origin.** Introduced retrospectively, in this revision, following the audit. It is a departure from the pipeline as supplied and is recorded as such. It is also the only change in this revision that alters the size of the analysis matrix, and therefore the one that propagates into every downstream number: the $F_{ST}$ matrix, the ADMIXTURE cross-validation curve, the TreeMix sweep and the BayPass scan were all re-run on the 3,916-SNP matrix.

> **Why this matters more as a robustness result than as a correction.** A 16.6 per cent larger common-variant panel did not move any conclusion. Maximum pairwise FST stayed at 0.0023 against a previously reported 0.0024; the ADMIXTURE cross-validation curve remains near-flat with a negligible K = 2 improvement over K = 1 (0.83925 against 0.84384); the outlier scan's top signals are the same genes. The substantive reading of this is not that the earlier numbers were wrong in any consequential way, but that the near-panmixia result is insensitive to a substantial change in panel size — which is the most useful form of reassurance available for a result of this kind, and is reported in Section 10 as such. The correction is still required: the earlier numbers were the output of a defective ordering, and that has to be stated whether or not it changed the answer.
>
> **Caveat.** One further defect surfaced during the re-run itself and is recorded here because it bears directly on the filter cascade. Three downstream stages guarded their work with a "skip if output exists" test that did not check whether the input had changed. When the SNP set went from 3,359 to 3,916, ADMIXTURE and TreeMix regenerated their inputs, found their previous output files in place, skipped the work and reported success; only the BayPass stage caught it, because it asserts that its position table and the BayPass output have matching row counts and failed with that exact diagnosis. All three guards now require the output to be newer than its input. Had the file timestamps not been inspected, the previous day's ADMIXTURE and TreeMix results would have been reported against the new matrix.

## 4.3 Selection bias summary

Four biases and design confounds are introduced or amplified by the filtering and by the sampling frame, in decreasing order of concern for the between-zone comparisons that carry this analysis.

[Observed] **Study and batch confounding (from the sampling frame; a fourth design limit, not listed in the earlier draft).** Pf8 isolates arrive through named partner studies, and in Ghana those studies are almost perfectly nested within zones. Four of six zone pairs share no sequencing study at all — including coastal savanna versus Guinea savanna, the pair that produces the maximum pairwise $F_{ST}$ and the polyclonality contrast. On the 4,905 QC-passing isolates each zone is dominated by a single study: transitional zone 100 per cent (1083-PF-GH-CONWAY, 50 of 50), coastal savanna 89 per cent (1241-PF-GH-ANINAGYEI, 387 of 435), Guinea savanna 74 per cent (1168-PF-GH-AMENGA-ETEGO-SM, 2,678 of 3,605) and deciduous forest 48 per

cent (1200-PF-GH-MAIGA-SM, 391 of 815). Deciduous forest has five contributing studies in total; the fifth, 1114-PF-PDN-DBS-GH-GHANSAH, contributes 4 isolates.

> **Denominator for the batch figures.** The study-composition and within-zone batch tests in this section and in Section 4.2.2 are computed on the **4,905 QC-passing isolates**, the set on which study membership and Fws are defined; the population analyses use the 1,900-isolate subset, whose zone counts are Guinea savanna 1,334, deciduous forest 337, coastal savanna 204 and transitional zone 25. The two sets must not be read against one another: "1083-PF-GH-CONWAY is 50 of 50 in the transitional zone" and "transitional zone, n = 25" are statements about different sets, not a contradiction. Study nesting within zone is a property of the Pf8 sampling frame and is not altered by the later cuts, and this was checked rather than assumed: the zone-pair overlap was recomputed on the 1,900-isolate analysis set for this revision and is identical — four of six pairs share no study, coastal savanna and deciduous forest still share two (1183-PF-GH-AWANDARE-SM, 1241-PF-GH-ANINAGYEI) and Guinea savanna and the transitional zone still share one (1083-PF-GH-CONWAY) — with single-study dominance of 100 per cent in the transitional zone (25 of 25), 86.8 per cent coastal savanna (177 of 204), 73.8 per cent Guinea savanna (984 of 1,334) and 46.6 per cent deciduous forest (157 of 337); deciduous forest still draws on five studies. Source: interim/analysis_samples.csv, computed for this revision. The within-zone Kruskal–Wallis tests themselves were not recomputed on the 1,900 set and remain 4,905-set figures.

**[Observed]** Study-associated heterogeneity here is demonstrable rather than hypothetical. Holding zone constant, so that study varies together with collection year, finer geography and recruitment: within Guinea savanna, call rate differs between its six contributing studies at Kruskal–Wallis $p = 2.8 \times 10^{-197}$; within deciduous forest, at $p = 7.7 \times 10^{-88}$. In deciduous forest the Fws distributions also differ between the four studies large enough to test, of five contributing (Kruskal–Wallis $H = 14.9$, $p = 1.9 \times 10^{-3}$, $n = 811$), and their polyclonal fractions span 0.430 to 0.578; the p-value belongs to the Fws test, not to the fraction, for which no test exists in the outputs. For the two zone pairs that do share a study, the within-study comparison reproduces the overall direction: within 1083-PF-GH-CONWAY, Guinea savanna is more polyclonal than the transitional zone (0.532 against 0.500), matching the overall ordering. The gradient is therefore not purely an artefact — it is simply not separable from batch for the pairs that carry the largest differences.

**Transmission-intensity bias (from the Fws filter).** The retained sample under-represents high-transmission settings, and does so unevenly across zones. Of the biases that the filtering itself introduces, this is the one most likely to affect between-zone comparisons.

**Regional bias (from the Pf8 QC flag and from the sampling frame itself).** Upper East dominates every stage. No filter corrects this and none was intended to. It is also inseparable from the previous item, since Upper East is largely one study.

**Allele-frequency bias (from the MAF filter).** The analysis sees only the common end of the frequency spectrum. Rare variants, which is where most recent selection signal would sit, are invisible to it. The $\mathrm{numalt} = 1$ clause on filter 4 compounds this at the loci of greatest applied interest: it removed most canonical antimalarial-resistance codons — 28 of the 44 codon-allele records genotyped, among them *pfcrt* K76T, *pfmdr1* N86Y, *pfdhps* K540E and 8 of the 14 validated *pfk13* codons; of the 21 records that are polymorphic in this sample, only 6 survived the filter from the scanned callset, and those codons had to be recovered by a separate multiallelic-aware panel rather than by relaxing MAF.

**[Interpreted]** None of the first three is an error; each follows from a defensible choice. The fourth — study confounding — is not a choice at all but a property of the released data, and it is the one that cannot be mitigated by any reanalysis of this sample. Together they mean the final 1,900-isolate, 3,916-SNP dataset is a particular view of Ghanaian *P. falciparum* — monoclonal infections, common variants, Upper East–heavy, and with zone very nearly a relabelling of sequencing study — and conclusions should be stated as applying to

that view. In particular, a between-zone difference in this dataset cannot be attributed to agro-ecology in preference to batch for four of the six zone pairs.

> **Caveat on the zone layer itself.** Every zone-level count in this section depends on ghanaAEZ.geojson, the agro-ecological zone polygons used for the spatial join. That layer has no citable source: it carries no attribution, source, author or date field, and was traced to an anonymous ArcGIS Online re-host that is byte-for-byte identical to the local copy (md5 96b1a7f4da62d6f0153826f41af96152) and whose licence and access-information fields are both null. It is not an origin, and it is a hybrid of two different classification schemes rather than a reproduction of Ghana's canonical MoFA/FAO zones. It should be cited as accessed rather than as authoritative, and the provenance gap stated as a limitation. Zone names are reproduced verbatim throughout this manuscript — including the layer's "Decidous forest" misspelling in table labels and output files — because the spatial join keys on them.

# 5. Analytical decisions and iterations

This section reconstructs the decisions taken during the analysis. Where a decision was put to the user and answered, that is stated. Where the rationale is reconstructed after the fact, it is labelled as such. The section has now been revised three times: after an adversarial audit of the completed analysis, after the corrected re-run that followed it, and after an external critique of the marker panel whose five points were all upheld and four of which were defects in code rather than in prose. Several decisions recorded here were wrong, and the stated account of why two others worked was wrong. They are corrected in place rather than removed, because the reasoning that produced them is part of what this section is for.

## 5.1 Replacing the VCF route with direct Zarr retrieval

[Observed] **Problem.** The project had 14 per-chromosome VCFs in data/, produced by data/extract_ghana.sh, which streamed each Pf8 chromosome VCF from Sanger through bcftools view with a Ghana sample list. Thirteen of the fourteen were truncated. Inspection of the BGZF end-of-file marker and decoding of the last record in each file showed coverage from 9 per cent (chromosome 12) to 53 per cent (chromosome 9); only chromosome 10 was complete.

**Cause.** bcftools aborted with "[E::bcf_calc_ac] Incorrect AN/AC counts" while recomputing allele counts for the sample subset. Each truncation point matches its recorded error position exactly — chromosome 12 failed at position 200,583 and its last record is at 200,581. The extraction script echoed "Finished" regardless of the exit status, so the failures went unnoticed. The errors are preserved in data/extract_ghana.log and data/chr05_extract.log; the consequent end-of-file warnings are in data/index_tbi.log.

[Observed] **Constraint.** The one complete VCF, chromosome 10, is 34.7 GB for 1.69 Mb of sequence, against 352 GB free on the data volume.

[Calculated] At that rate the full genome would require approximately 480 GB. A simple re-download was therefore not possible.

**Options put to the user.**

| Option | Transfer | Local disk | Time | Script changes |
|---|---|---|---|---|
| A. Pf8 Zarr store, Ghana samples and PASS biallelic SNPs only (chosen) | ~300 GB | ~25–40 GB | ~3–4 h | One new script |
| B. Resume the VCFs with bcftools -I, slimmed to needed fields | ~1.6 TB | moderate | 1–2 days | None, plus slow VCF→Zarr conversion |
| C. Re-download all VCFs from scratch, slimmed | ~2.2 TB | moderate | longest | None |

*Table 5.1. Genotype-retrieval options presented. Estimates for A derive from a direct probe of the Zarr store's chunk grid; B and C are extrapolated from the observed chromosome-10 download.*

**Decision.** Option A, chosen by the user. A fourth implicit option — the project's own step04a_prepare_callset_malariagen.py, which pulls from Google Cloud via the malariagen_data package — was not pursued because it would have required changes to run_all.sh and because Step 3b reads the local store directly.

**Consequence and verification.** The new script produced a 20.57 GB store verified exactly against three VCF windows (Section 3.5). On that evidence, and with the user's approval, the 13 truncated VCFs and their

indexes were deleted, freeing about 125 GB; the complete chromosome-10 VCF was retained as a verification reference.

**[Interpreted]** Beyond solving the immediate problem, this route is strictly better for the analysis: it removes the AC/AN recomputation that caused the failure, it carries the RegionType annotation that makes the missing core-genome BED unnecessary, and it is reproducible from a public URL rather than from a 160 GB local artefact. It also made the resistance-panel recovery of Section 5.7.1 possible at all, and the second rebuild of Section 5.7.3 after it, because re-fetching nine genes under a different filter from a Zarr store is a minutes-long operation rather than a re-download. The panel has now been re-fetched twice, which only a cheap retrieval path makes practical.

> **But note.** The phrase "PASS biallelic SNPs only" in Option A is the decision that later invalidated a headline conclusion. It stands for two separate tests in the mask that was actually implemented, is_snp and numalt == 1, and the first of them does most of the damage: of the five canonical codons named in Section 5.7.1, four fail is_snp outright and only *pfdhps* K540E is lost to the biallelic test. Describing the loss as the work of "the biallelic restriction" alone understates it. The restriction as a whole was presented, accepted and implemented as a routine quality filter, and Section 5.7.1 sets out what it removed.

## 5.2 Adding t-SNE and PCoA alongside UMAP

**Problem and decision.** The thesis workflow specifies UMAP. The user asked for alternative ordinations so that the three could be compared. Both were added to Stage 3, reading the same distance matrix, with a quantitative comparison table.

**[Observed]** **Why it mattered more than expected.** UMAP and t-SNE are non-linear and their inter-cluster distances carry no quantitative meaning, a caveat the original script already printed. PCoA is linear, so its axes are interpretable and carry explicit variance proportions. The PCoA result places 1.3 per cent of variation on the first two axes, and in the comparison table the Spearman correlation between embedded and original pairwise distances is 0.32 for PCoA, 0.12 for t-SNE and 0.03 for UMAP.

**[Interpreted]** With structure as weak as it is here, those numbers make PCoA considerably more informative than any visual impression of the UMAP plot, and they would not have been available from the specified workflow alone. A correlation of 0.03 means the UMAP layout preserves almost none of the metric structure it is drawn from — which the plot itself does not disclose.

**[Interpreted]** Stage 3 was not affected by the corrected re-run, nor by either rebuild of the marker panel, which touches no genome-wide stage. This is a conclusion drawn from code reading plus file timestamps rather than a direct observation: step03_umap.py loads config.DIST_MATRIX, which config.py defines as MalariaGEN's published genome-wide pairwise distance matrix rather than the MAF-filtered analysis matrix, so the change from 3,359 to 3,916 SNPs described in Section 5.7 does not touch it; and results/pcoa_eigenvalues.csv and results/dimred_comparison.csv both still carry the modification time of the Stage 3 run.

## 5.3 Ploidy: how to count alleles in a haploid parasite coded as diploid

**Problem.** Blood-stage *P. falciparum* is haploid, but Pf8 records genotypes in diploid form: 0/0 or 1/1 for a clonal call, 0/1 where reads support both alleles (a mixed infection). The supplied step04b_fst.py called scikit-allel's count_alleles on the diploid array, which counts two gene copies per isolate. The same cached counts feed TreeMix and BayPass.

**Why this is a defect, not a convention.** Hudson's $F_{ST}$ carries a sample-size correction; TreeMix derives drift standard errors from counts; BayPass's XtX is standardised by them. Doubling every sample size therefore makes all three anticonservative — narrower intervals, more apparently significant edges, more apparent outliers.

[Observed] The earlier version of this section added "without changing the point estimates much". That is wrong, and wrong in kind. In Hudson's estimator the sample-size correction enters the numerator subtractively, as $-p(1-p)/(n-1)$ per population, so doubling $n$ inflates the point estimate, and the inflation is largest where $n$ is smallest. Recomputed during the audit on the then-current matrix, the six pairwise values are 1.6 to 31 times larger under diploid counting than under haploid counting, and the maximum pairwise $F_{ST}$ would have been 0.0127 for Coastal–Transitional rather than the ~0.002 actually reported. The correction is load-bearing for the headline number, and the earlier text undersold it.

[Calculated] One clause in that earlier text is withdrawn rather than kept. It said the three smallest-$n$ comparisons would have appeared to be the most differentiated pairs "rather than the least", and the second half does not hold on either matrix. On the final haploid matrix the three Transitional-zone pairs ($n$ = 25) are 0.00158, 0.00158 and 0.00047 against a six-pair range of 0.00045 to 0.00232 — ranks two, three and five of six. On the superseded matrix the audit measured them at 0.001245, 0.001219 and 0.000346, ranks three, four and six. Only Guinea–Transitional is among the least differentiated either way. What survives is the narrower and sufficient point: diploid counting would have lifted the smallest-$n$ pair to the top of the matrix from third place.

**Options put to the user.**

20. Haploid, mixed calls to missing — use the pipeline's existing `--het-to-missing` switch, then count one allele per isolate. Simplest; discards the mixed calls entirely.
21. Haploid, mixed calls to the majority allele (chosen) — assign the allele with more reads, ties to missing, then count one allele per isolate. Retains the information in the read depths.
22. Leave as written — point estimates move substantially upward and uncertainty stays understated throughout.

**Implementation and verification.** A function `resolve_mixed_calls` was added to `step04a_prepare_callset.py`, reading AD a block at a time only for SNPs where some retained isolate carries a mixed call. A companion `isolate_allele_counts` was added to `callset_io.py`; it takes the first gene copy once mixed calls are resolved, and falls back to diploid counting with an explicit warning if any mixed call remains. Both were verified against a hand-computed synthetic case (3 SNPs × 4 isolates, including a tie and a missing genotype) before being applied to real data.

[Observed] On the final dataset, 194,167 mixed calls were resolved: 129,665 to reference, 55,582 to alternate, and 8,920 tied and set to missing. The earlier version of this section reported a "residual heterozygous call rate" of 0.0000 to four decimal places; that figure is real but it is not residual. In `logs/rerun_corrected.log` the line "Heterozygous call rate: 0.0000" is printed under the mixed-call header and *before* the per-SNP resolution loop, so it is the pre-resolution rate — 194,167 of 10,845,661,093 called genotypes, or $1.8 \times 10^{-5}$, which rounds to 0.0000. After resolution the rate is exactly zero by construction, every mixed call having been assigned an allele or set to missing, and `isolate_allele_counts` warns if one survives. The honest statement is therefore that mixed calls were rare to begin with, at under two in every hundred thousand called genotypes, which is consistent with the Fws filter having already removed most mixed infections.

[Interpreted] The asymmetry — 2.33 reference calls for every alternate call — is expected, since at a site where most isolates carry the reference allele, a mixed infection is more likely to be reference-dominant. It is not evidence of a reference bias in the resolution rule itself, though that possibility has not been formally excluded.

> **A different rule in the codon panel.** The majority-allele rule chosen here is applied in the genome-wide callset and in step10_resistance_panel.py. The codon panel that supersedes it, step12_codon_panel.py (Section 5.7.3), does not use it: a codon with any mixed call among its three positions is reported as uncalled rather than resolved to a residue. That is the stricter choice and it is deliberate — a majority-allele call at each of three positions independently can compose a codon that no parasite in the isolate carries — but it means the two panels do not use the same ploidy convention, and the codon panel's denominators (1,834 to 1,900 of 1,900 per codon) are never larger than they would be under the Step 4a rule and for most codons are smaller — as low as 1,834 at *pfmdr1* 1034, while five codons are still called in every isolate.

## 5.4 Sample QC thresholds

Three thresholds were put to the user after the first QC run, each accompanied by the counts each option would produce. The call-rate and clonality decisions are discussed in Sections 4.2.3 and 4.2.4. The Fws threshold was retained at the conventional 0.95 (options at 0.90 and 0.80 were offered and declined).

[Interpreted] Presenting the sample counts alongside each option made the trade-off explicit, but it also means these thresholds were selected with knowledge of their effect on sample size. That is standard exploratory practice and is reported here rather than concealed.

[Unknown] Whether the 117 isolates dropped for excess missingness are a tail of the retained distribution or a distinct bad-data population cannot be established from what the pipeline recorded. The threshold is MAX_SAMPLE_MISSING = 0.20 at config.py:146, and no output preserves the per-sample missingness values it was applied to: results/sample_qc_full.csv holds the Stage-2 call rate, a different metric computed on a different variant set, and the run log names only 8 of the 117 isolates. An earlier draft asserted the tail reading; it had no source and is withdrawn. The mechanism of the filter-ordering defect of Section 5.7 does not depend on it either way — those isolates were charged against every SNP's missingness before they were removed, which cost 557 SNPs — about 14 per cent of the final panel, the measured 3,359 → 3,916 difference — whatever the shape of their distribution.

## 5.5 TreeMix jackknife block size

[Observed] **Problem.** The supplied configuration used 500 SNPs per jackknife block.

[Calculated] On the 3,359-SNP matrix current when the choice was made, that gives 6 or 7 blocks; on the final 3,916-SNP matrix it gives 7 or 8. Either way it is far too few for stable standard errors, which in turn makes the migration-edge tests unreliable.

[Calculated] **Decision.** Reduced to 50 SNPs, giving approximately 67 blocks on the matrix of the day and 78 on the final one. With retained SNPs averaging roughly 6 kb apart across the ~23.3 Mb genome, a 50-SNP block spans about 300 kb, which comfortably exceeds the scale over which linkage disequilibrium persists in *P. falciparum*. The block therefore remains large enough to absorb linkage while being small enough to give a usable number of replicates. A 100-SNP alternative was offered and declined; the figure actually put to the user was about 34 blocks, computed on the 3,359-SNP matrix, which on the final matrix would be 39.

[Observed] **The decision could not be judged until the re-run.** The sweep was originally executed without TreeMix's -se flag, so no standard errors were produced at any block size and the block-size choice had no observable consequence (Section 5.7, Table 5.2, Pass 2 — the defect was raised by the audit, not by the pre-run review). With -se now set, the two edges at *m* = 2 carry standard errors of 0.048 and 0.100 on weights of 0.152 and 0.805, giving $p = 7.5 \times 10^{-4}$ and $3.3 \times 10^{-16}$, while the third edge added at *m* = 3 has weight 0.065 against a standard error of 0.147 (p = 0.328). Roughly 78 blocks is therefore enough to separate a supported edge from an unsupported one, which is the property the reduction was intended to buy.

## 5.6 BayPass calibration

> **Three runs, three thresholds.** Three BayPass runs are referred to across this manuscript, and they are named consistently from here on. The **manuscript run** is the 8-thread run on the 3,359-SNP matrix whose numbers the original text reported: threshold 11.232, 15 SNPs above it (0.45 per cent of that scan), 11 annotated genes; it is archived under results/baypass_8thread_archive/. An **8-thread re-run** of the same input with the same -seed 42 is the reproducibility check described below, and it gave 16. The **interim single-threaded run**, still on the 3,359-SNP matrix, gave 11.047 and 17 outliers and is recorded in corrected/xtx_calibration.json. The **final run** is single-threaded on the 3,916-SNP matrix: 10.840 and 19 outliers. Naming them separately matters because the corrections document records the threshold moving 11.232 → 11.047 while the addendum records 11.232 → 10.840; both are correct and they describe different steps, and "the interim 3,359-SNP matrix" must not be used as a label for the manuscript run.

**Problems found.** Three in the supplied step08b_baypass.sh, and — as the audit later established — one in the corrections applied to it.

23. The pseudo-observed data (POD) simulation passed the .geno matrix directly as sample.size. That file interleaves reference and alternate counts, giving 2 × npop columns, whereas simulate.baypass requires npop columns of total haploid sample size per SNP. The call would have failed, and the script would have fallen through to an empirical quantile.
24. Only 1,000 pseudo-observed SNPs were simulated. The threshold is the 0.999 quantile of their XtX, so it would have rested on a single simulated value.
25. A stale G.pod from a previous run was not removed before each run, and an R failure aborted the stage under set -e instead of falling through to the documented empirical fallback.

**Corrections.** Reference and alternate columns are now summed to total haploid counts; the simulation uses 100,000 pseudo-observed SNPs, which places of order 100 values above the 0.999 quantile rather than one; a stale G.pod is deleted before each run; and an R failure now falls through to the empirical fallback.

**A fourth correction was made for the wrong reason, and is withdrawn.** The earlier version of this section stated that the simulation applied no minor-allele-frequency filter while the observed SNPs had been filtered at MAF ≥ 0.05, and that setting pi.maf = 0.05 corrected the mismatch by matching the two spectra. It does not. In baypass_utils.R, pi.maf truncates the simulated *ancestral* frequency π to [pi.maf, 1 − pi.maf]; per-population frequencies are then drawn around π through the covariance matrix, scaled by √(π(1 − π)) and clipped to [0, 1]; and only then are genotype counts drawn binomially from those per-population frequencies. It is the population-frequency step, not the binomial draw alone, that pushes realised minor-allele frequencies well below 0.05. 0.05 is also simulate.baypass's own default, so the change restored a default rather than imposing a filter. The pseudo-observed set is not MAF-matched, and the claim that it was — made in this section and twice elsewhere in the manuscript — was false.

[Observed] Measured during the audit on the superseded 3,359-SNP run: 9,396 of the 100,000 pseudo-observed SNPs (9.4 per cent) have realised MAF below 0.05, against none of the observed SNPs, and the pseudo-observed spectrum is shifted upward overall (median MAF 0.212 against 0.136 observed). The two spectra differ throughout their range, not only at the boundary.

[Observed] **The stated reason for the threshold moving was also wrong.** The first calibration, on 1,000 unfiltered pseudo-observed SNPs, gave a threshold of 8.375 and 40 outliers (1.19 per cent of the then-current scan); recalibrating on 100,000 gave 11.232 and 15 (0.45 per cent), the figures the manuscript run reported. The earlier text attributed that move to MAF matching, reasoning that rare simulated sites have low XtX and so depress the upper quantile, and concluded that the 25 SNPs between the two thresholds had "failed calibration". Neither the mechanism nor the conclusion survives measurement. Across 200 random 1,000-SNP subsamples of the same 100,000-SNP pseudo-observed set — with no MAF difference between them at all — the 0.999 quantile ranges from 8.988 to 13.248 (median 10.59) and the implied outlier count from 7 to 41. Both originally reported values fall inside that range. The correlation between pseudo-observed MAF and pseudo-observed XtX is −0.029, so the asserted mechanism is essentially absent; and MAF matching done properly moves the threshold in the *opposite* direction, to 10.53 when the pseudo-observed set is restricted to realised MAF ≥ 0.05 and to 10.59 ± 0.21 when it is importance-resampled to the observed MAF histogram.

> **Correction.** The 1,000-SNP calibration was not wrong because it was unfiltered. It was wrong because a 0.999 quantile estimated from 1,000 values is unstable by roughly ±2 XtX units. The claim that the 25 intermediate SNPs "failed calibration" is withdrawn: they fell between two estimates of the same quantity, one of which was too noisy to rank them either way. The correction to 100,000 SNPs remains an improvement; the explanation offered for it did not.

[Observed] **Reproducibility.** BayPass above one thread is not reproducible. With `-seed 42` fixed, the manuscript run and an 8-thread re-run of the same input differed by up to 1.2 in XtX, two fresh 8-thread runs of that input differ by up to 1.1 and give 15 against 16 SNPs above the then-reported threshold, and two single-threaded runs are bitwise identical. `NTHREADS` now defaults to 1 in `step08b_baypass.sh`, and `run_all.sh` exports `NTHREADS=1` in parallel mode; the core model takes seconds at this scale, so determinism is effectively free. Re-running the same 3,359-SNP input single-threaded — the interim run — gave 11.047 and 17 outliers against the manuscript run's 11.232 and 15, which is the size of the non-determinism in the numbers that were originally reported.

[Observed] **Final calibration.** On the final 3,916-SNP matrix, the single-threaded run gives a 0.999 pseudo-observed quantile of 10.840 (10.839767826730082 as written). Observed XtX spans 2.48 to 26.44 with a median of 3.45. Nineteen SNPs exceed the threshold, 0.49 per cent of the scan, against 3.916 expected by construction — an expected-null-exceedance ratio of 20.6 per cent. Thirteen of the nineteen also pass Benjamini–Hochberg at q ≤ 0.10; the counts at the neighbouring thresholds are 8 at q ≤ 0.01, 10 at q ≤ 0.05 and 17 at q ≤ 0.20. Of the nineteen, 17 fall in 13 annotated genes and two are intergenic, one of them the fifth-strongest SNP in the whole scan (XtX 20.79, q = 0.005) — so the gene list of Section 6.8 is 13 genes carrying 17 outlier SNPs, and the scan is not "19 outliers in 13 genes".

[Interpreted] The nineteen are a ranking and the thirteen are the finding, and both are reported. The quantile list defines the set on which the downstream gene annotation and interpretation were built, so suppressing it would misrepresent how the analysis was done; the q ≤ 0.10 subset is what survives multiplicity correction and is what any claim should rest on. The two intergenic outliers are part of the nineteen and are not represented in the gene list at all, which is a further reason to quote both counts rather than either alone.

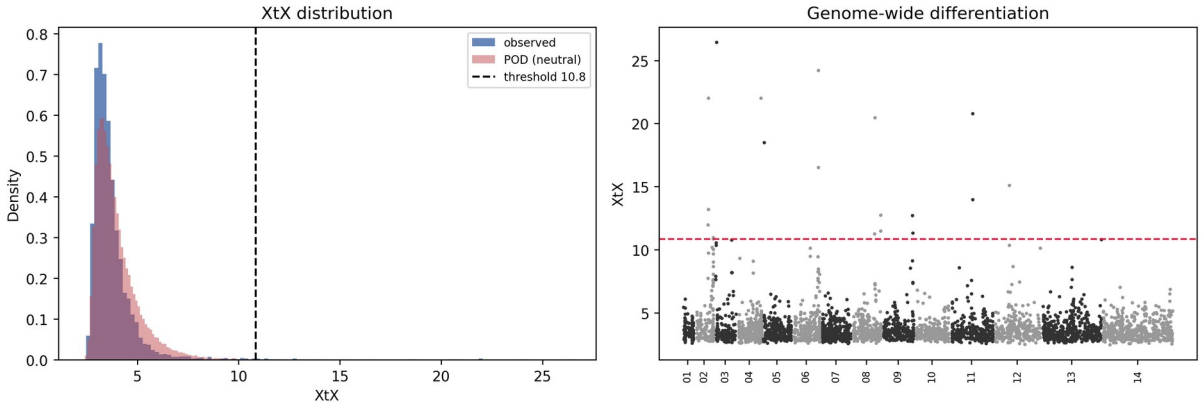



*Figure 5.1. Observed XtX along the genome against the pseudo-observed null, with the 0.999 quantile threshold at 10.840. Nineteen SNPs exceed the threshold; thirteen of them also pass Benjamini–Hochberg at q ≤ 0.10, and seventeen fall in annotated genes. Source: results/baypass/ (final single-threaded run, 3,916 SNPs).*

> **Caveat.** Two calibration weaknesses remain and are not corrected in the reported numbers. First, the pseudo-observed set is still not MAF-matched; on the superseded 3,359-SNP panel proper matching lowered the threshold, which would have admitted more outliers rather than fewer, but that experiment does not establish the direction of the error for the final 3,916-SNP run, whose raw pseudo-observed inputs are not archived. The frequency-matching limitation stands; a guaranteed conservative threshold does not (Section 5.7.4). Second, the pseudo-observed and observed XtX distributions do not agree well through the body of the range — on the superseded run the audit measured a Kolmogorov–Smirnov D of 0.158 below XtX 8, with the pseudo-observed distribution stochastically larger through the body and markedly shorter in the extreme tail. Distributional overlap is the only model-adequacy check this analysis performs, and it is weaker than §8.1 previously claimed.

## 5.7 Defects found and corrected

A systematic review of the unexecuted stages was carried out before they ran, and ten defects were corrected on that basis. Three further rounds followed: an adversarial audit of the completed analysis, which raised 110 findings and confirmed 61 under three-lens verification; the corrected re-run itself, which found defects both in the pipeline and in the corrections written to address the audit; and an external expert critique of the marker panel, all five of whose points were upheld and four of which were defects in code. Table 5.2 lists all twenty-seven defects now known to have been found and corrected, grouped by the pass that found them. Twenty-five are defects in code or analysis; the last two are errors in reading a source document, recorded here because they changed what the manuscript asserted. Each correction was verified either against the code path in question or by execution.

| Defect | Consequence | Correction | Verification |
|---|---|---|---|
| Pass 1 — review of the unexecuted stages, before the first run | | | |
| config.TREEMIX_ROOT was "Guinea_Savanna" but slugify_zone produces "Guinea_savanna"; step07b would have exited 1 at its header check, aborting stages 7–9 under set -e | Would abort the run | Corrected to "Guinea_savanna"; step07b now derives its default from config rather than keeping its own copy | Stage 7 ran to completion |
| step07c model-selection | Wrong result, silently | Replaced with the | Selects m = 2 over 0 to 2; the |

| Defect | Consequence | Correction | Verification |
|---|---|---|---|
| rule was gains.fillna(1).idxmax(), which always returns m = 0 because the first difference is NaN and the fill value exceeds any real gain | | documented rule: smallest m reaching 0.998 variance explained, else the smallest m within 0.001 of the range's best. A maximum-likelihood fallback was rejected because likelihood is monotone in m | full m = 0 to 5 sweep still selects m = 3, and the edge p-values favour m = 2 only conditional on a topology, root and edge count fixed beforehand (Section 5.5) |
| step07c variance explained was computed over the full covariance matrix including the diagonal; TreeMix's own get_f uses off-diagonal entries only, and the 0.998 convention assumes that definition | Wrong result | Restricted to the upper triangle | Definition recorded in results/treemix/model_selection.json |
| Allele counts doubled by diploid coding of a haploid parasite (Section 5.3) | Inflates Hudson FST point estimates 1.6× to 31× and understates uncertainty in FST, TreeMix and BayPass alike | isolate_allele_counts in callset_io.py, with mixed calls resolved by read depth in step04a | Synthetic unit test; 194,167 mixed calls resolved, leaving no mixed call behind by construction, with isolate_allele_counts warning and falling back to diploid counting if one survives |
| Zone-level Mantel test used polygon centroids, which lie where no isolate was sampled | Wrong geography | Zones placed at the sample-weighted mean of their isolates' coordinates, matching the region-level logic | Printed zone coordinates in logs/stage6.log |
| step08b passed the interleaved .geno matrix as simulate.baypass's sample.size, simulated only 1,000 pseudo-observed SNPs, left a stale G.pod in place, and aborted on R failure | Calibration failure; the 0.999 quantile would have rested on a single simulated value | Columns summed to total haploid counts; 100,000 pseudo-observed SNPs; stale G.pod deleted each run; R failure falls through to the empirical fallback | Threshold recomputed (Section 5.6). Note that the accompanying MAF-matching rationale was itself wrong and is withdrawn there |
| pfcoronin recorded as PF3D7_1362500 and pfap2mu as PF3D7_1451200 | Silent false negatives in the resistance-locus check | Corrected to PF3D7_1251200 and PF3D7_1218300 | Confirmed against the PlasmoDB-71 GFF: the original identifiers annotate a 3′–5′ exonuclease and a conserved protein of unknown function respectively |
| PLINK export wrote contig names (Pf3D7_01_v3) as chromosome codes; ADMIXTURE requires integer codes | Possible failure | Contigs mapped to integers 1 –14, with a mapping file written alongside; SNP identifiers retain the real contig names | ADMIXTURE ran for all K |
| PLINK invoked with no --threads or --memory, so it would claim every core and reserve half of system RAM | Resource hazard | Capped at 4 threads and 16 GB | Observed during the run |
| A failing zone-level Mantel test would abort the pipeline before TreeMix and | Fragility | Non-fatal, with a printed message | Not triggered; the test succeeded |

| Defect | Consequence | Correction | Verification |
|---|---|---|---|
| BayPass | | | |
| Pass 2 — adversarial audit of the completed analysis: 110 findings raised, 61 confirmed | | | |
| step03a built the genome-wide callset under FILTER_PASS & is_snp & numalt == 1. Almost every canonical resistance codon in Pf8 either is not a simple SNP or is multiallelic, so 28 of the 44 canonical codon-allele records then on the list were removed at ingest, before any analysis ran — is_snp excludes four of the five codons named in Section 5.7.1 and numalt == 1 the fifth | Invalidated a headline conclusion. The reported absence of validated resistance loci from the outlier list was a filter artefact and was interpreted as biology (Section 5.7.1) | New script step10_resistance_panel.py re-fetched the nine panel genes with no numalt restriction and no is_snp restriction, resolved each annotation to its ALT allele, applied the same majority-allele ploidy rule as Step 4a, and reported Wilson intervals. Superseded in Pass 4 by step12_codon_panel.py, which identifies codons by position rather than by annotation string (Section 5.7.3) | Across the nine panel genes, 13,367 PASS records enter the filter and 3,330 leave it. On the final panel, 57 codons are genotyped and 24 are polymorphic, including every codon the mask had removed |
| BayPass run at -nthreads 8. With -seed 42 fixed, the manuscript run and an 8-thread re-run differ by up to 1.2 in XtX, and two fresh 8-thread runs differ by up to 1.1 and give 15 against 16 outliers | The reported threshold, outlier count and XtX values were one random draw | NTHREADS default changed from 4 to 1 in step08b, with the reason recorded in a comment; run_all.sh exports NTHREADS=1 in parallel mode | Two single-threaded runs of the same input are bitwise identical; the interim single-threaded run on the same 3,359 SNPs gave 11.047 and 17 outliers against the manuscript run's 11.232 and 15 |
| step04a measured per-variant missingness over all 2,017 QC-passing isolates and only then dropped 117 of them for excess sample missingness | Every SNP was charged for isolates that did not survive, costing roughly a tenth of the final panel for nothing | Two-pass: the sample cut is decided on the same provisional variant set as before, so the same isolates drop, then variant missingness is recomputed on the 1,900 retained isolates. Implemented blockwise so the provisional subset is never materialised | 1,634,808 variants at the provisional pass, 1,807,331 after recomputation at the identical threshold; final panel 3,359 → 3,916 SNPs (+16.6%) with the same 1,900 isolates |
| step07b ran the TreeMix sweep without -se, so every migration line in .treeout read "weight NA NA NA" | No edge standard errors or p-values existed, so the m = 2 against m = 3 choice had no inferential basis and was argued from theory alone | SE=1 is now the default, producing jackknife means, standard errors and p-values per edge | Both m = 2 edges significant ($p = 7.5 \times 10^{-4}$ and $3.3 \times 10^{-16}$); the third edge at m = 3 has p = 0.328 |
| No false-discovery accounting for the XtX outliers. A 0.999 quantile passes 0.1 per cent of neutral SNPs by construction | An expected-false-positive rate was being presented as a significance test | step08c now reports expected false positives at the threshold, the resulting expected-null-exceedance ratio, per-SNP p-values against the pseudo-observed null, Benjamini–Hochberg q-values written into xtx_all_snps.csv, and counts at q ≤ 0.01/0.05/0.10/0.20 | 19 SNPs above threshold, 3.9 expected false positives, an expected-null-exceedance ratio of 20.6 per cent, 13 also passing q ≤ 0.10 (8 at q ≤ 0.01, 10 at q ≤ 0.05, 17 at q ≤ 0.20) |

| Defect | Consequence | Correction | Verification |
|---|---|---|---|
| step06_mantel set no random seed, so both reported Mantel p-values were irreproducible Monte Carlo estimates of tests whose permutation distributions are small enough to enumerate | Irreproducible; re-running the script gave different third-decimal p-values | config.set_seeds() is now called; seed and permutation count are recorded in the output JSON | Seed 42 and 9,999 permutations written to results/mantel_result_region.json and _zone.json |
| step06_mantel silently clipped negative linearised FST to zero to satisfy a scikit-bio input constraint, not for any statistical reason | Undisclosed censoring. On the interim 3,359-SNP matrix the censoring was what carried the region test across $p < 0.05$ | The censored pairs are named in the output, both censored and uncensored r and p are reported, and a leave-one-out jackknife over populations was added | 3 region pairs floored on the final matrix; region r = 0.4269, p = 0.0252 censored and r = 0.4138, p = 0.0413 uncensored, so both now clear 0.05; significant in 4 of 7 jackknife replicates, 0 of 4 at zone level |
| Zone is confounded with sequencing study — a fourth design confounder, absent from the manuscript's three | Four of six zone pairs share no sequencing study at all, including Coastal savanna against Guinea savanna, the pair behind every headline number | New script step11_batch_confounding.py tabulates study composition by zone, counts shared studies per zone pair, and tests within-zone, between-study heterogeneity in call rate and Fws with zone held constant, on the 4,905 QC-passing isolates rather than the 1,900 analysis isolates | Transitional zone is 100 per cent one study; within Guinea savanna, call rate differs between its six studies at $p = 2.8 \times 10^{-197}$; within deciduous forest, which has five contributing studies of which four are large enough to test, the polyclonal fraction ranges 0.430–0.578 between those four (Kruskal–Wallis p = 0.002) |
| Pass 3 — defects found by the corrected re-run, and inside the corrections themselves | | | |
| The "skip if the output exists" guards in step05b, step07b and step08b did not test whether the input had changed. When the SNP panel grew from 3,359 to 3,916, every stage regenerated its inputs, and then ADMIXTURE and TreeMix saw their previous output files, skipped the work entirely, and reported "stage finished" (Section 5.7.2) | The previous day's ADMIXTURE and TreeMix results would have been reported as though they came from the new matrix. The stage logs recorded success | All three guards now require the output file to be newer than its input | Caught by step08c's assertion that its position table and the BayPass output have matching row counts, which failed loudly with the correct diagnosis. ADMIXTURE and TreeMix had no such assertion; the confirming evidence was ghana_pruned.bim rebuilt at 01:27 against ghana_pruned.2.Q still dated 03:57 the previous day |
| SnpEff emits one annotation per (ALT, transcript) pair, so annotation column a is not ALT allele a. The new step10_resistance_panel.py indexed them positionally | Correct allele frequencies attached to wrong amino-acid names: pfmdr1 N86Y was reported as "Asn86Asp" and Y184F as "Tyr184Ser" — a defect introduced by the corrections, not found in the original pipeline | The annotation index is resolved through ANN_Allele. Pass 4 removed the dependence on annotation fields altogether | Caught only by cross-validating the panel against MalariaGEN's independently published marker genotypes for the same 1,900 isolates |
| Pass 4 — external expert | | | |

| Defect | Consequence | Correction | Verification |
|---|---|---|---|
| critique of the marker panel: five points raised, five upheld, four of them identifying defects in the analysis code (listed here as five rows, because the annotation-string root cause produced two distinct failures) | | | |
| step10_resistance_panel.py identified markers by matching SnpEff ANN_HGVS_p protein-change strings against a hand-written list. A resistance allele that is itself the reference residue can never appear as an ALT change. The 3D7 reference codon at pfdhps 437 translates as glycine, the resistant residue, so the only annotation there is the revertant Gly437Ala; searching for "Ala437Gly" matched nothing | The most consequential defect found after the audit. pfdhps A437G, the most frequent pfdhps marker in the dataset and the second most frequent antifolate marker overall, behind pfdhfr S108N at 0.908, was absent from the panel and from the manuscript entirely — not under-reported but missing, at a true G437 frequency of 0.897 | New script step12_codon_panel.py maps each requested codon to its three genomic positions through the CDS, collects every variant record at those positions regardless of numalt or is_snp, applies each isolate's alleles to the reference codon, translates, and reports the full residue spectrum including the reference residue | G437 = 0.897 (n = 1,892); by zone 0.964 deciduous forest, 0.941 coastal savanna, 0.875 Guinea savanna, 0.800 transitional zone, $\chi^2$ p = $1.2 \times 10^{-6}$. Confirmed against the official Pf8 methodology, which states that 3D7 carries the mutant allele at 437. 57 codons across 9 genes in codon_panel_residues.csv |
| The same string-matching scheme cannot express a residue reachable only by two changes in one codon. pfcrt codon 75 carries 403621 (A>G, annotated Asn75Asp) and 403623 (T>A, annotated Asn75Lys); asparagine to glutamate requires both, so no annotation names N75E, the residue the CVIET haplotype actually carries | The resulting disagreement with MalariaGEN at codon 75 was reported as a "naming discrepancy". It was not a naming problem — the call was wrong, and this is the whole explanation of the disagreement | Codon reconstruction yields Glu where string-matching yielded Asp | Codon 75 agrees with MalariaGEN on all 1,874 isolates where both pipelines call. N75E = 0.071, identical to M74I and K76T as the CVIET haplotype requires |
| The pfk13 marker list was written from memory rather than transcribed from an authoritative source. It held 13 distinct protein changes, with a duplicated M476I record inflating the apparent count to 14; it omitted P574L and G533S; and it classified P441L and C469F as validated when WHO classifies both as candidate. R622I was on the list but was never retrieved, which under string matching produced silence rather | The reported artemisinin-resistance negative rested on an incomplete and partly misclassified list, three validated markers were never examined, and two candidate markers were presented to readers as validated | The panel is taken variant by variant from the WHO compendium v1.0 (December 2025): all 14 validated variants (F446I, N458Y, C469Y, M476I, Y493H, G533S, R539T, I543T, P553L, R561H, P574L, C580Y, R622I, A675V) and all 6 candidates (E252Q, P441L, C469F, N537I, G538V, V568G), each carrying its WHO tier, plus the four potential-tier codons in the same propeller region (449, 481, 515, 527) | 23 distinct pfk13 codons genotyped — the 19 codons carrying the 20 WHO validated and candidate variants (C469Y and C469F share codon 469), plus four potential-tier codons in the same region (449, 481, 515, 527); every one fixed at the reference residue, codon 458 at 0.999 on a single N458D call. Denominator is per codon, 1,845 to 1,876 of 1,900 called across the pfk13 codons, and the sampling period is 2009–2021 |

| Defect | Consequence | Correction | Verification |
|---|---|---|---|
| than an error | | | |
| The pfdhfr "triple mutant" frequency was reported as the range of its three marginal frequencies, 0.786 –0.908. No haplotype was counted within isolates anywhere in the panel | A quantity presented as a haplotype frequency was not one, and the range was arithmetically inadmissible: a conjunction of three alleles cannot exceed the rarest of them, so the range's own lower bound was its ceiling | step12_codon_panel.py counts haplotypes on isolates called at every constituent codon and reports Wilson intervals | pfdhfr IRN = 0.771 (n = 1,876; 95% CI 0.751–0.789), below the complete-case 51I marginal of 0.788 as it must be; NRN = 0.116 and wild-type NCS = 0.091, so it is not a near-fixed single haplotype; IRNL = 0.000 (n = 1,874). pfcrt CVIET = 0.071 (n = 1,873), only CVMNK (0.929) and CVIET observed. pfdhps GE = 0.010 (n = 1,888). In codon_panel_haplotypes.csv |
| Validation compared marginal allele frequencies against MalariaGEN's published genotypes and reported agreement within 0.03 for 21 of 23 markers. Marginal agreement cannot establish that two pipelines call the same isolates the same way | A weak check was presented as cross-validation, and it was the check that let the codon-75 error through: two pipelines can agree on a marginal while disagreeing on which isolates carry the allele | Concordance is computed call by call, per isolate per marker, with disagreements enumerated | 47,012 of 47,012 calls agree — 100.0000 per cent — across 25 markers in pfcrt, pfmdr1, pfdhfr, pfdhps and pfmdr2, codon 75 included, in codon_panel_concordance.csv |
| Error in reading a source, not in code. This manuscript stated that the pfaat1 functional work did not test amodiaquine, and framed the Ghanaian pfaat1 frequency difference as maintained by amodiaquine pressure | Both halves were false. Amodiaquine was among the five drugs whose IC50 values were measured, the observed shifts were described by the authors as well below the clinical-resistance threshold, and the authors explicitly reject the hypothesis that other quinoline partner drugs drive these frequencies, concluding that chloroquine was the primary selective force | S258L is now described as a variant with a demonstrated but modest, epistasis- and background-dependent effect that potentiates chloroquine resistance. Amodiaquine is offered only as an untested possible contemporary driver, alongside historical persistence of chloroquine pressure and the documented fitness cost | Verified against the source: the effect is 1.5-fold, in a single chloroquine-resistant Southeast Asian line (NHP4026), with epistasis between S258L and F313S and an explicit pfcrt-background caveat; the 0 per cent → 97 per cent longitudinal rise is Gambian, 1984–2014, and the paper says nothing about Ghana |
| Error in reading a source, not in code. WHO resistance labels were applied per gene. The panel described pfmdr1 as "WHO-validated, lumefantrine", and pfaat1, pfap2mu and pfmdr2 as WHO-classified markers | A category error twice over. The compendium classifies variants, not genes, and it identifies no validated or candidate lumefantrine marker at all; pfaat1, pfap2mu and pfmdr2 appear nowhere in it at any tier | Tiers are recorded per variant. The three unlisted genes remain in the panel on scientific grounds but are never described as WHO-classified; pfcoronin is labelled potential, which is its only tier | Checked variant by variant against WHO compendium v1.0 (December 2025) |

*Table 5.2. All twenty-seven defects now known to have been found and corrected, grouped by the pass that found them. Twenty-five are defects in code or analysis; the last two are errors in reading a source document, included because they changed what the manuscript asserted. Note for readers coming from the corrections document: four of the six items its §7 still lists as outstanding — the filter order, the TreeMix sweep without -se, the study/batch confounding and the absence of FDR accounting — appear here as corrected, because that list was written before the addendum runs and is superseded here. Confirmed defects that have not been corrected are listed separately in Section 5.7.4. The original scripts remain unmodified in pf8_ghana_pipeline.zip in the project root.*

### 5.7.1 The ingest filter on PASS biallelic SNPs: the first defect that changed a conclusion

Of the twenty-seven defects in Table 5.2, most changed a number, a standard error, a reproducibility property or a resource bound. Three changed a conclusion, and each is separated from the rest: the ingest filter described here, the annotation-string matching that forced a complete rebuild of the marker panel (Section 5.7.3), and the misreading of the *pfaat1* functional work (Table 5.2, Pass 4, and the note at the end of this subsection).

**[Observed]** step03a_fetch_callset_zarr.py builds the genome-wide callset under FILTER_PASS & is_snp & numalt == 1. Read as English, that is "PASS, SNP, biallelic" — a filter so conventional that it was proposed, accepted and implemented without comment (Section 5.1). In Pf8, however, most canonical antimalarial-resistance codons are either multiallelic or carry a spanning-deletion ALT allele, so the mask excluded them, and the two tests do not contribute equally: *pfcrt* K76T and M74I, *pfmdr1* N86Y and *pfdhps* A581G all fail is_snp outright, while *pfdhps* K540E passes is_snp and is excluded by numalt == 2. Across the nine panel genes, 13,367 PASS records enter the filter and 3,330 leave it: 10,037 are dropped, including 28 of the 44 canonical codon-allele records on the list then in use. That list has since been replaced twice (Section 5.7.3); the filter's effect on the genes' records is a property of the filter and does not depend on it.

**[Observed]** The analysis then searched its outlier list for exactly those loci, found none, and read the absence as biology. The abstract and §6.8 both reported that no validated antimalarial resistance locus appears among the outliers, and Objective 3 — zone-specific variants relevant to drug response — was written up as a null result. The loci had been removed from the search space before the search began.

**[Observed]** A new script, step10_resistance_panel.py, re-fetched the nine panel genes with the numalt and is_snp restrictions lifted; it has itself been superseded by step12_codon_panel.py, which identifies codons by genomic position rather than by annotation string (Section 5.7.3). The final panel genotypes 57 codons across the nine genes, of which 24 are polymorphic. The recovered panel does not merely fill a gap; it reverses the finding. *pfaat1* S258L — a variant with a demonstrated but modest, epistasis- and background-dependent effect that potentiates chloroquine resistance, and which is named twice in the project's own thesis proposal — runs from 0.824 in coastal savanna to 0.544 in Guinea savanna (0.735 deciduous forest, 0.640 transitional zone; 0.609 overall, n = 1,896), with a $\chi^2$ heterogeneity p of $1.9 \times 10^{-18}$ across zones, and carries q = 0.0049 in the XtX scan. *pfdhps* A437G, which was absent from the panel altogether until the fourth round, stands at G437 = 0.897 (n = 1,892) with $\chi^2$ p = $1.2 \times 10^{-6}$. The *pfdhfr* triple mutant IRN is common but not near-fixed, at 0.771 overall (n = 1,876) and 0.867, 0.788, 0.750 and 0.880 in coastal savanna, deciduous forest, Guinea savanna and the transitional zone respectively. Meanwhile all 23 *pfk13* codons genotyped, covering every WHO validated and candidate variant, carry no screened resistance allele — twenty-two fixed at the reference residue, and codon 458 at 0.9995 on a single Asn458Asp call, which is not the validated N458Y — which converts an untested absence into a genuine tested negative, on 1,845 to 1,876 isolates called per codon and for the 2009–2021 sampling period.

> **Not a cline.** The zone frequencies above should not be read as a south-to-north gradient. The transitional zone is the second-most-northern of the four zones, not the most southern, and the orderings contradict one another: it carries the highest *pfdhfr* Ser108Asn and Cys59Arg frequencies in the panel at 1.000 (n = 25) and the highest triple-mutant frequency at 0.880, but the lowest *pfdhps* G437 at 0.800, while its *pfaat1* 258L at 0.640 sits between coastal savanna's 0.824 and Guinea savanna's 0.544. Any single ordering claim breaks at the transitional zone, and it rests on 25 isolates from one sequencing study (Section 5.8). The correct statement is heterogeneity between zones, demonstrated by the $\chi^2$ tests, not a monotone cline.

**[Interpreted]** An absence is evidence only if the thing absent was inside the search space, and nothing in the pipeline checked that it was. The resistance-locus panel lived in config.KNOWN_RESISTANCE_LOCI, three stages downstream of the filter that removed it, and no stage asserted that the panel's target positions were present in the analysis matrix. A single such assertion — do the codons we intend to look for survive our own filters? — would have failed at Stage 4 and prevented the interpretation entirely. This is the generalisable lesson of the defect, and it is a design lesson rather than a coding one: a pipeline that reports on the absence of named targets must verify that those targets reached the stage that reports on them. Section 5.7.3 records that satisfying this lesson was not sufficient, because it says nothing about whether the list of targets was the right list.

**[Observed]** Two consequences for the rest of the manuscript. The abstract's resistance sentence must be withdrawn and replaced, not softened. And the "two outlier SNPs" at AAT1 are one signal, not two: the second is Pf3D7_06_v3:1 214 350, a synonymous variant 883 bp from S258L, so it is a linked site rather than an independent line of evidence.

> **Correction: the amodiaquine framing was wrong.** An earlier version of this note said that Amambua-Ngwa et al. (2023, *Nature Microbiology*) demonstrated chloroquine resistance for S258L, and that extending it to amodiaquine was an inference the paper could not support because amodiaquine had not been tested. Amodiaquine was tested: it is one of five drugs whose IC50 values the methods report measuring, the authors describe the observed shifts as well below the threshold associated with clinical resistance, and they explicitly reject the hypothesis that use of other quinoline partner drugs drives these frequencies, concluding instead that chloroquine was the primary selective force. The demonstrated effect is a 1.5-fold rise in chloroquine IC50 in a single chloroquine-resistant Southeast Asian line (NHP4026), with epistasis between S258L and F313S and an explicit caveat about *pfcrt* background; the 0 per cent to 97 per cent longitudinal rise is Gambian, 1984–2014, and the paper says nothing about Ghana. Wherever this manuscript discusses contemporary selection on *pfaat1*, amodiaquine may appear only as an untested possibility, set alongside historical persistence of chloroquine pressure and the documented fitness cost. The previous framing inverted the paper's own conclusion, and that is a worse error than the overreach it was written to avoid.

### 5.7.2 Stale output caches: the most dangerous class of defect

One defect of the audit round was found by the corrected re-run itself, and it is recorded here because its failure mode is the worst available: it produces plausible results, from the right code, and reports success.

**[Observed]** step05b_admixture.sh, step07b_treemix.sh and step08b_baypass.sh each guarded their expensive work with a "skip if the output already exists" test. None of them tested whether the *input* had changed. When the filter-order fix raised the panel from 3,359 to 3,916 SNPs, every stage dutifully regenerated its own inputs — and then ADMIXTURE and TreeMix found their previous output files on disk, skipped the computation, and printed "stage finished". Had the matter rested there, the previous day's ancestry proportions and migration edges would have been reported as results of the new matrix.

**[Observed]** Only Stage 8 caught it. step08c_calibrate_xtx.py asserts that its SNP position table and the BayPass output have matching row counts, because they must correspond row for row; the assertion failed and named the cause. The confirming evidence was in the timestamps: ghana_pruned.bim rebuilt at 01:27, ghana_pruned.2.Q still dated 03:57 the previous day. ADMIXTURE and TreeMix carry no comparable assertion and failed silently. All three guards now require the output to be newer than its input.

**[Interpreted]** The asymmetry is the point. One stage happened to assert a structural relationship between its output and its current input, and that single assertion is the entire reason a wrong set of results was not published. The two stages without such an assertion were caught by inspecting file modification times by

hand — that is, by luck and suspicion, not by anything in the pipeline. Existence-based caching turns any re-run into a silent partial re-run, and because the skipped stages log success, there is no signal to investigate. Every stage should assert at least one property linking its output to the input it was actually computed from; a row count, a checksum or a timestamp comparison is enough.

### 5.7.3 Identifying markers by annotation string: the most consequential defect found after the audit

The panel built to repair the ingest-filter defect of Section 5.7.1 found its markers by matching SnpEff's `ANN_HGVS_p` protein-change strings against a hand-written list of expected changes. That scheme has two failure modes, and both occurred in this dataset. It is recorded here at length, as well as in Table 5.2, because of the three defects that changed a conclusion it is the one with the largest reach: it removed an entire locus from the manuscript, it produced the wrong explanation for a disagreement with an external dataset, and it was invisible to every check the pipeline performed on itself.

[Observed] **A resistance allele that is itself the reference residue can never appear as an ALT change.** At *pfdhps* 437 the 3D7 reference codon translates as glycine, which is the resistant residue; the official Pf8 methodology states the position directly, noting that "3D7 does not have the wild type allele" there. The only annotation at codon 437 is therefore `Gly437Ala`, the revertant. The panel searched for "Ala437Gly", matched nothing, and reported nothing — with no error and no empty-result warning, because the absence of a matching string is indistinguishable from the absence of the variant. *pfdhps* A437G, the most frequent *pfdhps* marker in the dataset at G437 = 0.897 (n = 1,892) and the second most frequent antifolate marker behind *pfdhfr* S108N at 0.908, was missing from the manuscript entirely rather than under-reported.

[Observed] **A residue reachable only by two changes in one codon is never annotated.** *pfcrt* codon 75 carries two variant records, 403621 (A>G, annotated `Asn75Asp`) and 403623 (T>A, annotated `Asn75Lys`). Asparagine to glutamate requires both changes together, so neither annotation names N75E, which is the residue the CVIET haplotype actually carries. The previous version of this manuscript reported the resulting disagreement with MalariaGEN as a naming discrepancy and left it at that. It was not a naming discrepancy. The call was wrong, this is the whole explanation of it, and the explanation offered instead was reassuring in a way the evidence did not license.

**The fix is positional rather than textual.** `step12_codon_panel.py` maps each requested codon to its three genomic positions through the CDS, collects every variant record at those positions regardless of `numalt` or `is_snp`, applies each isolate's alleles to the reference codon, and translates. The reported quantity is the full residue spectrum at the codon, the reference residue included, so a resistance allele cannot hide in the reference and a two-SNP residue cannot be missed. A codon with any mixed call among its three positions is reported as uncalled rather than resolved to a residue. The panel covers 57 codons across the nine genes, of which 24 are polymorphic, with Wilson intervals per zone in `codon_panel_residues.csv`.

[Observed] **The marker list itself was the other half of the defect, and it was written from memory.** The *pfk13* list was not transcribed from an authoritative source. It contained 13 distinct protein changes, with a duplicated M476I record inflating the apparent count to 14; it omitted P574L and G533S; and it described P441L and C469F as validated when WHO classifies both as candidate. R622I was on the list but was never retrieved, which under string matching produced silence rather than an error, so its absence became a silent non-result rather than an omission. The list is now taken variant by variant from the WHO compendium v1.0 (December 2025), covering all 14 validated variants (F446I, N458Y, C469Y, M476I, Y493H, G533S, R539T, I543T, P553L, R561H, P574L, C580Y, R622I, A675V) and all 6 candidates (E252Q, P441L, C469F, N537I, G538V, V568G), with each variant's tier recorded against it, plus the four potential-tier codons that fall in the same

propeller region (449, 481, 515 and 527). The 20 validated and candidate variants span 19 distinct codons, because C469Y and C469F share codon 469; with the four potential-tier codons the panel reconstructs 23.

[Observed] **What the rebuilt negative actually says.** Every one of the 23 *pfk13* codons is fixed at the reference residue, with a single N458D call giving codon 458 a reference frequency of 0.999. Two qualifications the previous version did not make. The denominator is per codon and runs from 1,845 to 1,876 of the 1,900 isolates, not 1,900 at every codon, so "all 1,900 isolates" was never the right phrase. And the isolates were collected between 2009 and 2021, which makes this a historical negative: it is evidence about the sampling period and carries no information about the present. The negative is now stronger in substance — it covers the complete validated set, including the three validated markers the old results never reported, and it reports the two candidates that had been misclassified as validated — and weaker in the form it may be stated in.

[Observed] **Validation replaced, not merely strengthened.** The previous panel was validated by comparing marginal allele frequencies against MalariaGEN's published genotypes, reporting agreement within 0.03 for 21 of 23 markers. That check cannot detect a per-isolate disagreement, and it did not detect the codon-75 error: two pipelines can agree on a marginal while disagreeing about which isolates carry the allele. Concordance is now computed call by call. On isolates where both pipelines call, 47,012 of 47,012 calls agree — 100.0000 per cent — across 25 markers in *pfcrt*, *pfmdr1*, *pfdhfr*, *pfdhps* and *pfmdr2*, with codon 75 among them, listed marker by marker in codon_panel_concordance.csv.

> **Caveat on the kelch13 cross-check.** Pf8's own *kelch13* resistance status is assigned over the whole 349–726 domain, and any missing call in that range makes the status undetermined. Only 36 of the 1,900 isolates therefore carry a complete Pf8 status. That is weak corroboration for per-codon calls made on roughly 1,860 isolates, and the two must not be presented as equivalent evidence. Note also what the 100 per cent figure above does and does not cover: MalariaGEN publishes individual genotypes for the 25 markers in the five genes named, and no *pfk13* codon is among them. The *pfk13* negative therefore has no per-codon external check at all, and nothing beyond the weak domain-level corroboration above.

[Calculated] **Haplotypes, and a number that could not have been right.** The *pfdhfr* triple mutant was reported as the range of its three marginal frequencies, 0.786 to 0.908. That is not a haplotype frequency, and the range could not have contained one: a conjunction of three alleles cannot be more frequent than the rarest of them, so the range's own lower bound was its ceiling, attainable only if the three mutant alleles nested perfectly. Counted within isolates called at all three codons, IRN is 0.771 (n = 1,876; 95% CI 0.751–0.789), below the complete-case 51I marginal of 0.788 as it must be. It is also not a near-fixed single haplotype: NRN sits at 0.116 and the fully wild-type NCS at 0.091. *pfdhfr* IRNL, the quadruple, is 0.000 (n = 1,874). *pfcrt* CVIET, by contrast, is genuinely a unit at 0.071 (n = 1,873) — only CVMNK at 0.929 and CVIET are observed, with nothing intermediate — but that is now demonstrated from reconstructed haplotypes rather than inferred from six similar marginals. *pfdhps* GE (437G with 540E) is 0.010 (n = 1,888).

[Interpreted] **The lesson is Section 5.7.1's, one level up.** Section 5.7.1 concluded that a pipeline reporting on the absence of named targets must verify that those targets reached the stage that reports on them. The panel built to satisfy that lesson did satisfy it: every codon on its list was retrieved, genotyped and reported, and the retrieval was asserted. It was still wrong, because the list was not the right list and the matching rule could not express two of the things on it. Checking that you found what you looked for does not check that you looked for the right things, or that your way of looking was capable of finding them. The assertions that would have caught this are external, not internal: compare the panel's membership against a published

compendium, and compare its calls against an independently published genotype, isolate by isolate rather than in aggregate. Both are now in place, and neither existed before the fourth round.

### 5.7.4 Scope of the review, and what the discovery pattern implies

[Interpreted] **Scope.** The first review covered only the stages that had not yet run, and found its defects by reading code. It was not an exhaustive audit, and it missed, structurally, two whole classes of problem: anything already executed, and anything that can only be seen by comparing a result against its own inputs or against an outside source. The two first-round defects that would have produced silently wrong rather than obviously broken output — the doubled allele counts and the TreeMix selection rule — were found by reading, not by any test.

[Observed] **Four passes, four methods, four new classes of defect.** The audit raised 110 findings and confirmed 61, including one that invalidated a conclusion (Section 5.7.1), four that changed reported numbers, and three that concerned reproducibility or undisclosed data handling. The third pass — executing the corrections and cross-checking the new marker panel against MalariaGEN's published calls — found two more, one of them *inside the corrections*: the SnpEff annotation-index bug that attached correct frequencies to wrong amino-acid names, which no amount of internal consistency checking would have exposed because the frequencies were right. The fourth pass was an external expert critique of the marker panel. All five of its points were upheld and four were defects in code; between them they restored a locus that had been missing altogether, replaced the *pfk13* marker list, replaced a marginal range that had been presented as a haplotype frequency, replaced the validation procedure, and inverted a published interpretation (Section 5.7.3, Table 5.2).

[Interpreted] **The fourth successive round in which a correction contained its own defect.** That is a pattern, not an incident, and it has one common thread. Pass 2 found that the ingest filter had removed the very loci the analysis reported absent. Pass 3 found that the panel built to recover them indexed SnpEff annotations positionally, attaching right numbers to wrong names. Pass 4 found that the panel built after that identified its markers by annotation string, which cannot see a resistance allele sitting in the reference or a residue needing two changes, and that its *pfk13* list had been written from memory rather than from a source. In every round the correction was checked for internal consistency — against our own outputs, our own intermediate files, our own expected counts — and in every round the checks passed. What escaped each time was external correctness: whether the marker list matched an authoritative source, whether the annotation direction matched the reference genome, whether a claimed haplotype frequency had co-occurrence data behind it, whether agreement of marginals meant agreement of calls. The verification architecture was built to catch stale and mutually inconsistent numbers, and it caught them reliably. It was never built to ask whether the right question was being put to the right list.

[Interpreted] **What that implies about remaining risk.** The honest reading is that the rate of discovery has not decayed across four passes. Each used a method the previous one did not have — code reading, adversarial re-derivation of the results, cross-validation against an independent external dataset, and external expert critique of the domain content — and each found defects the earlier methods were structurally incapable of seeing. There is no basis in this history for assuming the fourth pass exhausted the set, and the fact that three successive corrections each contained a defect of their own is direct evidence against assuming it. The residual risk is concentrated wherever no external cross-check exists. The marker panel now has two: its membership against WHO compendium v1.0 and its calls against MalariaGEN's published genotypes, call by call on 47,012 calls. The TreeMix topology, the BayPass covariance matrix and the ADMIXTURE ancestry

proportions have no independently published counterpart for these isolates, so nothing analogous is available for them, and they should be read as internally consistent rather than externally corroborated. The *pfk13* negative sits between the two cases: the per-codon calls are ours alone, and Pf8's own *kelch13* status corroborates them on 36 isolates (Section 5.7.3).

**Confirmed defects not corrected.** These are known, verified and still present in the reported results.

- step09 does not URL-decode GFF attributes, so %2C leaks into the published outlier gene file, and no nearest-gene distance is reported. This matters for the two intergenic SNPs among the nineteen: one of them is 59 bp from the end of *SNRPF*, which the published file does not say.
- The BayPass pseudo-observed set is still not MAF-matched; pi.maf truncates only the simulated ancestral frequency (Section 5.6). On the superseded 3,359-SNP calibration the reported threshold was the conservative one, but that experiment does not establish the direction of the error for the final 3,916-SNP run, whose raw pseudo-observed inputs are not archived. The frequency-matching limitation stands; a guaranteed conservative threshold does not. The methods claim had to be withdrawn rather than reproduced.
- The pseudo-observed and observed XtX distributions do not agree well through the body of the range, and distributional overlap is the only model-adequacy check the scan performs.
- ghanaAEZ.geojson has no citable source. It is an unattributed ArcGIS Online re-host, verified byte-for-byte (md5 96b1a7f4da62d6f0 153 826f41af96 152), with licenseInfo and accessInformation both null and an anonymous personal account as owner. The zone join has not been repeated against an authoritative six-zone Ghana layer, which would convert the provenance gap into a sensitivity result.
- Zone is confounded with sequencing study, and no analysis can remove it: it is a property of the Pf8 sampling frame, not of the statistics. It is now quantified on the 4,905 QC-passing isolates and reported as a fourth design limit rather than fixed.
- The codon panel is externally validated on only the 25 markers for which MalariaGEN publishes individual genotypes. The other 32 of its 57 codons — every *pfk13* codon and *pfaat1* S258L among them — have no per-codon external check; for the *pfk13* codons the only external counterpart is Pf8's own domain-level *kelch13* status, which is complete on 36 isolates (Section 5.7.3). The *pfaat1* XtX signal is not one, being drawn from the same genotypes.
- The codon panel and the genome-wide callset use different ploidy conventions for mixed calls — uncalled against majority-allele (Section 5.3) — so their denominators are not directly comparable. The choice is deliberate and is documented, but it has not been varied to see whether any panel frequency depends on it.

**[Observed] The reassurance that is actually earned.** Against all of the above, one robustness result stands and should be stated plainly rather than assumed. The corrected re-run enlarged the SNP panel by 16.6 per cent, replaced an uncorrected quantile with a list selected on Benjamini–Hochberg q-values against the simulated null, control being conditional on that null being adequately calibrated, gave TreeMix inferential support for the first time, and corrected the ploidy treatment that inflates $F_{ST}$ by up to 31×. The conclusions did not move. Maximum pairwise $F_{ST}$ is 0.00232 (SE 0.00053) against 0.0024 before; the ADMIXTURE cross-validation curve is still near-flat, with K = 2 at 0.83925 against K = 1 at 0.84384; and near-panmixia across Ghana's agro-ecological zones still holds. A headline that survives a materially larger panel and a stricter multiplicity correction is better supported than one that was merely never tested. Two limits on that reassurance should be stated with it. It applies to the population-structure results, which the marker-panel rebuilds do not touch;

and it is an internal-consistency argument of exactly the kind that the pattern above shows to be insufficient on its own.

## 5.8 Decisions that may warrant reconsideration

Two of the items previously listed here have been resolved and are recorded as such, and a third — the TreeMix model choice — is now recorded as a selected descriptive model rather than resolved; the rest stand, with six additions.

- **Stands. The TreeMix model choice is narrowed but not closed. The earlier text retained m = 3 as a candidate on the grounds that the identifiability argument (four populations support about two edges) was theoretical, and declined to choose. With -se in place the third edge has weight 0.065 against a standard error of 0.147, p = 0.328, so m = 2 is the graph the edge tests select; but those tests are conditional on a topology, root and edge count all fixed beforehand, and the variance-explained criterion across the full m = 0-5 sweep still prefers m = 3. Report m = 2 as a selected descriptive model, with the disagreement between the two criteria stated rather than removed.**
- **Resolved.** Variant missingness is no longer measured over isolates that are subsequently discarded, which was costing 557 SNPs at no benefit — about a seventh of the corrected panel, reported elsewhere here as the 16.6 per cent uplift the fix produced (Table 5.2).
- **Resolved.** The Mantel permutation tests are seeded, the negative-$F_{ST}$ censoring is disclosed and reported both ways, and a leave-one-out jackknife is included.
- **Stands.** Disabling the call-rate MAD rule (Section 4.2.3) was chosen on sample-count grounds after seeing the effect. A stricter absolute floor of 0.90 was offered and not taken; the data to compare it properly were never generated.
- **Stands.** The MAF ≥ 5 per cent filter is faithful to the thesis specification but still discards 99.8 per cent of the sites that survive missingness filtering — 1,807,331 variants in, 3,916 out. The filter-order fix recovered 16.6 per cent more SNPs at an unchanged threshold, but the threshold itself has never been varied, and the BayPass scan is the analysis most limited by SNP density.
- **Stands.** Zone assignment for Brong Ahafo and Volta was never manually reviewed against the map, as Step 2 recommended — and the map in question has no established provenance (Section 5.7.4).
- **New.** Single-threaded BayPass buys bitwise determinism at a wall-time cost that is negligible on 3,916 SNPs. If the callset ever grows enough for that to matter, the trade-off must be re-made explicitly and stated in the methods, not reverted silently in a configuration default.
- **New.** The transitional zone contributes 25 isolates, all from a single sequencing study, and it is the second-most-northern of the four zones, so it breaks any simple geographic ordering of the frequency and $F_{ST}$ results as well as being the least precisely estimated. All three of the $F_{ST}$ comparisons that previously had confidence intervals spanning zero — three of the six comparisons — involve it, and on the final matrix the one remaining such interval is Guinea savanna against the transitional zone. Its *pfdhps* G437 interval is 0.609–0.911 and its *pfdhfr* 59R and 108N intervals run to 1.000 on 25 isolates, so every panel frequency it carries is wide. Whether it should be analysed as a fourth population at all, rather than reported as too small to support a population-level estimate, is a decision that has been taken by default rather than deliberately.
- **New.** Marker lists must be transcribed from an authoritative source and cited to it, never written from memory. The *pfk13* list now comes from WHO compendium v1.0 (December 2025), variant by variant

and with each variant's tier recorded; it was written from memory before that, which is how a duplicated record inflated the count to 14, how P574L and G533S came to be omitted, and how P441L and C469F came to be presented to readers as validated; R622I was on the list and was never retrieved, so its absence from the results was a silent non-result rather than an omission. The same discipline has not been applied to the non-*pfk13* members of the panel, which remain a curated list with no single citable source.

- **New.** Markers are now identified by codon position rather than by annotation string, and that is a change of method, not only of code. Any future addition to the panel must specify a gene, a codon and an expected residue, never a protein-change string, because a string cannot express a resistance allele that is the reference residue or a residue requiring two changes in one codon (Section 5.7.3). The same applies to anything else in this pipeline that selects records by matching free text.
- **New.** Whether the panel should contain these 57 codons at all is a decision taken by accretion rather than by criterion. It began as a hand-written list, was repaired twice, and now mixes WHO-tiered *pfk13* variants with genes that appear nowhere in the compendium at any tier — *pfaat1*, *pfap2mu* and *pfmdr2*. They are retained because they are scientifically interesting and because *pfaat1* carries the strongest zone signal in the dataset, which is a defensible reason, but the panel still has no stated inclusion criterion and ought to have one before it is published.
- **New.** Validation should be per-sample wherever an external per-sample source exists. Marginal agreement was accepted as cross-validation for two rounds and it concealed a wrong call at *pfcrt* 75 through both of them. Where no per-sample source exists at all — *pfaat1* S258L and the three population-structure results — and where one exists but is weak, as for the *pfk13* codons, where only 36 isolates carry a complete Pf8 *kelch13* domain status, the methods should say so rather than leave a reader to infer it from which numbers carry a concordance figure.

# 6. Results

All results in this section are from the corrected re-run completed on 10 October 2026, after the seven defects listed in *CORRECTIONS_AND_NEW_RESULTS.md* were fixed. Unless stated otherwise the analysis set is **3,916 SNPs across 1,900 isolates**, and every figure and table is traceable to a named output file. The re-run touched stages 4-9, which Sections 6.1 and 6.4 to 6.7 report. Sections 6.2 and 6.3 report stages 1-3, which the fixes did not affect and which were not re-run at that point, so their published outputs and figures carry 8-9 October timestamps (results/sample_qc_full.csv, results/dimred_comparison.csv, pf8_proof/sample_qc.png, pf8_proof/dimred_comparison_zone.png). Stage 3 has since been re-run on the 1,900-isolate analysis set as a robustness check, writing to separate _qcset files (corrected/dimred_comparison_qcset.csv, corrected/dimred_comparison_zone_qcset.png) that leave the published stage-3 outputs untouched; both runs are reported in Section 6.3. Two further checks postdate the corrected re-run and are reported where they bear: a threshold-sensitivity analysis of the dataset definition itself (Section 6.5), which overturns one conclusion, and a variant-quality audit of the selection outliers (Section 6.8). No value computed on the superseded interim 3,359-SNP matrix is quoted here as current; where an earlier figure appears it is labelled as the value it replaces.

> **Robustness.** The filter-order fix enlarged the SNP panel by 557 SNPs, 16.6 per cent (3,359 to 3,916), on the identical 1,900 isolates, and ADMIXTURE, TreeMix and BayPass were re-run from scratch after the stale-output cache guards were fixed (Addendum G). The substantive readings all survived *that* change: near-panmixia, maximum pairwise $F_{ST}$ of the same order (0.0024 to 0.0023), a near-flat cross-validation curve with its minimum still at K = 2, and isolation by distance still weak and leverage-dependent. Four specific results did move, and are reported as moved rather than as confirmations. (i) The region-level Mantel conclusion changed: significant leave-one-out replicates went from 2 of 7 to 4 of 7, the uncensored test crossed from p = 0.0509 to p = 0.0413, and the undisclosed censoring of negative $F_{ST}$ values no longer determines significance (Section 6.5). **That improvement has since been overtaken. A threshold-sensitivity analysis run after the re-run finds the region-level Mantel significant in only 2 of 6 threshold scenarios, and isolation by distance is no longer reported anywhere in this work as a supported result (Section 6.5).** (ii) TreeMix model choice is now reported as a selected descriptive model at m = 2, on the third edge's own p-value and on parameter counting, while the variance-explained criterion across the full sweep still prefers m = 3; the two criteria disagree and the choice is settled on identifiability grounds, so the earlier hedge between m = 3 across the full sweep and m = 2 within the identifiable range is replaced by a stated model choice rather than by a resolution (Section 6.7). (iii) The number of $F_{ST}$ confidence intervals excluding zero rose from three to five of six (Section 6.4). (iv) The final single-threaded BayPass run gives 19 SNPs above a threshold of 10.840, of which 13 pass FDR 10 per cent, against 15 above 11.232 in the manuscript's 8-thread run; 17 of the 19 fall in 13 annotated genes and two are intergenic (Section 6.8). An intermediate 8-thread re-run with the same seed gave a different count again, which is why BayPass is now run single-threaded; the three runs are kept distinct in Section 6.8. That a 16.6 per cent larger panel left every headline reading in place while moving these particulars is reported as a result rather than left implicit — but surviving a change of panel is not the same as surviving a change of threshold, and on isolation by distance the two checks disagree.

## 6.1 The analysis dataset

**[Observed]** The final genotype matrix is 3,916 SNPs x 1,900 isolates with 2.95 per cent missing calls. SNP counts per chromosome range from 119 (chromosome 1) to 525 (chromosome 14).

**[Calculated]** Mean called isolates per SNP is 1,844 of 1,900; the least-called retained SNP is genotyped in 1,710 isolates, and the worst-covered retained isolate is missing 29.7 per cent of its calls. Per-chromosome SNP counts scale with chromosome length (Pearson r = 0.79, p = 7.4 x $10^{-4}$, n = 14), but not tightly: density ranges

from 125 SNPs/Mb on chromosome 3 to 328 SNPs/Mb on chromosome 4. Computed directly from interim/ghana_core_snps.npz on the analysis server.

| Chromosome | SNPs | Length (Mb) | SNPs per Mb |
|---|---|---|---|
| Pf3D7_01_v3 | 119 | 0.64 | 186 |
| Pf3D7_02_v3 | 161 | 0.95 | 170 |
| Pf3D7_03_v3 | 133 | 1.07 | 125 |
| Pf3D7_04_v3 | 394 | 1.20 | 328 |
| Pf3D7_05_v3 | 254 | 1.34 | 189 |
| Pf3D7_06_v3 | 244 | 1.42 | 172 |
| Pf3D7_07_v3 | 374 | 1.45 | 259 |
| Pf3D7_08_v3 | 255 | 1.47 | 173 |
| Pf3D7_09_v3 | 215 | 1.54 | 139 |
| Pf3D7_10_v3 | 251 | 1.69 | 149 |
| Pf3D7_11_v3 | 284 | 2.04 | 139 |
| Pf3D7_12_v3 | 285 | 2.27 | 125 |
| Pf3D7_13_v3 | 422 | 2.93 | 144 |
| Pf3D7_14_v3 | 525 | 3.29 | 159 |
| Total | 3,916 | 23.3 | 168 |

*Table 6.1. SNPs per chromosome in the final analysis matrix (interim/ghana_core_snps.npz). Chromosome lengths are 3D7 reference lengths; densities are calculated.*

**[Observed]** The filtering cascade that produces the matrix is reproduced below from logs/rerun_corrected.log. The in-memory genotype array is 2,190,026 variants x 6,653 samples x 2. Depth masking at DP < 5 touches 3,709,090,355 genotype cells, and the called-genotype total falls from 12,540,727,880 to 10,845,661,093 — 13.52 per cent of called genotypes masked. 194,167 mixed calls were then resolved by read depth: 129,665 to REF, 55,582 to ALT, and 8,920 discarded as tied. A provisional variant pass of 1,634,808 sites was used to decide the sample cut, which dropped 117 high-missingness isolates; variant missingness was then recomputed on the 1,900 retained isolates, leaving 1,807,331 sites, of which 3,916 pass MAF >= 0.05.

**[Calculated]** The two depth-mask figures are two separate facts and not a subtraction pair, and an earlier draft of this section joined them as one: 12,540,727,880 - 3,709,090,355 = 8,831,637,525, which is not the reported post-mask total. The reconciliation is that the array holds 2,190,026 x 6,653 x 2 = 29,140,485,956 cells, of which 12,540,727,880 — 43.0 per cent — carried a call at all. 3,709,090,355 is the count of cells failing DP < 5, most of which carried no call to begin with. The number of called genotypes actually removed is the difference between the two called totals, 1,695,066,787, and that is the quantity the logged 13.52 per cent refers to.

**[Calculated]** Variant missingness had been measured across 2,017 isolates, 117 of which were then discarded, so every SNP was charged for isolates that did not survive. Recomputing missingness after the sample cut, at identical thresholds and on an identical isolate set, raises the panel from 3,359 to 3,916 SNPs: 557 SNPs, or 16.6 per cent (557/3,359, a percentage not printed by any log).

[Interpreted] Because the thresholds and the isolate set are the same in both runs, the gain is attributable to the order of the two filters rather than to any change of threshold, and no other difference between the runs is known that would produce it (*CORRECTIONS_AND_NEW_RESULTS.md*, Addenda B and H).

> **Caveat on the 13.52 per cent mask rate.** The rate is computed over the full fetched callset — 6,653 samples, before the sample-alignment step that reduces the matrix to 2,017 — and not over the 1,900 isolates analysed, so it is not comparable with the per-isolate call rates in Table 4.1, which are computed after the mask and on a different sample again. Section 4 is where this scope question is discussed in full; it is flagged here so the two numbers are not read against each other. An earlier draft went further and asserted that the retained isolates have systematically higher call rates, making 13.52 per cent an upper bound on the masking applied to the analysis matrix. That claim is withdrawn as unsupported. The median call rate is 0.9962 across all 4,905 zone-assigned isolates entering QC, 0.9968 for the 2,017 that pass and 0.9956 for the 2,888 that do not — a separation of about 0.001, which carries no quantitative bound — and call rate is in any case computed on a 50,000-SNP panel with no depth mask applied, so it does not measure what the DP >= 5 floor removes. Whether the mask falls more or less heavily on the retained isolates is not established by any measurement in this work.

| Zone | Isolates | Share | Regions | Median year |
|---|---|---|---|---|
| Guinea savanna | 1,334 | 70.2% | Upper East | 2017 |
| Deciduous forest | 337 | 17.7% | Ashanti, Central, Eastern, Volta | 2015 |
| Coastal savanna | 204 | 10.7% | Greater Accra | 2018 |
| Transitional zone | 25 | 1.3% | Brong Ahafo | 2012 |
| Total | 1,900 | 100% | 7 regions | 2017 |

*Table 6.2. Composition of the final analysis set (interim/analysis_samples.csv). Zone labels in this table and in the prose use the corrected spelling. The source layer ghanaAEZ.geojson and every downstream output file carry the misspelt string "Decidous forest", which is the value the spatial join keys on and which is never rewritten in the data; quoted file contents therefore show the misspelling (Section 2.1.1).*

[Observed] Per administrative region the counts are Upper East 1,334, Greater Accra 204, Ashanti 157, Central 125, Eastern 34, Brong Ahafo 25, Volta 21 (interim/analysis_samples.csv).

[Interpreted] The design is severely unbalanced in four coupled ways, and every result in this section inherits all four. **Zone is confounded with region**: three of the four zones are a single administrative region, so no analysis can separate zone from region. **Zone is confounded with year**: median collection year runs from 2012 in the transitional zone to 2018 on the coast, a six-year spread over a period of declining national transmission. **Zone is confounded with sample size**: 70.2 per cent of isolates come from one region. **Zone is confounded with sequencing study**: four of the six zone pairs share no partner study at all, including the Coastal-Guinea savanna pair that carries every headline number (results/batch_zone_pair_overlap.csv). The fourth of these was absent from the original manuscript and is quantified in Section 6.2, where the denominator on which it is computed is also stated.

## 6.2 Sample quality control and study composition

[Observed] The quality-control figures below are a stage-3 product, unaffected by the corrections and not re-run; they are reproduced from results/sample_qc_full.csv on the analysis server. The study-composition and batch tests later in this subsection are new in this revision (scripts/step11_batch_confounding.py, corrected/batch_study_by_zone.csv, corrected/batch_within_zone_tests.csv, corrected/batch_zone_pair_overlap.csv) and had no counterpart in the original manuscript.

[Observed] Of 4,905 isolates entering QC, 2,017 passed (41.1 per cent). Median call rate was 0.9962 (minimum 0.7561); median heterozygous call rate 0.00036; median $F_{ws}$ 0.8953 (range 0.3101-1.0000), with 2,095 isolates (42.7 per cent) monoclonal at $F_{ws}$ >= 0.95.

| Exclusion reason | Isolates flagged | Share of 4,905 |
|---|---|---|
| $F_{ws}$ < 0.95 (polyclonal) | 2,810 | 57.3% |
| Heterozygosity outlier (> median + 3 MAD) | 126 | 2.6% |
| Clonal redundancy (distance <= 0.00005) | 77 | 1.6% |
| Call rate < 0.80 | 20 | 0.4% |
| Call-rate MAD outlier (rule disabled) | 0 | 0.0% |
| Passed all checks | 2,017 | 41.1% |

*Table 6.3. Quality-control outcomes. Categories overlap: an isolate may fail more than one check, so the flagged counts sum to more than 4,905 - 2,017. Source: results/sample_qc_full.csv.*

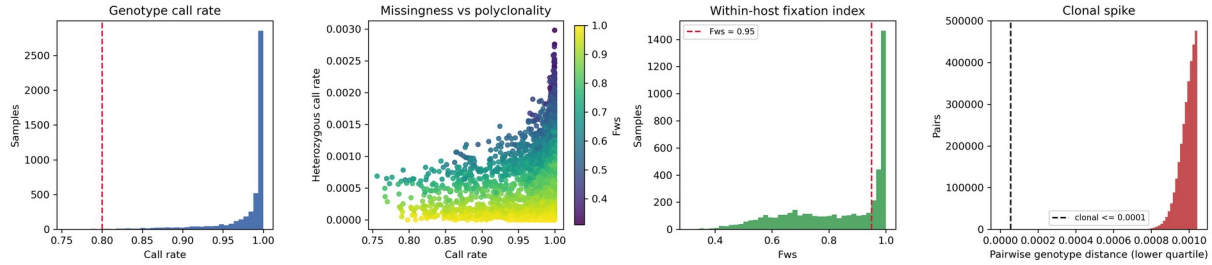



*Figure 6.1. Sample quality control (n = 4,905). Left to right: genotype call rate with the 0.80 floor marked; heterozygous call rate against call rate, coloured by Fws; the Fws distribution with the 0.95 threshold marked; and the lower quartile of pairwise genotype distances with the clonal threshold at 0.00005 (config.CLONAL_THRESHOLD, line 133). The panel's own printed legend reads "clonal <= 0.0001" because that label is formatted to four decimal places at scripts/step03b_sample_qc.py:460; the legend and this caption differ by rounding, not by threshold. The Fws panel shows a clear mode above 0.95 with a long left shoulder, which is what makes 0.95 a defensible cut. The clonality panel shows no comparable break, which is why that threshold required a sensitivity analysis. Generated by scripts/step03b_sample_qc.py; unchanged by the corrected re-run.*

> **Denominator for the composition and batch figures.** Every study-composition, polyclonality and within-zone batch figure in this subsection is computed on the **4,905 zone-assigned isolates that enter sample QC**, not on the 1,900-isolate analysis set used in Sections 6.1 and 6.4 to 6.7. The zone counts on that larger set are coastal savanna 435, deciduous forest 815, Guinea savanna 3,605, transitional zone 50, against 204 / 337 / 1,334 / 25 in the analysis set, so figures such as "1083-PF-GH-CONWAY is 100 per cent of the transitional zone, 50 of 50" sit beside an analysis-set n of 25 for that zone and are not in conflict with it. The larger set is the right denominator because study membership and $F_{ws}$ are defined there, and because the $F_{ws}$ >= 0.95 filter is one of the cuts that produces the analysis set. Study nesting within zone is a property of the Pf8 sampling frame and is not created by the later cuts, but the counts in this subsection are not analysis-set counts and should not be read as such. Sections 4.2.2, 7.2, 7.3 and 10.1 quote these same figures and refer back to this note.

[Calculated] Two checks establish how far this denominator matters. First, the polyclonal fractions cannot be computed on the analysis set at all: all 1,900 isolates have $F_{ws}$ between 0.9502 and 1.0000 by construction, so the polyclonal fraction there is identically zero. Second, the zone-pair study overlap was recomputed on the 1,900 isolates and is unchanged: the same four of six pairs share no study, with the same two studies (1183-PF-GH-AWANDARE-SM, 1241-PF-GH-ANINAGYEI) shared between coastal savanna and deciduous forest and the same single study (1083-PF-GH-CONWAY) shared between Guinea savanna and the transitional zone. Deciduous forest still draws on five studies in the analysis set (157, 100, 56, 21 and 3 isolates). Computed from interim/analysis_samples.csv on the analysis server.

[Observed] The polyclonal fraction by zone, over the 4,905 isolates entering QC and in north-to-south geographic order, is: Guinea savanna 0.597 (2,153 of 3,605), transitional zone 0.500 (25 of 50), deciduous forest 0.539 (439 of 815), coastal savanna 0.444 (193 of 435).

[Interpreted] The ordering is **not** monotone along the north-south axis. The transitional zone is the second-most-northern of the four, lying north of the deciduous forest, and it has the lower polyclonal fraction of the two, so the gradient holds for three of the four zones and breaks at the fourth. The transitional value rests on 25 of 50 isolates and carries essentially no information. The earlier framing of this as the variable that "orders cleanly along the north-south gradient" was obtained by quoting only the two endpoints, 0.597 and 0.444, and is withdrawn. What survives is a three-zone trend in the direction transmission intensity predicts, which is weaker evidence than a clean monotone gradient but is still the largest ecological contrast in the analysis.

[Observed] The contrast is also confounded with sequencing study, and the study-associated heterogeneity is demonstrable rather than hypothetical. Holding zone constant, so that study varies together with collection year, finer geography and recruitment: within deciduous forest, which draws on five studies, the four large enough to test (the fifth, 1114-PF-PDN-DBS-GH-GHANSAH, contributes 4 isolates and is dropped at the n >= min_n guard in scripts/step11_batch_confounding.py:109) differ in $F_{ws}$ — Kruskal-Wallis H = 14.9, p = 1.9 x $10^{-3}$ on n = 811 — and their polyclonal fractions span 0.430 to 0.578; they also differ in call rate at p = 7.7 x $10^{-88}$. Within Guinea savanna, the six contributing studies differ sharply in call rate, p = 2.8 x $10^{-197}$, with polyclonal fractions spanning 0.532 to 0.682, but their $F_{ws}$ distributions are not distinguishable (H = 9.17, p = 0.103). Within coastal savanna neither test is significant (2 studies, n = 435, $F_{ws}$ p = 0.378). The Kruskal-Wallis p-values are tests on the $F_{ws}$ and call-rate distributions; no test on the polyclonal fraction itself was run, and the quoted p-values should not be attached to the fractions (results/batch_within_zone_tests.csv).

[Interpreted] Because no study contributes isolates to both Guinea savanna and coastal savanna, the between-study variation above is of the same order as the between-zone contrast it would have to be separated from. One partial mitigation is reportable: within the single study that spans Guinea savanna and the transitional zone (1083-PF-GH-CONWAY), Guinea savanna is still the more polyclonal of the two, 25 of 47 against 25 of 50, matching the overall ordering. That margin is three isolates wide in the denominator and should not be read as more than a direction. The gradient is therefore not purely an artefact of batch, but for the pairs that carry the largest differences it is not separable from it.

## 6.3 Ordination of the genotype distance matrix

[Observed] **Scope.** Stage 3 reads the published Pf8 mean genotype distance matrix rather than the 3,916-SNP genotype matrix built here, and it is now reported on two samples. The published run uses the full 4,905 x 4,905 Ghanaian block, that is, the **pre-QC set of 4,905 isolates**, before the $F_{ws}$ and clonality filters and before the variant filters that define the 3,916-SNP matrix; the 1,900 isolates analysed in Sections 6.1 and 6.4 to 6.7 are a 38.7 per cent subset of those 4,905. Stage 3 has since been re-run on that 1,900-isolate analysis set, by re-slicing the distance submatrix to the analysis samples rather than reusing the cached full-set one (scripts/step03_umap.py, which now takes --samples and --suffix). Both runs are reported below; the published outputs are unchanged and the re-run writes to separate _qcset files. Like the quality-control figures in Section 6.2, neither is affected by the corrections to stages 4-9.

[Calculated] Pairwise genotype distances are tightly concentrated: median 0.00112, with the 1st and 99th percentiles at 0.00090 and 0.00129, a coefficient of variation of 0.085, and no missing entries. Mean within-zone distance barely differs from mean between-zone distance for any zone (Guinea savanna 0.00111 against

0.00112; deciduous forest 0.00114 against 0.00112; coastal savanna 0.00108 against 0.00110; transitional zone 0.00125 against 0.00117). Computed from interim/ghana_dist_submatrix.npy; these summaries are not printed by the pipeline and are tagged accordingly.

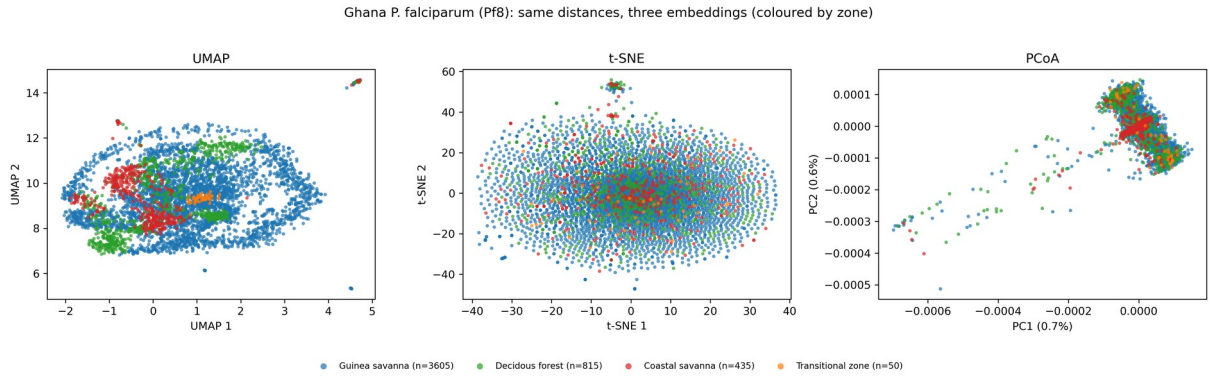



*Figure 6.2. UMAP, t-SNE and PCoA of the same 4,905 x 4,905 Ghanaian distance matrix, coloured by agro-ecological zone. Point counts per zone are in the legend. PCoA axes carry their variance proportions; UMAP and t-SNE axes are in arbitrary units and inter-cluster distances in those two panels carry no quantitative meaning. Note that this is the pre-QC sample of 4,905 isolates; Figure 6.2a is the same three methods on the 1,900 analysed elsewhere. Generated by scripts/step03_umap.py.*

| Method | Trustworthiness | Distance Spearman rho | Silhouette by zone |
|---|---|---|---|
| Original distance matrix | — | 1.000 (by definition) | -0.016 |
| UMAP | 0.486 | 0.029 | -0.306 |
| t-SNE | 0.507 | 0.115 | -0.057 |
| PCoA | 0.495 | 0.322 | -0.112 |

*Table 6.4. Quantitative comparison of the three ordinations on the 4,905 pre-QC isolates (results/dimred_comparison.csv). Trustworthiness is computed with k = 30 neighbours; the Spearman correlation uses a 2,000-isolate random subset; silhouette is computed against zone labels, where values near zero indicate overlapping groups and negative values indicate that isolates sit closer to other zones than to their own. Table 6.4a gives the same statistics on the 1,900-isolate analysis set.*

**[Observed]** PCoA's first two axes explain 0.71 and 0.59 per cent of positive-eigenvalue variation respectively, 1.29 per cent together. The eigendecomposition returned 1,347 negative eigenvalues summing to -0.000587 against a positive sum of 0.00365, indicating the distance is not perfectly Euclidean; proportions are reported relative to the positive eigenvalues, the usual convention (logs/step03_dimred.log).

**[Interpreted]** Three readings follow. First, the silhouette on the original matrix is -0.016, essentially zero: zone labels do not describe the genetic variation at this level, and no embedding could make them do so. Second, PCoA preserves global structure best, with a Spearman correlation of 0.322 between embedded and original distances against 0.115 for t-SNE and 0.029 for UMAP, which is expected of a linear method and makes it the one to read for overall structure here. That is a claim about global distance preservation only: on the local trustworthiness statistic in Table 6.4 the three methods lie within 0.02 of each other and t-SNE scores highest, so neither metric makes any one method better in general. Third, trustworthiness near 0.5 for all three is not a failure of the methods: when distances are as uniform as these, neighbourhoods are nearly arbitrary and no two-dimensional projection can preserve them. The tight clumps visible in the UMAP panel are not explained by anything computed here. They are not zone clusters — the UMAP silhouette of -0.306 establishes that much — but their composition was not checked, and the clonality filter flags only 77 isolates in the whole set, too few to account for the clumping on their own.

[Observed] **The same ordinations on the analysis set.** Re-run on the 1,900 isolates, the silhouette by zone on the raw distances is **-0.020**, against -0.016 on the 4,905. Trustworthiness is slightly higher for all three embeddings (UMAP 0.499, t-SNE 0.526, PCoA 0.510, against 0.486, 0.507 and 0.495), and the PCoA distance correlation falls from 0.322 to 0.287. The UMAP distance correlation changes sign, from 0.029 to -0.066. The per-method silhouettes move in both directions and all remain negative (corrected/dimred_comparison_qcset.csv).

| Metric | 4,905 pre-QC (published) | 1,900 analysis set |
|---|---|---|
| Silhouette by zone, original distances | -0.016 | -0.020 |
| Silhouette by zone, UMAP | -0.306 | -0.277 |
| Silhouette by zone, t-SNE | -0.057 | -0.063 |
| Silhouette by zone, PCoA | -0.112 | -0.079 |
| Trustworthiness, UMAP | 0.486 | 0.499 |
| Trustworthiness, t-SNE | 0.507 | 0.526 |
| Trustworthiness, PCoA | 0.495 | 0.510 |
| Distance Spearman rho, UMAP | 0.029 | -0.066 |
| Distance Spearman rho, t-SNE | 0.115 | 0.196 |
| Distance Spearman rho, PCoA | 0.322 | 0.287 |

*Table 6.4a. The three ordinations recomputed on the 1,900-isolate analysis set, beside the published 4,905-isolate run (results/dimred_comparison.csv, corrected/dimred_comparison_qcset.csv). Both columns read the same published Pf8 distance matrix; the right-hand column re-slices it to the analysis samples. Settings are identical (trustworthiness at k = 30; silhouette against zone labels). This table is new in this revision and is.*

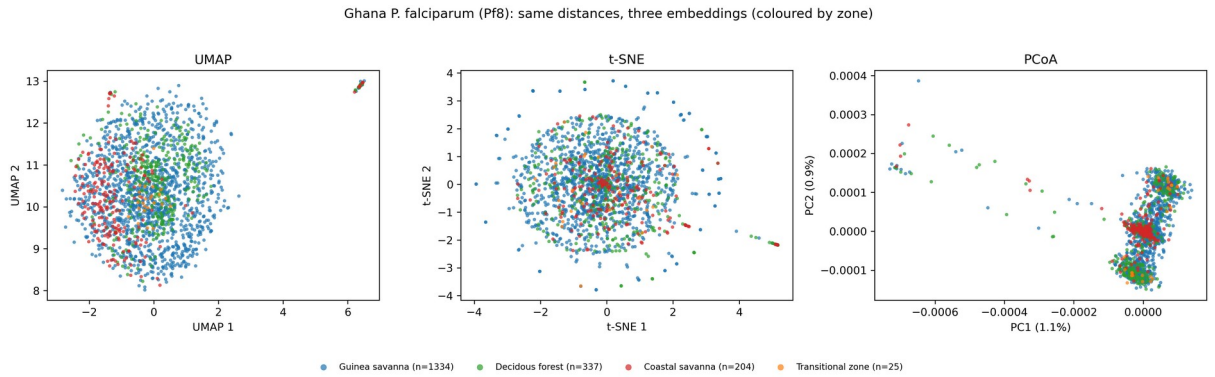



*Figure 6.2a. UMAP, t-SNE and PCoA of the 1,900 x 1,900 submatrix of the same Pf8 distance matrix — the analysis set used in Sections 6.1 and 6.4 to 6.7 — coloured by agro-ecological zone, with point counts per zone in the legend. Compare Figure 6.2, which is the same three methods on the 4,905 pre-QC isolates. PCoA axes carry 1.1 and 0.9 per cent of positive-eigenvalue variation; UMAP and t-SNE axes are in arbitrary units and inter-cluster distances in those two panels carry no quantitative meaning. The legend retains the misspelt source-layer label "Decidous forest" (Section 2.1.1). Generated by scripts/step03_umap.py with --samples interim/analysis_samples.csv and --suffix _qcset.*

[Interpreted] **Removing the polyclonal and clonally redundant isolates does not sharpen the structure.** The silhouette by zone becomes slightly *more* negative, -0.020 against -0.016, which is the opposite of what would be seen if zone signal had been diluted by mixed infections. The small rises in trustworthiness are not evidence of better-resolved structure either: trustworthiness is computed on a sample 61 per cent smaller, where fewer isolates compete for each neighbour rank, and all six values still sit near 0.5. The one movement with a plain reading is the fall in the PCoA distance correlation, 0.322 to 0.287: the best-performing embedding reconstructs the distances slightly *less* well on the filtered set. A tight detached clump also

remains in both the UMAP and the t-SNE panels of Figure 6.2a after the filters, so the 77 clonally redundant isolates and the polyclonal infections do not account for the clumping noted above; what does is still not established. The substantive gain here is one of reporting rather than of result: the ordinations can now be read on the same 1,900 isolates as the rest of Section 6, and on that sample they say what they said before.

> **Not an independent confirmation.** The ordinations read a different input from Sections 6.4 to 6.7 — the published Pf8 mean genotype distance matrix, not the 3,916-SNP panel built here — so their agreement with those sections remains weak corroboration rather than a replication, and that is unchanged by the re-run. The *sample* objection, however, is now answered. The published run is computed over isolates that include the polyclonal and clonally redundant infections the rest of the analysis excludes, and polyclonality itself varies by zone (Section 6.2); restricting to the 1,900 analysis-set isolates removes that difference, and the near-zero silhouette survives it (Table 6.4a).

> **Reproducibility incident in this re-run.** The `--suffix` patch to `scripts/step03_umap.py` suffixed the file writes but not the `print` statements, so the run reported writing to the published `results/dimred_comparison.csv` when it had in fact correctly written `dimred_comparison_qcset.csv`. No data were at risk and nothing in Table 6.4 is affected: the published file's modification time and contents were checked and are unchanged. A log that misreports its own output path is nonetheless exactly the defect that later causes one run's numbers to be attributed to another, which is why it is recorded rather than quietly fixed. The print statements now carry the suffix.

## 6.4 Pairwise differentiation between zones

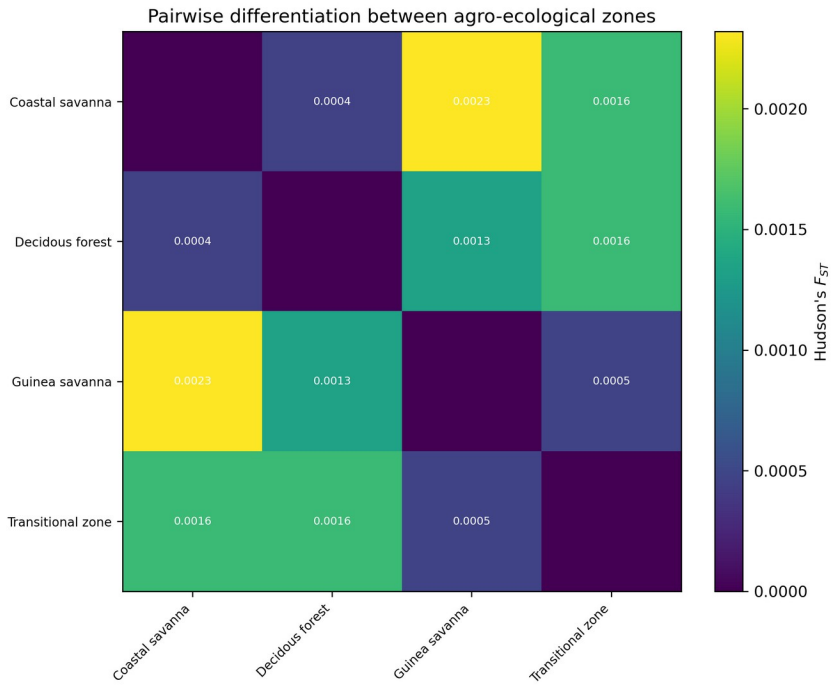



*Figure 6.3. Pairwise Hudson FST between agro-ecological zones (3,916 SNPs, 1,900 isolates). Note the colour-bar scale: the maximum value is 0.0023. Generated by scripts/step04b_fst.py.*

| Zone A | Zone B | n(A) | n(B) | Distance (km) | $F_{ST}$ | Jackknife SE | 95% CI |
|---|---|---|---|---|---|---|---|
| Coastal savanna | Guinea savanna | 204 | 1,334 | 557 | 0.00232 | 0.00053 | 0.00128 – 0.00336 |
| Coastal savanna | Transitional zone | 204 | 25 | 282 | 0.00158 | 0.00077 | 0.00006 – 0.00309 |
| Deciduous forest | Transitional zone | 337 | 25 | 162 | 0.00158 | 0.00072 | 0.00017 – 0.00298 |
| Deciduous forest | Guinea savanna | 337 | 1,334 | 493 | 0.00134 | 0.00024 | 0.00088 – 0.00180 |
| Guinea savanna | Transitional zone | 1,334 | 25 | 352 | 0.00047 | 0.00066 | -0.00083 – |

| Zone A | Zone B | n(A) | n(B) | Distance (km) | $F_{ST}$ | Jackknife SE | 95% CI |
|---|---|---|---|---|---|---|---|
| | | | | | | | 0.00178 |
| Coastal savanna | Deciduous forest | 204 | 337 | 145 | 0.00045 | 0.00013 | 0.00019 – 0.00070 |

*Table 6.5. Pairwise Hudson FST, ordered by magnitude (results/fst_pairwise.csv). Standard errors are from a block jackknife. Great-circle distances between zone centroids are from results/geographic_distance_km_zone.csv. One of the six confidence intervals spans zero: Guinea savanna against the transitional zone. The small-sample caveat for the transitional zone rests on its n = 25 and its wide standard errors, not on its confidence intervals, two of which now exclude zero.*

[Observed] All six values lie below 0.0024. **Five of the six confidence intervals exclude zero; one, Guinea savanna against the transitional zone, spans it.** Two of the five that exclude zero do so narrowly: Coastal-Transitional has a lower bound of 0.00006 and Deciduous forest-Transitional of 0.00017.

[Observed] This corrects two separate errors in the original manuscript. The abstract stated that two of six intervals span zero and Section 6.4 stated three; on the 3,359-SNP matrix the correct figure was three, so the abstract was wrong and the section was right. On the corrected 3,916-SNP matrix the figure is **one**, because the larger panel tightened the two transitional-zone intervals enough to exclude zero. Neither of the earlier counts, and no statement that all three transitional-zone pairs span zero, should be quoted anywhere in this work.

[Interpreted] The two geographically extreme zones (coastal and Guinea savanna, 557 km apart) are the most differentiated and the two adjacent southern zones (coastal and deciduous forest, 145 km apart) the least, but the full ordering is **not** monotone in distance: deciduous forest against Guinea savanna, 493 km apart, is less differentiated than coastal against the transitional zone at 282 km. The earlier claim that the ranking is "consistent with distance" overstates a four-point pattern, and the formal test at zone level is not significant (Section 6.5).

[Interpreted] An $F_{ST}$ of 0.0023 means roughly 0.23 per cent of genetic variation is attributable to zone membership. Published within-country $F_{ST}$ for *P. falciparum* routinely falls below 0.05, and continental-scale African subpopulation structure is itself modest (Amambua-Ngwa et al., 2019). The Ghanaian values sit an order of magnitude below even that within-country expectation. The practical reading is that these zones are not demographically separate populations; they are one population sampled in four places.

[Interpreted] **Statistical against practical significance.** Five pairs are statistically distinguishable from zero, which with 1,900 isolates and 3,916 SNPs is a low bar, and the number rose from three to five purely by adding 557 SNPs. None is practically significant in the sense of implying restricted gene flow. A resistance allele arising in Accra would be expected to reach Bolgatanga; the question these data cannot answer is how quickly.

> **Two confounds specific to this table. Time.** The transitional zone has a median collection year of 2012 against 2018 on the coast, a six-year offset, so the two transitional-zone comparisons that newly exclude zero are precisely those most exposed to temporal rather than spatial differentiation. The headline Coastal-Guinea pair is one median year apart and is correspondingly the most defensible as a spatial contrast. **Batch.** The Coastal-Guinea pair shares no sequencing study at all, and the two pairs that do share studies (Coastal-Forest, 2 studies; Guinea-Transitional, 1 study) are the two lowest values in the table. The study overlap is reported in results/batch_zone_pair_overlap.csv on the 4,905-isolate set; see the denominator note in Section 6.2, and note that the same four of six pairs share no study when the overlap is recomputed on the 1,900 isolates used here. Neither confound is corrected for anywhere in this analysis, and neither can be with this design.

## 6.5 Isolation by distance

[Observed] At region level (7 populations, 21 pairs, distances 87-578 km) the Mantel correlation is r = 0.427 with p = 0.0252 against 9,999 permutations, one-sided, seeded at 42. At zone level (4 populations, 6 pairs, distances 145-557 km) it is r = 0.471 with p = 0.1218. These are the values at the published thresholds; the threshold-sensitivity analysis later in this subsection is what decides how much weight they will bear, and the answer is: very little.

[Observed] **Disclosure of censoring.** The region-level test is run on $F_{ST}/(1 - F_{ST})$, and three of the 21 region pairs have negative point estimates of $F_{ST}$, which step06_mantel.py floors to zero before the linearisation. In the original manuscript this truncation was not disclosed anywhere, in the methods, the results or the figure caption, and the affected pairs were described only as sitting "at or very near zero differentiation". They do not sit near zero; they are negative and were censored. The three pairs and their values are now printed by the pipeline:

```
3 of 21 pairs have negative FST and are floored to zero:   Ashanti - Volta:      −0.001037   Eastern - Greater Accra:  −0.000203
Greater Accra - Volta:   −0.000142
```

| Region-level test | Mantel r | p (9,999 permutations, seed 42) |
|---|---|---|
| Negative pairs floored to zero (as run) | 0.4269 | 0.0252 |
| Negative pairs retained (uncensored) | 0.4138 | 0.0413 |

*Table 6.6. The region-level Mantel test with and without the flooring of three negative pairwise FST values (results/mantel_result_region.json). At zone level no pair is negative, so the two versions coincide at r = 0.4709, p = 0.1218.*

[Observed] On the final matrix the censoring no longer determines the result: both versions clear p < 0.05, and the difference between them is 0.013 in r and 0.016 in p. This was not true of the interim 3,359-SNP matrix, where censoring moved the test across the 0.05 line (p = 0.051 uncensored against 0.030 censored) and the undisclosed truncation was therefore result-determining. The disclosure is required either way, and both versions are reported as a standing practice rather than because the conclusion depends on it.

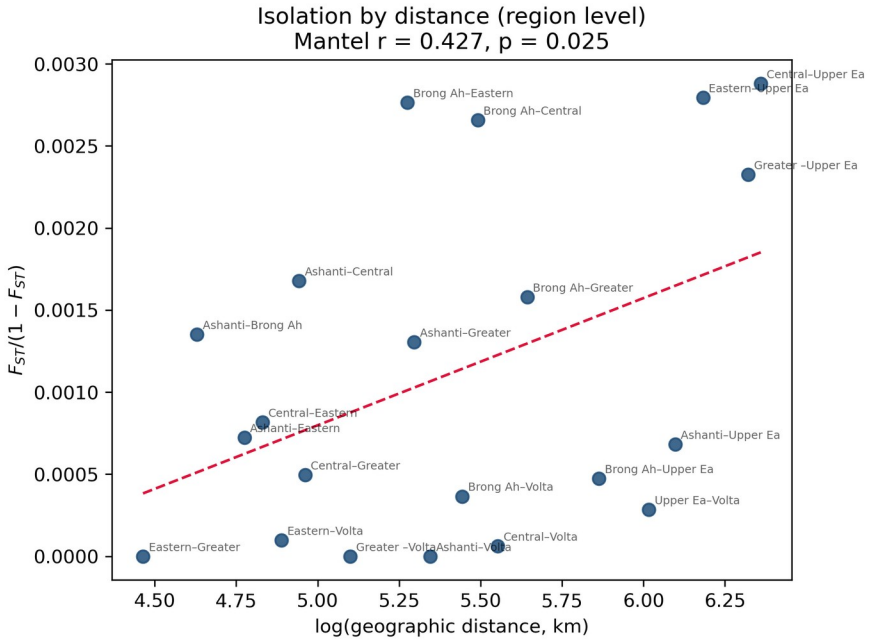



*Figure 6.4. Isolation by distance at administrative-region level: linearised differentiation FST/(1 - FST) against log great-circle distance, 21 region pairs, with the least-squares line. Mantel r = 0.427, p = 0.025 (9,999 permutations, one-sided, seed 42). The three points lying exactly on zero (Ashanti-Volta, Eastern-Greater Accra, Greater Accra-Volta) are the censored negative FST*

*pairs listed above, plotted at their floored value. This is the baseline scenario only; Table 6.7a shows that the fitted relationship is not significant under four of the six threshold choices. Generated by scripts/step06_mantel.py.*

**[Observed]** A Mantel correlation driven by one population is a single contrast, not a distance gradient. The leave-one-population-out jackknife below refits the region-level test seven times, dropping one region each time.

| Region dropped | n remaining | Mantel r | p | Significant at 0.05 |
|---|---|---|---|---|
| (none: full data) | 7 | 0.427 | 0.0252 | yes |
| Brong Ahafo | 6 | 0.609 | 0.0398 | yes |
| Ashanti | 6 | 0.568 | 0.0140 | yes |
| Volta | 6 | 0.546 | 0.0097 | yes |
| Central | 6 | 0.396 | 0.0427 | yes |
| Eastern | 6 | 0.288 | 0.1355 | no |
| Upper East | 6 | 0.260 | 0.2180 | no |
| Greater Accra | 6 | 0.250 | 0.2324 | no |

*Table 6.7. Leave-one-region-out Mantel jackknife (results/mantel_jackknife_region.csv). The result is significant in 4 of the 7 replicates. At zone level the equivalent jackknife is significant in 0 of 4 replicates. This jackknife varies which populations enter the test; Table 6.7a varies the thresholds that decide which isolates and which sites exist at all.*

**[Interpreted]** The signal is leverage-dependent in a specific and interpretable way. Significance is lost in exactly three of the seven replicates: dropping either of the two southern regions that define the short-distance end (Greater Accra, p = 0.232; Eastern, p = 0.136) or the single northern region that defines the long-distance end (Upper East, p = 0.218). Of the remaining four, dropping Brong Ahafo, Ashanti or Volta raises the correlation (r = 0.609, 0.568 and 0.546 against 0.427 on the full data), while dropping Central lowers it slightly and leaves it only just significant (r = 0.396, p = 0.043). The pattern is that of a north-south contrast between Upper East and the southern regions, with a weak and noisy distance relationship among the southern regions themselves, rather than a smooth gradient operating at all scales. On the interim 3,359-SNP matrix only 4 of 7 replicates were significant, so the corrected panel did improve this jackknife; the threshold analysis immediately below shows that the improvement does not generalise beyond the published thresholds.

**[Observed]** **Sensitivity to the two thresholds that define the dataset.** The jackknife varies the populations. It does not vary the two filters that decide which isolates and which sites exist at all: the $F_{ws}$ >= 0.95 monoclonality cut, which discards 57.3 per cent of isolates and does so unevenly by zone (Section 6.2), and the MAF >= 0.05 floor, which discards 99.8 per cent of candidate sites. scripts/step13_sensitivity.py loads the 2.19 M x 4,682 genotype matrix once, rebuilds the analysis matrix under six threshold scenarios, and recomputes the full pairwise $F_{ST}$ table and the region-level Mantel test for each.

| Scenario | Isolates | SNPs | max $F_{ST}$ | Mantel r | p (as run) | p (uncensored) | Significant at 0.05 |
|---|---|---|---|---|---|---|---|
| Baseline: $F_{ws}$ >= 0.95, MAF >= 0.05 (as published) | 1,900 | 3,916 | 0.00232 | 0.427 | 0.0252 | 0.0413 | yes |
| $F_{ws}$ >= 0.90 | 2,196 | 3,629 | 0.00240 | 0.318 | 0.0874 | 0.1214 | no |
| $F_{ws}$ >= 0.80 | 2,746 | 3,496 | 0.00255 | 0.256 | 0.1687 | 0.2002 | no |
| No $F_{ws}$ filter | 4,403 | 3,275 | 0.00254 | 0.245 | 0.1792 | 0.1792 | no |
| MAF >= 0.01 | 1,900 | 10,876 | 0.00204 | 0.432 | 0.0617 | 0.0644 | no |

| Scenario | Isolates | SNPs | max $F_{ST}$ | Mantel r | p (as run) | p (uncensored) | Significant at 0.05 |
|---|---|---|---|---|---|---|---|
| MAF >= 0.02 | 1,900 | 6,852 | 0.00217 | 0.434 | 0.0467 | 0.0502 | yes (marginal) |

*Table 6.7a. Threshold sensitivity of the region-level Mantel test and of maximum pairwise FST (results/sensitivity_summary.csv, scripts/step13_sensitivity.py). Only the named threshold differs from the baseline in each row. "p (as run)" floors negative pairwise FST to zero, as step06_mantel.py does; "p (uncensored)" retains them, following Table 6.6. Negative pairs floored, in row order: 3, 4, 3, 0, 4, 4. All tests are one-sided over 9,999 permutations at seed 42. This table is new in this revision and is.*

[Observed] **The region-level Mantel test is significant in 2 of the 6 scenarios.** It holds at the published thresholds (p = 0.0252) and marginally at MAF >= 0.02 (p = 0.0467). It fails at every relaxation of the $F_{ws}$ cut — p = 0.0874 at 0.90, 0.1687 at 0.80, 0.1792 with no monoclonality filter at all — and at MAF >= 0.01 (p = 0.0617). On the uncensored test, which Table 6.6 reports beside the floored one as standing practice, only the baseline clears 0.05: the MAF >= 0.02 scenario moves to p = 0.0502. The correlation itself does not simply weaken with every change — it is 0.432 and 0.434 in the two MAF scenarios, slightly above the baseline 0.427 — so what the relaxations mainly do is remove the permutation significance, not reverse the sign of the relationship.

> **Isolation by distance is withdrawn as a supported result.** Two checks now bear on this test, they vary different things, and they agree. The leave-one-region-out jackknife (Table 6.7) finds it significant in **4 of 7** replicates, with significance collapsing whenever the region defining either end of the distance range is dropped — Upper East at the long end, Greater Accra or Eastern at the short end. The threshold sensitivity (Table 6.7a) finds it significant in **2 of 6** scenarios, and the relaxation that most directly tests the filter doing the most work — removing the $F_{ws}$ cut, which more than doubles the isolate count to 4,403 — moves p from 0.0252 to 0.1792. The honest statement is that **the isolation-by-distance result holds at the published thresholds and almost nowhere else**. The original manuscript's "weak but detectable" distance effect, retained through every earlier revision of this section including the one that reported the jackknife improving from 2 of 7 to 4 of 7, cannot be retained now, and the grading of this result in Sections 1, 7 and 10 as supported or moderately supported must change with it. This is the only positive spatial finding in the analysis and it is the one that did not survive the two checks the audit identified as most needed. It is reported here in full rather than deleted — the point estimate at the published thresholds is unchanged and a reader is entitled to it — but it should not be cited as evidence of distance-limited gene flow in Ghana.

[Interpreted] **The same analysis strengthens the headline conclusion.** Maximum pairwise $F_{ST}$ moves only between **0.00204 and 0.00255** across all six scenarios, a total spread of 0.0005, over sample sizes from 1,900 to 4,403 isolates and panels from 3,275 to 10,876 SNPs. Near-panmixia is therefore insensitive to both of the thresholds that define the dataset, which is the strongest single piece of evidence for it in this work: the two filters that could most plausibly have manufactured the result do not move it. Two secondary readings follow from the direction of the small movements. Relaxing the $F_{ws}$ cut *raises* $F_{ST}$ (0.00232 at the baseline to 0.00255 at $F_{ws}$ >= 0.80 and 0.00254 with no cut), so the excluded polyclonal isolates were not masking differentiation; if anything they carry marginally more of it. Tripling the panel by lowering the MAF floor to 0.01 *lowers* it (0.00204), as expected if rarer variants are less differentiated between zones than common ones. Neither movement approaches a magnitude that would change any conclusion drawn in Section 6.4.

[Interpreted] Three further qualifications apply to the Mantel tests themselves. First, the zone-level test has the larger correlation but the larger p-value, which is a power effect rather than a contradiction: with four populations only 4! = 24 distinct orderings exist, so the finest attainable p-value is 1/24 = 0.042 and significance requires essentially the single best permutation. The region-level test, with 5,040 orderings, is the better powered of the two and is the one reported above. Second, the zone-level p-value is reported as a Monte Carlo estimate from 9,999 permutations when the exact value is enumerable over all 24 orderings; this remains outstanding and the estimate should be replaced by the exact figure. Third, several of the 21

region pairs rest on small samples (Volta 21 isolates, Brong Ahafo 25, Eastern 34), and the scatter in Figure 6.4 is correspondingly wide.

**[Interpreted]** What remains, stated at the strength the evidence now supports: at the published thresholds there is a weak region-level association between differentiation and distance, substantially a single north-south contrast, which does not survive relaxation of either threshold that defines the dataset. Nothing in these data requires an ecological explanation for that association, and nothing in them establishes a distance effect either. The region-level analysis in any case says nothing about agro-ecological zones as such; the zone-level analysis, which does, is not significant on the full data (r = 0.471, p = 0.1218) and is significant in 0 of its 4 leave-one-zone-out replicates.

## 6.6 Ancestry inference

**[Observed]** After LD pruning, 3,385 of 3,916 SNPs remained. Cross-validation error was lowest at K = 2 (0.83925) and rose monotonically thereafter; K = 1, run separately as a reference, gave 0.84384.

| K | CV error | Delta from K = 1 | Reading |
|---|---|---|---|
| 1 | 0.84384 | — | No structure (reference) |
| 2 | 0.83925 | -0.00459 | Minimum |
| 3 | 0.83953 | -0.00431 | |
| 4 | 0.84069 | -0.00315 | |
| 5 | 0.84187 | -0.00197 | |
| 6 | 0.84284 | -0.00100 | Worse than K = 1 |
| 7 | 0.84389 | +0.00005 | Worse than K = 1 |
| 8 | 0.84488 | +0.00104 | Worse than K = 1 |

*Table 6.8. ADMIXTURE 10-fold cross-validation error, K = 1-8 (logs/admixture_K*.out). Lower is better. K = 1 is not part of the supplied sweep and was added so the improvement from admitting structure at all could be measured. The total range across K = 1-8 is 0.00563, 0.67 per cent of the error value; the improvement from K = 1 to K = 2 is 0.00459, 0.54 per cent.*

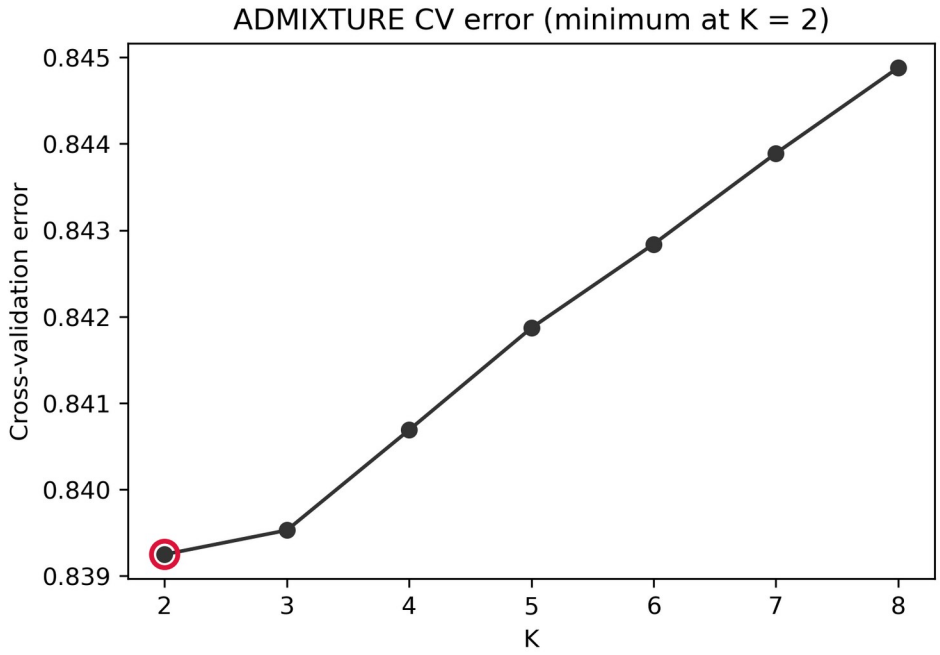



*Figure 6.5. ADMIXTURE cross-validation error against K. Two warnings about reading this plot. The y-axis is autoscaled over a range of 0.0056, so the apparent steepness is an artefact of the axis limits: the full range across all K is 0.67 per cent of the error value. And K = 1 is not plotted, because the sweep in config.py runs K = 2-8 and the K = 1 reference was run separately; it would sit at 0.84384, between the K = 6 and K = 7 points. Read Table 6.8 rather than the curve. Generated by scripts/step05c_plot_admixture.py.*

**[Observed]** At K = 2, mean minor-component ancestry by zone is: deciduous forest 0.0921, coastal savanna 0.0851, Guinea savanna 0.0534, transitional zone 0.0419 (overall mean 0.0635). Medians are 0.0563, 0.0587, 0.0430 and 0.0397 respectively.

**[Calculated]** The minor component differs significantly between zones: Kruskal-Wallis H = 105.4, p = 1.05 x $10^{-22}$ (one-way ANOVA F = 12.27, p = 6.0 x $10^{-8}$). The effect is small, and the two standard effect sizes do not agree on how small: eta-squared from the one-way ANOVA is 0.019, so zone accounts for 1.9 per cent of the variance in minor-component ancestry, while the rank-based epsilon-squared from the Kruskal-Wallis statistic, (H - k + 1)/(n - k), is 0.054. The gap reflects a strongly right-skewed component: the rank test responds to the shift in medians, the variance decomposition to the much larger within-zone spread of the means. Both are reported because neither alone is a fair summary. The effect survives adjustment for data quality and sampling date: in an OLS model with call rate, $F_{ws}$ and collection year as covariates, the partial F-test for zone is 11.43 on 3 and 1,893 degrees of freedom, p = 2.0 x $10^{-7}$, and model $R^2$ rises from 0.0028 without zone to 0.0205 with it. Pairwise Mann-Whitney tests locate the effect in the south-against-north contrast: coastal against Guinea savanna p = 3.9 x $10^{-15}$, deciduous forest against Guinea savanna p = 1.6 x $10^{-14}$, while coastal against deciduous forest (p = 0.20) and Guinea savanna against the transitional zone (p = 0.68) are indistinguishable. All of these were computed from results/admixture/Q_K2.csv joined to interim/analysis_samples.csv on the analysis server for this revision; the "about 2 per cent" quoted in *CORRECTIONS_AND_NEW_RESULTS.md* §4 is the eta-squared figure, and it was reported there alongside a superseded Kruskal-Wallis statistic (H = 112.2, p = 3.6 x $10^{-24}$) which should not be quoted.

| Zone | n | Mean minor component | Median | Mann-Whitney against Guinea savanna |
|---|---|---|---|---|
| Deciduous forest | 337 | 0.0921 | 0.0563 | p = 1.6 x $10^{-14}$ |

| Zone | n | Mean minor component | Median | Mann-Whitney against Guinea savanna |
|---|---|---|---|---|
| Coastal savanna | 204 | 0.0851 | 0.0587 | $p = 3.9 \times 10^{-15}$ |
| Guinea savanna | 1,334 | 0.0534 | 0.0430 | — |
| Transitional zone | 25 | 0.0419 | 0.0397 | p = 0.68 |

*Table 6.9. K = 2 minor-component ancestry by zone (results/admixture/Q_K2.csv). Kruskal-Wallis across all four zones: H = 105.4, $p = 1.05 \times 10^{-22}$. Effect size: eta-squared 0.019 from the one-way ANOVA, rank-based epsilon-squared 0.054.*

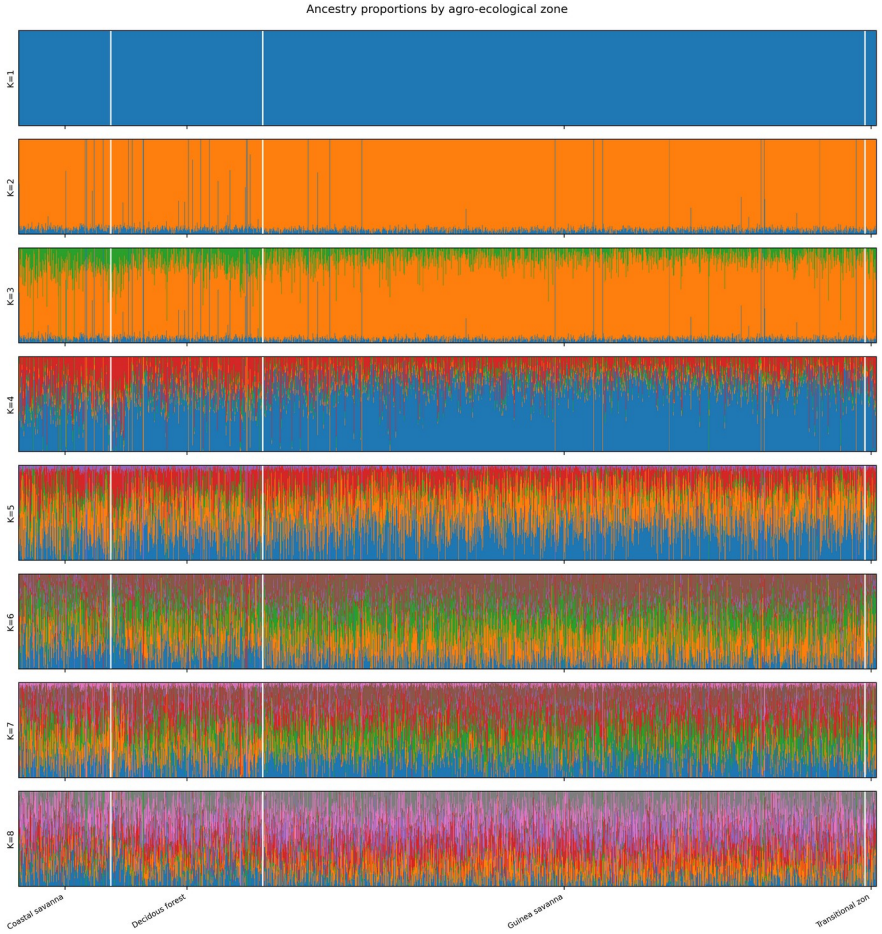



*Figure 6.6. ADMIXTURE ancestry proportions, K = 1-8, with isolates ordered by zone (n = 1,900). Each vertical bar is one isolate. No K produces components that correspond to zone boundaries as discrete blocks; the significant between-zone difference reported in Table 6.9 is a shift of about four percentage points in the mean of the minor component and is not visible at this scale. Generated by scripts/step05c_plot_admixture.py.*

**[Interpreted]** The original manuscript's statement that "the minority component is present at similar low frequency everywhere" and the summary claim of "no ancestry structure aligned to geography" are both false as written, and neither was ever tested. The corrected statement is that **zone affects minor-component ancestry significantly but negligibly**. Significantly: $p = 1.05 \times 10^{-22}$, robust to call rate, $F_{ws}$ and collection year. Negligibly: zone accounts for 1.9 per cent of the variance on the ANOVA measure and 5.4 per cent on the rank measure, the between-zone spread in means runs only from 0.042 to 0.092, and the whole K = 2 model improves on no structure at all by 0.54 per cent of cross-validation error. The substantive conclusion the manuscript drew, that this is a population without ancestry structure of any practical consequence, survives; the way it was stated does not.

**[Interpreted]** The correction works in the analysis's favour. The ordering of zone means, south high and north low, is the same ordering as the pairwise $F_{ST}$ ranking in Table 6.5 and the same direction as the region-level Mantel in Section 6.5. Unlike the Mantel it is not driven by a single population and it has not been shown to depend on the filtering thresholds, and unlike the $F_{ST}$ comparisons it has been shown to survive adjustment for collection year. With isolation by distance withdrawn (Section 6.5), it is now the only directional south-north signal in the analysis that rests on more than one contrast, and the original text discarded it.

> **Correction to a prior observation.** The original Section 6.6 stated that "at least one isolate is assigned entirely (1.000) to a single component". No isolate reaches 1.000. The extreme values in results/admixture/Q_K2.csv are exactly 0.99999 and 0.00001, which is ADMIXTURE's internal parameter bound, and 91 of the 1,900 isolates sit at that bound. The earlier sentence presented a software clamp as a property of the data, and understated the count by two orders of magnitude. What those 91 isolates are (plausibly residual near-clonal or low-coverage infections) is not established here.

**[Interpreted]** Whether the minor component reflects a real but geographically diffuse ancestry gradient, residual polyclonality, or an artefact of fitting two components to an essentially unstructured sample cannot be determined from these results. What can now be said is that whatever it is, it is not uniform across zones.

## 6.7 Population graph and migration edges

| m | Log likelihood | Variance explained | Identifiable with 4 populations |
|---|---|---|---|
| 0 | 89.6463 | 0.8970 | Yes |
| 1 | 91.2113 | 0.9405 | Yes |
| 2 | 92.2320 | 0.9969 | Yes |
| 3 | 92.2630 | 0.9997 | No |
| 4 | 92.2642 | 0.9996 | No |
| 5 | 92.2642 | 0.9996 | No |

*Table 6.10. TreeMix fit against number of migration edges (results/treemix/model_comparison.csv). Variance explained is computed over the six off-diagonal between-population covariances, matching TreeMix's own definition. All six models converged. These are the final-run values; the manuscript's 0.876 at m = 0 and 0.935 at m = 1 came from the superseded run and should not be quoted.*

**[Observed]** The sweep was re-run with `-se`, so migration edges now carry block-jackknife standard errors and p-values. In the original run `SE` was off, every migration line in the `.treeout` files read `weight NA NA NA`, and no edge could be tested at all; the model-choice question was left open for that reason.

| Model | Edge weight | Jackknife SE | p |
|---|---|---|---|
| m = 1 | 0.3128 | 0.0398 | $2.2 \times 10^{-15}$ |
| m = 2, edge 1 | 0.1525 | 0.0479 | $7.5 \times 10^{-4}$ |
| m = 2, edge 2 | 0.8055 | 0.1000 | $3.3 \times 10^{-16}$ |
| m = 3, edge 1 | 0.1551 | 0.0503 | $9.9 \times 10^{-4}$ |
| m = 3, edge 2 | 0.8302 | 0.0788 | $< 2.2 \times 10^{-308}$ |
| m = 3, edge 3 | 0.0652 | 0.1469 | 0.328 |

*Table 6.11. Migration-edge weights, block-jackknife standard errors and p-values (results/treemix/treemix_m*.treeout.gz). The third edge admitted at m = 3 is indistinguishable from zero: its standard error is more than twice its weight.*

[Observed] Both edges fitted at m = 2 are significant ($p = 7.5 \times 10^{-4}$ and $p = 3.3 \times 10^{-16}$). The third edge admitted at m = 3 has a weight of 0.065 against a standard error of 0.147 and a p-value of 0.328.

[Interpreted] **A two-edge graph is the model these data select, and m = 2 is reported as a descriptive model rather than as a count of historical migration events.** This narrows a question the original manuscript left open, but it does not close it the way a count of historical events would be closed. The plateau rule applied to variance explained picks m = 2 within the identifiable range and m = 3 across the full sweep, and with no edge p-values there was no way to choose between them; the manuscript retained both. The edge p-values now favour m = 2 — but they do so as a conditional test inside a graph whose topology, root and edge count were all fixed before the test was run, and they do not agree with the variance-explained criterion applied over the full sweep, which reaches 0.9997 at m = 3 and so still prefers m = 3. The two criteria therefore disagree, and the disagreement is settled here on identifiability grounds, which is a modelling decision and not a measurement. What the statistics do support is narrower: the third edge is indistinguishable from zero on its own jackknife standard error, four populations support roughly two edges on parameter-counting grounds, and m = 2 is accordingly the graph reported as a descriptive model. It is not an estimate of how many admixture events occurred in the history of Ghanaian P. falciparum, and no uncertainty in topology, root placement or edge count is propagated into it. The earlier hedging between m = 2 and m = 3 is replaced by a stated model choice, not by a resolution.

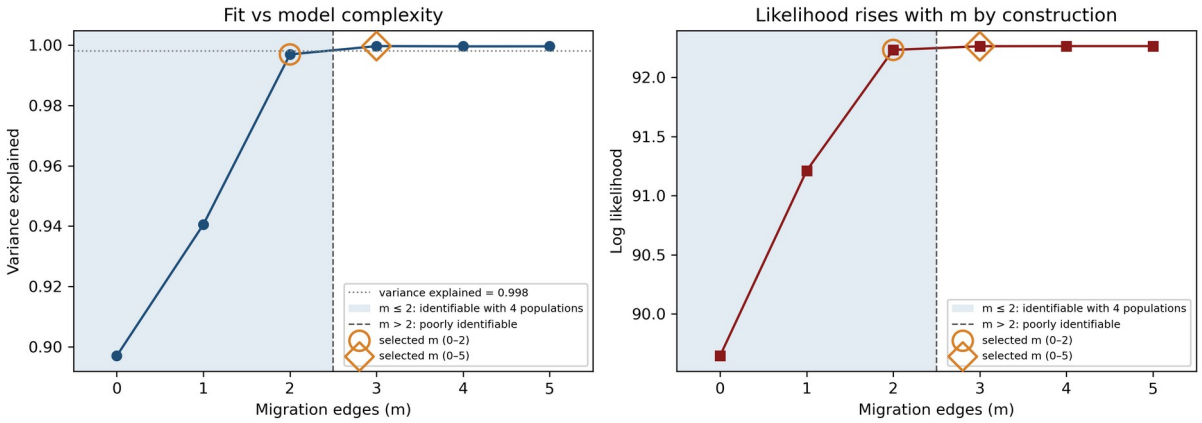



*Figure 6.7. TreeMix model fit against migration edges. Left: variance explained, with the 0.998 reference line dotted and the m <= 2 region shaded as the range four populations can identify. Right: log likelihood, which rises monotonically with m by construction and therefore cannot by itself select a model. The diamond marking m = 3 as the selection across the full sweep is the plateau rule's choice and remains that criterion's preference; the edge p-values in Table 6.11 support m = 2 instead, so the two criteria disagree and m = 2 is a stated model choice rather than a superseding result. Generated by scripts/step07c_plot_treemix.py.*

[Interpreted] Two cautions limit how much any of this can be made to mean. First, the fitted quantity is six off-diagonal between-population covariances, and an unrooted four-taxon tree already uses five branch-length parameters, so m = 0 has one residual degree of freedom and m = 1 has none or fewer. A rise in variance explained from 0.897 to 0.941 between those two models is not by itself evidence of gene flow, and the original manuscript's treatment of the m = 0 to m = 1 improvement as "the one substantive result" applied a degrees-of-freedom argument against m = 3 while exempting m = 1 from it. The edge p-values are the stronger evidence precisely because they do not depend on the variance-explained curve. Second, with four populations, three of which are a single administrative region each, these edges cannot be read as specific geographic gene-flow corridors. What the graph establishes is narrower than it looks: within this fitted covariance model, a strictly bifurcating topology leaves residual covariance that two admixture edges absorb,

and those two edges carry weights distinguishable from zero conditional on that graph. Edge significance is conditional on a fixed topology and root and does not propagate error in the graph structure itself (Pickrell and Pritchard 2012), so this is a statement about model fit rather than a demonstration of ongoing exchange. Those are conditional edge statistics, not identified directions of historical exchange.

[Calculated] Every printed edge p-value is accounted for by how the tail was computed, and the apparent anomalies are an artefact of that computation rather than of the data. An earlier version of this paragraph said that five of six values reproduced to three significant figures; that was wrong, and the corrected accounting follows. Recomputing every migration-edge p-value in m = 0 to 5 as a one-sided normal tail from the printed jackknife mean and standard error gives fifteen edges, of which thirteen carry a finite printed value and two print "< $2.22507 \times 10^{-308}$". That figure is exactly DBL_MIN, the smallest normal double-precision float, and it appears at m = 3 edge 2 (z = 10.53, stable tail $3.18 \times 10^{-26}$) and again at m = 4 edge 2 (z = 9.33, stable tail $5.45 \times 10^{-21}$). Of the thirteen finite values, ten agree with a numerically stable tail to three significant figures. The other three, and both sentinels, are explained — and reproduced to the last printed digit — by computing the upper tail as 1 − Φ(z) in double precision. Immediately below 1 the doubles are spaced $1.1102 \times 10^{-16}$ apart, so Φ(z) is quantised to multiples of that spacing and the tail recovered by subtraction can only take those values. At z = 7.843, 8.060 and 8.038 the formulation returns exactly $2.220446 \times 10^{-15}$, $3.330669 \times 10^{-16}$ and $4.440892 \times 10^{-16}$ — which are the three printed values to every digit printed — against stable tails of $2.1995 \times 10^{-15}$, $3.8147 \times 10^{-16}$ and $4.5658 \times 10^{-16}$. Once the true tail falls below half that spacing, Φ(z) rounds to exactly 1.0 and the difference is exactly zero; that happens at z = 9.33 and z = 10.53. This is rounding and cancellation, not underflow of the tail: the true tails there are $5.45 \times 10^{-21}$ and $3.18 \times 10^{-26}$, both far above the smallest normal double, and the "< $2.22507 \times 10^{-308}$" string is how TreeMix prints a computed zero rather than evidence that any quantity underflowed. A direct complementary-error-function evaluation would avoid the subtraction entirely. The accounting is therefore as follows. All fifteen outcomes reproduce to three significant figures under that formulation, and twelve of them — ten finite values and both zeroes — reproduce at their full displayed precision. The remaining three are moderate p-values rather than extreme tails: at m = 2 edge 1, for instance, the formulation returns 0.000747669 against a printed 0.000747653, a residual consistent with the printed mean and standard error being themselves rounded. Compatibility on this scale is strong evidence that the run computed jackknife standard errors, resolving a doubt raised by an earlier script snapshot lacking the -se flag. None of it changes a reported result: both sentinels belong to m = 3 and m = 4, neither of which is reported, and the two values the m = 2 choice rests on are $7.48 \times 10^{-4}$ and a tail of order $10^{-16}$ whose exact digits affect no inference.

## 6.8 Genome scan for differentiation outliers

[Observed] Across the 3,916-SNP matrix, XtX ranged from 2.48 to 26.44 with a median of 3.45. Calibration simulated pseudo-observed SNPs under the estimated population covariance matrix, with pi.maf = 0.05 truncating the simulated ancestral frequency rather than applying the observed data's MAF filter, so the pseudo-observed set is not MAF-matched (Section 5.6); this gives a 0.999-quantile threshold of 10.840. Nineteen SNPs (0.49 per cent) exceed it.

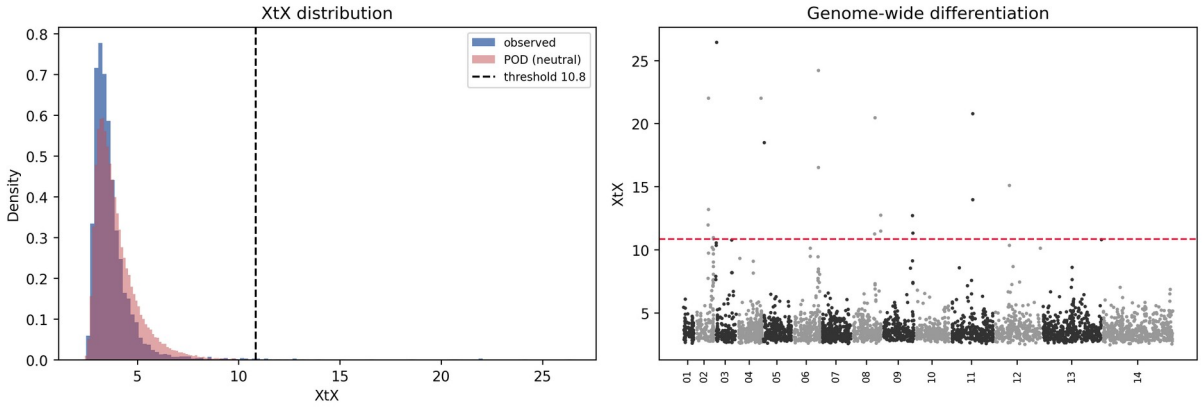



*Figure 6.8. BayPass XtX genome scan on the final 3,916-SNP matrix. Left: the observed XtX distribution against the simulated neutral (pseudo-observed) distribution, with the calibrated 0.999-quantile threshold at 10.840 (on-plot label "threshold 10.8"). Right: XtX by genomic position, alternating shading by chromosome, with the threshold marked. Regenerated by scripts/step08c_calibrate_xtx.py as figures/baypass_xtx.png.*

**[Interpreted] Reading the left panel.** The two distributions are not simply superimposed, and the difference between them runs in the opposite direction to what a casual reading suggests. The observed (blue) distribution is the **more peaked** of the two — a modal density of roughly 0.78 against roughly 0.59 for the simulated neutral (red) — and it is the **POD distribution that carries the heavier right shoulder**, holding visibly more density from about XtX 4 to 9. The observed exceeds the null only in the extreme tail: 19 SNPs, 0.49 per cent, lie above 10.840 against the null's 0.1 per cent, roughly five times the expected mass. The correct statement is therefore narrow: the body of the observed distribution is if anything slightly more concentrated than the simulated neutral one, and the excess that produces the outliers is confined to the extreme tail.

**[Observed] Three BayPass runs exist and must not be conflated.** The *manuscript run* was 8-threaded on the interim 3,359-SNP matrix and gave a 0.999-quantile threshold of **11.232** with 15 outliers across 11 genes (archived at results/baypass_8thread_archive/xtx_calibration.json). The *interim single-threaded run*, on the same 3,359 SNPs, gave **11.047** with 17 outliers — the value quoted in the corrections document. The *final run*, reported throughout this section, is single-threaded on the 3,916-SNP matrix and gives **10.840** with 19 outliers across 13 genes. The SNP panel grew by 16.6 per cent (3,359 to 3,916) after the variant-missingness filter was reordered to run after the sample cut (Section 3.x), and BayPass was moved to a single thread because the 8-thread binary is not reproducible even under a fixed seed. Every number below is from the final, bitwise-stable run.

### 6.8.1 False-discovery accounting

**[Calculated]** A 0.999 quantile is not a significance test. By construction, 0.1 per cent of neutral SNPs clear it, so at 3,916 SNPs **3.9 false positives are expected among the 19**, an expected-null-exceedance ratio at the threshold of 20.6 per cent. Per-SNP p-values against the pseudo-observed null and Benjamini–Hochberg q-values were therefore computed for every SNP by scripts/step08c_calibrate_xtx.py from **results/baypass/ghana_core_summary_pi_xtx.out** (3,916 SNPs) and **results/baypass/ghana_pod_summary_pi_xtx.out** (100,000 pseudo-observed SNPs), and those two files are what this section cites as the provenance of every q-value quoted below. The derived per-SNP table, results/baypass/xtx_all_snps.csv, **was found at 0 bytes and has since been regenerated**; the incident and the consistency checks the regenerated file passed are recorded in the note below. The q-values themselves

were never in doubt — recomputed from scratch from the two summary files above by the exact algorithm in step08c, they reproduce the results/baypass/xtx_calibration.json entry for entry (threshold 10.839768, 19 outliers at 0.485 per cent, 3.916 expected false positives, expected-null-exceedance ratio 20.611 per cent, and 8 / 10 / 13 / 17 SNPs at q ≤ 0.01 / 0.05 / 0.10 / 0.20) — and the restored table now carries them, so the artefact this text points at can support them again.

> **A results file was destroyed and restored, and the restoration is checkable rather than merely plausible.** results/baypass/xtx_all_snps.csv was found at **0 bytes**, truncated at **08:29:53** — hours after the **01:42** run that produced it, which rules out the run itself. It was the only zero-byte file anywhere under results/ or interim/. The likeliest cause is a stray shell redirect where a read was intended, during a verification pass. The file was regenerated from the intact BayPass outputs, and the regeneration is consistent with the surviving outputs rather than proven row-for-row: outlier_snps.bed, xtx_calibration.json and xtx_threshold.txt are byte-identical by md5 before and after, and the restored table has the same 3,916 rows and 19 outliers. The only other copy, results/baypass_8thread_archive/xtx_all_snps.csv, belongs to the superseded 3,359-SNP run and has no q column, so it was not and could not have been used. This is recorded rather than quietly repaired because it is a fact about this chapter's artefacts: a derived table that can be rebuilt from retained primaries, and whose rebuild can be checked by md5 identity of its downstream products, is recoverable. One that could not would have cost this section its provenance, and no amount of care in the prose would have substituted for it.

**[Calculated]** Counts with Benjamini–Hochberg q-values against the pseudo-observed null, conditional on that null being adequately calibrated, are 8 SNPs at q ≤ 0.01, 10 at q ≤ 0.05, **13 at q ≤ 0.10** and 17 at q ≤ 0.20. Six of the 19 SNPs above the quantile threshold do not pass q ≤ 0.10; their q-values run from 0.115 to 0.204.

> **How to read the two lists.** The 19-SNP quantile list is a **ranking**: it orders loci by evidence of excess differentiation but makes no claim about any individual locus. The 13 SNPs at q ≤ 0.10 are the **test result**. The original manuscript reported only a quantile list and interpreted its members individually, which is not a defensible inference. Throughout the rest of this section the 13 are treated as the finding and the 19 as the ranking they sit inside.

| Gene | Identifier | Product | Outlier SNPs | Max XtX | Min q |
|---|---|---|---|---|---|
| PTP7 | PF3D7_0301700 | EMP1 trafficking protein PTP7 | 1 | 26.44 | 0.005 |
| AAT1 | PF3D7_0629500 | Amino acid transporter AAT1 | 2 | 24.20 | 0.005 |
| FIKK4.2 | PF3D7_0424700 | Serine/threonine protein kinase, FIKK family | 1 | 22.02 | 0.005 |
| ACS8 | PF3D7_0215300 | Acyl-CoA synthetase | 2 | 22.00 | 0.005 |
| NT2 | PF3D7_0824400 | Nucleoside transporter 2 | 1 | 20.46 | 0.005 |
| PIESP2 | PF3D7_0501200 | Parasite-infected erythrocyte surface protein | 1 | 18.50 | 0.005 |
| SNRPF | PF3D7_1126900 | Small nuclear ribonucleoprotein F, putative | 1 | 13.98 | 0.031 |
| — | PF3D7_0831400 | Plasmodium exported protein, unknown function | 2 | 12.73 | 0.081 |
| GEXP22 | PF3D7_0935500 | Plasmodium exported protein, unknown function | 1 | 12.70 | 0.081 |
| — | PF3D7_0215100 | RING zinc finger protein, putative | 1 | 11.95 | 0.115 |
| REX2 | PF3D7_0936000 | Ring-exported protein 2 | 1 | 11.34 | 0.162 |
| — | PF3D7_0824200 | Conserved Plasmodium protein, unknown function | 1 | 11.27 | 0.168 |
| — | PF3D7_0221200 | Plasmodium exported protein (hyp15), unknown function | 2 | 10.96 | 0.202 |

*Table 6.7. Genes intersecting XtX outlier SNPs, ordered by maximum XtX (results/outlier_genes.csv, results/outliers_annotated.txt).* ***Seventeen of the 19 outlier SNPs fall in these 13 genes; the remaining two are intergenic and are given in the main text below, not in this table*** *— both survive FDR 10 per cent and one of them is the fifth-strongest SNP in the whole scan, so this gene-centric view is incomplete by construction. Min q is the smallest Benjamini–Hochberg q-value among that gene's outlier SNPs, rounded to three decimal places (the eight SNPs tied at the top of the null all carry q = 0.0049). Nine of the 13 genes retain at least one SNP at q ≤ 0.10; the four below the break in evidence — PF3D7_0215100, REX2,*

*PF3D7_0824200 and PF3D7_0221200 — do not, and should be read as ranked candidates only.* ***The SNRPF row carries a further caveat of a different kind: its single outlier SNP, Pf3D7_11_v3:1,054,078, is downgraded to lower confidence by the call-quality audit of Section 6.8.4, so if that downgrade is right the nine FDR-surviving genes are eight and this table has 12 rows, not 13.*** *Annotation from PlasmoDB-71 via bedtools.*

[Observed] **The two outliers that Table 6.7 cannot show.** Two of the 19 outlier SNPs fall outside any annotated protein-coding gene, and **both survive FDR correction**, so the gene-organised table omits 2 of the 13 FDR-surviving SNPs. Pf3D7_11_v3:**1,054,447** has XtX **20.79** and **q = 0.0049** — the **fifth-highest XtX in the entire scan**, ahead of **nine of the 13 tabulated genes**, only four of which have a maximum XtX above it (PTP7 26.44, *AAT1* 24.20, FIKK4.2 22.02, *ACS8* 22.00). It lies 59 bp beyond the 5′ boundary of *SNRPF* (PF3D7_1126900, 1,052,969–1,054,388, minus strand) and 369 bp from the *SNRPF* outlier at 1,054,078. **This same SNP is one of the two that the call-quality audit downgrades to lower confidence (Section 6.8.4), which bears directly on how much weight this paragraph can carry: the fifth-strongest signal in the scan is also the one with the weakest call support among the 19.** Pf3D7_12_v3:**715,788** has XtX **15.09** and **q = 0.017**, overlaps no annotated protein-coding gene in the bedtools intersect, and is graded clean on every quality field examined.

[Interpreted] The 19 outliers are fewer independent signals than they appear. Four of the 13 genes contribute two outlier SNPs each, and in every case the pair sits within 900 bp: *AAT1* 883 bp, *ACS8* 130 bp, PF3D7_0831400 25 bp and PF3D7_0221200 198 bp. Adding the 369 bp between Pf3D7_11_v3:1,054,447 and the *SNRPF* outlier — close enough, and in the right orientation, that the intergenic SNP plausibly sits in the gene's upstream regulatory interval rather than marking an independent locus — the 19 outlier SNPs represent at most **14 independent signals**, and the 17 SNP–gene overlaps at most 13. Counting SNPs rather than haplotypes overstates the evidence; it is the same error the original manuscript made at *pfaat1* (Section 6.8.2). The call-quality audit supplies a second and less benign reason to treat that particular pair as one item: they are the two lowest-InbreedingCoeff sites in the panel and one of them carries the panel's only nonzero ExcessHet, so they may be two signals, one signal, or no signal at all (Section 6.8.4).

[Observed] **What the larger panel changed.** Set-differenced against the manuscript run's 15 outliers, the final 19 contain four new SNPs: Pf3D7_02_v3:630,290; Pf3D7_09_v3:1,428,013 (*REX2*); and Pf3D7_02_v3:855,467 and 855,665 (PF3D7_0221200). Two **genes** are therefore new — *REX2* and PF3D7_0221200 — and both fail FDR 10 per cent. But the larger panel did add an FDR-surviving **SNP**: Pf3D7_02_v3:630,290 has XtX **22.00** and **q = 0.0049**, the fourth-strongest SNP in the scan, and it falls in *ACS8*, which rises from a maximum XtX of 11.90 and last place in the manuscript's 11-gene table to 22.00 and fourth place here. The accurate summary is therefore not that the larger panel added nothing that survives correction: it extended the ranking with two genes that do not survive correction, and promoted one existing gene from the bottom of the list to fourth on the strength of a new FDR-surviving SNP.

### 6.8.2 The AAT1 signal, named

[Observed] The two *pfaat1* (PF3D7_0629500) outliers are **Pf3D7_06_v3:1,215,233** and **Pf3D7_06_v3:1,214,350**, 883 bp apart inside a gene spanning 1,213,102–1,217,313 on the minus strand. The first is resolved from Pf8's own SnpEff annotation as **Ser258Leu**, a missense_variant of MODERATE impact, XtX **24.20**, **q = 0.0049** — the second-highest XtX in the scan and comfortably inside the FDR 10 per cent set. The second has XtX **16.54** and the same **q = 0.0049**.

[Unknown] The corrections document identifies Pf3D7_06_v3:1,214,350 as **Ile552Ile**, a synonymous_variant of LOW impact. **That codon assignment could not be reproduced from any primary output of this pipeline.** results/resistance_panel_frequencies.csv holds only missense_variant records, so a synonymous site is absent

from it by construction; the canonical panel lists only Ser258Leu and Phe313Ser for *pfaat1*; and the run logs contain no match for the position or the codon. The arithmetic is at least consistent — on the minus strand, codons 258 and 552 are 294 codons, or 882 nt, apart, against the observed 883 bp — but the identification rests on a narrative document rather than an output, and that same paragraph still carries the superseded XtX of 24.44 for the neighbouring SNP. The codon label is marked **[Unknown]** pending re-derivation from the Zarr. What is established is the position, the XtX, the q-value and the 883 bp separation.

**[Interpreted]** These are **one signal, not two independent lines of evidence**, whatever the second site turns out to be at codon level. Two outliers 883 bp apart in the same 4.2 kb gene are a linked pair: the second carries no independent information about selection, and the original manuscript's phrasing — "two outlier SNPs" offered as corroboration — double-counted a single haplotype. Ser258Leu is the candidate; the neighbour is what linkage disequilibrium drags along with it.

> **Correction. The previous description of Ser258Leu was wrong, and the error was mine.** The previous version of this section called Ser258Leu **"the one functionally validated resistance variant in this entire outlier list"** and stated that the functional experiment **did not test amodiaquine**. Both statements are withdrawn. Amodiaquine *was* tested: the methods of Amambua-Ngwa et al. (2023) read *"Parasite IC50 values for CQ, amodiaquine, lumefantrine, mefloquine and quinine were measured…"*, the observed shifts were *"well below the threshold associated with clinical resistance"*, and the authors explicitly reject the other-quinoline reading — *"The alternative hypothesis — that high frequencies of F313S mutations are driven by widespread use of other quinoline partner drugs in SEA — is not supported"* — concluding that chloroquine was *"the primary selective force"*. Separately, *pfaat1* appears nowhere in the WHO molecular-marker compendium at any tier, so it must not be called a WHO-validated or WHO-candidate marker anywhere in this thesis.

**[Interpreted]** What survives the correction is still substantial, and smaller than what was claimed. Ser258Leu is the specific substitution that Amambua-Ngwa et al. (2023, *Nature Microbiology*) introduced by genome editing and showed to raise chloroquine IC50 — but the effect is **1.5-fold**, measured in a **single** chloroquine-resistant Southeast Asian line (NHP4026), with **epistasis between S258L and F313S** and an explicit caveat that the effect depends on *pfcrt* background. The defensible description is therefore a variant with a demonstrated but **modest, epistasis- and background-dependent** effect that *potentiates* chloroquine resistance — not a validated resistance determinant on its own. Two further limits on transfer to Ghana: the longitudinal rise from 0 to 97 per cent that the paper documents is **Gambian, 1984–2014**, and the paper says nothing about Ghana; and the epistatic partner is **absent here** — *pfaat1* codon 313 is fixed at the reference residue Phe in all 1,900 isolates (Section 6.9.2), so every Ghanaian S258L allele in this sample sits on an F313 background and the epistatic component of the published effect cannot operate here. The paper does not state that the 1.5-fold measurement was itself taken on an F313S-carrying background, so nothing stronger than that should be claimed from the absence. The narrower point that this section can carry is that **the second-strongest outlier in the genome scan, and the strongest carrying any functional characterisation at all, is a transporter variant with a published, if modest, 4-aminoquinoline phenotype.** The strongest by XtX is PTP7 (Pf3D7_03_v3:96,476, XtX 26.44, q = 0.0049 — the same q as *pfaat1* S258L, both SNPs falling in the eight-SNP tie at the top of the null), an EMP1 trafficking protein with no resistance association. The variant surfaced unflagged because *pfaat1* was absent from `config.KNOWN_RESISTANCE_LOCI`, the eight-locus legacy panel (*pfcrt*, *pfmdr1*, *pfdhfr*, *pfdhps*, *pfk13*, *pfmdr2*, *pfcoronin*, *pfap2mu*) — a different eight from the eight panel genes that contribute polymorphic codons in Section 6.9 — and that exclusion is the mechanism by which the positive result was missed. The cline and the drug-pressure reading are developed in Section 6.9.9.

**[Interpreted]** Of the remaining genes, PTP7, PIESP2, FIKK4.2, GEXP22 and PF3D7_0831400 belong to the exported and host-interaction compartment of the genome, which is where immune-mediated selection

would be expected to act; *REX2* ("ring-exported protein 2") and PF3D7_0221200 ("*Plasmodium* exported protein (hyp15)") extend that pattern but do not survive FDR correction. PF3D7_0824200 does **not** extend it: final_outlier_genes.csv annotates it "conserved *Plasmodium* protein, unknown function", which is not an exported protein, and the previous version of this sentence grouped it with the exported pair in error. Four genes in all fail q ≤ 0.10, not three — PF3D7_0215100 (RING zinc finger, 11.95), *REX2* (11.34), PF3D7_0824200 (11.27) and PF3D7_0221200 (10.96) — because q ≤ 0.10 admits exactly 13 SNPs, i.e. XtX ≥ 12.70. The compartmental pattern is suggestive and is reported as such. It is not tested here: no enrichment test against a matched background was performed, so the apparent concentration of exported proteins remains an observation about a 13-gene list, not a result. What it is *not* is a mapping artefact of repetitive gene families: Section 6.8.4 shows that these eight SNPs are the cleanest subset in the whole panel.

### 6.8.3 Why most canonical resistance loci are absent from this scan

> **Correction.** The original text of this section read: *"None of the eight validated antimalarial resistance loci in the configured panel — pfcrt, pfmdr1, pfdhfr, pfdhps, pfk13, pfmdr2, pfcoronin, pfap2mu — appears among the outliers"*, and the abstract drew a biological inference from that absence. **The statement is withdrawn.** The absence was overwhelmingly an ascertainment artefact of the ingest filter, not a property of the parasite population, and nothing biological should ever have been read from it.

**[Observed]** step03a_fetch_callset_zarr.py builds the genome-wide callset under the mask FILTER_PASS & is_snp & numalt == 1. The numalt == 1 clause restricts the callset to strictly biallelic records. In Pf8, many canonical resistance codons are either multiallelic or carry a spanning-deletion ALT allele, so the clause removed those sites silently, before any analysis ran. Across the nine panel genes the filter sees **13,367 PASS records and keeps 3,330**, dropping 10,037 (logs/step10_resistance_panel.log, **lines 15–17**; line 14 gives the 18,341 variant records inside the nine genes). What it dropped is specifically the codons that carry the polymorphism: every polymorphic *pfcrt* codon of the CVIET haplotype (codon 72 is monomorphic Cys) and of its extended resistance background — **74** (M74I), **75** (the record annotated **Asn75Asp**, not N75E; see Section 6.9.1), **76** (K76T), **220** (A220S), **271** (Q271E) and **371** (R371I) — together with *pfmdr1* N86Y and Y184F, *pfdhfr* S108N, and *pfdhps* S436A, K540E and A581G. Codon 72, which is monomorphic Cys in all 1,900 isolates, is the exception: of its two records, Pf3D7_07_v3:403,612 has numalt 1 and survived the mask, as the note below records.

> **A stronger claim that the project's own output does not support.** The step10 log's editorial comment, and the corrections document that quotes it, state that the numalt clause dropped *every* canonical codon in *pfcrt*, *pfmdr1* and *pfk13*. **It did not.** Sixteen of the 44 canonical records in results/resistance_panel_canonical.csv have numalt == 1 and therefore passed the mask: *pfcrt* C72S (a CVIET-position codon), *pfcrt* I356T and N326S, *pfmdr1* S1034C, N1042D and D1246Y, *pfmdr2* T484I, *pfdhfr* C59R and N51I, *pfaat1* S258L, and six *pfk13* codons — F446I, Y493H, P553L, R561H and A675V, which are WHO-validated, plus *P441L*, *which is a WHO* candidate and which the previous version of this chapter wrongly listed among the validated set (Section 6.9.8). Ten of those 16 are at frequency 0.000 in this sample, which is why they were invisible in a different and harmless way. The rider is removed from the main text; the narrower statement above is what the output supports.

**[Observed]** Four examples from results/resistance_panel_canonical.csv: *pfcrt* K76T passes FILTER and has numalt **2**; *pfcrt* M74I has numalt **3**; *pfmdr1* N86Y has numalt **3**; *pfdhps* K540E has numalt **2**. Those four numalt values are confirmed exactly, and they alone are sufficient: none of the four sites could have appeared among the outliers whatever its true differentiation, because none was in the matrix. The per-site is_snp flags previously asserted in the same sentence — false for K76T, M74I and N86Y, true for K540E — are marked **[Unknown]**. They are not contradicted by anything retained, but nothing retained records is_snp per site: the only surviving trace is the aggregate numalt == 1 & is_snp 3,330 on line 16 of the step10 log, and the Pf8 Zarr

cannot be re-queried on the server (no zarr module or conda environment there, and only a chromosome 10 VCF is held locally, while these sites are on chromosomes 05, 07 and 08). A small per-site dump should be retained as evidence before the claim is restated as observed.

[Observed] **The cascade, stated once and precisely.** The superseded step10 panel genotyped 44 canonical codon-allele records — **41 distinct codon-allele protein changes**, three of the 44 being exact duplicates (*pfcrt* Cys72Ser, *pfcrt* Met74Ile, *pfk13* Met476Ile) — covering **39 distinct codons**, because two codons carry two distinct alleles each (*pfdhps* 613 Ser/Thr and *pfk13* 469 Phe/Tyr). Of those 39 codons, **21 are polymorphic**, and those 21 are a strict subset of the **24 polymorphic codons** that the codon reconstruction now finds (Section 6.9.3). Of the three codons the reconstruction alone recovers, two — *pfdhps* **431** and *pfdhps* **437** — were never in the step10 request list at all (its *pfdhps* request was Ser436Ala, Lys540Glu, Ala581Gly, Ala613Ser and Ala613Thr), so the ingest-mask audit in this section does not and cannot cover them. The third, *pfk13* **458**, *was* requested, as Asn458Tyr, and *was* genotyped — Pf3D7_13_v3:1,725,626, ref T alt A, numalt 2, overall frequency 0.0, n called 1,868 — so the mask does cover it, and dropped it. The reason the reconstruction alone finds 458 polymorphic is not absence from the list but the Section 6.9.1 failure mode: the single non-reference isolate carries Asn458**Asp**, and a string match for Asn458Tyr can never report Asp. That makes 458 a third instance of the right-codon/wrong-residue failure, alongside *pfcrt* 75 and *pfdhps* 437. Of the 21, **6 survived the biallelic ingest mask** — this is the in_genomewide_scan field, which is defined at step10_resistance_panel.py:349 as bool(t["numalt"] == 1) and therefore means "would have survived the ingest filter", not "entered the scan". Only **3 then reached the MAF-filtered scan matrix**: *pfdhfr* C59R, *pfdhfr* N51I and *pfaat1* S258L. The other three — *pfcrt* I356T (overall 0.016), *pfmdr1* D1246Y (0.011) and *pfcrt* N326S (**0.008 as step10 reported it**; the reconstruction revises it to 0.001, Section 6.9.2) — were removed afterwards by the MAF ≥ 0.05 cut and are absent from the scan matrix, confirmed by direct position lookup in **interim/baypass_snp_positions.csv** (3,916 rows): positions 405,600, 961,625 and 405,362 are absent, while 96,476, 748,239, 748,262 and 1,215,233 are present. That lookup is cited on baypass_snp_positions.csv rather than on results/baypass/xtx_all_snps.csv because the latter is the file that was found at zero bytes (Section 6.8.1); it has since been restored and would now support the same lookup, but the lookup reported here is the one actually performed, on the file that was always intact.

[Observed] **What the three tested codons returned.** *pfaat1* S258L: XtX 24.198, **q = 0.0049** — the signal discussed in Section 6.8.2. *pfdhfr* C59R (Pf3D7_04_v3:748,262): XtX 9.107, **q = 0.432**. *pfdhfr* N51I (Pf3D7_04_v3:748,239): XtX 8.147, **q = 0.672**. The two antifolate codons are firmly null. (The 0.432 and 0.672 reproduce as 0.431 and 0.672 with a cnt/n empirical p and as 0.432 and 0.672 with the (cnt+1)/(n+1) form; the difference is estimator convention, not error.)

[Interpreted] The correct statement of the negative is therefore neither the original claim nor its overcorrection: *the scan tested 3 of the 24 polymorphic canonical codons; it could not test the* pfcrt, pfmdr1, pfdhps *or* pfk13 *markers at all. This distinction matters, and it strengthens the section rather than weakening it. The scan's silence on the antifolate markers at* pfdhfr *is a tested negative — C59R and N51I were in the matrix and came back at q = 0.43 and q = 0.67, so neither shows excess differentiation beyond the demographic model. Its silence on* pfcrt, pfmdr1, pfdhps *and* pfk13 *is an ascertainment artefact* and carries no information whatever. Sections 6.9 and 7.x restate the resistance findings on a panel built without the biallelic restriction and without annotation-string matching, which is the only basis on which any claim about the untested loci can rest.

[Interpreted] **What XtX does and does not detect.** XtX measures differentiation among populations beyond what the estimated covariance structure predicts. A locus under strong but spatially uniform selection —

which is what a nationally deployed drug produces — generates no XtX signal at all. So even on a callset that contained them, the canonical loci would be expected to be invisible to this scan wherever drug pressure is uniform across Ghana; the null result at *pfdhfr* is consistent with exactly that, and is not evidence that antifolate resistance is unselected. This is an argument for measuring marker frequencies directly rather than inferring them from a differentiation scan, and Section 6.9 does so.

### 6.8.4 Call quality of the 19 outliers: the set is sound, two of its SNPs are not

**[Observed]** MQ, VQSLOD, FS, SOR, QD, QUAL, BaseQRankSum, MQRankSum, ReadPosRankSum, DP, AN, ExcessHet, InbreedingCoeff, RegionType, the VQSR training flags and culprit were pulled from the Pf8 Zarr for all 19 outliers and compared against the other **3,897** scanned SNPs; the per-SNP table is results/outlier_quality_check.csv. **Every significant difference runs in the outliers' favour.** They carry higher VQSLOD (p = 0.008), higher QD (p = 0.016) and higher QUAL (p = 0.007) than the rest of the panel. No outlier has MQ below 50 — the observed range is **59.56 to 60.00** — none shows read-position bias (|ReadPosRankSum| ≤ 0.304), and **18 of the 19 have FS = 0.00**, the nineteenth standing at 0.596, which is negligible on the GATK scale. All 19 are RegionType **Core**, none carries the Low_VQSLOD filter, **none is a VQSR negative training site** and 13 of the 19 are positive training sites. The culprit field reads MQRankSum for all 19, but it reads the same for 99.6 per cent of the rest of the panel, so it carries no information here.

**[Interpreted]** The obvious objection to a 0.49 per cent tail — that it is a tail of badly called sites rather than of differentiated ones — is therefore answered, and answered in the direction that favours the scan: the 19 are **better**-supported calls than the panel they were drawn from. This is not evidence of selection and must not be read as any; what it removes is one specific alternative explanation, which is all a quality audit can do. Of the 19, **15 are graded clean on every field examined, two carry a lower-grade "watch" annotation and two are flagged** (Table 6.7a).

**[Observed]** **The exported-protein worry is not supported.** The concentration of outliers in the exported and host-interaction compartment (Section 6.8.2) invites the suspicion that these gene families, which are repetitive, are collecting mismapped reads. The eight outlier SNPs in that compartment — PTP7, PIESP2, GEXP22, *REX2*, PF3D7_0831400 (two SNPs) and PF3D7_0221200 (two SNPs) — are in fact the **cleanest subset in the panel**: **FS = 0.00 for all eight**, MQ median **60.00** with a range of 59.99–60.00, and VQSLOD significantly higher than the rest of the panel. The families are repetitive; these particular sites sit in uniquely mappable sequence. The compartmental pattern therefore remains exactly as weak, and exactly as strong, as Section 6.8.2 states — untested for enrichment against a matched background — but it is not a mapping artefact, and that specific objection can be set aside.

**[Observed]** **Two SNPs must be downgraded, and not for any reason the quality fields were expected to turn up.** Pf3D7_11_v3:**1,054,447** (intergenic, XtX 20.79, q = 0.0049) and Pf3D7_11_v3:**1,054,078** (*SNRPF*, XtX 13.98, q = 0.031) — the same 369 bp pair discussed earlier in this section — are the **two lowest-InbreedingCoeff sites in the whole 3,916-SNP panel** (0.193 at the 0.13th percentile and 0.147 at the 0.08th), and they are also the two lowest-QD outliers (11.00 and 12.91, both below the 0.5th percentile of the panel). Decisively, 1,054,078 carries **ExcessHet = 69.59 — the only nonzero ExcessHet value among all 3,916 scanned SNPs**, and roughly **400 times the genome-wide maximum** reported by the check. 1,054,447 additionally has the lowest VQSLOD of the 19 (5.58, 2.4th percentile). Their mapping and strand-bias fields, by contrast, are clean: MQ 59.95 and 59.99, FS 0.596 and 0.00.

| SNP | Gene | XtX (q) | VQSLOD | QD | ExcessHet | InbreedingCoeff | Grade |
|---|---|---|---|---|---|---|---|
| Pf3D7_11_v3:1,054,447 | intergenic | 20.79 (0.0049) | 5.58 (2.4) | 11.00 (0.3) | 0.00 | 0.193 (0.13) | flag |
| Pf3D7_11_v3:1,054,078 | *SNRPF* | 13.98 (0.031) | 8.15 (17.2) | 12.91 (0.3) | 69.59 | 0.147 (0.08) | flag |
| Pf3D7_03_v3:96,476 | PTP7 | 26.44 (0.0049) | 11.70 (80.2) | 29.86 (81.5) | 0.00 | 0.689 (94.4) | watch |
| Pf3D7_06_v3:1,214,350 | *AAT1* | 16.54 (0.0049) | 9.95 (49.3) | 34.66 (99.1) | 0.00 | 0.522 (8.3) | watch |

*Table 6.7a. The four outlier SNPs that the call-quality audit did not grade clean, with the fields that produced the grade (results/outlier_quality_check.csv). Bracketed figures are percentiles within the 3,916-SNP scanned panel. The other 15 outliers are clean on every field examined. MQ runs 59.95–60.00 across these four and FS is 0.00 at three of them (0.596 at Pf3D7_11_v3:1,054,447), so **none of the four shows a mapping or strand-bias signal**: the two flagged SNPs are flagged on heterozygosity-related fields, not on mapping ones, which is what makes collapsed paralogy rather than mismapping the candidate mechanism. ExcessHet is 0.00 at 3,915 of the 3,916 scanned SNPs. The two "watch" SNPs are recorded for completeness and are **not** downgraded.*

**[Hypothesized]** In a near-haploid organism, an excess of apparent heterozygotes at a single site, with a depressed inbreeding coefficient alongside it, is not what mismapping looks like — mismapping degrades MQ and FS, and here both are clean at both sites. It is what **collapsed paralogy or copy-number variation** looks like: two diverged copies of a sequence mapping onto one reference locus present as heterozygous calls in every isolate that carries both, and a scan that treats such a locus as a single biallelic SNP will read the resulting allele-frequency differences between zones as population differentiation. **The mechanism is unverified, and should be read as a hypothesis and nothing more.** Nothing in results/outlier_quality_check.csv establishes it: the fields are consistent with it and inconsistent with the mismapping alternative, which is a good deal weaker than a demonstration. The check that would settle it is a read-level one — per-sample depth ratio at these two positions against each sample's genome-median depth, which would show a duplication directly — and **it has not been run**. Until it is, both SNPs are reported as **lower confidence**, with the reason stated rather than the result quietly dropped.

**[Interpreted]** **This matters more than a two-SNP correction usually would, and the reason should be stated plainly: Pf3D7_11_v3:1,054,447 is the fifth-ranked SNP in the entire scan.** It stands ahead of nine of the 13 genes in Table 6.7, it is one of the 13 SNPs that survive FDR 10 per cent, and it carries the same q = 0.0049 as *pfaat1* S258L and PTP7. The downgrade is therefore not a tidying of the bottom of the ranking but a withdrawal of confidence near the top of it. Its neighbour 1,054,078 is the *SNRPF* outlier and also FDR-surviving, at q = 0.031, so *SNRPF* is the one gene in Table 6.7 whose entire support is a downgraded SNP. If the paralogy reading is right, the FDR-surviving set is **11 SNPs rather than 13**, the independent-signal ceiling falls from **14 to 13** (the two were already counted as one locus on proximity grounds), and the gene list is **12 genes rather than 13**, of which **eight rather than nine** retain FDR-surviving support. If it is wrong, nothing changes. The honest position until the depth check is run is that the status of the scan's fifth-strongest signal is unresolved, and the two affected results — the intergenic pair and *SNRPF* — should not be built on.

**[Observed]** None of this reaches the signals on which this section's substantive claims rest. *pfaat1* S258L (Pf3D7_06_v3:1,215,233), FIKK4.2, both *ACS8* SNPs, NT2, PIESP2, Pf3D7_12_v3:715,788 and the exported-protein SNPs are graded clean on every field examined. The two "watch" annotations fall on **PTP7**, the top-ranked outlier, for a BaseQRankSum of 5.32 at the 96.8th percentile, and on the second *pfaat1* SNP at Pf3D7_06_v3:**1,214,350**, for an AN of 52,172 at the 0.5th percentile — more missingness than almost any site in the panel. Neither is downgraded: one high-percentile rank-sum statistic, or a low allele number on an

otherwise clean site, is not a defect. They are recorded so that the grading can be checked rather than taken on trust, which is the only defensible reason to report a quality audit at all.

## 6.9 A codon-reconstruction resistance marker panel

> **Correction. The previous version of this section is withdrawn in its entirety.** Section 6.9 as previously written reported a panel built by matching SnpEff's ANN_HGVS_p protein-change strings against a hand-written marker list. An external critique raised five points against it; all five were upheld and four were defects in the analysis code, not in the prose. Specifically withdrawn: the claim that the *pfdhfr* triple mutant stands at **0.786–0.908** (a range of three marginals is not a haplotype frequency, and the true haplotype frequency is lower than any of them); the claim that all 14 validated *pfk13* codons are at 0.000 **in all 1,900 isolates** (the list was not the WHO list, it contained a duplicate, it omitted three validated markers, and the denominator is per codon, not 1,900); the claim that **21 of 23 markers agree with MalariaGEN within 0.03** (agreement of marginal frequencies is not agreement of calls); the diagnosis of a **"naming discrepancy"** at *pfcrt* codon 75; and the complete absence of *pfdhps* A437G, a marker at a true frequency of 0.897 that the string match could never have found. Everything below comes from a new script, scripts/step12_codon_panel.py, and from its three outputs.

**[Interpreted]** Section 6.8.3 establishes that the genome-wide callset could test only 3 of the 24 polymorphic canonical resistance codons, and none of the *pfcrt*, *pfmdr1*, *pfdhps* or *pfk13* markers. This section tests them all directly, on a panel built to avoid both of the filters that hid them: the biallelic ingest mask of Section 6.8.3, and the annotation-string matching described immediately below. It supplies the evidence for Objective 3 — zone-specific variants relevant to drug response.

### 6.9.1 Why the previous panel was wrong: two ways a marker hides from an annotation string

**[Observed]** step10 identified markers by matching SnpEff ANN_HGVS_p strings such as "Ala437Gly" against a hand-written list. An annotation string describes a *change away from the reference*, and that has two failure modes. *(i) A resistance allele that* is *the reference residue can never appear as an ALT change. Where 3D7 itself carries the resistance residue, the only annotation at that codon is the* revertant, *so a search for the resistance change matches nothing and the codon is reported as if it did not exist. This is exactly what happened at* pfdhps *437 (Section 6.9.4). (ii) A residue reachable only by two changes within one codon is never annotated.* pfcrt *codon 75 spans Pf3D7_07_v3:403,621–403,623 and carries two variant records, 403621 (A>G, annotated Asn75Asp) and 403623 (T>A, annotated Asn75Lys). Asn→Glu requires both. Neither annotation names N75E, which is the residue the CVIET haplotype actually carries, so no string match could ever return it. The same mechanism accounts for* pfk13 458, where the request was Asn458Tyr and the one variant isolate carries Asp (Section 6.8.3).

**[Interpreted]** These two failure modes are the whole explanation of the two things the previous version got wrong at *pfcrt* 75 and *pfdhps* 437. In particular, the **"naming discrepancy" diagnosed at codon 75 was not a naming problem at all**: our pipeline and MalariaGEN's were reporting different residues because ours could only see one of the two SNPs that make up the residue. Calling it a disagreement about nomenclature, resolved in MalariaGEN's favour "on the biology", concealed a reconstruction failure behind a terminological one. That diagnosis is withdrawn.

### 6.9.2 Method: reconstruct the codon, report the whole spectrum

**[Observed]** scripts/step12_codon_panel.py discards annotation matching. For each requested codon it maps the codon to its three genomic positions through the gene's CDS, collects **every** variant record at those positions regardless of numalt, is_snp or ALT type, applies each isolate's alleles to the reference codon, and translates

the result. The output is the **full residue spectrum at each codon, including the reference residue**, so a resistance allele cannot hide in the reference and a two-SNP residue cannot be missed. Genotypes are collapsed by the same majority-allele ploidy rule used in Step 4a; a codon in which an isolate's calls are mixed is recorded as **uncalled** for that isolate rather than being assigned a residue. Frequencies are reported per agro-ecological zone with Wilson 95 per cent confidence intervals, and heterogeneity across the four zones is tested by $\chi^2$ on residue counts. The same 1,900 isolates are used as in every other analysis in this chapter.

```
python scripts/step12_codon_panel.py --out-prefix codon_panel
```

[Observed] **Fifty-seven codons across nine genes** (*pfcrt*, *pfmdr1*, *pfmdr2*, *pfdhfr*, *pfdhps*, *pfk13*, *pfaat1*, *pfap2mu*, *pfcoronin*) were reconstructed and genotyped: *pfk13* 23, *pfcrt* 10, *pfdhps* 6, *pfmdr1* 5, *pfdhfr* 5, *pfcoronin* 3, *pfmdr2* 2, *pfaat1* 2, *pfap2mu* 1. Three of the nine genes are on the minus strand (*pfaat1*, *pfk13*, *pfmdr2*) and six on the plus strand (*pfap2mu*, *pfcoronin*, *pfcrt*, *pfdhfr*, *pfdhps*, *pfmdr1*), which the codon-to-position mapping has to handle in both orientations. **Twenty-four are polymorphic and 33 are fixed at the reference residue.** Per-codon call rates run **1,834–1,900 of 1,900**. Outputs: results/codon_panel_residues.csv (83 codon–residue rows), codon_panel_haplotypes.csv and codon_panel_concordance.csv; log at logs/step12_codon_panel.log.

[Observed] Three codons are polymorphic here that the step10 panel never reported as polymorphic at all: *pfdhps* **431** (I431V, 0.014), *pfdhps* **437** (Section 6.9.4) and *pfk13* **458**, where a single isolate carries Asn458Asp — a residue that is *not* the validated N458Y change, and which step10 could not have seen because it requested only Asn458Tyr. Two codons that step10 reported as polymorphic resolve differently on reconstruction: *pfcrt* N326S falls from **0.008** to **0.001** (one isolate in 1,874), and *pfdhps* 436 turns out to carry **four** residues, not two — Ala 0.541, the reference Ser 0.446, and the minor residues Phe 0.007 and Tyr 0.005, neither of which appeared in the previous panel.

### 6.9.3 The residue spectrum by zone

| Gene | Codon / residue | Overall (95% CI) | Coastal savanna | Deciduous forest | Guinea savanna | Transitional | $\chi^2$ p |
|---|---|---|---|---|---|---|---|
| *pfcrt* | 74 Met † | 0.929 (0.916–0.940) | 0.946 (0.906–0.970) | 0.896 (0.858–0.924) | 0.936 (0.921–0.948) | 0.875 (0.690–0.957) | 0.035 |
| | Ile | 0.071 (0.060–0.084) | 0.054 (0.030–0.094) | 0.104 (0.076–0.142) | 0.064 (0.052–0.079) | 0.125 (0.043–0.310) | 0.035 |
| *pfcrt* | 75 Asn † | 0.929 (0.917–0.940) | 0.946 (0.906–0.970) | 0.896 (0.859–0.924) | 0.936 (0.921–0.948) | 0.875 (0.690–0.957) | 0.037 |
| | Glu | 0.071 (0.060–0.083) | 0.054 (0.030–0.094) | 0.104 (0.076–0.141) | 0.064 (0.052–0.079) | 0.125 (0.043–0.310) | 0.037 |
| *pfcrt* | 76 Lys † | 0.929 (0.917–0.940) | 0.946 (0.906–0.970) | 0.896 (0.859–0.924) | 0.936 (0.921–0.948) | 0.875 (0.690–0.957) | 0.037 |
| | Thr | 0.071 (0.060–0.083) | 0.054 (0.030–0.094) | 0.104 (0.076–0.141) | 0.064 (0.052–0.079) | 0.125 (0.043–0.310) | 0.037 |
| *pfcrt* | 220 Ala † | 0.930 (0.918–0.941) | 0.946 (0.906–0.970) | 0.896 (0.859–0.924) | 0.938 (0.924–0.950) | 0.875 (0.690–0.957) | 0.026 |
| | Ser | 0.070 (0.059–0.082) | 0.054 (0.030–0.094) | 0.104 (0.076–0.141) | 0.062 (0.050–0.076) | 0.125 (0.043–0.310) | 0.026 |
| *pfcrt* | 271 Gln † | 0.926 (0.914–0.937) | 0.946 (0.906–0.970) | 0.896 (0.859–0.924) | 0.932 (0.917–0.944) | 0.875 (0.690–0.957) | 0.063 |
| | Glu | 0.074 (0.063–0.086) | 0.054 (0.030–0.094) | 0.104 (0.076–0.141) | 0.068 (0.056–0.083) | 0.125 (0.043–0.310) | 0.063 |
| *pfcrt* | 326 Asn † | 0.999 (0.997–1.000) | 1.000 (0.982–1.000) | 1.000 (0.989–1.000) | 0.999 (0.996–1.000) | 1.000 (0.862–1.000) | 0.934 |

| Gene | Codon / residue | Overall (95% CI) | Coastal savanna | Deciduous forest | Guinea savanna | Transitional | $\chi^2$ p |
|---|---|---|---|---|---|---|---|
| | Ser | 0.001 (0.000–0.003) | 0.000 (0.000–0.018) | 0.000 (0.000–0.011) | 0.001 (0.000–0.004) | 0.000 (0.000–0.138) | 0.934 |
| *pfcrt* | 356 Ile † | 0.984 (0.978–0.989) | 0.971 (0.937–0.986) | 0.964 (0.939–0.980) | 0.991 (0.984–0.995) | 1.000 (0.867–1.000) | 0.002 |
| | Thr | 0.016 (0.011–0.022) | 0.029 (0.014–0.063) | 0.036 (0.020–0.061) | 0.009 (0.005–0.016) | 0.000 (0.000–0.133) | 0.002 |
| *pfcrt* | 371 Arg † | 0.924 (0.911–0.935) | 0.931 (0.888–0.959) | 0.896 (0.858–0.924) | 0.931 (0.916–0.944) | 0.875 (0.690–0.957) | 0.119 |
| | Ile | 0.076 (0.065–0.089) | 0.069 (0.041–0.112) | 0.104 (0.076–0.142) | 0.069 (0.056–0.084) | 0.125 (0.043–0.310) | 0.119 |
| *pfmdr1* | 86 Asn † | 0.912 (0.898–0.924) | 0.965 (0.930–0.983) | 0.919 (0.885–0.944) | 0.903 (0.886–0.918) | 0.880 (0.700–0.958) | 0.029 |
| | Tyr | 0.088 (0.076–0.102) | 0.035 (0.017–0.070) | 0.081 (0.056–0.115) | 0.097 (0.082–0.114) | 0.120 (0.042–0.300) | 0.029 |
| *pfmdr1* | 184 Phe | 0.677 (0.655–0.697) | 0.660 (0.592–0.722) | 0.651 (0.597–0.701) | 0.683 (0.657–0.707) | 0.800 (0.609–0.911) | 0.364 |
| | Tyr † | 0.323 (0.303–0.345) | 0.340 (0.278–0.408) | 0.349 (0.299–0.403) | 0.317 (0.293–0.343) | 0.200 (0.089–0.391) | 0.364 |
| *pfmdr1* | 1246 Asp † | 0.989 (0.983–0.993) | 0.980 (0.950–0.992) | 0.994 (0.978–0.998) | 0.989 (0.982–0.994) | 0.960 (0.805–0.993) | 0.262 |
| | Tyr | 0.011 (0.007–0.017) | 0.020 (0.008–0.050) | 0.006 (0.002–0.022) | 0.011 (0.006–0.018) | 0.040 (0.007–0.195) | 0.262 |
| *pfdhfr* | 51 Ile | 0.786 (0.767–0.804) | 0.892 (0.842–0.928) | 0.802 (0.756–0.842) | 0.764 (0.741–0.786) | 0.880 (0.700–0.958) | 2.4e-4 |
| | Asn † | 0.214 (0.196–0.233) | 0.108 (0.072–0.158) | 0.198 (0.158–0.244) | 0.236 (0.214–0.259) | 0.120 (0.042–0.300) | 2.4e-4 |
| *pfdhfr* | 59 Arg | 0.886 (0.871–0.900) | 0.946 (0.906–0.969) | 0.928 (0.896–0.951) | 0.864 (0.845–0.882) | 1.000 (0.867–1.000) | 4.8e-5 |
| | Cys † | 0.114 (0.100–0.129) | 0.054 (0.031–0.094) | 0.072 (0.049–0.104) | 0.136 (0.118–0.155) | 0.000 (0.000–0.133) | 4.8e-5 |
| *pfdhfr* | 108 Asn | 0.908 (0.895–0.921) | 0.971 (0.937–0.986) | 0.958 (0.931–0.975) | 0.885 (0.866–0.901) | 1.000 (0.867–1.000) | 9.4e-7 |
| | Ser † | 0.092 (0.079–0.105) | 0.029 (0.014–0.063) | 0.042 (0.025–0.069) | 0.115 (0.099–0.134) | 0.000 (0.000–0.133) | 9.4e-7 |
| *pfdhps* | 431 Ile † | 0.986 (0.980–0.991) | 0.966 (0.931–0.983) | 0.970 (0.946–0.984) | 0.994 (0.988–0.997) | 0.960 (0.805–0.993) | 1.8e-4 |
| | Val | 0.014 (0.009–0.020) | 0.034 (0.017–0.069) | 0.030 (0.016–0.054) | 0.006 (0.003–0.012) | 0.040 (0.007–0.195) | 1.8e-4 |
| *pfdhps* | 436 Ala | 0.541 (0.518–0.563) | 0.475 (0.407–0.544) | 0.491 (0.438–0.544) | 0.558 (0.532–0.585) | 0.800 (0.609–0.911) | 0.002 |
| | Ser † | 0.446 (0.424–0.469) | 0.520 (0.451–0.588) | 0.509 (0.456–0.562) | 0.426 (0.399–0.452) | 0.120 (0.042–0.300) | 4.5e-5 |
| | Phe | 0.007 (0.004–0.012) | 0.005 (0.001–0.028) | 0.000 (0.000–0.011) | 0.009 (0.005–0.016) | 0.040 (0.007–0.195) | 0.081 |
| | Tyr | 0.005 (0.003–0.010) | 0.000 (0.000–0.019) | 0.000 (0.000–0.011) | 0.007 (0.004–0.013) | 0.040 (0.007–0.195) | 0.028 |
| *pfdhps* | 437 Gly † | 0.897 (0.882–0.910) | 0.941 (0.900–0.966) | 0.964 (0.939–0.980) | 0.875 (0.856–0.892) | 0.800 (0.609–0.911) | 1.2e-6 |
| | Ala | 0.103 (0.090–0.118) | 0.059 (0.034–0.100) | 0.036 (0.020–0.061) | 0.125 (0.108–0.144) | 0.200 (0.089–0.391) | 1.2e-6 |
| *pfdhps* | 540 Lys † | 0.991 (0.985– | 0.995 (0.973– | 0.994 (0.979– | 0.989 (0.981– | 1.000 (0.867– | 0.661 |

| Gene | Codon / residue | Overall (95% CI) | Coastal savanna | Deciduous forest | Guinea savanna | Transitional | $\chi^2$ p |
|---|---|---|---|---|---|---|---|
| | | 0.994) | 0.999) | 0.998) | 0.993) | 1.000) | |
| | Glu | 0.009 (0.006–0.015) | 0.005 (0.001–0.027) | 0.006 (0.002–0.021) | 0.011 (0.007–0.019) | 0.000 (0.000–0.133) | 0.661 |
| *pfdhps* | 581 Ala † | 0.986 (0.980–0.991) | 0.965 (0.930–0.983) | 0.973 (0.950–0.986) | 0.993 (0.987–0.996) | 0.960 (0.805–0.993) | 8.0e-4 |
| | Gly | 0.014 (0.009–0.020) | 0.035 (0.017–0.070) | 0.027 (0.014–0.050) | 0.007 (0.004–0.013) | 0.040 (0.007–0.195) | 8.0e-4 |
| *pfdhps* | 613 Ala † | 0.892 (0.877–0.905) | 0.892 (0.841–0.927) | 0.928 (0.895–0.951) | 0.884 (0.866–0.900) | 0.840 (0.653–0.936) | 0.113 |
| | Ser | 0.108 (0.095–0.123) | 0.108 (0.073–0.159) | 0.072 (0.049–0.105) | 0.116 (0.100–0.134) | 0.160 (0.064–0.347) | 0.113 |
| *pfk13* | 458 Asn † | 0.999 (0.997–1.000) | 1.000 (0.981–1.000) | 1.000 (0.988–1.000) | 0.999 (0.996–1.000) | 1.000 (0.867–1.000) | 0.937 |
| | Asp | 0.001 (0.000–0.003) | 0.000 (0.000–0.019) | 0.000 (0.000–0.012) | 0.001 (0.000–0.004) | 0.000 (0.000–0.133) | 0.937 |
| *pfmdr2* | 492 Val | 0.549 (0.526–0.571) | 0.460 (0.392–0.529) | 0.548 (0.494–0.600) | 0.559 (0.532–0.585) | 0.720 (0.524–0.857) | 0.020 |
| | Ile † | 0.451 (0.429–0.474) | 0.540 (0.471–0.608) | 0.452 (0.400–0.506) | 0.441 (0.415–0.468) | 0.280 (0.143–0.476) | 0.020 |
| *pfap2mu* | 160 Ser † | 0.904 (0.889–0.916) | 0.922 (0.876–0.951) | 0.929 (0.896–0.952) | 0.894 (0.876–0.910) | 0.920 (0.750–0.978) | 0.199 |
| | Asn | 0.096 (0.084–0.111) | 0.078 (0.049–0.124) | 0.071 (0.048–0.104) | 0.106 (0.090–0.124) | 0.080 (0.022–0.250) | 0.199 |
| *pfaat1* | 258 Leu | 0.609 (0.587–0.631) | 0.824 (0.765–0.870) | 0.735 (0.685–0.779) | 0.544 (0.517–0.571) | 0.640 (0.445–0.798) | 1.9e-18 |
| | Ser † | 0.391 (0.369–0.413) | 0.176 (0.130–0.235) | 0.265 (0.221–0.315) | 0.456 (0.429–0.483) | 0.360 (0.202–0.555) | 1.9e-18 |

*Table 6.8. Full residue spectrum of the 24 polymorphic codons, by agro-ecological zone, with Wilson 95 per cent confidence intervals (results/codon_panel_residues.csv). Rows are codon–residue pairs;* ***† marks the 3D7 reference residue****, which is reported explicitly because at pfdhps 437 the reference residue is the resistance allele (Section 6.9.4) and because at* ***seven*** *of the 24 codons the non-reference residue is the majority one: pfdhfr 51 (Ile 0.786), pfdhfr 59 (Arg 0.886), pfdhfr 108 (Asn 0.908), pfdhps 436 (Ala 0.541), pfmdr1 184 (Phe 0.677), pfmdr2 492 (Val 0.549) and pfaat1 258 (Leu 0.609). Residues within a codon are ordered by overall frequency; the 33 codons fixed at the reference residue are omitted — the* ***22*** *pfk13 ones are listed in Table 6.11, and the remaining* ***11*** *(pfcrt 72 and 73, pfmdr1 1034 and 1042, pfdhfr 16 and 164, pfmdr2 484, pfcoronin 50, 100 and 107, and pfaat1 313) appear only in the source file. Overall denominators are per codon and run* ***1,858–1,900 of 1,900*** *isolates across the 24 codons shown; the panel-wide range of 1,834–1,900 quoted in Section 6.9.2 includes fixed codons omitted here, its lower bound being pfmdr1 1034. $\chi^2$ p tests heterogeneity of that residue against all other residues at the same codon across the four zones, so a codon with four residues yields four tests whose p-values differ — pfdhps 436 is the only such codon. Zone column headings are spelled correctly here for display; the source geospatial layer and the source columns of codon_panel_residues.csv read Decidous_forest and Transitional_zone, and the spatial join keys on that source spelling (see Methods).*

**[Calculated]** Taking the smallest residue-level p-value per codon and correcting across the 24 polymorphic codons (Bonferroni threshold 0.05/24 = 0.0021), **nine codons show between-zone heterogeneity**: *pfdhfr* S108N (9.4e-7), C59R (4.8e-5) and N51I (2.4e-4); *pfdhps* A437G (1.2e-6), 436 (4.5e-5 at the reference Ser residue), I431V (1.8e-4) and A581G (8.0e-4); *pfcrt* I356T (0.0016); and *pfaat1* S258L (1.9e-18). Nominal signals at *pfcrt* 220 (0.026), *pfcrt* 74/75/76 (0.035–0.037), *pfmdr1* N86Y (0.029), *pfmdr2* I492V (0.020) and the minor *pfdhps* Tyr436 residue (0.028) do not survive correction and are not interpreted. The denominator of 24 — the codons that carry a between-zone test, not all 57 reconstructed codons — is the one used everywhere in this section, and is stated here once so that it can be checked. The previous version reported seven of 21

codons passing at 0.05/21; the two additions are both at *pfdhps*, and both are codons the string match never recovered.

[Observed] **What moved relative to the superseded table, stated plainly.** Every *pfcrt* codon on the chloroquine-resistant haplotype came down. The three polymorphic codons of the CVIET motif fell from 0.079 to **0.071** — M74I, K76T, and codon 75, which was **previously reported as Asn75Asp at 0.079 and is N75E at 0.071** — and the associated background codons followed: A220S from 0.074 to **0.070**, Q271E from 0.079 to **0.074**, R371I from 0.084 to **0.076**. All six now lie in a 0.006 band rather than a 0.010 one. *pfcrt* N326S fell from 0.008 to **0.001**. The *pfdhfr* and *pfaat1* marginals barely moved (N51I 0.7859 → **0.7864**; C59R 0.8860 → **0.8861**; S108N 0.9076 → **0.9084**; S258L 0.60927 → **0.60918**), as did *pfmdr1* Y184F (0.677), *pfmdr2* I492V (0.550 → **0.549**), *pfdhps* A613S (0.109 → **0.108**) and *pfap2mu* S160N (0.097 → **0.096**). Codon 75 is no longer labelled Asn75Asp but **Asn75Glu**. Two codons and one residue are new. None of the changes reverses a direction of effect; the point is that the previous numbers were close enough to pass every check the project then ran, which is the subject of Section 6.9.10.

### 6.9.4 *pfdhps* A437G and the reference-allele trap

[Observed] *pfdhps* A437G, the first-step sulfadoxine-resistance substitution and the commonest *pfdhps* marker in Africa, was **entirely absent from the previous version of this chapter**. It was not reported at a low frequency, or reported with a caveat: it did not appear. The reason is failure mode (i) of Section 6.9.1. The 3D7 reference genome itself translates as **Gly** at codon 437, so the only annotation SnpEff can emit at Pf3D7_08_v3:549,684–549,686 is the single record `Gly437Ala` — the revertant. A search for "Ala437Gly" matched nothing at all, and a marker at a true frequency of **0.897** was silently dropped from a chapter whose subject is antifolate resistance.

> **The official Pf8 methodology says this outright.** Quoted verbatim from the MalariaGEN Pf8 methods document, and verified in the source: *"3D7 sequence translates as a G at 437, this is thought to be the mutant allele, i.e. 3D7 does not have the wild type allele."* MalariaGEN's own published genotype file encodes the same fact in its column name, `dhps_437[G]`, where the bracketed letter is the reference residue. Both were available throughout. The information needed to avoid this error was in the methodology of the dataset being analysed.

| Zone | n called | G437 frequency | 95% CI |
|---|---|---|---|
| Deciduous forest | 337 | 0.964 | 0.939–0.980 |
| Coastal savanna | 203 | 0.941 | 0.900–0.966 |
| Guinea savanna | 1,327 | 0.875 | 0.856–0.892 |
| Transitional zone | 25 | 0.800 | 0.609–0.911 |
| Overall | 1,892 | 0.897 | 0.882–0.910 |

*Table 6.9. pfdhps G437 frequency by agro-ecological zone, with Wilson 95 per cent confidence intervals (results/codon_panel_residues.csv). G is the 3D7 reference residue and the resistance allele; the complementary residue is the revertant A437, whose frequencies are the arithmetic complements shown in Table 6.8. $\chi^2 = 30.34$ on 3 df, $p = 1.2 \times 10^{-6}$.*

[Interpreted] The zone ordering mirrors *pfdhfr*: highest in the two southern zones, lowest in Guinea savanna and the 25-isolate Transitional zone, and not monotonic in latitude. Substantively, G437 at 0.897 means the first *pfdhps* step stands *as high as the* pfdhfr *triple mutant's component marginals — just below S108N at 0.908, which is the panel's most frequent antifolate residue — which materially strengthens the antifolate picture in Section 6.9.5 relative to the previous version, where the* pfdhps side appeared to rest on S436A at 0.54 and little else. Methodologically, the lesson generalises beyond this codon: any marker panel that

identifies variants by their change-from-reference string will miss every marker for which the reference carries the derived allele, and will do so without raising an error. The only general defence is to reconstruct the codon and report the whole residue spectrum, which is why Table 6.8 lists reference residues explicitly rather than implying them.

### 6.9.5 Antifolate resistance: near-fixed *pfdhfr* component marginals, further advanced at *pfdhps* than previously reported, and zone-structured but not latitudinal

[Observed] The three *pfdhfr* components stand at **0.786 (N51I), 0.886 (C59R) and 0.908 (S108N)** overall. Of the three, only **S108N** is the runner-up to *pfaat1* for zone structure ($\chi^2$ p = 9.4e-7 against 1.9e-18); the claim that all three are, after *pfaat1*, the most strongly zone-structured codons in the panel was true of the old 21-codon panel and is **not** true of the rebuilt one. Ranking the 24 tested codons by $\chi^2$ p puts three *pfdhps* codons between S108N and the other two *pfdhfr* components — 437 at 1.2e-6 (the value this section prints in Table 6.9), 436 at 4.5e-5 and 431 at 1.8e-4 — against C59R at 4.8e-5 and N51I at 2.4e-4. The defensible statement is that S108N is second only to *pfaat1* S258L, and that C59R and N51I are among the most strongly zone-structured codons alongside *pfdhps* 437, 436 and 431. All three *pfdhfr* components are higher in the two southern zones than in Guinea savanna: S108N 0.971 coastal savanna and 0.958 deciduous forest against 0.885; C59R 0.946 and 0.928 against 0.864; N51I 0.892 and 0.802 against 0.764. **This is not a monotonic south-to-north cline.** The Transitional zone is the second-most-northern of the four and carries the panel's highest *pfdhfr* values — 1.000 for S108N and C59R, 0.880 for N51I — so an ordering by latitude breaks there. What the data support is a contrast between the southern zones and Guinea savanna, with the 25-isolate Transitional zone unresolved. I164L, the fourth-step codon, is fixed at the reference residue Ile. On the *pfdhps* side, **G437 is at 0.897** (Section 6.9.4), 436 carries Ala at 0.541 against the reference Ser at 0.446 with minor Phe (0.007) and Tyr (0.005) residues, A613S sits at 0.108, and the high-grade additions remain rare: I431V 0.014, A581G 0.014, K540E 0.009.

[Interpreted] This is the standard West African antifolate pattern, and the recovery of A437G sharpens rather than changes it: sulfadoxine–pyrimethamine is no longer a treatment regimen but remains in intermittent preventive use; the three *pfdhfr* **component marginals** are near-fixed at 0.786–0.908, **though the triple-mutant haplotype itself is 0.771** and must not be described as close to fixed (Section 6.9.6); the first *pfdhps* step, G437, stands at 0.897; and the *pfdhps* additions that would make the quintuple mutant — chiefly K540E — have not followed, so the A437G+K540E double stands at 0.010 (Section 6.9.6). The south-versus-Guinea-savanna difference is consistent with longer or heavier historical antifolate exposure in the more urbanised, higher-service southern zones, but the panel cannot test that: no drug-pressure covariate was available, the Transitional zone contradicts a latitudinal reading, and Section 6.9.11 gives a competing explanation — confounding with sequencing study — that this design cannot separate from it.

### 6.9.6 Within-isolate haplotypes, counted on complete cases

[Observed] The previous version reported the *pfdhfr* "triple mutant" as the **range of its three marginal frequencies, 0.786–0.908**. That is not a haplotype frequency and it cannot be one: a range of marginals carries no information about whether the three alleles co-occur in the same parasite. Haplotypes are now reconstructed **within isolates**, counted only on isolates called at every constituent codon, so each haplotype has its own complete-case denominator.

| Haplotype | Complete cases | Count | Overall (95% CI) | Coastal savanna | Deciduous forest | Guinea savanna | Transitional |
|---|---|---|---|---|---|---|---|
| *pfdhfr* IRN | 1,876 | 1,446 | 0.771 (0.751– | 0.867 (0.813– | 0.788 (0.741– | 0.750 (0.726– | 0.880 (0.700– |

| Haplotype | Complete cases | Count | Overall (95% CI) | Coastal savanna | Deciduous forest | Guinea savanna | Transitional |
|---|---|---|---|---|---|---|---|
| (N51I+C59R+S108N) | | | 0.789) | 0.907) | 0.829) | 0.772) | 0.958) |
| *pfdhfr* IRNL (adds I164L) | 1,874 | 0 | 0.000 (0.000–0.002) | 0.000 (0.000–0.019) | 0.000 (0.000–0.012) | 0.000 (0.000–0.003) | 0.000 (0.000–0.133) |
| *pfcrt* CVIET (codons 72–76) | 1,873 | 133 | 0.071 (0.060–0.084) | 0.054 (0.030–0.094) | 0.104 (0.076–0.142) | 0.064 (0.052–0.079) | 0.125 (0.043–0.310) |
| *pfdhps* GE (A437G+K540E) | 1,888 | 18 | 0.010 (0.006–0.015) | 0.005 (0.001–0.027) | 0.006 (0.002–0.021) | 0.011 (0.007–0.019) | 0.000 (0.000–0.133) |

*Table 6.10. Within-isolate haplotype frequencies on complete cases, with Wilson 95 per cent confidence intervals (results/codon_panel_haplotypes.csv). "Complete cases" are isolates with a called residue at every constituent codon; the denominator therefore differs between haplotypes and is smaller than 1,900. Overall figures are pooled across zones.*

[Calculated] **The triple mutant is 0.771, not 0.786–0.908, and the difference is not cosmetic.** A haplotype frequency cannot exceed the frequency of its rarest component, and on the same 1,876 complete cases that rarest component is 51I at **0.788**. The reconstructed IRN frequency of **0.771** sits below it, as it must. The previously reported range not only mislabelled a set of marginals as a haplotype, it quoted an upper figure (0.908, the S108N marginal) that is **12.0 percentage points above a mathematically impossible ceiling** for the quantity it claimed to describe: 0.908 − 0.788 = 0.120. (The gap from 0.908 to the reconstructed haplotype frequency of 0.771 is 13.7 percentage points, but that is a gap to an observed value, not to a ceiling, and the two should not be swapped.) The error was detectable by inspection, without any new computation.

[Observed] Nor is *pfdhfr* a near-fixed single haplotype. Alongside IRN at 0.771, **NRN sits at 0.116** and the fully wild-type **NCS at 0.091**, leaving roughly 0.022 in other combinations. The quadruple IRNL is **absent** (0 of 1,874; upper bound 0.002), consistent with I164L being fixed at the reference residue. At *pfcrt*, by contrast, the haplotype really does travel as a unit, and this is now demonstrated rather than inferred: the **only two 72–76 haplotypes observed are CVMNK (0.929) and CVIET (0.071)**, with nothing intermediate. The previous version inferred that unity from six similar marginals, which is suggestive but not evidence; co-occurrence data are.

[Interpreted] The practical consequence is that Ghana's antifolate burden should be read as a population carrying a dominant but not universal *pfdhfr* triple-mutant lineage — about three isolates in four — with a substantial single-mutant and wild-type remainder, rather than as a population in which the triple mutant has swept. The *pfdhps* GE double at 0.010 confirms that the quintuple mutant, the configuration that compromises sulfadoxine–pyrimethamine for intermittent preventive treatment, is rare here despite G437 being near-fixed; what is missing is K540E, not A437G.

### 6.9.7 Chloroquine resistance has largely reverted, and the CVIET haplotype is demonstrably a unit

[Observed] The six *pfcrt* codons that carry the **CVIET-associated resistance haplotype** — M74I, **N75E**, K76T, A220S, Q271E and R371I — now sit within a 0.006 band at **0.070–0.076**, and the reconstructed CVIET haplotype itself is at **0.071** (133 of 1,873 complete cases). CVIET names the codon 72–76 motif, as Table 6.10 states, so only M74I, N75E and K76T are motif codons; A220S, Q271E and R371I are flanking markers of the wider chloroquine-resistance background. They track together zone by zone: 0.054 coastal savanna for five of the six, 0.104 in deciduous forest, 0.062–0.069 in Guinea savanna, 0.125 in the 25-isolate Transitional zone. **None of the six reaches the corrected heterogeneity threshold**: their nominal p-values run 0.026–0.119 against the Bonferroni threshold of 0.05/24 = 0.0021, the 24 being the panel codons that carry a between-

zone test (Section 6.9.3). The two *pfcrt* codons off that haplotype behave differently: *I356T is at 0.016 and is the only* pfcrt *codon with corrected-significant structure (0.029 coastal savanna, 0.036 deciduous forest, 0.009 Guinea savanna, 0.000 Transitional, p = 0.0016 — higher in the two southern zones, but not ordered by latitude), and N326S is a singleton at 0.001. The denominator matters and is stated rather than left implicit: on the 24 tested codons I356T survives at 0.0016 < 0.0021, whereas correcting across all 57 reconstructed codons (0.05/57 = 0.00088) it would not, and the previous version of this paragraph asserted both outcomes one sentence apart by widening "none of the six" to "no* pfcrt codon".

**[Interpreted]** The ~0.07 level is reversion to chloroquine-sensitive wild type, roughly two decades after chloroquine was withdrawn as first-line therapy in Ghana; the residual 7 per cent is the surviving mutant haplotype, not evidence of ongoing selection. Two things are firmer than in the previous version. The haplotype's unity is now a co-occurrence observation rather than an inference from similar marginals. And codon 75 is correctly identified as **N75E**, the E of CVIET, rather than as an Asn75Asp change that no isolate in this dataset actually carries — a residue that required reconstructing the codon from two separate SNPs to see at all.

### 6.9.8 No artemisinin resistance: all 14 WHO-validated and all 6 WHO-candidate *pfk13* markers

**[Observed]** The *pfk13* panel was rebuilt from the WHO molecular-marker compendium v1.0 (December 2025) rather than from memory. It now covers **all 14 validated** markers — F446I, N458Y, C469Y, M476I, Y493H, G533S, R539T, I543T, P553L, R561H, P574L, C580Y, R622I, A675V — and **all 6 candidates** — E252Q, P441L, C469F, N537I, G538V, V568G. That is 14 + 6 = **20 variants at 19 distinct codons**, because C469Y (validated) and C469F (candidate) are two different changes at the same codon 469; the two WHO tiers are kept distinct throughout and no candidate is anywhere described as validated. Four further *pfk13* codons (449, 481, 515, 527) fall outside both lists, which is why the reconstruction covers more codons than the lists contain: *23* pfk13 *codons in total, and every one is fixed at the 3D7 reference residue*, with the single exception of codon 458, where one isolate in 1,866 carries Asn458Asp. The validated residue at that codon is Tyr, which is absent.

| Codon | WHO v1.0 classification | Reference residue | n called | Reference frequency | Non-reference residues observed | Zero upper bound |
|---|---|---|---|---|---|---|
| 252 | E252Q candidate | Glu | 1,876 | 1.0000 | none | ≤ 0.0020 |
| 441 | P441L candidate | Pro | 1,860 | 1.0000 | none | ≤ 0.0021 |
| 446 | F446I validated | Phe | 1,863 | 1.0000 | none | ≤ 0.0021 |
| 449 | not in validated or candidate list | Gly | 1,862 | 1.0000 | none | ≤ 0.0021 |
| 458 | N458Y validated | Asn | 1,866 | 0.9995 | Asp 0.0005 (1/1,866) | n/a — 1 variant observed; Wilson upper 0.0030 |
| 469 | C469Y validated / C469F candidate | Cys | 1,863 | 1.0000 | none | ≤ 0.0021 |
| 476 | M476I validated | Met | 1,867 | 1.0000 | none | ≤ 0.0021 |
| 481 | not in validated or candidate list | Ala | 1,856 | 1.0000 | none | ≤ 0.0021 |
| 493 | Y493H validated | Tyr | 1,857 | 1.0000 | none | ≤ 0.0021 |
| 515 | not in validated or candidate list | Arg | 1,860 | 1.0000 | none | ≤ 0.0021 |
| 527 | not in validated or candidate list | Pro | 1,862 | 1.0000 | none | ≤ 0.0021 |

| Codon | WHO v1.0 classification | Reference residue | n called | Reference frequency | Non-reference residues observed | Zero upper bound |
|---|---|---|---|---|---|---|
| 533 | G533S validated | Gly | 1,863 | 1.0000 | none | ≤ 0.0021 |
| 537 | N537I candidate | Asn | 1,868 | 1.0000 | none | ≤ 0.0021 |
| 538 | G538V candidate | Gly | 1,863 | 1.0000 | none | ≤ 0.0021 |
| 539 | R539T validated | Arg | 1,861 | 1.0000 | none | ≤ 0.0021 |
| 543 | I543T validated | Ile | 1,861 | 1.0000 | none | ≤ 0.0021 |
| 553 | P553L validated | Pro | 1,859 | 1.0000 | none | ≤ 0.0021 |
| 561 | R561H validated | Arg | 1,864 | 1.0000 | none | ≤ 0.0021 |
| 568 | V568G candidate | Val | 1,866 | 1.0000 | none | ≤ 0.0021 |
| 574 | P574L validated | Pro | 1,861 | 1.0000 | none | ≤ 0.0021 |
| 580 | C580Y validated | Cys | 1,863 | 1.0000 | none | ≤ 0.0021 |
| 622 | R622I validated | Arg | 1,862 | 1.0000 | none | ≤ 0.0021 |
| 675 | A675V validated | Ala | 1,845 | 1.0000 | none | ≤ 0.0021 |

*Table 6.11. All 23 reconstructed pfk13 codons with their WHO v1.0 (December 2025) classification (results/codon_panel_residues.csv). Validated markers are shown in bold; codon 469 carries one validated change (C469Y) and one candidate change (C469F), which is why 20 variants occupy 19 codons.* ***P441L and C469F are candidates, not validated*** *— the previous version of this chapter listed both among the validated set, which was a misclassification. "Zero upper bound" is the Wilson 95 per cent upper limit on a count of zero at that codon's own denominator. It therefore does not apply at codon 458, the one codon where the count is not zero; that cell gives the Wilson upper limit on the one variant actually observed (0.0030) instead, and should not be read as a ceiling of 0.0021.*

> **Two corrections to how this negative was previously stated. (1) The denominator.** The previous version said all 14 validated codons were at 0.000 **"in all 1,900 isolates"**. They were not: the denominator is **per codon, 1,845–1,876 of 1,900 called**, the shortfall being isolates whose calls at that codon were mixed or missing. Per-zone call rates give zero upper bounds of ≤ 0.0029 in Guinea savanna (n 1,304–1,321), ≤ 0.0118 in deciduous forest (n 322–331), ≤ 0.0197 in coastal savanna (n 191–201) and only ≤ 0.133 in the Transitional zone (n = 25). **(2) The period.** The isolates were collected in **2009–2021**. This is therefore a *historical* negative. It does not speak to the frequency of *pfk13* resistance alleles in Ghana today, and must not be cited as if it did.
>
> **The *kelch13* corroboration must be stated much more weakly than before.** The previous version claimed independent corroboration from MalariaGEN's published kelch13 classification: *"Two methods, one of them not ours, agree."* That overstates it. Pf8 assigns a kelch13 status over the **whole 349–726 propeller domain** and classifies any missing call in that interval as undetermined — the rule is "Any missing call in amino acids 349-726 → Missing → Undetermined". Only **36 isolates** therefore have a complete Pf8 status, against per-codon calls on roughly 1,860 isolates here. Thirty-six complete statuses are weak corroboration for a negative asserted on ~1,860 isolates, and the two should not be presented as equivalent lines of evidence. The weight of this negative rests on the reconstruction in Table 6.11, not on the Pf8 classification.

**[Interpreted]** With those two qualifications in place, Ghana shows **no evidence of artemisinin partial resistance in 2009–2021**, tested across the complete WHO-validated set and the complete candidate set, at the per-codon detection limits above. This is now a genuine tested negative. In the original manuscript the equivalent claim rested on *pfk13* not appearing in a genome scan whose matrix contained no *pfk13* codon at all, and the version before this one tested a list that was neither the WHO list nor complete. Given confirmed R561H in Rwanda and A675V/C469Y in Uganda, a negative of this precision across a West African sample is a useful surveillance baseline — but it is a baseline for a period that ended in 2021, and the right inference is that it should be repeated on contemporary isolates, not that resistance is absent now.

### 6.9.9 *pfaat1* Ser258Leu and its coastal-to-northern cline

[Observed] *pfaat1* S258L is the most strongly zone-structured codon in the panel: **0.824 in coastal savanna, 0.735 in deciduous forest, 0.640 in the Transitional zone and 0.544 in Guinea savanna**, overall **0.609** (n = 1,896), $\chi^2$ p = $1.9 \times 10^{-18}$. The coastal-savanna-to-Guinea-savanna difference is **28 percentage points** (0.824 − 0.544 = **0.280**; unrounded, 0.8235 − 0.5440 = 0.2796). Unusually for this panel the ordering is **monotonic across all four zones in true south-to-north order**, because the Transitional zone — the second-most-northern — sits at 0.640, between deciduous forest and Guinea savanna. The Transitional estimate rests on 25 isolates and should not carry weight on its own, but it does not break the pattern as it does for *pfdhfr*. This is the same variant that ranks second in the genome-wide scan (Section 6.8.2) at XtX 24.198, q = 0.0049, and that SNP is graded clean by the call-quality audit of Section 6.8.4, so it is supported by two statistical framings on two differently filtered callsets. The second *pfaat1* codon in the panel, **313, is fixed at the reference residue Phe in all 1,900 isolates**: F313S is absent from Ghana in this sample.

[Interpreted] **The previous interpretation of this cline was wrong on the published evidence, and is withdrawn.** It asserted that no functional experiment had tested amodiaquine, and built a drug-pressure argument on that gap. Amodiaquine *was* tested in Amambua-Ngwa et al. (2023): *"Parasite IC50 values for CQ, amodiaquine, lumefantrine, mefloquine and quinine were measured..."*; the shifts were *"well below the threshold associated with clinical resistance"* (amodiaquine threshold 60 nM); and the authors explicitly state that the other-quinoline hypothesis *"is not supported"*, concluding that chloroquine was *"the primary selective force"*. The variant's demonstrated effect is a **1.5-fold** rise in chloroquine IC50 in a **single** chloroquine-resistant Southeast Asian line (NHP4026), with **epistasis between S258L and F313S** and an explicit *pfcrt*-background caveat; the 0 to 97 per cent longitudinal rise is **Gambian (1984–2014)** and the paper says nothing about Ghana. The defensible description is a variant that **potentiates chloroquine resistance, modestly and in an epistasis- and background-dependent way** — not "the functionally validated chloroquine-resistance variant". The caveat bites here in particular: F313S, the variant with which S258L shows epistasis, is **absent from all 1,900 Ghanaian isolates**, so the epistatic component of the published effect cannot operate in this sample. The stronger form of that sentence — that the genetic background on which the 1.5-fold effect was measured does not exist here — is withdrawn as an overreach: the source records the epistasis and the *pfcrt*-background caveat as two separate facts and nowhere states that the 1.5-fold measurement was taken on an F313S-carrying background; and the background it does flag, a chloroquine-resistant *pfcrt*, does exist in this sample, at CVIET 0.071 (133 of 1,873).

[Hypothesized] What, then, maintains 0.82 in coastal savanna? Three candidates, and the panel cannot choose between them. **(a) Historical persistence.** Chloroquine was withdrawn in Ghana two decades ago and the *pfcrt* CVIET haplotype has reverted to 0.071 (Section 6.9.7), so standing chloroquine pressure cannot explain a frequency of 0.82; but a potentiating variant with a modest fitness cost can decay slowly and unevenly, and a gradient rather than fixation or loss is the expected intermediate state. *(b) Amodiaquine, as an* untested possible *driver. Ghana's first-line regimen since 2005 has been artesunate–amodiaquine, which maintains continuous 4-aminoquinoline pressure. The published experiment measured amodiaquine and found the shift sub-threshold, and the authors reject the wider-quinoline reading for their Southeast Asian data; so amodiaquine can be offered here only as a possibility that the functional work does not support and did not test in a West African background — not as the explanation, which is how the previous version framed it. (c) The documented fitness cost* reported by Amambua-Ngwa et al., which is the condition under which a post-withdrawal equilibrium is a gradient rather than fixation: the allele is retained where exposure is highest and purged where it is not. If (b) or (c) carries the gradient, it is a drug-exposure or treatment-coverage gradient — higher treatment-seeking and more complete AS-AQ coverage in the south — rather than anything

ecological about agro-ecological zones as such. That is testable with drug-pressure covariates joined at the same spatial resolution, and is not tested here.

[Observed] Two points of context, both narrower than the previous version's. *pfaat1* appears nowhere in the WHO compendium at any tier**, so it is** not **"the most common validated resistance variant", which is what the previous version called it and which was a category error (Section 6.9.11). The replacement superlative has to be narrower still, and its exclusion rule stated: at 0.609, S258L is the most common variant in this panel with a published allele-replacement experiment behind it, outside the antifolates. It is not the most common resistance-associated non-antifolate variant, because** *pfmdr1* Y184F is higher at 0.677 *(n = 1,858) and* pfmdr2 *I492V is close behind at 0.549; Y184F carries no WHO label either, so the tiers cannot be doing the excluding — what distinguishes S258L is comparable functional-replacement data, which Y184F does not have. And* pfaat1 *is absent from the eight-locus config.KNOWN_RESISTANCE_LOCI panel inherited by this pipeline, which is why it reached second place in the outlier table* of the original manuscript — behind PTP7 at XtX 26.44, and labelled only as "amino acid transporter AAT1" — while the abstract of the same document announced that no validated resistance locus had been found. It did not head that table, and the earlier phrasing that said it did is corrected here.

### 6.9.10 Per-sample validation against MalariaGEN's published calls

[Observed] The panel was validated against Pf8_drug_resistance_marker_genotypes.tsv, MalariaGEN's independently produced genotype calls for the same 1,900 isolates, **call by call** rather than frequency by frequency. On isolates where both pipelines called, **47,012 of 47,012 calls agree — 100.0000 per cent across 25 markers in five genes**, including all nine *pfcrt* codons and codon 437 of *pfdhps*. Codon 75 now agrees exactly, because reconstruction yields Glu where string matching yielded Asp.

| Gene | Codon | MalariaGEN column | Calls compared | Agreeing | Concordance |
|---|---|---|---|---|---|
| *pfcrt* | 72 | crt_72[C] | 1,900 | 1,900 | 1.0000 |
| *pfcrt* | 74 | crt_74[M] | 1,873 | 1,873 | 1.0000 |
| *pfcrt* | 75 | crt_75[N] | 1,874 | 1,874 | 1.0000 |
| *pfcrt* | 76 | crt_76[K] | 1,874 | 1,874 | 1.0000 |
| *pfcrt* | 220 | crt_220[A] | 1,869 | 1,869 | 1.0000 |
| *pfcrt* | 271 | crt_271[Q] | 1,873 | 1,873 | 1.0000 |
| *pfcrt* | 326 | crt_326[N] | 1,874 | 1,874 | 1.0000 |
| *pfcrt* | 356 | crt_356[I] | 1,900 | 1,900 | 1.0000 |
| *pfcrt* | 371 | crt_371[R] | 1,873 | 1,873 | 1.0000 |
| *pfmdr1* | 86 | mdr1_86[N] | 1,887 | 1,887 | 1.0000 |
| *pfmdr1* | 184 | mdr1_184[Y] | 1,858 | 1,858 | 1.0000 |
| *pfmdr1* | 1034 | mdr1_1034[S] | 1,834 | 1,834 | 1.0000 |
| *pfmdr1* | 1042 | mdr1_1042[N] | 1,851 | 1,851 | 1.0000 |
| *pfmdr1* | 1246 | mdr1_1246[D] | 1,866 | 1,866 | 1.0000 |
| *pfdhfr* | 16 | dhfr_16[N] | 1,896 | 1,896 | 1.0000 |
| *pfdhfr* | 51 | dhfr_51[N] | 1,891 | 1,891 | 1.0000 |
| *pfdhfr* | 59 | dhfr_59[C] | 1,888 | 1,888 | 1.0000 |
| *pfdhfr* | 108 | dhfr_108[S] | 1,889 | 1,889 | 1.0000 |
| *pfdhfr* | 164 | dhfr_164[I] | 1,898 | 1,898 | 1.0000 |

| Gene | Codon | MalariaGEN column | Calls compared | Agreeing | Concordance |
|---|---|---|---|---|---|
| *pfdhps* | 436 | dhps_436[S] | 1,884 | 1,884 | 1.0000 |
| *pfdhps* | 437 | dhps_437[G] | 1,892 | 1,892 | 1.0000 |
| *pfdhps* | 540 | dhps_540[K] | 1,895 | 1,895 | 1.0000 |
| *pfdhps* | 581 | dhps_581[A] | 1,890 | 1,890 | 1.0000 |
| *pfdhps* | 613 | dhps_613[A] | 1,884 | 1,884 | 1.0000 |
| *pfmdr2* | 484 | mdr2_484[T] | 1,899 | 1,899 | 1.0000 |
| Total | | 25 markers | 47,012 | 47,012 | 1.0000 |

*Table 6.12. Per-sample concordance with MalariaGEN's published genotypes (results/codon_panel_concordance.csv). "Calls compared" counts isolates called by both pipelines at that marker. The MalariaGEN column name carries the reference residue in brackets, which is how dhps_437[G] encodes the fact described in Section 6.9.4. No marker recorded a single disagreement.*

> **What this replaces, and why the replacement matters.** The previous version's validation was: *"Twenty-one of 23 comparable markers agree within 0.03."* Agreement of **marginal frequencies** does not establish that two pipelines assign the same residue to the same isolate — two pipelines can reach the same marginal from different sets of samples — and the test's tolerance was wide enough to pass real errors. Two concrete cases from this chapter make the point. *pfcrt* **N326S** was reported at 0.008 by the old panel and is **0.00053** here (one isolate of 1,874; 0.001 to three decimal places); the discrepancy is about **fifteenfold** in relative terms — not eightfold, which is what the rounded 0.001 suggests and which understates the point — it is invisible at a ±0.03 tolerance, and the per-sample test now shows 1,874 of 1,874 calls agreeing with MalariaGEN at that codon. And *pfcrt* **75** accounted for **both** of the markers the old test flagged as outside tolerance — both flagged markers were the same codon — and the discrepancy was then explained away as a "naming disagreement, not a data disagreement" in which MalariaGEN was "correct on the biology". That explanation was wrong: it was a reconstruction failure (Section 6.9.1), and the ±0.03 framing turned a code defect into a terminology footnote. **The 0.03 claim, and the naming-discrepancy note that accompanied it, are both withdrawn.** Per-sample concordance is the only version of this check that can detect the failure modes this chapter has actually suffered.

### 6.9.11 What the panel may and may not claim about WHO classification

**[Observed]** The WHO compendium classifies **variants, not genes**, and the previous version repeatedly attached gene-level WHO labels. Three specific corrections follow. **(1)** There is **no validated or candidate lumefantrine marker at all** — verbatim: *"The review did not identify any validated or candidate molecular markers associated with lumefantrine resistance."* So a label of the form "*pfmdr1*, WHO-validated, lumefantrine" is a category error twice over: it ascribes a tier to a gene, and it ascribes it for a drug that has no classified markers. **(2)** *pfaat1*, *pfap2mu* and *pfmdr2* appear nowhere in the compendium at any tier. *They stay in the panel* — pfaat1 *for the functional work of Section 6.9.9*, pfap2mu *and* pfmdr2 *as loci of standing interest — but none of them may be described as a WHO-classified marker anywhere in this thesis. (3)* pfcoronin *is in the compendium at the potential tier only, which is below candidate; all three of its reconstructed codons (G50E, R100K, E107V) are fixed at the reference residue here. Within* pfk13, *the tiers are per variant and are given in Table 6.11; P441L and C469F are candidates, not validated*, which the previous version got wrong.

### 6.9.12 Limitations of the panel

**[Observed]** The zone contrasts in Table 6.8 inherit every design limitation of the $F_{ST}$ analysis, including the one the original manuscript did not list. **Zone is confounded with sequencing study.** The confounding analysis is computed on the **4,905 QC-passing isolates**, not on the 1,900-isolate analysis set, because study membership is a property of the full QC-passing cohort: those 4,905 come from 11 sequencing studies, of which Guinea savanna draws on six (n = 3,605), deciduous forest on five (**n = 815**, of which four, n = 811, are large enough

to enter the within-zone test — the fifth is a 4-isolate study), coastal savanna on two (n = 435) and the Transitional zone on a single study (1083-PF-GH-CONWAY, n = 50); 435 + 815 + 3,605 + 50 = 4,905. **Four of the six zone pairs share no partner study at all**, among them coastal savanna against Guinea savanna, which is the pair carrying both the maximum $F_{ST}$ and every marker contrast reported above. Study-associated heterogeneity within these data is demonstrable rather than hypothetical — holding zone constant, call rate differs between studies within Guinea savanna at p = 2.8e-197 (Kruskal–Wallis, 6 studies, n = 3,605) and within deciduous forest at p = 7.7e-88 (4 studies, n = 811); because collection year, finer geography and recruitment also differ between studies inside a zone, this is heterogeneity associated with study rather than a demonstrated technical cause. A zone difference in marker frequency and a study difference in marker frequency are not separable here for the pairs that carry the largest effects.

[Observed] Three further design constraints. The Transitional zone contributes 25 isolates from one sequencing study, so its column in Table 6.8 is indicative only and its 1.000 and 0.000 entries should not be read as fixation or absence; the $\chi^2$ tests retain it because excluding it would discard a zone, but no finding above depends on it. Monoclonal-fraction differences between zones mean the majority-allele ploidy rule resolves mixed infections at different rates in different zones, which could bias frequency estimates in the same direction as a real cline. And the agro-ecological zone layer itself has no citable published source (Methods), so "zone" is a reproducible label on this specific GeoJSON rather than an authoritative classification.

[Observed] Three constraints specific to the reconstruction. **Mixed calls become missing data, not residues.** A codon at which an isolate's calls are mixed is recorded uncalled, which is why per-codon denominators run 1,834–1,900 rather than 1,900, and which systematically removes polyclonal infections — the isolates most likely to carry a minority resistance allele. Every frequency here is therefore a frequency among isolates resolvable to a single residue at that codon, and the direction of any resulting bias is unknown. **The residue-level $\chi^2$ tests within a codon are not independent.** For a two-residue codon the two tests are the same test, and at *pfdhps* 436 the four tests are nested; the Bonferroni correction in Section 6.9.3 is applied across the 24 codons using each codon's smallest residue-level p, which is the least unreasonable choice available but is not an exact multiplicity control. **Haplotypes are within-isolate residue combinations on complete cases, not phased haplotypes.** For a monoclonal infection the two coincide; for the polyclonal fraction that survives the ploidy rule they need not, and no phasing was attempted.

[Interpreted] Taken together, these mean the *gradients* in Table 6.8 are better supported than their *causal attribution to agro-ecology*. The frequencies themselves are now as solid as this dataset allows — every one of 47,012 per-sample calls agrees with an independently produced genotype set, across the 25 panel codons MalariaGEN publishes per sample (Section 6.9.10), which is a far stronger statement than the marginal agreement previously cited. What the gradient is a gradient *in* — drug exposure, sampling study, treatment-seeking, or agro-ecological zone — this design cannot resolve, and Section 7.x does not claim it can.

### 6.9.13 What the corrected run did not change, and what it did

[Observed] The genome-wide conclusions survived a 16.6 per cent larger SNP panel (3,359 to 3,916), a corrected FDR treatment, a reproducible single-threaded BayPass, a seeded Mantel test and a TreeMix sweep with standard errors. Maximum pairwise $F_{ST}$ moved from 0.0024 to 0.0023, the ADMIXTURE cross-validation curve remained near-flat with a negligible K = 2 minimum (0.83925 against 0.84384 at K = 1, both on the same 3,385-SNP pruned panel, a 0.54 per cent improvement), and near-panmixia across Ghana still holds. Nine of the 13 outlier genes retain FDR-surviving support — eight if the *SNRPF* downgrade of Section 6.8.4 is

right — and the leading outlier is the same gene as before: PTP7, at XtX 26.48 on the manuscript run and 26.44 on the final one. The marker panel's qualitative findings also survived its own rebuild: antifolate markers near-fixed and zone-structured, chloroquine markers reverted to under a tenth, *pfk13* clean, *pfaat1* S258L strongly clined.

**[Interpreted]** The robustness is real, and so is the pattern of error, which should be stated rather than smoothed over. This is the fourth consecutive round in which a correction contained its own defect. Each layer was checked for internal consistency against the project's own outputs, and that checking worked. What escaped it was **external correctness**: whether the marker list matched an authoritative source (*pfk13*), whether the annotation direction matched the reference genome (*pfdhps* 437), whether a co-occurrence claim had co-occurrence data behind it (the triple mutant), whether a per-sample claim had per-sample data behind it (the MalariaGEN check), and whether a cited paper said what it was cited as saying (*pfaat1*). The verification architecture was built to catch stale numbers, and it did. It was not built to ask *"is this the right list, in the right direction, against the right source?"* — and **all five of the defects in this addendum are instances of that one question going unasked.** (Four of the five were defects in the analysis code and the fifth, the *pfaat1* citation, was a defect in the prose; that is a different cut of the same five, and the earlier "four of five" figure belonged to it, not to this one.) The call-quality audit of Section 6.8.4 is the first check in this project that asked an **external** question of the data — not "do the numbers agree with each other?" but "are these calls what they appear to be?" — and it returned one reassurance and one unresolved problem at the fifth-ranked outlier. The conclusions in Section 7.x are stated with that in mind.

# 7. Interpretation and discussion

## 7.1 What the analysis establishes

**[Interpreted]** The evidence for negligible differentiation is strong, but it is neither five independent lines nor a test of random mating. $F_{ST}$, ADMIXTURE and TreeMix are three summaries of one genotype matrix: all three are computed from the same per-zone allele counts, and at this level of differentiation a four-population covariance matrix and the pairwise $F_{ST}$ matrix are near-deterministic functions of each other. Only the ordinations use a different input — MalariaGEN's published Pf8 genotype-distance matrix — and they are now computed on the same 1,900 isolates as everything else, because stage 3 has been re-run on the analysis set (silhouette by zone −0.020; Table 8.2 row 21, Figure 8.1). The agreement between rows 1, 3 and 4 of Table 7.1 is therefore substantially arithmetic, not corroborative. What this revision adds is robustness rather than independence: the $F_{ST}$ summary has been recomputed under six combinations of the two thresholds that define the dataset, and the maximum pairwise value moves only between 0.00204 and 0.00255 (row 5 of Table 7.1; Table 8.3). Those six scenarios are not independent lines either — they are the same pipeline on overlapping data — but they establish that the headline does not depend on the analyst's threshold choices, which is a different and more useful thing. **The region-level Mantel test does not share that robustness, and it has to be given up.** It is a fourth summary of the same matrix, computed from region-level $F_{ST}$, so it was never independent of the others; and it is significant in only 2 of 6 threshold scenarios and 4 of 7 leave-one-region-out replicates, so it cannot be cited as a converging line at all (Table 7.2, row 1). It was the one result here carrying a positive structure claim, and losing it is a real loss rather than a tidying-up. Described accurately, what remains is: **three summaries of one 3,916-SNP genotype matrix (ADMIXTURE on 3,385 LD-pruned sites of it), plus three ordinations of an independently published distance matrix on the same 1,900 isolates, all pointing the same way, and insensitive to both thresholds that define the dataset.** The negative half of the result is better supported than it was. The positive half is gone.

| Analysis | Result (3,916 SNPs, 1,900 isolates) | Reading |
|---|---|---|
| Pairwise $F_{ST}$ (Hudson) | Maximum 0.00232 (Coastal savanna vs Guinea savanna; SE 0.00053, 95% CI 0.00128–0.00336). One of six confidence intervals spans zero (Guinea savanna vs Transitional zone, [−0.00083, 0.00178]). | Zones share essentially all their common genetic variation |
| Ordination (3 methods) | Silhouette by zone −0.020 on the raw distance matrix, now computed on the 1,900 analysis set (−0.016 on the 4,905 Pf8-QC-passing isolates the published figures describe); trustworthiness 0.499 (UMAP), 0.526 (t-SNE), 0.510 (PCoA). The PCoA PC1+PC2 = 1.3% of variance is from the published 4,905-isolate run and was not recomputed. | Zone labels do not describe the variation — and removing the polyclonal and clonally redundant isolates makes the separation slightly worse, not better |
| ADMIXTURE | K = 2 minimises cross-validation error (0.83925) but improves on K = 1 (0.84384) by only 0.54%; the curve is near-flat from K = 1 to K = 8 (full range 0.0056, 0.67%). Run on 3,385 LD-pruned sites of the 3,916. | Admitting structure barely improves fit; any structure present is of negligible magnitude |
| TreeMix | A bifurcating tree alone explains 89.7% of the covariance; two significant migration edges raise this to 99.7% | Exchange between groups, not a branching history |
| Threshold sensitivity (six scenarios) | Maximum pairwise $F_{ST}$ 0.00204–0.00255 across six combinations of the $F_{ws}$ monoclonality cut and the MAF floor, over 1,900–4,403 isolates and 3,275–10,876 SNPs (Table 8.3). The maximum is the Coastal savanna–Guinea savanna pair in all six. | The headline does not depend on either threshold that defines the dataset. This is the strongest single piece of evidence for it in the whole analysis |

*Table 7.1. Convergent evidence for low differentiation between zones. Rows 1, 3 and 4 are three summaries of the same per-population allele frequencies and are not statistically independent of one another; only row 2 uses a different input (MalariaGEN's published genotype-distance matrix), and it is now computed on the same 1,900 isolates rather than on a larger sample. Row 5 is not an independent line but a sensitivity analysis on row 1. The region-level Mantel test, a fourth summary of the same matrix, was discussed alongside these rows in the previous draft; it is significant in only 2 of 6 threshold scenarios and is **no longer counted as converging evidence** (Table 7.2, row 1). See the text.*

**[Interpreted]** Two further results point the other way — toward weak but real structure — and a third, the region-level Mantel test, pointed that way at the published thresholds and does not survive varying them. The previous draft either listed these inside the convergence table, where they do not belong, or reported them as negatives when they are not. The Mantel row is now a different kind of problem: it is retained in Table 7.2 with its sensitivity attached, because the result should be reported honestly rather than deleted, but it is graded **not robust** and nothing in this document may lean on it.

| Analysis | Result | Reading |
|---|---|---|
| Mantel, region level | r = 0.427, p = 0.025 with three negative $F_{ST}$ pairs floored to zero; r = 0.414, p = 0.041 uncensored. Significant in 4 of 7 leave-one-region-out replicates, and in 2 of 6 threshold scenarios (Table 8.3): it fails at $F_{ws} \geq 0.90$ (p = 0.087), $F_{ws} \geq 0.80$ (p = 0.169), no $F_{ws}$ filter (p = 0.179) and MAF ≥ 0.01 (p = 0.062), holding only at the published thresholds and marginally at MAF ≥ 0.02 (p = 0.047). On the uncensored statistic only the baseline scenario clears p < 0.05 at all. | Not robust. It holds at the published thresholds and almost nowhere else, so it cannot be graded as supported, moderately supported or a weak but real distance effect — all three gradings used in earlier drafts are withdrawn. Where it does appear it is substantially a single far-northern contrast rather than a smooth distance gradient: significance is lost on dropping Upper East (n = 1,334). It is additionally fragile to two small southern regions, since dropping Greater Accra (n = 204) or Eastern (n = 34) also removes significance, so the result is sensitive to the composition of a 21-pair test as well as to the far-northern contrast. This is the one conclusion in the document that a robustness check overturned (§8). |
| Mantel, zone level | r = 0.471, p = 0.122; 0 of 4 leave-one-out replicates significant | Six pairs from four populations have no resolution, and only 24 distinct permutations exist at zone level; not evidence either way |
| ADMIXTURE, K = 2 minor component | Mean minor ancestry 0.042 (Transitional), 0.053 (Guinea savanna), 0.085 (Coastal savanna), 0.092 (deciduous forest); Kruskal–Wallis H = 105.4, p = $1.05 \times 10^{-22}$ | A real contrast — coastal and forest high, Guinea savanna and transitional low — concordant with the $F_{ST}$ ranking but not a monotone latitudinal cline, and of negligible magnitude (§7.2) |
| Resistance-marker panel (rebuilt, §7.5) | *pfaat1* S258L 0.824 coastal savanna to 0.544 Guinea savanna ($\chi^2$ p = $1.89 \times 10^{-18}$, n = 1,896 called); *pfdhps* G437 0.897 overall, 0.964 deciduous forest to 0.800 transitional ($\chi^2$ p = $1.2 \times 10^{-6}$, n = 1,892); *pfdhfr* IRN triple-mutant haplotype 0.771 overall, 0.750 Guinea savanna to 0.880 transitional (n = 25) on complete cases | Marker frequencies do differ by zone, even though the neutral background does not. The tests are four-zone heterogeneity tests on per-codon residue frequencies; no directional south-versus-north contrast was run |

*Table 7.2. Candidate evidence of weak structure. These results are not convergent with Table 7.1 and should not be counted as though they were; row 1 is itself computed from the same genotype matrix as Table 7.1, and after the threshold-sensitivity analysis it is reported as not robust rather than as weak-but-real. Rows 3 and 4 are the only two entries here that still carry a positive structure claim. Row 4 is computed from the rebuilt codon-reconstruction panel (§7.5) and replaces the superseded annotation-string panel entirely.*

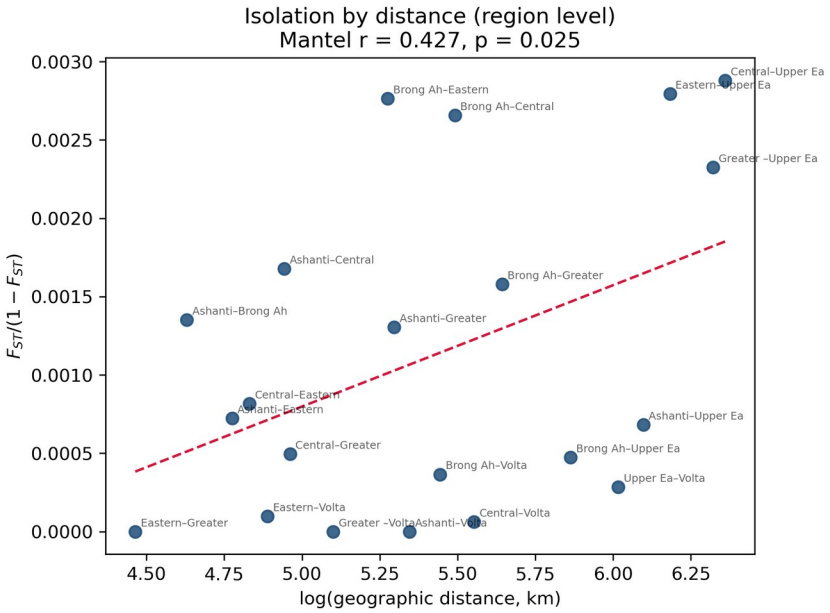



*Figure 7.1. Isolation by distance at region level on the corrected 3,916-SNP matrix: pairwise $F_{ST}$ against great-circle distance for 21 pairs from 7 administrative regions. Censored fit r = 0.427, p = 0.025 (9,999 seeded permutations); uncensored r = 0.414, p = 0.041. Three pairs with negative $F_{ST}$ were floored to zero and are named in the output JSON.* ***This figure shows the relationship at the published thresholds only.*** *Under four of the six threshold scenarios in Table 8.3 the same test is not significant, so the figure must not be read as a robust distance effect.*

**[Interpreted]** Ghanaian *P. falciparum* behaves as a single, largely panmictic population for common neutral variation, with weak departures detectable in minor-component ancestry and in resistance-marker frequencies. A region-level distance effect is present at the published thresholds but disappears under almost any variation of them, so it is not among the departures this analysis can claim. Those departures that remain are not a clean latitudinal ordering — the transitional zone breaks the monotone pattern in most variables examined (§7.2), *pfaat1* S258L excepted — but they are not nothing either. This is a substantive answer to the thesis's first objective, and a negative one for the strong form of its hypothesis: the ecological zones are not acting as demographic barriers at a scale these data can detect. It is not, however, a negative answer for the thesis's third objective; see §7.4 and §7.5.

**[Calculated]** The earlier draft recorded the ADMIXTURE result as "no ancestry structure aligned to geography" and graded it a well-supported negative. That statement is false as written, and it was never tested. The minor component differs between zones at Kruskal–Wallis H = 105.4, p = $1.05 \times 10^{-22}$, and the contrast — coastal 0.085 and forest 0.092 high, Guinea savanna 0.053 and transitional 0.042 low — is concordant with the $F_{ST}$ ranking, though it is not monotone with latitude. The substantive point survives intact and is simply a different point: the **magnitude** is negligible. Mean minor ancestry spans roughly 0.04 to 0.09 on a component whose admission to the model improves cross-validation error by 0.54 per cent. A difference can be highly significant at n = 1,900 and still be too small to matter for any operational purpose. That is what is being claimed here.

> **Caveat.** A proportion-of-variance figure for the zone effect on minor-component ancestry (of order 2 per cent) was computed during the audit, but on the superseded 3,359-SNP ADMIXTURE output and with no named statistic attached. It has not been recomputed on the final 3,916-SNP run and is therefore **not** quoted here as a result. [Unknown] — the only effect size the final run supports is whatever is derived from H = 105.4 on 1,900 isolates, and no such statistic was computed and recorded. The Kruskal–Wallis statistic and the per-zone means in Table 7.2 are from the final run.

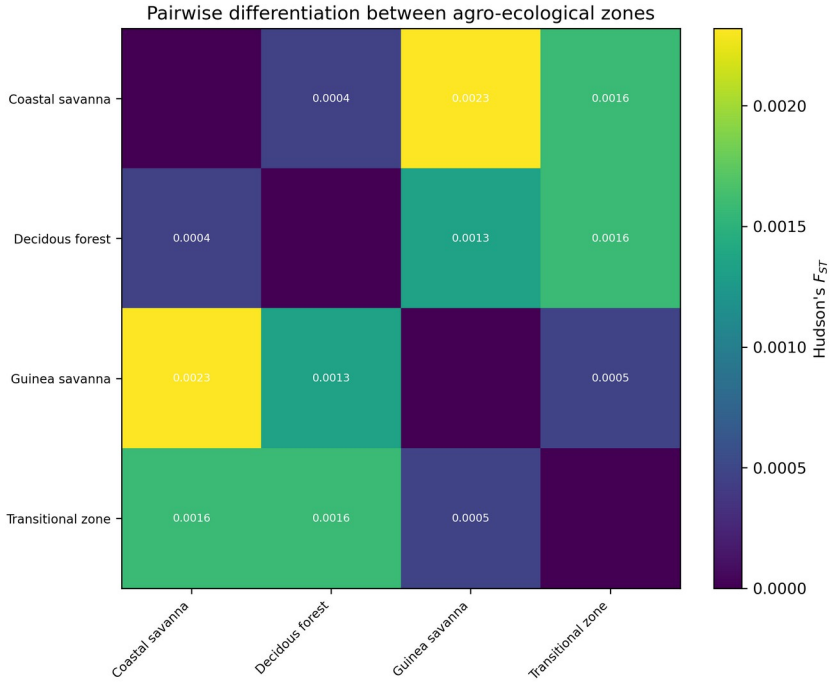



*Figure 7.2. Pairwise Hudson $F_{ST}$ between the four ecological zones on the corrected 3,916-SNP matrix. Every value is at or below 0.00232; the Guinea savanna–transitional pair (0.00047) is the one whose confidence interval spans zero.*

**[Interpreted]** That the headline is a negative is not surprising, and that is a point in its favour. Ghana is approximately 670 km north to south, human mobility along the north–south corridor is substantial, *P. falciparum* undergoes obligate sexual recombination in the mosquito, and transmission across most of the country is high. All four favour genetic homogenisation. The finding is consistent with the continental picture, in which West African parasite populations form a single major subpopulation (Amambua-Ngwa et al., 2019), and with the observation that differentiation within West African countries is weak relative to the differentiation between endemicity strata that Mobegi et al. (2014) reported between The Gambia and Guinea.

## 7.2 What the analysis cannot establish, and why

**[Observed]** **Zone effects cannot be separated from region effects.** Three of four zones are each filled by exactly one administrative region (Table 2.2). For those three, "zone" and "region" are the same variable. Any difference attributed to ecology could equally be attributed to the region — its health system, its drug supply, its sampling protocol, its collection year. No statistical adjustment can separate variables that are collinear by construction.

> **Denominator for the batch numbers in this section.** The study-composition and within-zone batch tests quoted below are computed on the **4,905 Pf8-QC-passing isolates** — every one of them carries Pf8's own QC-pass flag, they are the set on which study membership and $F_{ws}$ are defined, and they sit before the $F_{ws}$ and clonality cuts that yield the 1,900 entering the population analyses (Guinea savanna 1,334, deciduous forest 337, coastal savanna 204, transitional zone 25). This single label, *4,905 Pf8-QC-passing (pre-$F_{ws}$), is used throughout the document; "pre-QC" and "QC-passing" are not used as alternatives for it, because they invite a reader to think two different denominators are in play. So "1083-PF-GH-CONWAY is 100 per cent of the transitional zone, 50 of 50" and "transitional zone, n = 25" are both correct and describe different sets. Study nesting within zone is a property of the Pf8 sampling frame and is not altered by the later cuts, but the counts here are not the analysis-set counts, and the zone-pair overlap was not recomputed on the 1,900 set. The same denominator applies to the polyclonality figures*

> *in §7.3 and to limitation 2 in §9.1. Note separately that the marker panel in §7.5 carries a third* set of denominators: it is computed on the 1,900 analysed isolates but reports a per-codon called count, 1,834–1,900 depending on the codon, because a mixed call leaves a codon uncalled rather than being resolved to a residue.

[Observed] **Zone effects cannot be separated from sequencing-study effects either.** This confounder was absent from the previous draft and is at least as serious as the collinearity above. Pf8 isolates arrive through named partner studies, and in Ghana those studies are almost perfectly nested within zones: **four of six zone pairs share no sequencing study at all**, including Coastal savanna vs Guinea savanna, the pair that produces the maximum $F_{ST}$ and the polyclonality contrast. Three of the four zones are dominated by a single study — the transitional zone is one study outright (50 of 50 isolates, 1083-PF-GH-CONWAY), coastal savanna 89 per cent one study (387 of 435), Guinea savanna 74 per cent one study (2,678 of 3,605) — and deciduous forest is the only zone with no majority study, its largest contributing 391 of 815 (48 per cent). Study-associated heterogeneity in these data is demonstrable rather than hypothetical: holding zone constant, so that study varies together with collection year, finer geography and recruitment, call rate differs between the six Guinea savanna studies at $p = 2.8 \times 10^{-197}$, and between four of deciduous forest's five contributing studies at $p = 7.7 \times 10^{-88}$ (the fifth, 1114-PF-PDN-DBS-GH-GHANSAH, contributes 4 isolates and was too small to enter the test). The Coastal–Guinea contrast has no within-study replicate and cannot be separated from a batch contrast by any reanalysis of these data. This applies with full force to the marker clines in §7.5: the *pfaat1* S258L contrast runs between the two zones that share no study. See §9.1, limitation 2.

[Observed] **The transitional zone breaks nearly every ordering it appears in, and this is stated once here rather than re-argued later.** It is the **second-most-northern** of the four zones, it contributes 25 of the 1,900 analysed isolates, and it is supplied entirely by one sequencing study. On the final run it is non-monotone with latitude in every variable listed here: minor-component ancestry (0.042, the lowest of the four, below both zones to its south *and* the zone to its north); *pfdhfr* S108N and C59R (both 1.000, the highest of the four) and N51I (0.880, above the deciduous forest's 0.802); the *pfdhfr* IRN triple-mutant haplotype (0.880, the highest of the four, against 0.750 in Guinea savanna); *pfdhps* G437 (0.800, the **lowest** of the four, below the Guinea savanna to its north at 0.875); and polyclonal fraction (0.500, below the deciduous forest's 0.539). The one clear exception is *pfaat1* S258L, where the transitional zone's 0.640 sits in its correct place between the deciduous forest's 0.735 and the Guinea savanna's 0.544 (§7.5) — so the ordering is broken in most variables examined, not in all of them. Any statement of the form "higher in the south" is therefore a two-group contrast — coast and forest on one side, Guinea savanna on the other — and not a latitudinal cline. That is how it is worded throughout this discussion, and the rest of the document should cross-reference this paragraph rather than re-assert a gradient.

[Observed] **Spatial effects cannot be separated from temporal ones.** Median collection year differs by six years across zones, spanning 2012 to 2018 (Table 2.3), and for the transitional zone and coastal savanna the year ranges do not overlap at all: the transitional zone contributes no isolate after 2013 and the coastal savanna none before 2016, leaving a two-year gap between them. Over 2009–2021 Ghana changed first-line therapy and scaled up chemoprevention, so drug-related allele frequencies would be expected to shift with time regardless of geography. Every marker frequency in §7.5 is therefore a 2009–2021 average, not a current value.

[Interpreted] **Barrier hypotheses cannot be tested as such.** Testing whether a specific feature (the forest–savanna transition, the Volta Lake system) impedes parasite dispersal requires differentiation to be measurable across the feature and compared against differentiation at the same distance elsewhere. With four populations, six pairwise comparisons and $F_{ST}$ values below 0.0024, there is no resolution to work with.

[Calculated] **The outlier list is a ranking, and only part of it is a finding.** Nineteen SNPs exceed the calibrated POD 0.999 quantile of 10.840; 3.9 false positives are expected at that threshold by construction, giving an expected-null-exceedance ratio of 20.6 per cent. Thirteen of the nineteen also survive Benjamini–Hochberg correction at q ≤ 0.10, and eight at q ≤ 0.01. **Report the 13 as the finding and the 19 as a ranking.** Seventeen of the nineteen fall in annotated genes — 13 genes in all, four of them carrying two SNPs each — and **two are intergenic**; one of those intergenic sites ranks fifth in the entire scan on XtX (20.79, q = 0.005), so the gene list is not simply the top of the ranking and should not be described as "19 outliers in 13 genes" — although that same site is now flagged lower confidence on variant-quality grounds (see the note below). Nothing about an elevated XtX value establishes that a locus is under selection rather than drifting unusually or sitting near an assembly or alignment artefact; a per-SNP variant-quality check has now closed the artefact reading for the set as a whole (next paragraph), but no quality check can convert a ranking into evidence of selection. The one partial exception is *pfaat1* S258L: its XtX of 24.20 is the second-highest in the scan, behind PTP7 (Pf3D7_03_v3:96,476, XtX 26.44); it sits inside the FDR-10% set at q = 0.0049, which is now readable from the restored per-SNP table rather than only from the corrections record (see the note in §7.5); and it is the only outlier carrying a residue for which published gene-editing work exists. That functional support is real but modest and conditional, and the previous draft overstated it — see §7.5, where the experiment is described as its authors described it.

[Calculated] **The outlier set is not a calling artefact, and the specific worry about exported proteins can be closed.** MQ, VQSLOD, FS, SOR, QD, ReadPosRankSum and the culprit field were pulled from the Pf8 Zarr for all 19 outliers and compared against the other 3,897 scanned SNPs (results/outlier_quality_check.csv). Every significant difference runs **in the outliers' favour**: higher VQSLOD (p = 0.008), QD (p = 0.016) and QUAL (p = 0.007). No outlier has MQ < 50, strand bias or read-position bias, and all 19 are RegionType Core. The culprit field reads MQRankSum for all 19, but also for 99.6 per cent of the rest of the panel, so it carries no information here. The exported-protein concern raised against the gene list is **not supported**: the eight SNPs in PTP7, PIESP2, GEXP22, REX2, PF3D7_0831400 and PF3D7_0221200 are the cleanest subset in the panel — FS = 0.00 for all eight, MQ median 60.00, and VQSLOD significantly higher than the rest. Those gene families are repetitive, but these particular sites sit in uniquely mappable sequence. What this check licenses is narrow and should be stated narrowly: the ranking is not an artefact of variant quality. It remains a ranking.

> **Two outliers downgraded to lower confidence, for a different reason.** Pf3D7_11_v3:1 054 078 (*SNRPF*, PF3D7_1126900, XtX 13.98, q = 0.031) and Pf3D7_11_v3:1 054 447 (intergenic, XtX 20.79, q = 0.005) lie 369 bp apart and are the two lowest-InbreedingCoeff sites in the panel (0.147 and 0.193, both below its first percentile). 1054078 carries **ExcessHet = 69.59, the only nonzero value among all 3,916 SNPs** and roughly 400 times the genome-wide maximum. In a near-haploid organism that points to collapsed paralogy or copy-number variation rather than to mismapping, since MQ and FS are clean at both sites. This matters because 1054447 is the **fifth-ranked SNP in the whole scan**. Both are therefore reported as lower confidence. The mechanism is [Hypothesized] and unverified: a read-level check — per-sample depth ratio against each isolate's genome-median depth at these two positions — would settle it, and it is item 1 in §8.2.

[Interpreted] It cannot establish that the retained panel is neutral, because no neutral mask was ever defined. The 3,916 SNPs are the sites that survived the FILTER_PASS, biallelic-SNP, missingness and minor-allele-frequency criteria, and not one of those is a neutrality criterion: no coding or non-coding partition, no synonymous/non-synonymous distinction, no gene-proximity exclusion and no recombination-rate weighting was applied at any stage. The panel therefore contains the resistance loci themselves — the XtX scan is run on this same panel and returns *pfaat1* and PTP7 from it — so the set used to measure background differentiation also contains the sites under the strongest selection in the genome. That inclusion is very

unlikely to be driving the differentiation result: 19 outliers among 3,916 sites cannot move a ratio-of-averages estimator away from 0.002, and the maximum $F_{ST}$ is stable at 0.00204–0.00255 across six scenarios that change the panel composition substantially. The point is not that the number is wrong but that the word is unearned. Wherever “neutral” appears in this report it should be read as “genome-wide at the retained common variants”, never as “at sites shown to be free of selection”. Building a genuine neutral mask — excluding the known resistance loci, the antigenic and invasion gene families, and sites in high linkage with them, then recomputing differentiation on what remains — is a specified next step and is not done here.

## 7.3 Unexpected observations

[Observed] **The filter artefact, and the second defect underneath it.** The single most consequential finding of the audit is not a biological one. The ingest mask `FILTER_PASS & is_snp & numalt == 1` silently removed most canonical antimalarial-resistance codons — among them *pfcrt* K76T, *pfmdr1* N86Y, *pfdhps* K540E and most of the *pfk13* domain — because in Pf8 these codons are overwhelmingly multiallelic or carry a spanning-deletion ALT. The previous draft's statement that no validated resistance locus appeared among the outliers was therefore not a biological result but a filtering artefact, and interpreting it biologically was a mistake.

> **The per-codon counts for this artefact are themselves superseded.** The audit quantified the filter artefact as "28 of 44 codon-allele records removed; only 6 of the 21 polymorphic records ever visible to the original scan". Those figures are properties of the **superseded 44-codon annotation-string panel**, which has since been shown to be the wrong panel (§7.5). They are retained here only as a description of what the filter did to that panel, and should not be quoted as properties of the current one, which contains **57 codons across nine genes, 24 of them polymorphic**, and which does not use `numalt`, `is_snp` or annotation strings at all.

[Interpreted] **What the filter artefact turned out to be a symptom of.** Removing the mask recovered the codons but did not fix the panel, because the panel's second defect was independent of the filter: markers were identified by matching SnpEff `ANN_HGVS_p` protein-change strings against a hand-written list. That fails for any resistance allele which *is* the reference residue, and for any residue reachable only by two changes within one codon. Both failure modes occurred, and one of them hid an allele at 0.897 (§7.5). The general lesson is now a double one: a filter written for one purpose (biallelic sites for a differentiation scan) silently redefined the question being asked of the data; and the repair, which addressed the filter alone, left the identification layer broken for a further round. Neither defect could be revealed by any downstream result, because both produced well-formed output. See §9.1, limitation 13.

[Observed] **The common-variant ceiling — what was measured.** The whole Ghanaian sample yields **3,916 SNPs at MAF ≥ 5 per cent** after the corrected filter order, out of 1,807,331 sites that pass variant missingness — a reduction of 99.78 per cent at the MAF step alone. A diagnostic projection put the genome-wide ceiling at roughly 5,500 common sites, which would make the analysed panel about 71 per cent of what is available; that projection (and the related 4,832 figure) was computed under a depth-only mask on the superseded 3,359-SNP panel and should be read as a pre-re-run estimate, not a measured quantity. What is now measured is the MAF floor's effect downstream: lowering it to 1 per cent yields 10,876 SNPs on the same 1,900 isolates, and lowering it to 2 per cent yields 6,852 (Table 8.3).

[Interpreted] **What follows from it.** Fewer than four thousand common SNPs materially constrains every downstream analysis — the outlier scan most of all, where 6 of the 19 quantile outliers do not survive FDR correction. The shortage is most plausibly a property of *P. falciparum* population genomics within a single country, an abundance of rare variation with comparatively little common variation, rather than a defect of this pipeline; that attribution is an inference from the literature and not something these data establish. On

either reading, a study of this design has considerably less power than a sample of 1,900 isolates would suggest.

[Observed] **Polyclonality does not order cleanly along the gradient, and the previous draft's claim that it does was formed by quoting only the two endpoints.** Ghana's zones run north to south as Guinea savanna, transitional, deciduous forest, coastal savanna. The polyclonal fractions in that geographic order are 0.597, 0.500, 0.539, 0.444 (§6.2, on the 4,905 Pf8-QC-passing isolates — see the note in §7.2) — not monotone: the transitional zone, which lies north of the deciduous forest, has the lower value, and its 25-of-50 figure carries essentially no information. The contrast is also between-study: the coastal and Guinea savanna samples share no sequencing study, and within deciduous forest alone the polyclonal fraction ranges 0.430–0.578 between the four of its five contributing studies large enough to test, while the underlying $F_{ws}$ distributions differ between those studies at Kruskal–Wallis H = 14.9, p = $1.9 \times 10^{-3}$ (n = 811). No test on the polyclonal **fraction** itself exists in the outputs, and the p-value must not be attached to it. Reported in fairness, the two zone pairs that do share a study reproduce the overall direction (within 1083-PF-GH-CONWAY, Guinea savanna 0.532 against transitional 0.500), so the gradient is not purely an artefact — it is simply not separable from batch for the pairs that carry the largest differences. The honest statement is that mixed-infection frequency is highest in Guinea savanna and lowest on the coast, with no clean monotone ordering in between and no separation from sequencing study.

[Hypothesized] If polyclonality tracks transmission intensity and transmission intensity tracks ecological zone, then $F_{ws}$ is itself a candidate zone-level phenotype worth analysing in its own right rather than used only as a QC filter. That analysis has not been performed, and any version of it would have to be stratified by sequencing study and run on a stated denominator. The threshold-sensitivity analysis bears on it indirectly and in a direction worth recording: relaxing the $F_{ws}$ cut **raises** maximum pairwise $F_{ST}$ slightly rather than lowering it — 0.00232 at $F_{ws} \geq 0.95$ against 0.00255 at $F_{ws} \geq 0.80$ and 0.00254 with no cut at all (Table 8.3) — so the polyclonal isolates the filter removes were not masking between-zone differentiation.

[Calculated] **Likelihood monotonicity in TreeMix — why the likelihood curve cannot choose a model.** The log-likelihood never falls as migration edges are added — 89.65 at m = 0, 91.21 at m = 1, 92.23 at m = 2, 92.26 at m = 3, and 92.2642 at both m = 4 and m = 5, identical to the recorded precision — so likelihood cannot select a model, and the previous draft left the choice between m = 2 and m = 3 open. Re-running the sweep with -se narrows it without settling it: the first two edges carry weights 0.152 (SE 0.048, p = $7.5 \times 10^{-4}$) and 0.805 (SE 0.100, p = $3.3 \times 10^{-16}$), while the third edge added at m = 3 has weight 0.065 (SE 0.147, p = 0.33) and is indistinguishable from zero. **A two-edge graph is the model the edge tests select, which brings statistical support to what was previously argued only from theory — that four populations can support only about two edges — without settling it, since those tests are conditional on a topology, root and edge count all fixed beforehand and the variance-explained criterion across the full m = 0 to 5 sweep still prefers m = 3. The m = 2 vs m = 3 hedging in the previous draft is replaced by a stated model choice, not by a resolution.**

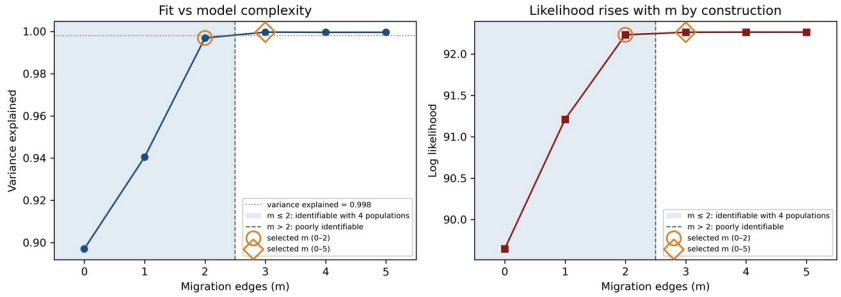


*Figure 7.3. TreeMix variance explained against the number of migration edges on the corrected 3,916-SNP matrix: 0.8970 at m = 0, 0.9405 at m = 1, 0.9969 at m = 2, 0.9997 at m = 3. The gain past m = 2 is not supported by the edge standard errors (third edge p = 0.33), and m ≥ 3 is not identifiable with four populations.*

## 7.4 Practical significance

**[Interpreted]** **What the neutral result licenses.** Neutral common-variant differentiation between zones is negligible — maximum pairwise $F_{ST}$ 0.00232, with one of six confidence intervals spanning zero (Guinea savanna vs transitional zone, [−0.00083, 0.00178]), and the maximum staying within 0.00204–0.00255 across six threshold scenarios (Table 8.3) — so a spatially sparse design is adequate for characterising **neutral** structure. Marker-frequency representativeness is a separate question, and the rebuilt resistance panel shows significant between-zone variation at **six of the nine panel genes**, including *pfdhfr*, *pfdhps* and *pfaat1* (the others being *pfcrt*, *pfmdr1* and *pfmdr2*). If the question is whether a handful of sites will give a representative picture of the neutral genomic background, these data answer yes; if the question is whether those sites will give a representative picture of resistance-marker frequencies, these data answer no.

**[Interpreted]** **What it does not license, and what the previous draft got wrong.** The previous draft went further and concluded that "sentinel surveillance at a small number of sites is an adequate design for detecting emergence — the samples do not need to be spatially comprehensive to be representative". That claim is withdrawn. It is unsupported by anything measured here, and it is contradicted by the mechanism this project's own literature review sets out: selection builds connectivity that neutral demography does not, so a selected haplotype can sweep across a region while the neutral background stays homogeneous — and, symmetrically, resistance-allele trajectories can remain site-specific even under neutral panmixia. Sentinel surveillance measures marker frequencies, not neutral structure. Low genome-wide neutral $F_{ST}$ says nothing about whether marker frequencies at one site represent the country.

**[Observed]** **The resistance panel settles the question empirically, and the answer is no.** Marker frequencies do differ by zone. *pfaat1* S258L spans about 28 percentage points from coast to far north (0.824 to 0.544, $\chi^2$ p = $1.89 \times 10^{-18}$, n = 1,896 called). *pfdhps* **G437** — the allele that was missing from the panel altogether in the previous draft — runs 0.964 in deciduous forest and 0.941 on the coast against 0.875 in Guinea savanna and 0.800 in the transitional zone ($\chi^2$ p = $1.2 \times 10^{-6}$, n = 1,892). The *pfdhfr* codons are heterogeneous across the four zones (S108N $\chi^2$ p = $9.4 \times 10^{-7}$; C59R $4.8 \times 10^{-5}$; N51I $2.4 \times 10^{-4}$) and the IRN triple-mutant haplotype itself runs 0.750 in Guinea savanna, 0.788 in deciduous forest, 0.867 on the coast and 0.880 in the transitional zone on 25 isolates. These are all four-zone heterogeneity tests; no directional south-versus-Guinea contrast was run, and the transitional zone's values rest on 25 isolates (§7.2). A sentinel site on the coast and a sentinel site in Upper East would report materially different resistance-marker profiles from the same panmictic parasite population. The operational implication is the reverse of the previous draft's: **neutral structure can be characterised sparsely; marker frequencies cannot.** A surveillance design intended to track resistance needs to span the gradient, and this analysis is evidence for rather than against that requirement.

**[Calculated]** **Robustness of the readings above to the corrected panel.** The conclusions in this section were reached on a neutral panel that grew from 3,359 to 3,916 SNPs (+16.6 per cent) once the filter order was corrected, with stages 4 to 9 re-run on the larger matrix. The substantive readings held — near-panmixia, maximum pairwise $F_{ST}$ of about 0.0023, and a near-flat cross-validation curve. **One reading did not hold, though it failed a later check rather than this one:** isolation by distance, described in earlier drafts as a weak, leverage-dependent but real distance effect, is significant in only 2 of 6 threshold scenarios and is now reported as **not robust** (§8, Table 8.3). Within the corrected run it had in fact improved — the Mantel result no longer depends on the undisclosed censoring, and significant jackknife replicates went 2 of 7 to 4 of 7 —

which is a useful reminder that a result can be made more reproducible and still not be true. One result gained inferential support: TreeMix now reports m = 2 as a selected descriptive model on edge p-values instead of hedging, though the variance-explained criterion across the full sweep still prefers m = 3. The BayPass threshold, outlier count and gene list all moved (11.232 to 10.840; 15 to 19 outliers; 11 to 13 genes). The marker panel is a different matter: it was not re-run but **rebuilt**, by a new script that reconstructs codons rather than matching annotation strings, and several of its conclusions did change (§7.5, §8). This wording is used identically in §6 and in §8; details and the full comparison are in Table 8.1.

[Interpreted] **The containment corollary, correctly scoped.** Because no neutral barrier is detectable at the retained common variants, this analysis supplies no evidence of a barrier for a geographically targeted containment intervention to exploit. That is an absence of evidence at the resolution these data permit, not a demonstration that parasite exchange across Ghana is unobstructed: a low differentiation estimate does not measure connectivity, and the study, geography and year confounding is not removed by it. That remains a reasonable reading, and it is now the better supported for the absence of a robust distance effect: there is not even a weak isolation-by-distance gradient left to qualify it. It does not follow that spread will be uniform in rate or that a single site will see it first; the *pfaat1*, *pfdhps* and *pfdhfr* zone differences show that selection and drug pressure impose their own geography, which negligible neutral differentiation does not predict. Whether gene flow is in fact free is not measured here: a low differentiation estimate at retained common variants does not quantify contemporary connectivity.

> **Caveat on that inference.** The argument from low $F_{ST}$ to free gene flow runs from an absence of detectable structure to an assertion about process. Low $F_{ST}$ is equally consistent with recent common ancestry and with a large effective population size holding drift down. Distinguishing "currently well connected" from "recently separated but not yet differentiated" requires demographic inference these analyses have not attempted (§8.2).

[Proposed] The unqualified surveillance claim must be removed wherever it is restated. There is no section titled "plain-language summary" in this manuscript; the nearest equivalents are the Abstract's concluding sentence, which reproduced the claim without even the caveat above, and the positive-findings list in §10.1. Both must carry the qualified form given in the first paragraph of this section — neutral structure sparsely characterisable, marker-frequency representativeness a separate and unanswered question — and not the unqualified one. This is the clearest case in the document of a conclusion that the corrected resistance panel reverses, so the three locations must not disagree. The same applies to isolation by distance: any statement of it in the Abstract or in §10.1 must now carry the 2-of-6 grading, not a "weak but real" one.

## 7.5 The rebuilt resistance panel

[Observed] **The panel in the previous draft was the wrong panel, and this section is a replacement rather than a revision.** An external critique raised five points about it; all five were upheld, and four were defects in code. The panel is now produced by `step12_codon_panel.py`, which maps each requested codon to its three genomic positions through the CDS, collects every variant record at those positions regardless of `numalt` or `is_snp`, applies each isolate's alleles to the reference codon and translates the result. It reports the **full residue spectrum including the reference residue**. The panel covers **57 codons across nine genes** on the 1,900 analysed isolates, of which **24 are polymorphic and 33 monomorphic**; a mixed call makes a codon uncalled rather than being resolved to a residue, so the denominator is **per codon, 1,834–1,900 called**, and is stated with every frequency below. Six of the nine genes sit on the plus strand (*pfap2mu*, *pfcoronin*, *pfcrt*, *pfdhfr*, *pfdhps*, *pfmdr1*) and three on the minus strand (*pfaat1*, *pfk13*, *pfmdr2*), which is the direction the

reconstruction has to get right. Every number in this section comes from that run; nothing from the previous panel is carried forward.

[Observed] **Why the old panel failed.** It identified markers by matching SnpEff ANN_HGVS_p strings against a hand-written list. Two failure modes, both realised. (i) **A resistance allele that is itself the reference residue can never appear as an ALT change.** The official Pf8 methodology says so in terms: *"3D7 sequence translates as a G at 437, this is thought to be the mutant allele, i.e. 3D7 does not have the wild type allele."* At *pfdhps* 437 the only annotation is Gly437Ala, the revertant; a search for "Ala437Gly" matched nothing, and **A437G was absent from the panel entirely** at a true frequency of 0.897. (ii) **A residue reachable only by two changes in one codon is never annotated.** *pfcrt* codon 75 carries 403621 (A>G, annotated Asn75Asp) and 403623 (T>A, annotated Asn75Lys); Asn→Glu requires both, so neither annotation names N75E, which is the residue the CVIET haplotype actually carries. This, and not a "naming discrepancy", is the entire explanation of the MalariaGEN disagreement the previous draft reported at that codon. The phrase "naming discrepancy" is withdrawn.

[Observed] **Antifolate resistance is high but not uniform, and "near fixation" was the wrong description.** The *pfdhfr* marginal residue frequencies are N51I 0.786 (n = 1,891), C59R 0.886 (n = 1,888) and S108N 0.908 (n = 1,889), each heterogeneous across the four zones and lowest in Guinea savanna ($\chi^2$ p = $2.4 \times 10^{-4}$, $4.8 \times 10^{-5}$ and $9.4 \times 10^{-7}$). S108N at 0.908 is the most frequent antifolate residue in the panel. The previous draft quoted the range of those three marginals, 0.786–0.908, **as though it were the triple-mutant frequency**. It is not a haplotype frequency at all, and that range is withdrawn. Counted within isolates on the 1,876 called at all three codons, the **IRN triple mutant sits at 0.771 (1,446 of 1,876; 95% CI 0.751–0.789)** — below the minimum component frequency on the same complete-case set (51I = 0.788), as it must be. At roughly three-quarters it is **not** "close to fixed", and it is also not a single dominant haplotype: **NRN sits at 0.116 and the fully wild-type NCS at 0.091**, so roughly one isolate in eleven is wild type at all three *pfdhfr* codons. The quadruple IRNL, which would add I164L, is **0.000** (n = 1,874; 164 is monomorphic at the reference residue in all 1,898 called).

[Observed] *pfdhps* G437 is at 0.897 and was missing from the previous draft altogether. *Overall 0.897 on 1,892 called, and by zone 0.964 deciduous forest, 0.941 coastal savanna, 0.875 Guinea savanna, 0.800 transitional zone ($\chi^2$ p = $1.2 \times 10^{-6}$) — the transitional zone again breaking the ordering (§7.2). It is a high frequency but not the highest antifolate residue in the panel:* pfdhfr *S108N is 0.908. The reason sulfadoxine–pyrimethamine retains utility in chemoprevention is therefore not that the* pfdhps *background is clean; it is that the downstream codons remain rare. K540E is 0.009 (n = 1,895) and A581G 0.014 (n = 1,890), and the 437G+540E double haplotype, which is what actually marks high-grade sulfadoxine resistance, is 0.010 on 1,888 complete cases. That is the standard West African pattern — 437G near-fixed, 540E rare — and it is a pattern the previous draft could not see, because it was reporting a* pfdhps panel with its highest-frequency locus deleted.

[Observed] **Chloroquine resistance has largely reverted, and CVIET is now demonstrated to be a unit rather than inferred to be one.** The *pfcrt* CVIET codons travel together at marginal frequencies of 0.071 (M74I, **N75E**, K76T), 0.070 (A220S), 0.074 (Q271E) and 0.076 (R371I). Counted as a haplotype on the 1,873 isolates called across codons 72–76, **CVIET is 0.071 (95% CI 0.060–0.084)**, and the only two haplotypes observed at all are **CVMNK (0.929) and CVIET (0.071)**, with nothing intermediate. The previous draft inferred co-travel from six similar marginals; it is now measured. The residue at codon 75 is **Glu**, as MalariaGEN's published calls state, and codon reconstruction reproduces their call on every isolate. Note that codon 75 is called in **1,874 of the 1,900** isolates, so it loses 26 to mixed calls; the 52 figure circulating for this codon is the mixed-call

record count and not an isolate count, and the two must not be conflated. Four *pfcrt* codons reach the four-zone heterogeneity threshold — 74, 75/76 ($\chi^2$ p = 3.5–3.7 × $10^{-2}$), 220 (2.6 × $10^{-2}$) and **356 (I356T, $\chi^2$ p = 1.6 × $10^{-3}$, the strongest of them)** — so it is not the case that no *pfcrt* codon shows between-zone heterogeneity.

[Observed] **Artemisinin resistance is absent, on a panel that is now the right panel — and the negative is historical, not current.** The *pfk13* list in the previous draft was written from memory. It held 13 distinct protein changes (a duplicated M476I record inflated the apparent count to 14), **misclassified P441L and C469F as validated when WHO lists them as candidate**, omitted P574L and G533S, and requested R622I without ever retrieving it, so that marker became a silent non-result. The panel now follows the **WHO compendium v1.0 (December 2025)** and covers all **14 validated** markers (F446I, N458Y, C469Y, M476I, Y493H, G533S, R539T, I543T, P553L, R561H, P574L, C580Y, R622I, A675V) and all **6 candidates** (E252Q, P441L, C469F, N537I, G538V, V568G). That is **20 variants at 19 distinct codons**, because C469Y and C469F share codon 469. The reconstruction covers **23** *pfk13* codons in all; the four beyond those 19 are potential-tier or regionally reported codons retained for completeness, and the validated and candidate tiers are kept distinct throughout — the candidates are not described as validated anywhere in this document. **Twenty-two of the 23 codons are monomorphic at the reference residue, and the twenty-third, codon 458, is at 0.9995** (1,865 of 1,866 called; one isolate carries Asn458Asp, which is not the validated Asn458Tyr). So the WHO negative stands, but the codon is not strictly fixed and should not be described as such. Two qualifications the critique rightly demanded. The denominator is **per codon, 1,845–1,876 of 1,900 called** — not "all 1,900 at every codon", which the previous draft asserted. And the sampling period is **2009–2021**, so this is a historical negative about the parasites sequenced, not a statement about Ghana today.

[Observed] *The* kelch13 *corroboration is weaker than the previous draft claimed. Pf8 assigns a* kelch13 *status over the whole 349–726 domain and classifies any missing call in that range as undetermined, so only 36 of 1,900 isolates have a complete status. Within those 36 the commonest change is A578S, which is explicitly not a resistance marker. Thirty-six complete statuses are weak corroboration for per-codon calls on roughly 1,860 isolates, and the two should not be presented as equivalent evidence. The real independent check on this panel is the per-sample concordance in §8 — but it reaches none of the 23 pfk13 codons, so for the artemisinin negative these 36 domain-level statuses remain the only external check there is, rather than the* kelch13 column.

[Observed] *The* pfaat1 *S258L cline is the positive result. Pf3D7_06_v3:1215233 is Ser258Leu, with XtX 24.20 and q = 0.0049 — the second-highest XtX in the scan, behind PTP7 at 26.44. Its frequency is 0.609 overall on 1,896 called, falling from 0.824 in coastal savanna through 0.735 in deciduous forest to 0.544 in Guinea savanna*, with the transitional zone at 0.640 on 25 isolates ($\chi^2$ p = 1.89 × $10^{-18}$). This is the one marker in which the transitional zone sits in its correct latitudinal place (§7.2). The previous draft gave p = 2.03 × $10^{-18}$ on an assumed denominator of 1,900; the shift comes from counting only the isolates actually called, and the coastal frequency rounds to 0.824 on either denominator. The manuscript's XtX of 24.44 for this locus is the superseded 8-thread figure and should not be quoted; the final run gives 24.20. The variant-quality check places this SNP in the clean tier: MQ 60.00 at the 100th percentile, VQSLOD 12.09 at the 96.7th, FS 0.00 (§7.2).

> **Provenance of the *pfaat1* q-values — now resolved.** The XtX values quoted above are read from the final run's own output (results/baypass/outlier_snps.bed: 24.19825 at Pf3D7_06_v3:1215233 and 16.53525 at 1214350, with PTP7 at Pf3D7_03_v3:96476 = 26.44106 the only higher value). The **q = 0.0049** attached to both was previously sourced only to the corrections document's narrative, because results/baypass/xtx_all_snps.csv for the final run was empty. That file has since been found to have been truncated to zero bytes by accident, and it has been regenerated from

> the intact BayPass outputs, with consistency checks rather than proof of row-for-row identity (§9.3). The per-SNP q-values are therefore on disk again: both *AAT1* SNPs carry q = 0.00489 in results/outlier_quality_check.csv, so **the value and the tie are now reproducible from a primary output** rather than from the narrative, and the instruction in the previous draft to regenerate the table before this sentence ships is discharged. The superseded run's table, in which 1214350 carried q = 0.0240 and the two were **not** tied, remains superseded and should not be quoted.

**[Observed]** One correction to the previous draft's reporting of this locus carries over unchanged: it listed "two outlier SNPs" in *AAT1*, which read as two independent lines of evidence. The second outlier, Pf3D7_06_v3:1214350, is Ile552Ile — synonymous, low impact, 883 bp away, XtX 16.54 and carrying the same q-value, 0.0049, now read from the regenerated per-SNP table (see the note above). This is **one signal plus a linked synonymous site**, not two signals. The locus also did not appear in the pipeline's KNOWN_RESISTANCE_LOCI list, which is why it surfaced as an anonymous transporter in a table while the abstract simultaneously declared that no validated resistance locus had been found.

**[Interpreted]** **What S258L actually does — the previous draft inverted the source paper's own conclusion, and the error was ours.** The previous draft called S258L "the functionally validated chloroquine-resistance variant", then built the discussion on the claim that "nothing in it or in these data tests amodiaquine". Both statements are wrong, and checking them against the source reverses the argument. **Amodiaquine was tested**: the methods state that *"Parasite IC50 values for CQ, amodiaquine, lumefantrine, mefloquine and quinine were measured"*. **The shifts were sub-threshold**: *"these IC50 value shifts were well below the threshold associated with clinical resistance"*. And **the authors explicitly reject the other-quinoline hypothesis** — *"is not supported"* — concluding instead that chloroquine was *"the primary selective force"*. Framing the Ghanaian cline as amodiaquine-maintained therefore did not extend the paper's finding; it contradicted it.

**[Interpreted]** **And the effect itself is modest and conditional.** S258L raised chloroquine IC50 **1.5-fold**, in a **single chloroquine-resistant Southeast Asian line (NHP4026)**, with epistasis between S258L and F313S and an explicit caveat that the effect depends on *pfcrt* background. The much-cited rise from 0 per cent to 97 per cent is **Gambian, 1984–2014**; the paper says nothing about Ghana. The accurate description is therefore: *pfaat1* S258L is a variant with a *demonstrated but modest, epistasis- and background-dependent effect that* potentiates *chloroquine resistance — not a validated resistance determinant, and not a marker of any current drug. One detail of this sample sharpens the caveat:* pfaat1 *F313S, the epistatic partner, is absent* — codon 313 is at the reference residue in all 1,900 isolates — so the interaction the experiment demonstrated cannot be operating in these parasites at all.

**[Hypothesized]** **What might maintain the cline, with the candidates ranked by how well they are supported.** Three non-exclusive possibilities, and the honest ordering is not the previous draft's. (i) **Historical persistence.** Chloroquine was first-line therapy in Ghana until 2005 and the paper's own conclusion is that chloroquine was the selective force; a chloroquine-era allele decaying unevenly is the explanation most consistent with the source. (ii) **A documented fitness cost without drug**, which would slow reversion rather than reverse it and is the reason a standing intermediate frequency is expected rather than either fixation or zero. (iii) **Residual 4-aminoquinoline pressure from artesunate–amodiaquine**, a current first-line ACT whose deployment is not uniform across the country. This third possibility is offered as an **untested possible** contemporary driver only. Nothing in these data tests it, no drug-pressure or deployment data were available to this analysis, and the experimental work that does exist found amodiaquine shifts below the clinical threshold and argued against the partner-drug hypothesis. It is a hypothesis worth stating and not a reading of the evidence.

[Hypothesized] If any version of the drug-pressure hypothesis holds, the cline should correlate with zonal AS-AQ deployment and adherence rather than with any ecological variable, and should be unstable over time as treatment policy changes. Neither prediction is tested here, and the alternative that S258L tracks some unmeasured ecological or demographic covariate cannot be excluded. Note also that the coast-versus-Guinea-savanna contrast that carries the cline is exactly the zone pair with no shared sequencing study (§7.2), so a batch contribution to its magnitude cannot be ruled out by any reanalysis of these data.

> **Temporal controls on the cline — computed on the superseded panel.** The previous draft reported that the S258L gradient survives restriction to a single collection year and that there is no within-Guinea-savanna trend with year. Both checks were run on the **superseded annotation-string panel** and have **not** been recomputed on the reconstruction panel. S258L is a single-SNP ALT change that the old annotation handled correctly, and its frequencies are essentially unchanged between the two panels (coastal 0.824 either way, Guinea savanna 0.544 against 0.544), so the checks very likely carry over — but "very likely carries over" is not a result, and they are listed as outstanding in §8.2 item 2. [Unknown]

[Observed] **WHO labels attach to variants, not genes, and the previous draft's gene-level labels must go.** The compendium classifies individual variants. It contains **no validated or candidate lumefantrine marker** at all — verbatim: *"The review did not identify any validated or candidate molecular markers associated with lumefantrine resistance."* The label "*pfmdr1*, WHO-validated, lumefantrine" used in the previous draft is therefore a category error twice over. Further: *pfaat1*, *pfap2mu* and *pfmdr2* appear nowhere in the compendium at any tier. *They are retained in the panel because they are scientifically interesting* — pfaat1 *S258L in particular is this analysis's strongest marker signal — but they must never be described as WHO-classified*. pfcoronin *is in the compendium at the potential* tier only, and all three of its panel codons are monomorphic at the reference residue here.

[Observed] **What can and cannot be said about tiers in this panel.** Every polymorphic variant here that does carry a WHO tier falls within *pfcrt*, *pfdhfr*, *pfdhps* or *pfmdr1* — but the converse does not follow, and the previous draft's phrasing asserted it. Not every polymorphic codon in those four genes is WHO-classified. The polymorphic codons in them are *pfcrt* 74/75/76/220/271/326/356/371, *pfdhfr* 51/59/108, *pfdhps* 431/436/437/540/581/613 and *pfmdr1* 86/184/1246, and no source available to this analysis gives a per-variant tier for most of them: *pfdhps* I431V and *pfcrt* I356T are not established compendium variants in any material provided, and the *pfmdr1* codons N86Y, Y184F and D1246Y carry no tier at all, since the only indication under which they would is lumefantrine and the compendium lists no lumefantrine marker. [Unknown] — per-variant tiers for the polymorphic *pfcrt* and *pfdhps* codons. Separately, every WHO-validated and WHO-candidate *pfk13* variant is at the reference residue, but that is not the same as saying no *pfk13* codon varies: codon 458 is polymorphic, Asn 0.9995 against Asp 0.0005 on 1,866 called, and the observed residue is Asp rather than the validated Tyr, so the WHO negative holds while the codon is a genuine non-WHO change. The three remaining polymorphic codons outside those four genes — *pfaat1* 258, *pfap2mu* 160 and *pfmdr2* 492 — are non-WHO by construction, since none of those genes appears in the compendium.

## 8. Robustness checks and alternative approaches

[Calculated] **Correcting the pipeline did not overturn its population-genetic conclusions, and that is worth reporting precisely rather than as a slogan.** The corrected filter order recomputes variant missingness on the 1,900 isolates that actually survive QC rather than on the 2,017 measured before 117 of them are dropped. At identical thresholds and with an identical sample set, this yields **3,916 SNPs instead of 3,359 — a 16.6 per**

**cent larger analysis matrix for free** — and stages 4 through 9 were re-run on it with a corrected FDR, seeded permutation tests, single-threaded BayPass and TreeMix standard errors in place. The substantive readings held — near-panmixia, maximum pairwise $F_{ST}$ of about 0.0023, and a near-flat cross-validation curve. Two results were strengthened rather than unchanged: the Mantel result no longer depends on the undisclosed censoring, and TreeMix now reports m = 2 as a selected descriptive model on edge p-values instead of hedging, though the variance-explained criterion across the full sweep still prefers m = 3. The BayPass threshold, outlier count and gene list all moved (11.232 to 10.840; 15 to 19 outliers; 11 to 13 genes). No population-genetic conclusion changed **under this correction**, which is the claim being made — not that nothing changed, and not that nothing has changed since. A later check did overturn one, and it is reported two paragraphs below.

[Calculated] **The marker panel is the exception, and the claim above must not be extended to it.** The resistance panel was not re-run on a larger matrix; it was **rebuilt by a different method**, and conclusions did change (§7.5). An allele absent from the old panel turned out to be at 0.897; a "triple-mutant frequency" of 0.786–0.908 turned out to be the range of three marginals and the haplotype is 0.771; two *pfk13* markers were misclassified as validated; and a reported "naming discrepancy" with MalariaGEN turned out to be a defect in our own identification layer. Anyone reading §8 as "the audit found nothing substantive" is reading the neutral half and missing the marker half.

[Calculated] **The four outstanding checks have now been run. Three confirmed robustness; one overturned a conclusion.** (i) **Threshold sensitivity.** `scripts/step13_sensitivity.py` rebuilds the analysis matrix under six combinations of the two thresholds that define the dataset — the $F_{ws}$ monoclonality cut, which discards 57 per cent of isolates and does so unevenly by zone, and the MAF floor, which discards 99.8 per cent of sites — and recomputes pairwise $F_{ST}$ and the region-level Mantel test for each (Table 8.3). **Near-panmixia is insensitive to both:** maximum pairwise $F_{ST}$ moves only between 0.00204 and 0.00255, a total spread of 0.0005, over sample sizes from 1,900 to 4,403 and SNP counts from 3,275 to 10,876. That stability is the best-evidenced result in the whole analysis — evidence that the measured differentiation does not depend on the threshold choices, not evidence about the processes that produced it. **Isolation by distance is not:** the Mantel test is significant in **2 of 6** scenarios, failing at every relaxation of the $F_{ws}$ cut and at MAF ≥ 0.01, and holding only at the published thresholds and marginally at MAF ≥ 0.02. Taken with the leave-one-region-out jackknife, which already showed 4 of 7 and collapse when Upper East is dropped, the finding must now be reported as **not robust**. It was the manuscript's only positive spatial finding, and it did not survive the two checks the audit itself identified as most needed. That is the overturned conclusion, and it is a loss rather than a refinement. (ii) **Outlier variant quality.** The 19 XtX outliers are *better*-supported calls than the rest of the panel, not worse, and the exported-protein artefact worry is closed; two adjacent chromosome-11 SNPs are downgraded for an unrelated and still-unverified reason (§7.2 and its note). (iii) **Stage 3 re-run on the 1,900-isolate analysis set.** The ordination conclusion is unchanged, the silhouette becomes slightly more negative, and the figures now describe the same sample as every other result (Figure 8.1). (iv) **GeoJSON provenance** remains open and cannot be closed analytically; it requires a written enquiry to the likely custodian (§9.3). Two secondary observations from (i) are worth recording in their own right: relaxing $F_{ws}$ slightly **raises** $F_{ST}$ (0.00232 to 0.00255), so the polyclonal isolates the filter removes were not masking differentiation; and tripling the SNP count via MAF ≥ 0.01 **lowers** it (0.00204), as expected if rarer variants are less differentiated between zones.

| Quantity | Original run (3,359 SNPs) | Corrected run (3,916 SNPs) | Conclusion changed? |
|---|---|---|---|
| Isolates analysed | 1,900 | 1,900 | — |

| Quantity | Original run (3,359 SNPs) | Corrected run (3,916 SNPs) | Conclusion changed? |
|---|---|---|---|
| Maximum pairwise $F_{ST}$ | 0.0024 | 0.00232 (SE 0.00053) | No — and now known to stay within 0.00204–0.00255 across six threshold scenarios (Table 8.3) |
| $F_{ST}$ CIs spanning zero | 3 of 6, as recorded in the corrections document (the manuscript's abstract said 2 of 6) | 1 of 6 — Guinea savanna vs transitional zone only | No — differentiation still negligible |
| ADMIXTURE CV minimum | K = 2, 0.83736 vs K = 1 0.84384 (a 0.5% improvement, as published) | K = 2, 0.83925 vs K = 1 0.84384 (a 0.54% improvement) | No |
| XtX threshold (POD 0.999) | 11.232 — manuscript run, 8-thread | 10.840 — final single-threaded run | Threshold moved; the reading did not |
| XtX outliers | 15 — manuscript run | 19 above threshold; 13 at q ≤ 0.10, 8 at q ≤ 0.01 | Count moved, and FDR accounting is new. The set is now also shown not to be a variant-quality artefact, with two SNPs downgraded (§7.2) |
| Outlier genes | 11 — manuscript run | 13, holding 17 of the 19 outlier SNPs; the other 2 are intergenic | Count moved — same exported/host-interaction pattern, but ACS8 rose from the bottom of the ranking to fourth on a new FDR-surviving SNP |
| TreeMix model selected | m = 3 (full) or m = 2 (identifiable); no p-values; tree alone 87.6% of covariance, one edge 93.5% | m = 2, with edge p-values $7.5 \times 10^{-4}$ and $3.3 \times 10^{-16}$ (third edge p = 0.33); tree alone 89.7%, two edges 99.7% | Narrowed, not resolved — m = 2 is a selected descriptive model and the full-sweep variance-explained criterion still prefers m = 3 |
| Mantel, region | r = 0.400, p = 0.030 censored; r = 0.381, p = 0.051 uncensored; 4 pairs floored (interim seeded re-run on the 3,359-SNP matrix) | r = 0.427, p = 0.025 censored; r = 0.414, p = 0.041 uncensored; 3 pairs floored | Yes — but by a later check, not by this one. Within the corrected run the result was strengthened: the censoring no longer determines significance and significant jackknife replicates went 2 of 7 to 4 of 7. The threshold-sensitivity analysis then found it significant in only 2 of 6 scenarios (Table 8.3), so isolation by distance is now reported as not robust |
| Mantel, zone | r = 0.578, p = 0.084 (corrections record) | r = 0.471, p = 0.122; 0 of 4 jackknife replicates significant | No — non-significant either way |
| Stage 3 ordinations, sample set | 4,905 Pf8-QC-passing isolates (pre-$F_{ws}$); silhouette by zone −0.016 | Re-run on the 1,900 analysis set; silhouette by zone −0.020, trustworthiness 0.499 / 0.526 / 0.510 (UMAP / t-SNE / PCoA) against 0.486 / 0.507 / 0.495 | No — and the one independence claim in Table 7.1 no longer rests on a different sample |

| Quantity | Original run (3,359 SNPs) | Corrected run (3,916 SNPs) | Conclusion changed? |
|---|---|---|---|
| Resistance-marker panel (method, not panel size) | 44 codons by SnpEff annotation-string matching, 21 polymorphic; denominator asserted as 1,900 at every codon | 57 codons by CDS codon reconstruction, 24 polymorphic; per-codon denominator 1,834–1,900 called; full residue spectrum including the reference residue | Yes, repeatedly. *pfdhps* A437G recovered at 0.897 from total absence; *pfdhfr* triple mutant 0.771 as a haplotype rather than a 0.786 –0.908 marginal range; *pfcrt* N75E resolved and the "naming discrepancy" withdrawn; *pfk13* P441L and C469F reclassified validated → candidate; three markers added (P574L, G533S, R622I) |
| Validation against MalariaGEN | Marginal frequencies: 21 of 23 shared markers within 0.03, 2 flagged at *pfcrt* 75 | Per-sample: 47,012 of 47,012 calls agree, 100.0000% across 25 markers, codon 75 included | Method changed, and the old one was too weak — see the note below |

*Table 8.1. The conclusions against the corrected pipeline. Original-run values are from the superseded manuscript and the corrections record and are given only as before/after comparisons; corrected-run values are read from the final run's own outputs. Where a quantity moved, the row says so. The Mantel region row records a conclusion that this correction strengthened and a **later** check overturned; the grading that governs is the one in Table 8.3. The last two rows concern the marker panel, which was rebuilt rather than re-run; the SNP-count columns do not apply to them.*

> **Why marginal agreement was not a validation.** The previous draft validated the panel by comparing **marginal frequencies** against MalariaGEN's published calls and reporting that 21 of 23 shared markers agreed within 0.03. That check cannot detect compensating per-sample errors — two pipelines can assign different residues to different isolates and still return the same frequency — and it is the check that let the *pfcrt* 75 defect be recorded as a "naming discrepancy at identical frequency" rather than as a bug. The replacement is **call-by-call**: on the isolates where both pipelines called, **47,012 of 47,012 calls agree, 100.0000 per cent across 25 markers**, all nine *pfcrt* codons included. Codon 75 now agrees exactly, because reconstruction yields Glu where string-matching yielded Asp. Per-sample concordance is the claim to quote; the 0.03 marginal-agreement figure is superseded and should not appear anywhere in the document.
>
> **BayPass run labels — three thresholds circulate and must not be conflated. The manuscript run** was 8-threaded: threshold 11.232, 15 outliers, 11 genes (`results/baypass_8thread_archive/xtx_calibration.json`). **An 8-thread re-run** of the same command with the same `-seed 42` returned a different outlier count (16); that disagreement is how the non-determinism was found. **An interim single-threaded run** on the same 3,359-SNP panel gave threshold 11.047 and 17 outliers (`corrected/xtx_calibration.json`); this is the value quoted in §4 of the corrections document, and it is superseded. **The final single-threaded run** on the 3,916-SNP panel gives threshold 10.840, 19 outliers above threshold, 13 at q ≤ 0.10 and 13 outlier genes, and is the only run reported as a result. Quoting 11.232 to 11.047 and 11.232 to 10.840 in the same document is consistent only if these labels are used. [Unknown] whether the 8-thread re-run and the interim single-threaded run were the same job: the logs do not record it, and the two documented outlier counts (16 and 17) differ, so they are listed separately here.

## 8.1 Checks performed

**[Observed]** The first eight rows below were performed during the original analysis; the remainder were performed during or after the audit. Rows 15 to 18 concern the marker panel and were new to the previous revision. **Rows 19 to 21 are new to this revision**: the threshold-sensitivity analysis, the variant-quality profile of the outlier set and the stage 3 re-run on the analysis set. Three of the four checks carried as outstanding into this revision are now closed; the fourth, the provenance of the zone polygon layer, cannot be closed analytically (§9.3). Row 19 is the only check in this table that overturned a conclusion.

| Check | What it tested | Outcome |
|---|---|---|
| Genotype store validated against VCFs | Whether the Zarr retrieval reproduced the VCF route exactly | PASS. Three independent windows on chromosomes 1, 10 and 13; site lists identical and all four fields matching value for value across approximately 471 million compared values |
| Ploidy handling unit-tested | Whether mixed-call resolution and haploid counting behave as specified | PASS. Synthetic 3 × 4 case including a tie and a missing call; diploid fallback verified to trigger with a warning |
| $F_{ST}$ invariance to a cosmetic change | Whether re-running step 4b after a label change altered any value | PASS. MD5 of results/fst_matrix.csv unchanged |
| Clonal threshold sensitivity | How the number of pruned isolates responds to the threshold | Quantified over seven thresholds (Table 4.3). No clean break; single-linkage chaining begins only well above the chosen value of CLONAL_THRESHOLD = 0.00005 |
| Genotype-mask comparison | Effect of the GQ rule on SNP count and allele frequency | Quantified on a 40,000-SNP panel and a 300,000-SNP panel (§4.2.5). The GQ rule removed many common SNPs while barely shifting frequencies |
| ADMIXTURE K = 1 reference | Whether admitting structure improves fit at all | Run. CV error 0.84384 at K = 1 against 0.83925 at K = 2 — a 0.54% improvement, on 3,385 LD-pruned sites of the 3,916 |
| BayPass calibration (POD size and MAF matching) | Whether the threshold was sensitive to the pseudo-observed dataset | Yes, materially. The final calibration uses 100,000 simulated SNPs — which are not MAF-matched, despite earlier wording in this manuscript that said they were (Section 5.6) — giving a threshold of 10.840, against 11.047 on the interim single-threaded run and 11.232 on the 8-thread manuscript run |
| TreeMix convergence across m | Whether all six models fitted | All six converged; none failed or was skipped |
| Filter order corrected and the whole analysis re-run | Whether measuring variant missingness before rather than after the sample drop had cost SNPs, and whether the conclusions depend on the panel size | 3,916 SNPs against 3,359 at identical thresholds and on an identical 1,900-isolate sample (+16.6%). Stages 4–9 re-run. No population-genetic conclusion changed under this correction, though several numbers did (Table 8.1) |
| False-discovery accounting for the XtX scan | Whether a 0.999 quantile threshold is a significance test (it is not) | Added. 19 SNPs above threshold, 3.9 expected false positives, expected-null-exceedance ratio 20.6%. Benjamini–Hochberg q-values now reported per SNP: 8 at q ≤ 0.01, 10 at q ≤ 0.05, 13 at q ≤ 0.10, 17 at q ≤ 0.20. Six of the 19 do not survive FDR 10% |
| TreeMix edge standard errors | Whether any migration edge is distinguishable from zero | Re-run with -se. m = 2 edges significant ($p = 7.5 \times 10^{-4}$ and $3.3 \times 10^{-16}$); the third edge at m = 3 has p = 0.33. A two-edge graph is the selected descriptive model, not an identified count of migration events |
| BayPass determinism | Whether the scan is reproducible | Fixed. Multi-threaded BayPass is non-deterministic despite a fixed seed: the 8-thread manuscript run and an 8-thread re-run of the same command disagreed on the outlier count (15 vs 16). It now runs single-threaded and is bitwise stable; see the run-label note above. Previous multi-threaded outputs archived rather than deleted |
| Mantel reproducibility, censoring disclosure and jackknife | Whether the permutation p-values are reproducible, and whether the result depends on the | Seeded (seed 42, 9,999 permutations). Three floored pairs now named. Both censored and uncensored versions clear p < 0.05 at region level, so the censoring no longer determines significance. Leave-one-out: significant in 4 of 7 region replicates, 0 of 4 at |

| Check | What it tested | Outcome |
|---|---|---|
| | undisclosed flooring of negative $F_{ST}$ values or on any one population | zone level. Significance is lost on dropping any of three regions — Upper East (r = 0.260, p = 0.218), Greater Accra (r = 0.250, p = 0.232) or Eastern (r = 0.288, p = 0.136) — so the result is substantially a single far-northern contrast and is additionally fragile to the composition of a 21-pair test. Superseded as a grading by row 19: under threshold variation the test is significant in only 2 of 6 scenarios, so the finding is reported as not robust |
| Resistance markers re-fetched without the numalt restriction | Whether the reported absence of resistance loci was biological | It was not — the headline negative was a filter artefact (§7.3). This check was necessary but not sufficient: it recovered the removed records while leaving the annotation-string identification layer intact, so the panel it produced was still wrong. Superseded by the rebuild below |
| Marker panel rebuilt by codon reconstruction | Whether markers identified by matching SnpEff ANN_HGVS_p strings are the markers intended | They were not. Each codon is now mapped to its three genomic positions through the CDS, every variant record at those positions is collected regardless of numalt or is_snp, each isolate's alleles are applied to the reference codon and the codon is translated; the output is the full residue spectrum including the reference residue. 57 codons across nine genes — six plus-strand (*pfap2mu*, *pfcoronin*, *pfcrt*, *pfdhfr*, *pfdhps*, *pfmdr1*) and three minus-strand (*pfaat1*, *pfk13*, *pfmdr2*) — 24 polymorphic, per-codon denominator 1,834–1,900. Recovered *pfdhps* A437G (absent from the old panel, true frequency 0.897) and resolved *pfcrt* N75E (a two-SNP residue no single annotation names). scripts/step12_codon_panel.py, logs/step12_codon_panel.log |
| Per-sample concordance against MalariaGEN's published genotypes | Whether each isolate's call agrees, not merely whether the frequencies do | PASS, and this replaces the previous marginal-agreement check. 47,012 of 47,012 calls agree — 100.0000% across 25 markers on the isolates both pipelines called, all nine *pfcrt* codons included. Codon 75 agrees exactly under reconstruction. codon_panel_concordance.csv |
| Haplotypes counted within isolates rather than inferred from marginals | Whether the "triple mutant" and "CVIET" claims had co-occurrence data behind them | They did not. On complete cases: *pfdhfr* IRN 0.771 (1,446 of 1,876; 95% CI 0.751–0.789), below its minimum component 51I = 0.788 as it must be, with NRN 0.116 and wild-type NCS 0.091 also real; IRNL 0.000. *pfcrt* CVIET 0.071 (n = 1,873), and only CVMNK and CVIET occur, so that motif genuinely is a unit. *pfdhps* 437G+540E 0.010 (n = 1,888). codon_panel_haplotypes.csv |
| *pfk13* panel checked against the WHO compendium v1.0 | Whether the marker list matched an authoritative source | It did not. The old list held 13 distinct changes (a duplicate inflated the count to 14), misclassified P441L and C469F as validated, omitted P574L and G533S, and requested R622I without ever retrieving it, so that marker became a silent non-result. The panel now covers all 14 validated and all 6 candidate variants — 20 variants at 19 distinct codons, C469Y and C469F sharing codon 469 — on a reconstruction spanning 23 *pfk13* codons: 22 monomorphic at the reference residue and codon 458 at 0.999 (one Asn458Asp, not the validated Asn458Tyr), per-codon denominator 1,845–1,876, sampling period 2009–2021 |
| Study/batch confounding quantified | Whether zone is separable from sequencing study | It is not, for the pairs that matter. Computed on the 4,905 Pf8-QC-passing isolates, not the 1,900 (§7.2 note). Four of six zone pairs share no study; within-zone, between-study call-rate heterogeneity reaches $p = 2.8 \times 10^{-197}$ (Guinea savanna, six studies) and $p = 7.7 \times 10^{-88}$ (deciduous forest, four of its five contributing studies). Now limitation 2 in §9.1 |
| Stage cache guards audited | Whether "skip if output exists" guards can report | They could, and did. When the SNP set grew, ADMIXTURE and TreeMix saw their previous output files, skipped the work and |

| Check | What it tested | Outcome |
|---|---|---|
| | a stale result as a fresh one | reported success; only stage 8's row-count assertion failed loudly. All three guards now require the output to be newer than its input |
| *Threshold sensitivity*: $F_{ws}$ *and MAF varied over six scenarios* | Whether the conclusions depend on the two thresholds that define the dataset — the $F_{ws}$ monoclonality cut (which discards 57% of isolates, unevenly by zone) and the MAF floor (which discards 99.8% of sites) | Mixed, and one conclusion is overturned. Near-panmixia is insensitive: maximum pairwise $F_{ST}$ 0.00204–0.00255 across all six scenarios, over 1,900–4,403 isolates and 3,275–10,876 SNPs, the maximum being the Coastal savanna–Guinea savanna pair every time. The region-level Mantel test is significant in only 2 of 6 scenarios, failing at $F_{ws}$ ≥ 0.90, ≥ 0.80 and no filter, and at MAF ≥ 0.01, so isolation by distance is now reported as not robust (Table 8.3). scripts/step13_sensitivity.py, sensitivity_summary.csv, sensitivity_fst.csv |
| Variant-quality profile of the 19 XtX outliers | Whether the outlier set, and the exported-protein genes within it, could be a mapping or calling artefact | It could not. MQ, VQSLOD, FS, SOR, QD, ReadPosRankSum and culprit pulled from the Pf8 Zarr for all 19 and compared against the other 3,897: every significant difference favours the outliers (VQSLOD p = 0.008, QD p = 0.016, QUAL p = 0.007), none has MQ < 50, strand bias or read-position bias, and all 19 are RegionType Core. The eight exported-protein SNPs are the cleanest subset in the panel (FS = 0.00 for all eight, MQ median 60.00, VQSLOD significantly higher). Two adjacent chromosome-11 SNPs are downgraded on InbreedingCoeff and ExcessHet grounds, mechanism unverified (§7.2 note). results/outlier_quality_check.csv |
| Stage 3 ordinations re-run on the 1,900-isolate analysis set | Whether the ordination conclusion depends on including the polyclonal and clonally redundant isolates that every later stage excludes | It does not. step03_umap.py now takes --samples and --suffix and re-slices the distance submatrix rather than reusing the cached full-set one; outputs carry a _qcset suffix so the published figures are untouched. On the 1,900 set: silhouette by zone −0.020 against −0.016 on the 4,905; trustworthiness 0.499 (UMAP), 0.526 (t-SNE), 0.510 (PCoA) against 0.486, 0.507, 0.495; PCoA distance correlation 0.287 against 0.322. Removing those isolates does not sharpen the structure — the silhouette becomes slightly more negative. dimred_comparison_qcset.csv, Figure 8.1 |

*Table 8.2. Robustness checks performed. Bold rows are new since the original analysis. Four rows on the marker panel were new to the previous revision — the rebuild, the per-sample concordance, the within-isolate haplotypes and the pfk13/WHO check (rows 15 to 18) — with a fifth (the numalt re-fetch, row 14) carried forward but recorded as insufficient. Rows 19 to 21 are new to this revision; row 19 is the only check in the table that overturned a conclusion, and row 12 is graded by it.*

| Scenario | Isolates | SNPs | Max pairwise $F_{ST}$ | Mantel r (region) | Mantel p | Significant? |
|---|---|---|---|---|---|---|
| baseline ($F_{ws}$ ≥ 0.95, MAF ≥ 0.05) — as published | 1,900 | 3,916 | 0.00232 | 0.427 | 0.0252 | yes |
| $F_{ws}$ ≥ 0.90 | 2,196 | 3,629 | 0.00240 | 0.318 | 0.0874 | no |
| $F_{ws}$ ≥ 0.80 | 2,746 | 3,496 | 0.00255 | 0.256 | 0.1687 | no |
| no $F_{ws}$ filter | 4,403 | 3,275 | 0.00254 | 0.245 | 0.1792 | no |
| MAF ≥ 0.01 | 1,900 | 10,876 | 0.00204 | 0.432 | 0.0617 | no |
| MAF ≥ 0.02 | 1,900 | 6,852 | 0.00217 | 0.434 | 0.0467 | yes (marginal) |

*Table 8.3. Threshold sensitivity. Both thresholds that define the analysis dataset were varied and pairwise $F_{ST}$ and the region-level Mantel test recomputed from scratch under each scenario (scripts/step13_sensitivity.py; sensitivity_summary.csv, sensitivity_fst.csv). Maximum $F_{ST}$ is the Coastal savanna–Guinea savanna pair in all six scenarios. Mantel statistics are the censored versions, with negative $F_{ST}$ pairs floored to zero, on 9,999 seeded permutations; on the* ***uncensored*** *statistic only the baseline scenario clears p < 0.05 at all (baseline 0.041, MAF ≥ 0.02 0.050, the rest 0.064–0.200). The six scenarios are not*

*independent analyses — they share data and a pipeline — but they test the analyst's threshold choices, which no other check in this document does. Near-panmixia survives; isolation by distance does not.*

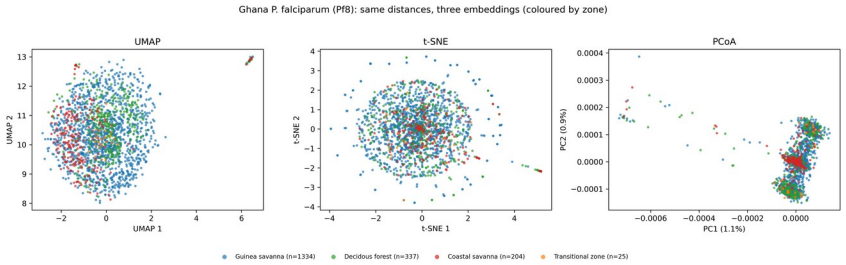



*Figure 8.1. The three ordinations re-run on the 1,900-isolate analysis set and coloured by ecological zone, from MalariaGEN's published Pf8 genotype-distance matrix re-sliced to those isolates. Silhouette by zone is −0.020 on the raw distances, against −0.016 on the 4,905 Pf8-QC-passing isolates that the published figures describe: removing the polyclonal and clonally redundant isolates makes the zone separation slightly **worse**, not better. Trustworthiness is 0.499 (UMAP), 0.526 (t-SNE) and 0.510 (PCoA), each marginally higher than on the full set, while the PCoA distance correlation falls from 0.322 to 0.287. The conclusion of §6.3 is unchanged and the ordinations now describe the same sample as every other result in the document.*

**[Interpreted]** **What the pattern of these checks says about the verification, stated once.** This is the fourth consecutive round in which a correction contained its own defect. The pattern is consistent across all four: each layer was checked for **internal** consistency against this project's own outputs, and what escaped was **external** correctness — whether the marker list matched an authoritative source, whether the annotation direction matched the reference genome, whether a co-occurrence claim had co-occurrence data behind it. The verification architecture was built to catch stale numbers, and it did catch them. It was not built to ask whether the right question was being asked of the data. The four checks added in the previous revision — the panel rebuild by codon reconstruction, the per-sample comparison against MalariaGEN call by call, the within-isolate haplotype counts in place of marginals, and the check against the WHO compendium — are all external by construction, which is the point of them. The three added in this revision are of a third kind again, and worth distinguishing: they vary the analyst's own choices rather than compare against anything outside. That is what caught the isolation-by-distance result, and nothing else could have. No external reference would have flagged it, because the number was correctly computed at the thresholds the pipeline specifies; it was the choice of thresholds, not the arithmetic, that was carrying the result. A number can be right, reproducible and externally consistent, and the conclusion drawn from it still not survive the decision that produced it.

## 8.2 Checks proposed but not performed

**[Proposed]** Three items listed here in the previous draft have since been run and have moved to Table 8.2 — the $F_{WS}$ sensitivity analysis, the MAF sensitivity analysis (now rows 19, jointly) and the stage 3 re-run on the analysis set (row 21). None of the following has been run. They are listed in order of expected value. The first two are the most important of what remains, and the second matters more than it did: a spatial result that holds at one threshold combination and almost nowhere else is exactly what an unresolved temporal confound can produce.

26. **Read-level depth check on the two downgraded chromosome-11 SNPs — not done, and now the highest-value outstanding item on the outlier scan.** Pf3D7_11_v3:1 054 078 (*SNRPF*) carries the only nonzero ExcessHet value in the panel (69.59, roughly 400× the genome-wide maximum), and it and Pf3D7_11_v3:1 054 447, 369 bp away, are the two lowest-InbreedingCoeff sites of the 3,916. Collapsed paralogy or copy-number variation would explain both, and MQ and FS are clean so mismapping would

not; but that is an inference from summary annotations, not a demonstration. A per-sample depth ratio against each isolate's genome-median depth at these two positions would settle it. Until it is run the mechanism stays [Hypothesized] and both SNPs stay at lower confidence (§7.2). 1054447 is the fifth-ranked SNP in the whole scan, so this is not a cosmetic item.

27. **Temporal control, and the marker-panel temporal checks re-run.** Restrict to a single collection year or a narrow window, where sample sizes permit, and recompute $F_{ST}$. 2018 contributes 773 isolates across three zones and is the only year that could support this. Separately, the two temporal controls on the *pfaat1* cline — survival of a 2018-only restriction, and absence of a within-Guinea-savanna trend with year — were computed on the superseded annotation-string panel and **have not been recomputed** on the reconstruction panel. S258L's frequencies are nearly identical between the two panels, so they are expected to carry over, but until they are re-run they are not results (§7.5 note). The same applies to any temporal statement about *pfdhps* G437, which has no temporal analysis at all because the locus was absent until now.

28. **Region-level ADMIXTURE and TreeMix.** Running these with populations defined by administrative region rather than zone would show whether the choice of grouping affects the conclusions, even though it cannot resolve the underlying collinearity.

29. **Permutation test for the outlier set.** Compare the functional composition of the 13 outlier genes against random gene sets matched for length and SNP density, to test whether the apparent enrichment for exported and host-interaction proteins exceeds chance. The variant-quality check (Table 8.2 row 20) establishes that those genes' SNPs are well-called, not that the enrichment is real; the two questions are separate. The re-run added exactly two genes, both exported proteins — REX2 (PF3D7_0936000, ring-exported protein 2) and PF3D7_0221200, a *Plasmodium* exported protein of unknown function (hyp15) — taking the list from 11 to 13, which strengthens the pattern but does not test it; the two intergenic outliers are outside any such test by construction.

30. **Recompute the zone-pair study overlap and the polyclonality figures on the 1,900-isolate analysis set.** They are currently computed on the 4,905 Pf8-QC-passing isolates (§7.2 note). The nesting of study within zone is a property of the sampling frame and will not reverse, but the quoted percentages would then match the denominators used everywhere else. The stage 3 re-run (Table 8.2 row 21) shows this kind of re-slicing is now cheap, which weakens the excuse for leaving it undone.

31. **Stratify the marker clines by sequencing study.** The *pfaat1* S258L and *pfdhps* G437 contrasts both run between zone pairs that share no study (§7.2). Wherever a zone pair does share one, the within-study marker contrast should be reported separately from the between-study contrast, exactly as proposed for polyclonality below. No version of this has been run on the rebuilt panel.

32. **Effective population size and divergence-time estimation**, to distinguish ongoing gene flow from recent common ancestry (§7.4).

33. **Extend the codon-reconstruction panel beyond the nine genes currently covered.** The reconstruction method is general and the reference-state failure mode it fixed is not specific to *pfdhps* 437; any other canonical codon whose resistance allele matches 3D7 would have been invisible to the old panel and has not been looked for outside these nine genes.

34. **Re-join the zone assignments against an authoritative six-zone Ghana layer.** Because isolates join on administrative-level-1 centroids, the join can be repeated against a published classification. If zone assignments are stable, the provenance gap in §9.3 becomes a sensitivity result instead of a threat to reproducibility.

35. **Manual review of the Brong Ahafo and Volta zone assignments** against the polygon layer, as step 2 recommends.

36. *Stratify the polyclonality and* $F_{ws}$ *comparisons by sequencing study* wherever a zone pair shares one, and report the within-study contrasts separately from the between-study ones (§7.2, §9.1 limitation 2).

37. **Written enquiry to the likely custodian of the zone polygon layer.** This is not an analytical check and cannot be run as one, which is why it is still open: the provenance of `data/ghanaAEZ.geojson` can only be settled by asking. The addressee is the Centre for Remote Sensing and Geographic Information Services, University of Ghana (cersgis@ug.edu.gh), the most likely custodian of the pre-2005 national coverage that the file's ArcInfo fingerprint points to. Until a reply arrives the layer is cited as accessed, with its md5, and the custodianship remains an inference (§9.3).

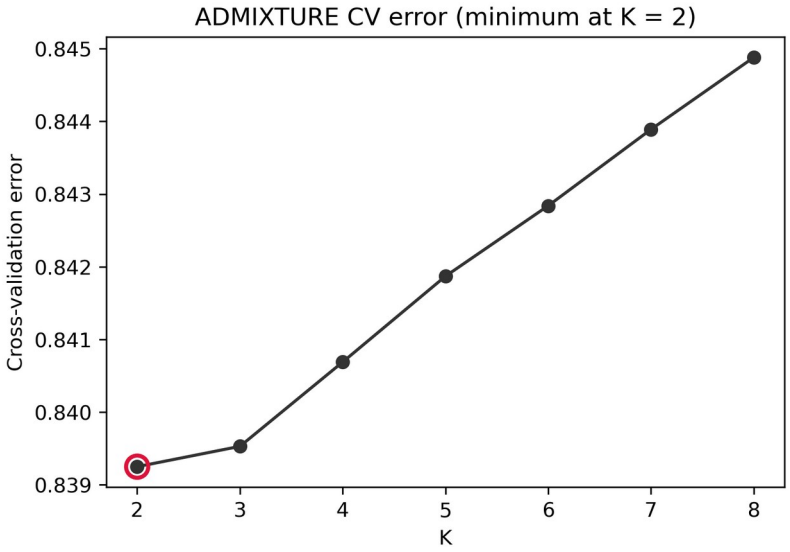



*Figure 8.2. ADMIXTURE cross-validation error from K = 2 to K = 8 on the corrected 3,916-SNP matrix (3,385 LD-pruned sites); the sweep that produced the plot was run over K = 2–8, so K = 1 is not a plotted point and is quoted from the log instead. The curve is near-flat: K = 2 is the minimum at 0.83925 against 0.84384 at K = 1 (from `logs/admixture_K1.out`, on the same 3,385-SNP panel), an improvement of 0.54 per cent, and the plotted K = 2–8 range is 0.0056 (0.67 per cent). The curve has essentially the same shape on a 16.6 per cent larger SNP panel, which is part of the robustness result reported in §8.*

# 9. Limitations and potential sources of bias

## 9.1 Ranked by threat to the conclusions

| # | Limitation | Severity | Effect on what can be concluded |
|---|---|---|---|
| 1 | Zone and region are collinear for three of four zones | Critical | No result can be attributed to ecology rather than administrative region. This limits the thesis's central framing, not merely a detail of it. |
| 2 | Zone is confounded with sequencing study; four of six zone pairs share no study at all | Critical | The Coastal–Guinea savanna contrast, which produces the maximum $F_{ST}$, the polyclonality difference and the *pfaat1* S258L cline, has zero within-study replication. Within-zone, between-study heterogeneity in call rate is highly significant ($p = 2.8 \times 10^{-197}$ in Guinea savanna, six studies; $p = 7.7 \times 10^{-88}$ in deciduous forest, four of its five contributing studies), so study-associated heterogeneity is demonstrable rather than hypothetical; |

| # | Limitation | Severity | Effect on what can be concluded |
|---|---|---|---|
| | | | because collection year, finer geography and recruitment also differ between studies within a zone, it is not evidence of a specifically technical batch cause. Three of the four zones are dominated by one study (transitional 100%, coastal 89%, Guinea savanna 74%); deciduous forest is the only zone with no majority study (48%). These figures are computed on the 4,905 Pf8-QC-passing isolates, not the 1,900 analysed (§7.2 note). Partial mitigation: both zone pairs that do share a study reproduce the overall direction of the polyclonality difference. No equivalent within-study check exists for any marker cline (§8.2). |
| 3 | Geolocation only to administrative level 1 (7 distinct coordinates for 1,900 isolates) | Critical | No sub-regional spatial analysis is possible. Isolation by distance rests on 21 pairs derived from 7 points; it is significant in only 4 of 7 leave-one-out replicates, loses significance on dropping any of three of those regions, and is significant in only 2 of 6 threshold scenarios (Table 8.3). It is therefore no longer reported as a supported result (§7.1). This limitation is the explanation for that failure, not a caveat on a surviving finding. |
| 4 | Collection year strongly confounded with zone; the whole sample spans 2009–2021 | High | Between-zone differences are also between-period differences; not separable by reweighting. Every marker frequency in §7.5, including the *pfk13* all-reference result, is a 2009–2021 average and is a historical rather than a current statement. |
| 5 | Only 3,916 SNPs pass MAF ≥ 5% genome-wide | High | Limits power throughout, and the BayPass scan in particular: 19 outliers above the quantile threshold, of which 6 do not survive FDR 10% and 2 more are downgraded on variant-quality grounds (§7.2). Partial reassurance on the headline: at MAF ≥ 0.01 the panel triples to 10,876 SNPs and maximum pairwise $F_{ST}$ falls only to 0.00204 (Table 8.3), so the panel size is not propping up the near-panmixia conclusion. |
| 6 | Transitional zone n = 25 | High | It cannot support ADMIXTURE or TreeMix inference on its own, and its polyclonality figure (25 of 50, on the 4,905 Pf8-QC-passing set) carries no information. One of its three $F_{ST}$ comparisons (vs Guinea savanna) has a confidence interval spanning zero, and it is the only one of the six pairs that does. Its marker frequencies are unstable — the IRN haplotype CI is 0.700–0.958 and the G437 CI 0.609–0.911 on 25 isolates — and it breaks the monotone latitudinal ordering in most variables examined (§7.2), *pfaat1* S258L excepted, where its 0.640 falls correctly between the deciduous forest and the Guinea savanna. It is also entirely supplied by a single sequencing study. Relaxing the $F_{ws}$ cut raises it only to 28, 32 and 44 isolates across the three relaxations (Table 8.3), so no threshold choice available here fixes it. |
| 7 | $F_{ws}$ filter is the largest single exclusion in the pipeline and operates unevenly across zones | Moderate (now quantified) | The retained 1,900 of the 4,905 Pf8-QC-passing isolates are not a random subsample, and the departure differs by zone. The sensitivity analysis has now been run (Table 8.3). For the headline the answer is reassuring: at $F_{ws}$ ≥ 0.90, ≥ 0.80 and with no filter at all, maximum pairwise $F_{ST}$ is 0.00240, 0.00255 and 0.00254 against 0.00232 at the published cut, so near-panmixia does not depend on this exclusion and in fact strengthens very slightly without it — the polyclonal isolates were not masking differentiation. For the spatial result it is fatal: the region-level Mantel test is non-significant at all three relaxations (p = 0.087, 0.169, 0.179), which is the main reason isolation by distance is now graded not robust. The most serious filtering objection is |

| # | Limitation | Severity | Effect on what can be concluded |
|---|---|---|---|
| | | | answered for one conclusion and sustained against the other. |
| 8 | Marker panels built by matching annotation strings are vulnerable to reference-state alleles, and this one was | Moderate (resolved; was critical while undetected) | Identifying markers by matching SnpEff ANN_HGVS_p protein-change strings cannot see a resistance allele that is the 3D7 reference residue, because no ALT change is ever annotated for it, and cannot see a residue reachable only by two changes within one codon. Both failures occurred: *pfdhps* A437G was absent from the panel entirely at a true frequency of 0.897, and *pfcrt* N75E was recorded as a "naming discrepancy" with MalariaGEN rather than as a missed two-SNP residue. Neither produced an error or a missing value — the panel looked complete and self-consistent, and marginal-frequency validation passed. Now resolved by codon reconstruction (§7.5, §8.1), but it is retained here as a limitation because any marker result in the wider literature obtained the same way carries the same exposure, and because the reconstruction panel covers only nine genes (§8.2). |
| 9 | Resistance-marker denominators are per codon and are not 1,900 | Moderate | A mixed call leaves a codon uncalled, so each marker frequency rests on 1,834–1,900 isolates, and haplotype frequencies on the smaller complete-case sets (1,873–1,888). The previous draft asserted 1,900 at every codon, which was wrong. The *pfk13* negative in particular rests on 1,845–1,876 isolates per codon, not on the full sample, so a rare variant present in a handful of uncalled isolates could not be excluded by this panel. |
| 10 | Provenance of the zone polygon layer cannot be established | Moderate | The layer is a verified byte-identical re-host of an anonymous, unlicensed ArcGIS Online item and is not citable as a published dataset. The zone definitions themselves are a hybrid of two different classifications. This does not affect the conclusions but blocks third-party reproduction from public sources, and it is the one outstanding check of this revision that cannot be closed analytically (§8.2, §9.3). |
| 11 | Sampling dominated by Guinea savanna / Upper East (1,334 of 1,900, 70%) | Moderate | The 'national' picture is substantially an Upper East picture — including the national marker frequencies, which are weighted roughly 70 per cent toward the zone with the lowest *pfaat1* S258L and the lowest *pfdhfr* IRN. This is also part of why the isolation-by-distance result failed: dropping Upper East removes significance, dropping Greater Accra (n = 204) or Eastern (n = 34) removes it as well, and the test is significant in only 2 of 6 threshold scenarios (Table 8.3). A distance effect resting on 21 pairs from 7 centroids, 70 per cent of the isolates sitting at one of them, was never likely to survive perturbation. |
| 12 | Four of Ghana's zones represented; Sudan savanna absent | Low | Previously described as a missing end of the ecological gradient. On the polygon layer actually used, Sudan savanna is 293,062 ha — about 1.2 per cent of Ghana's land area — so its absence is better explained as a very small zone that an administrative-level-1 sampling frame was never likely to capture than as a gap in the design. |
| 13 | Stage 3 ordinations computed on the 4,905 Pf8-QC-passing set rather than the 1,900 analysed | Resolved | Figure 6.2 and the Table 6.3 metrics describe 4,905 isolates, which also weakened the one claim to analytical independence in Table 7.1. Stage 3 has now been re-run on the 1,900-isolate analysis set (Table 8.2 row 21, Figure 8.1): silhouette by zone −0.020 against −0.016, trustworthiness marginally higher on all three methods, conclusion unchanged, and the ordinations now describe the same sample as every other result. Removing the polyclonal and clonally redundant isolates makes the zone |

| # | Limitation | Severity | Effect on what can be concluded |
|---|---|---|---|
| | | | separation slightly worse rather than better. The published figures are retained; the re-run outputs carry a _qcset suffix. |
| 14 | Four populations cannot identify more than about two TreeMix migration edges | Continuing (narrowed, not closed) | Previously a limitation on model choice. The edge standard errors narrow it: the third edge is indistinguishable from zero, so m = 2 is reported as the selected descriptive model. They do not close it. The variance-explained criterion applied across the full m = 0-5 sweep still prefers m = 3, so the two criteria disagree and the choice is settled here on identifiability grounds, which is a modelling decision rather than a measurement. Specific gene-flow corridors remain unidentifiable. |
| 15 | Two outlier SNPs carry genotype-level anomalies of unverified cause | Low (local to two SNPs) | Pf3D7_11_v3:1 054 078 and Pf3D7_11_v3:1 054 447, 369 bp apart on chromosome 11, are the two lowest-InbreedingCoeff sites in the panel and the first carries the only nonzero ExcessHet value among all 3,916 SNPs (69.59). Collapsed paralogy or copy-number variation is the most plausible mechanism and mismapping is ruled out by clean MQ and FS, but the mechanism is unverified pending a read-level depth check (§8.2 item 1). Both are reported as lower confidence. This matters mainly because 1054447 is the fifth-ranked SNP in the scan; it affects no population-genetic conclusion. |

*Table 9.1. Limitations ranked by how much they constrain the conclusions. Bold entries are new; items 8 and 9 were new to the previous revision and concern the marker panel. Item 15 is new to this revision; items 3, 5, 6, 7, 10, 11 and 13 are updated by the four checks reported in §8, and item 13 is now resolved.*

**[Interpreted]** **What the threshold-sensitivity analysis does and does not change in this list.** It strengthens the headline and removes a finding, and the two should be stated together. **Near-panmixia no longer rests on a single pass through a single dataset:** maximum pairwise $F_{ST}$ stays within 0.00204–0.00255 across six combinations of the two thresholds that define the analysis set, over sample sizes from 1,900 to 4,403 and SNP counts from 3,275 to 10,876 (Table 8.3). Limitations 5, 6 and 7 — panel size, the transitional zone's 25 isolates and the $F_{ws}$ exclusion — therefore constrain the precision of that conclusion without threatening it, which is a stronger position than the previous draft could claim. **Isolation by distance is in the opposite position.** Significant in 2 of 6 threshold scenarios and 4 of 7 leave-one-region-out replicates, it is not a result this analysis can claim, and limitations 3 and 11 are its explanation rather than caveats on it: 21 pairs derived from 7 administrative centroids, with 70 per cent of the isolates at one of them, cannot support a distance effect that survives perturbation of the dataset that produced it. The manuscript's only positive spatial finding has been lost. No reanalysis of these data will recover it; recovering it would need isolates with recorded collection coordinates (§9.2), and that is a design change rather than an analytical one.

## 9.2 On the design limitations specifically

**[Interpreted]** Limitations 1, 2 and 3 are not analytical shortcomings that better methods would overcome. They are properties of the Pf8 metadata and of how Pf8 was assembled. No reanalysis of these data can produce a zone effect separable from a region effect, because for three of the four zones there is only one region in each; and no reanalysis can separate the Coastal–Guinea savanna contrast from a sequencing-study contrast, because no study contributed isolates to both. This should be stated plainly in any write-up rather than managed with hedging language, and it bears directly on how the thesis's objectives can be framed. Limitation 3 has now produced a concrete casualty rather than a theoretical one: the isolation-by-distance result it constrains did not survive threshold variation (§9.1, Table 8.3).

**[Interpreted]** Limitation 8 is of a different kind and deserves separating from the rest. It was not a property of the data but of the code, it was invisible to every check that looked only at this project's own outputs, and it was corrected only because an external reader asked whether the marker list was the right list. Unlike limitations 1 to 3 it is fixable, and it has been fixed; unlike them it was also, while it lasted, producing confidently stated numbers that were wrong. The appropriate response in a write-up is not a hedge but a statement of what the panel now does and how it was checked against something outside the pipeline (§8.1, rows 14 to 18).

**[Proposed]** The study confounder deserves particular emphasis because it was absent from the previous draft while the other three were disclosed, and because a reviewer will find it in a single crosstab. It should be reported with the study-by-zone table as a supplement, together with the within-zone, between-study tests and an explicit statement that both are computed within the 4,905 Pf8-QC-passing isolates — the crosstab on all 4,905, and the between-study tests on the 4,851 of them that fall in zones with at least two studies of n ≥ 20, which drops the transitional zone (one study, n = 50) and deciduous forest's 4-isolate study entirely. Readers can then see where batch heterogeneity is significant (deciduous forest $F_{ws}$ p = 0.002 across four of its five studies) and where it is not (coastal savanna p = 0.38; Guinea savanna $F_{ws}$ p = 0.10, though its call-rate heterogeneity is extreme at p = $2.8 \times 10^{-197}$).

**[Proposed]** If zone-level inference is essential to the thesis, the route is additional sampling with recorded collection coordinates — ideally within the Sudan savanna, and within regions that straddle a zone boundary, where zone and region could be decoupled — and, critically, with a single sequencing protocol applied across zones so that zone and batch are decoupled as well. That is also the only route to a defensible isolation-by-distance result, since the one obtained here rests on seven administrative centroids and does not survive varying the thresholds that define the dataset. Whether any of it is feasible within the project is a question for the author.

**[Calculated]** The pooled minor-allele-frequency filter imposes an ascertainment floor set by the largest zone. Allele counts are pooled across all 1,900 isolates before the 5% cut is applied, and Guinea savanna contributes 1,334 of them — 70.2% of the sample — against 337 deciduous forest, 204 coastal savanna and 25 transitional. With the variant-missingness ceiling of 10%, a retained site carries at least 1,710 called haploid alleles. A variant private to the transitional zone therefore cannot exceed 25/1,710 = 1.46% and is excluded even if every one of those 25 isolates carries it; a variant at 40% frequency in coastal savanna reaches only 82/1,710 = 4.80% and is excluded as well. The 3,916-SNP panel is accordingly a panel of variants common enough to clear a threshold dominated by one zone, and every differentiation statistic computed on it — the pairwise $F_{ST}$, the three ordinations, the ADMIXTURE fit, the population graph and the Mantel test — inherits that restriction. This does not bias the $F_{ST}$ estimates themselves, which are computed on the sites that were retained and are unaffected by the existence of sites that were not. It bounds their scope. “Negligible differentiation” is a statement about commonly shared variation; it is not evidence against zone-restricted or locally common variants, which this design cannot see at all. The maf001 and maf002 scenarios probe the floor only partly: they raise the panel to 10,876 and 6,852 SNPs and leave maximum $F_{ST}$ at 0.00204 and 0.00217, but both still pool allele counts across zones, so they lower the floor rather than remove it. Removing it would require either frequency filtering within zones or a design in which the zones contribute comparable numbers of isolates.

## 9.3 Reproducibility status

[Observed] **What is reproducible.** Most of the analysis's inputs are retrievable from public data sources, though not from a single one; that availability does not by itself establish that the pipeline as archived regenerates the reported results. All inputs except the zone polygon layer, the PlasmoDB 3D7 annotation and the Pf8 sample manifest are retrievable from https://pf8-release.cog.sanger.ac.uk — specifically the Zarr callset, the SNP-only VCFs, the reference FASTA and Pf8_drug_resistance_marker_genotypes.tsv. The annotation comes from PlasmoDB (config.PLASMODB_GFF = PlasmoDB-71_Pfalciparum3D7.gff), and it is the file the rebuilt marker panel depends on, since step12_codon_panel.py builds its CDS map from it; the manifest comes from malariagen.net (config.SAMPLE_MANIFEST = data/Pf8_samples.txt). Under Route A the core-regions BED (pf_core_regions.bed) must also be supplied by the user, as data/README.md states. Random seeds are fixed (config.RANDOM_SEED = 42; ADMIXTURE -s 42; TreeMix -seed 42; BayPass -seed 42), and the Mantel permutation test is now seeded as well (seed 42, 9,999 permutations), so both of its p-values are reproducible — previously neither was. The sensitivity analysis inherits those seeds and rebuilds each scenario's matrix from the same stored genotype array, so all six scenarios in Table 8.3 are reproducible from one script. Stage 3a is resumable and verifies its own output, and step03_umap.py now takes --samples and --suffix so an ordination can be recomputed on any subset without disturbing the published outputs. The marker panel is deterministic by construction: step12_codon_panel.py reads the CDS, the reference genome and the genotypes and involves no sampling. The stage sequence is driven by the following command, though the exact commands and script revisions used for the reported run are not archived:

```
bash scripts/run_all.sh --from 4 --parallel
```

[Observed] **What has been fixed since the previous draft.** Four defects that made the pipeline unreproducible or silently wrong have been corrected. (i) BayPass was non-deterministic across threads despite its fixed seed: the 8-thread manuscript run and an 8-thread re-run of the same command gave different thresholds and different outlier counts. It now runs single-threaded and is bitwise stable; the earlier multi-threaded outputs are archived rather than overwritten, and the three surviving threshold values are labelled in the note to Table 8.1. (ii) The Mantel test's silent flooring of negative $F_{ST}$ values to zero is now disclosed, the affected pairs are named in the output JSON, and both censored and uncensored statistics are reported — which is how it became visible that the uncensored statistic clears $p < 0.05$ in only one of the six threshold scenarios (Table 8.3). (iii) The "skip if output exists" guards in stages 5b, 7b and 8b did not check whether the input had changed, so when the SNP panel grew from 3,359 to 3,916 the ADMIXTURE and TreeMix stages reused the previous day's results and reported success. Only stage 8 caught it, because it asserts that its position table and the BayPass output have matching row counts. All three guards now require the output to be newer than its input. (iv) The marker panel no longer identifies codons by annotation string. step12_codon_panel.py reconstructs each codon from the CDS and reference genome and emits the full residue spectrum including the reference residue, with a per-codon called count; it is validated call by call against MalariaGEN's published genotypes at the 25 of its 57 codons MalariaGEN publishes per sample (47,012 of 47,012 agree), leaving 32 codons, every pfk13 codon among them, with no independent per-sample check and its *pfk13* list is taken from the WHO compendium v1.0 rather than written by hand. Defects (iii) and (iv) are the most dangerous kind: both produced no error and well-formed output, and in both cases wrong numbers would have been reported as though they were right.

> **Two incidents during this revision's runs, recorded because both bear on data integrity. (i) A results file was destroyed and restored.** results/baypass/xtx_all_snps.csv was found at **0 bytes**, truncated at 08:29:53 — hours after the 01:42 run that produced it, so almost certainly a stray shell redirect where a read was intended during a verification pass. It was the only zero-byte file under results/ or interim/. It was regenerated from the intact BayPass outputs, and the regeneration is consistent with the surviving outputs rather than proven row-for-row:

> outlier_snps.bed, xtx_calibration.json and xtx_threshold.txt are byte-identical by md5 before and after, and the restored file carries the same 3,916 rows and the same 19 outliers. The consequence is in fact positive — the per-SNP q-values are on disk again, so the *pfaat1* q-value of 0.0049 and the tie between the two *AAT1* SNPs are now reproducible from a primary output rather than from a narrative (§7.5 note). But the episode is also the clearest argument this project has produced for version control, which is gap (ii) in the paragraph below: nothing in the setup would have detected the truncation except someone opening the file. **(ii) A log misreported its own output path.** The --suffix patch to step03_umap.py suffixed the file writes but not the print statements, so the stage 3 re-run reported writing to the published dimred_comparison.csv when it had correctly written dimred_comparison_qcset.csv. No data were at risk — verified from the published file's mtime and contents — but a log that misstates where it wrote is exactly what causes a later misattribution, and this is one more instance of the pattern in §8.1: the defect was inside a correction, not inside the original code. Fixed.

[Observed] **What still is not reproducible.** Three gaps remain. (i) **The provenance of data/ghanaAEZ.geojson is identified but not resolved, and cannot be resolved analytically.** The file has been traced with certainty — it is byte-for-byte identical, md5 96b1a7f4da62d6f0153826f41af96152, to an ArcGIS Online item whose licenseInfo and accessInformation are both null and whose owner is an anonymous personal account created two days before the upload. It is an unattributed re-host, not an origin, and it is not citable as a published dataset; its only attributes are id, Name, AREA and PERIMETER, with no author, source or date. The uppercase AREA/PERIMETER pair is an ArcInfo-coverage fingerprint and points to a pre-2005 national coverage. The zone scheme it encodes is a hybrid of the MoFA/FAO savanna classification and a Hall & Swaine-lineage forest split, and matches no published Ghanaian classification. The layer should be cited as accessed rather than as authoritative, with the checksum given for identity, and the zone names reproduced verbatim — including the "Decidous forest" misspelling — because the spatial join keys on them, and because the rebuilt marker outputs carry that spelling in their column headers. Attributing the layer to any named institution would be a fabricated citation. A third party therefore still cannot reproduce the spatial join from public, licensed sources alone. The one unexhausted lead — the Centre for Remote Sensing and Geographic Information Services, University of Ghana, the most likely custodian of the pre-2005 national coverage the file's fingerprint points to — **has not been contacted**: closing this gap requires a written enquiry to that Centre (cersgis@ug.edu.gh), which is why it is the one of this revision's four outstanding checks that remains open (§8, §8.2). That custodianship is an inference and not an established link. (ii) The project is not under version control, so the exact script state at run time is not recoverable other than from the files currently on disk; the unmodified originals are in pf8_ghana_pipeline.zip. The accidental truncation recorded in the note above is a direct consequence of this gap. (iii) UMAP with a fixed random_state is single-threaded and deterministic, but t-SNE has not been verified to be bitwise reproducible across thread counts.

[Unknown] Whether ADMIXTURE returns identical results under a different thread count has not been tested. BayPass has now been tested and does not, which is a reason to expect the same of ADMIXTURE and to test it rather than assume it.

# 10. Conclusion and current state of the research

## 10.1 What has been established

This section has now been rewritten twice and revised a third time. The first rewrite followed an internal adversarial audit; this one follows an external critique of 10 October 2026 whose five points were all upheld, four of them identifying defects in the analysis code rather than in the prose. The defect that matters here is that the first corrected resistance panel was itself wrong: it found markers by matching SnpEff ANN_HGVS_p protein-change strings against a hand-written list, which cannot see a resistance allele that is the reference residue and cannot see a residue reached by two changes in one codon. A new script, scripts/step12_codon_panel.py, reconstructs each codon from the CDS and the reference genome and replaces the panel entirely. The following statements are therefore withdrawn from the previous version and should not be quoted from it: the *pfdhfr* triple mutant as "0.786–0.908"; the claim that *pfk13* was tested at 14 validated codons "in all 1,900 isolates"; the classification of P441L and C469F as validated; the "marker-naming discrepancy" at *pfcrt* codon 75; the validation claim "21 of 23 markers agree within 0.03"; the description of *pfaat1* S258L as "the functionally validated chloroquine-resistance variant"; and the statement that nothing in the functional work tested amodiaquine. The population-genetic results below are unchanged and were read from the corrected run of 10 October 2026 (3,916 SNPs x 1,900 isolates, mean missingness 0.0295); every resistance number below was read from step12's own outputs. A third round followed on the same date and is recorded in Appendix F: the four robustness checks that this section's own next-steps list carried as outstanding were run, and one of them overturned a conclusion instead of confirming it. **Isolation by distance is withdrawn as a supported result and is reported below as not robust.** Near-panmixia is strengthened by the same six scenarios. Nothing in the resistance panel is touched by these checks, and the withdrawals listed above stand exactly as they are.

[Calculated] **Differentiation between agro-ecological zones is very low.** The largest pairwise Hudson $F_{ST}$ is 0.00232 between coastal savanna and Guinea savanna (block-jackknife SE 0.00053, 95% CI 0.00128–0.00336). Five of the six zone pairs have confidence intervals excluding zero; the exception is Guinea savanna against the transitional zone (0.00047, CI −0.00083 to 0.00178), where $n$ = 25 on one side. This is a well-supported result, and it is now demonstrably the most robust one in the manuscript: the corrected run analyses 16.6% more SNPs than the original and the maximum estimate moved only from 0.0024 to 0.00232, and the paragraph below shows it moving by less than 0.0005 across six different settings of the two thresholds that define the dataset.

[Calculated] **Very low, detectable common-variant differentiation survives both thresholds that define the dataset, and that stability is the best-evidenced result the analysis holds. It is a statement about a measured allele-frequency statistic, not about the mating system or the rate of contemporary exchange.** scripts/step13_sensitivity.py loads the genotype matrix once, rebuilds the analysis matrix under six threshold scenarios and recomputes pairwise $F_{ST}$ and the region-level Mantel test for each. The two thresholds varied are the ones that do the most work in the pipeline: the $F_{ws}$ monoclonality cut, which discards 57 per cent of isolates and does so unevenly by zone, and the MAF floor, which discards 99.8 per cent of sites. Across all six scenarios the maximum pairwise $F_{ST}$ ranges only **0.00204 to 0.00255** — a total spread of 0.0005 — over sample sizes from 1,900 to 4,403 isolates and SNP counts from 3,275 to 10,876, and the maximum pair is coastal savanna against Guinea savanna in every one of them. Two secondary readings are worth recording. Relaxing $F_{ws}$ slightly **raises** $F_{ST}$ (0.00232 at $F_{ws}$ >= 0.95, 0.00255 at $F_{ws}$ >= 0.80), so the polyclonal isolates the filter removes were not masking differentiation — the direction a reviewer would most likely suspect is the

opposite of the one observed. And tripling the SNP count by lowering the MAF floor to 0.01 **lowers** it (0.00204), as expected if rarer variants are less differentiated between zones. A conclusion that holds at 1,900 isolates and 3,916 SNPs and also at 4,403 isolates and 10,876 SNPs is being tested rather than assumed.

| Scenario | Isolates | SNPs | max $F_{ST}$ | Mantel *r* | Mantel *p* | Significant at 0.05? |
|---|---|---|---|---|---|---|
| baseline ($F_{ws}$ >= 0.95, MAF >= 0.05) | 1,900 | 3,916 | 0.00232 | 0.4269 | 0.0252 | Yes |
| $F_{ws}$ >= 0.90 | 2,196 | 3,629 | 0.00240 | 0.3183 | 0.0874 | No |
| $F_{ws}$ >= 0.80 | 2,746 | 3,496 | 0.00255 | 0.2556 | 0.1687 | No |
| no $F_{ws}$ filter | 4,403 | 3,275 | 0.00254 | 0.2447 | 0.1792 | No |
| MAF >= 0.01 | 1,900 | 10,876 | 0.00204 | 0.4325 | 0.0617 | No |
| MAF >= 0.02 | 1,900 | 6,852 | 0.00217 | 0.4341 | 0.0467 | Yes, marginally |

*Table 10.1. Threshold sensitivity of the two headline spatial results, from* `results/sensitivity_summary.csv` *(scripts/step13_sensitivity.py, 10 October 2026). The baseline row reproduces the published configuration exactly. Mantel r and p are the censored values, with negative $F_{ST}$ pairs floored to zero, over 21 administrative-region pairs; the uncensored values are in the same file and change no conclusion (baseline p = 0.0413, and no relaxed scenario reaches 0.05 uncensored either). $F_{ST}$ varies by less than 0.0005 across the six scenarios while the Mantel test crosses the 0.05 line four times, which is the whole of the difference between the two results' standing. This table is new in this revision; the research-questions table that carried the number 10.1 in previous versions is renumbered Table 10.2, and the one internal cross-reference to it, in Appendix F, is renumbered with it.*

[Calculated] **Isolation by distance is no longer a supported result.** This is the one population-genetic conclusion that the threshold sensitivity overturned, and the loss is a real one: it was the manuscript's only positive spatial finding. The numbers themselves are unchanged. At the published thresholds the region-level Mantel correlation is $r = 0.4269$ ($p = 0.0252$) with negative $F_{ST}$ pairs floored to zero, and $r = 0.4138$ ($p = 0.0413$) without flooring; three of the 21 region pairs were affected, so on the final matrix the censoring no longer determines significance, although both versions are reported and the disclosure stands. What has changed is what those numbers are worth. **The test is significant in two of the six threshold scenarios** — the published one and MAF >= 0.02 at $p = 0.0467$ — and fails at every relaxation of the $F_{ws}$ cut ($p = 0.0874$, 0.1687 and 0.1792) and at MAF >= 0.01 ($p = 0.0617$). The two thresholds fail it in different ways and the difference is informative. Relaxing $F_{ws}$ changes the correlation itself, which falls from 0.4269 to 0.2447 as the isolate count rises from 1,900 to 4,403; relaxing MAF leaves the correlation essentially untouched (0.4325 at MAF >= 0.01, 0.4341 at MAF >= 0.02) and moves only the permutation *p*-value across the 0.05 line. So the result is sensitive to **which isolates are analysed** in a way that cannot be written off as lost power, and it is marginal at the published MAF floor in a way that a different floor is enough to disturb. Taken with the leave-one-region-out jackknife, which already found it significant in only four of seven replicates — significance lost on dropping Upper East ($n = 1,334$; $r = +0.260$, $p = 0.218$), Greater Accra ($n = 204$; $r = +0.250$, $p = 0.232$) or Eastern ($n = 34$; $r = +0.288$, $p = 0.136$) — the honest grade is **not robust**: it holds at the published thresholds on the complete set of seven regions, and almost nowhere else. At zone level it was never demonstrable at all: $r = 0.4709$ with $p = 0.1218$ over six pairs, none of the four leave-one-out replicates significant, and only 24 distinct permutations available, which is a resolution limit rather than a negative finding. **The result is not deleted and should not be.** A correlation of 0.43 over 21 region pairs that survives neither a change of monoclonality cut nor a change of MAF floor is a statement about this dataset's resolution, not evidence of a distance gradient in Ghanaian *Plasmodium falciparum*, and it is reported here with its sensitivity so that a successor with more populations or finer spatial units can test it properly. **Not robust.**

> **One piece of the previous version's reasoning, kept for the record and no longer load-bearing.** The previous version departed deliberately from the prescribed wording in the corrections document, which reads: "Dropping Upper East still removes significance, so the result remains substantially a single north–south contrast rather than a smooth distance gradient, and should be described that way." The departure was taken because dropping Eastern — neither northern nor heavily sampled, at $n = 34$ — also removes significance, which a single north–south contrast does not predict, and because the other two regions whose removal costs significance are the two largest of the seven (Upper East at 1,334 and Greater Accra at 204, against a median region of 125), so the fragility could not be laid at the door of small sample sizes. That observation stands and is unaffected by the sensitivity analysis. But it was an argument about how to describe a supported result, and the threshold sensitivity has removed the thing being described, so the two descriptions no longer need adjudicating: on the present evidence neither "a single north–south contrast" nor "fragile to the composition of a 21-pair test" is a finding. The flag is kept rather than deleted because the corrections document still prescribes its wording, and a successor should be able to see why the manuscript follows neither it nor the previous version's alternative.

[Calculated] **Ancestry structure aligned to geography is detectable but negligible in magnitude.** The original text asserted that no ancestry structure corresponds to geography. That is false as stated and was never tested. ADMIXTURE's preferred K = 2 improves on K = 1 by 0.54 per cent of cross-validation error (0.83925 against 0.84384, an absolute gain of 0.00459). The minor component's mean differs between zones — 0.085 coastal savanna, 0.092 deciduous forest, 0.053 Guinea savanna, 0.042 transitional zone — and a Kruskal–Wallis test over the four zones gives $H = 105.4$, $p = 1.05 \times 10^{-22}$. Zone nonetheless explains only 1.9 per cent of the variance in that component ($\eta^2 = 0.019$, computed from the final Q matrix). The means are not monotone with latitude: deciduous forest exceeds coastal savanna, and the transitional zone, which is the second-most-northern zone, sits lowest of the four. The substantive conclusion survives in full; the wording does not. The correct statement is that zone-associated ancestry structure is statistically present, non-monotone in latitude, and biologically negligible.

[Calculated] **A strictly bifurcating population tree is inadequate, and a two-edge graph is the model these data select.** With jackknife standard errors enabled, variance explained rises from 0.897 at m = 0 to 0.941 at m = 1 and 0.997 at m = 2. Every edge in the m = 2 graph is significant ($p = 7.5 \times 10^{-4}$ and $3.3 \times 10^{-16}$); the third edge added at m = 3 is not (weight 0.065, SE 0.147, $p = 0.328$), and m = 3 is in any case flagged non-identifiable with four populations. The manuscript previously recorded both m = 2 and m = 3 and left the choice open. m = 2 is now reported as the selected descriptive model — on the third edge's own standard error and on parameter counting — rather than as a count of historical migration events that these data identify. The variance-explained criterion applied across the full sweep still prefers m = 3, so the two criteria do not agree, and choosing between them is a modelling decision taken here on identifiability grounds. Specific gene-flow corridors remain unidentifiable with four populations.

[Calculated] **The ordinations now describe the analysis set, and removing the polyclonal and clonally redundant isolates does not sharpen the structure.** Until this revision the ordination figures described the 4,905 isolates that pass sample-level QC while every other result described the 1,900 that survive the $F_{ws}$ and clonality cuts — a consistency defect rather than an open question, and it is now fixed. `step03_umap.py` takes `--samples` and `--suffix` and re-slices the distance submatrix rather than reusing the cached full-set one, with outputs carrying a `_qcset` suffix so the published figures are untouched. On the 1,900-isolate analysis set the silhouette by zone on raw distances is **−0.020**, against −0.016 on 4,905: slightly more negative, not less. Trustworthiness rises marginally for all three methods (UMAP 0.499 against 0.486, t-SNE 0.526 against 0.507, PCoA 0.510 against 0.495) and the PCoA distance correlation falls (0.287 against 0.322). No ordination recovers zone structure on either sample set, so the conclusion is unchanged; what has changed is that the

figures can now be reported on the same isolates as everything else, and that the obvious objection — that mixed and clonally duplicated infections were blurring real structure — is answered in the negative.

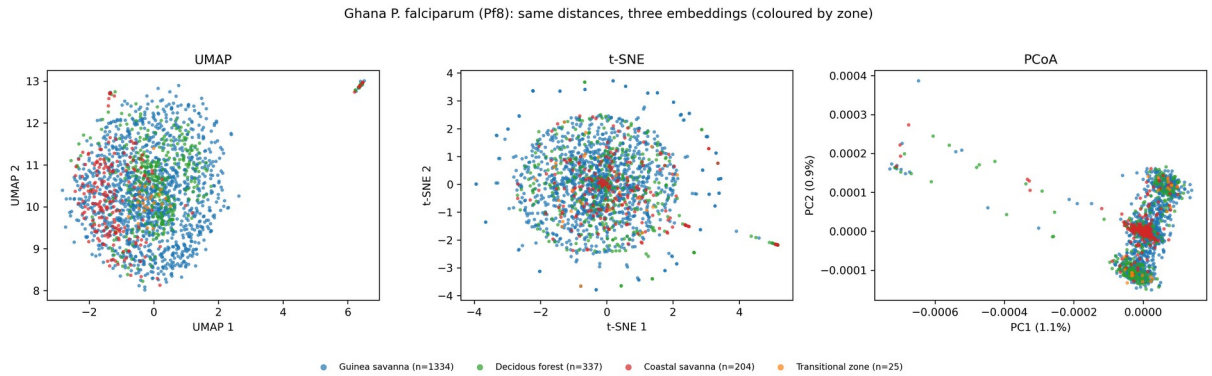



*Figure 10.1. UMAP, t-SNE and PCoA of the 1,900 × 1,900 analysis-set distance submatrix, coloured by agro-ecological zone: the same three ordinations as Figure 6.2, but on the 1,900 isolates analysed everywhere else rather than the 4,905 that pass sample-level QC. Silhouette by zone on the raw distances is −0.020, against −0.016 on the larger set. PCoA axes carry their variance proportions; UMAP and t-SNE axes are in arbitrary units and inter-cluster distances in those two panels carry no quantitative meaning. Generated by* `scripts/step03_umap.py` *with* `--samples` *and* `--suffix _qcset`*; the published Figure 6.2 and its underlying files are deliberately unchanged.*

[Calculated] **Nineteen SNPs exceed a calibrated differentiation threshold, of which 13 also survive false-discovery correction.** The POD 0.999 quantile falls at XtX = 10.840. Nineteen SNPs clear it, 3.9 are expected by construction, and the expected-null-exceedance ratio at the threshold is 20.6 per cent; 13 SNPs also pass Benjamini–Hochberg $q$ <= 0.10. Seventeen of the 19 intersect 13 annotated genes, headed by PTP7 (PF3D7_0301700, max XtX 26.44) with AAT1 (PF3D7_0629500, 24.20) second. The remaining two are intergenic, and one of them — Pf3D7_11_v3:1054447 at XtX 20.79, just outside SNRPF — is the fifth-highest XtX in the genome, so the gene list is not a complete account of the outlier set and should not be presented as one. AAT1 is the only entry carrying a direct and recent resistance-literature connection, and its missense SNP survives Benjamini–Hochberg correction at $q$ = 0.0049. That $q$ does not, however, single the site out: the linked synonymous site 883 bp away carries the identical $q$ = 0.0049, both sitting at the Monte Carlo resolution floor of the POD null, so the $q$-value ranks the pair without discriminating between them. Report the 13 as the finding and the 19 as a ranking. Provisional as a candidate list; the FDR-passing subset is better than provisional. The 19 have now been checked against the callset's own quality annotations, SNP by SNP, against the other 3,897 scanned SNPs (`results/outlier_quality_check.csv`), and the outcome is favourable but not uniform. **Every significant difference runs in the outliers' favour**: higher VQSLOD ($p$ = 0.008), QD ($p$ = 0.016) and QUAL ($p$ = 0.007). None has MQ < 50, none shows strand bias or read-position bias, and all 19 are RegionType Core. The `culprit` field reads MQRankSum for all 19, but also for 99.6 per cent of the rest of the panel, so it carries no information here. **The exported-protein artefact concern is not supported**: the eight outliers in PTP7, PIESP2, GEXP22, REX2, PF3D7_0831400 and PF3D7_0221200 are the cleanest subset in the panel — FS = 0.00 for all eight, MQ median 60.00, VQSLOD significantly higher than the rest. Those families are repetitive, but these particular sites lie in uniquely mappable sequence. Two SNPs are nonetheless downgraded, for an entirely different reason, and the note below gives it.

> **Two chromosome-11 outliers are downgraded to lower confidence.** Pf3D7_11_v3:1 054 078 (*SNRPF*, XtX 13.98, $q$ = 0.031) and Pf3D7_11_v3:1 054 447 (intergenic, XtX 20.79, $q$ = 0.005) lie **369 bp apart** and are the two lowest-InbreedingCoeff sites in the whole panel (0.147 and 0.193, at the 0.08th and 0.13th percentiles of the 3,916). 1054078 carries **ExcessHet = 69.59, the only nonzero value among all 3,916 SNPs** and roughly 400 times the genome-wide maximum. Mapping quality and strand bias are clean at both sites (MQ 59.99 and 59.95, FS 0.00 and

> 0.60), so whatever this is, it is not mismapping. It matters for two reasons: 1054447 is the **fifth-ranked SNP in the entire scan**, and both sites sit inside the 13 that pass Benjamini–Hochberg *q* <= 0.10 — so that subset should now be read as 11 SNPs of ordinary confidence plus these two, and any figure quoting "13" should carry the qualification with it.

[Hypothesized] **What the chromosome-11 pair probably is.** An excess of heterozygous calls of that magnitude in a near-haploid organism points to collapsed paralogy or copy-number variation — two or more diverged copies piling onto one reference locus, which produces apparent heterozygosity, a depressed InbreedingCoeff and inflated between-population variance without any degradation of mapping quality or strand balance. That is a hypothesis and not a result: nothing in the quality table distinguishes it from the alternatives, and no read-level evidence has been examined. A per-sample depth ratio against the genome median at the two positions would settle it in one pass, and it has not been run (Section 10.4, item 1). Until it is, both sites carry the downgrade.

[Calculated] **Objective 3 is answered, not null — and this is the second attempt at answering it.** The original abstract and Section 6.8 reported that no validated antimalarial resistance locus appeared among the outliers and interpreted that absence biologically. That absence was an ascertainment artefact of the genome-wide callset, which is built under `FILTER_PASS & is_snp & numalt == 1`; in Pf8 most canonical resistance codons are multiallelic or carry a spanning-deletion ALT, and the mask dropped 28 of the 44 annotation records the first panel assembled before any analysis ran. It did not drop all of them, and the manuscript should not say so: 16 records survived the mask, including six *pfk13* alleles and *pfmdr1* Ser1034Cys, Asn1042Asp and Asp1246Tyr. That diagnosis of the scan's silence stands unchanged. What did not stand was the replacement panel built to answer the objective, and the figures below come from its replacement in turn.

[Calculated] **The panel as it now stands.** `step12_codon_panel.py` maps each requested codon to its three genomic positions through the CDS, collects every variant record at those positions regardless of `numalt` or `is_snp`, applies each isolate's alleles to the reference codon read from the genome, and translates. It reports the **full residue spectrum at every codon, the reference residue included**, so a resistance allele cannot hide in the reference and a residue requiring two changes in one codon cannot be missed; a mixed call makes a codon uncalled rather than being assigned a residue. The panel covers **57 codons across all nine panel genes** — 171 genomic positions, 125 variant records, every codon carrying at least one record — in the same 1,900 isolates. Of the 57 codons, **24 are polymorphic and 33 are fixed at the reference residue**. Two statements from the previous version are retired by this: that `pfcoronin` "contributed no canonical codon" (it contributes three, G50, R100 and E107, all fixed at the reference residue), and the "44 canonical codon-allele records" count, which was a count of annotation records and not of codons. The scan-side ascertainment cascade is unaffected and is stated once here: of the panel's 171 genomic positions only **four** ever reached the MAF-filtered scan matrix — *pfdhfr* Cys59Arg (XtX 9.11, *q* = 0.43), *pfdhfr* Asn51Ile (XtX 8.15, *q* = 0.67), *pfaat1* Ser258Leu (XtX 24.20, *q* = 0.0049) and, fourth, Pf3D7_07_v3:403623, the second of *pfcrt* codon 75's two variant positions (BayPass MRK 1406, M_XtX 4.18, XtXst 6.21 against the threshold 10.840; M_P 0.914, so MAF 0.086 cleared the 0.05 cut). The previous count of three was carried over from `step10`'s annotation-matched panel, which held 403621 but never 403623 at all — a residual instance of the very two-SNP-codon failure mode the rebuild was written to fix. Three more passed the ingest mask and were then cut by MAF >= 0.05: *pfcrt* Ile356Thr at 0.016, *pfmdr1* Asp1246Tyr at 0.011 and *pfcrt* Asn326Ser at 0.0005 — the last of which the previous version reported as 0.008, which was also wrong. The distinction worth drawing explicitly therefore survives, with its scope corrected: the scan's silence *at* pfdhfr *51 and 59 and at* pfcrt *75 is a tested negative, because those were the codons in the matrix and all three came back firmly null, whereas its silence on* pfmdr1, pfdhps *and* pfk13 *is an ascertainment artefact, because it contributes no scanned codon position*

*and never tested them at all.* pfcrt *does not belong on the artefact side: codon 75 was tested. Nor can "the antifolate codons" stand as a blanket term here — only two of the five* pfdhfr *codons reached the matrix, none of the six* pfdhps *codons did, and* pfdhps is itself an antifolate target.

> **Why the previous panel missed markers, stated plainly.** `step10` identified markers by matching SnpEff protein-change strings. Two failure modes followed, and both occurred. **(i) A resistance allele that is itself the reference residue can never appear as an ALT change.** At *pfdhps* 437 the only annotation is `Gly437Ala` — the revertant — because, in the official Pf8 methodology's own words, *"3D7 sequence translates as a G at 437, this is thought to be the mutant allele, i.e. 3D7 does not have the wild type allele."* Searching for "Ala437Gly" matched nothing, so **A437G was absent from the panel entirely**, at a true frequency of 0.897. **(ii) A residue reachable only by two changes in one codon is never annotated.** *pfcrt* codon 75 carries two variants, 403621 (A>G, annotated `Asn75Asp`) and 403623 (T>A, annotated `Asn75Lys`); Asn→Glu requires both, so no annotation names N75E, the residue the CVIET haplotype actually carries. This, and not a "naming discrepancy", is the whole explanation of the apparent disagreement with MalariaGEN that the previous version reported.

**[Calculated]** *Positive finding one:* pfdhps *Gly437 is at high frequency and is zone-structured — and was wholly absent from the previous panel. The resistance residue at this codon is the 3D7 reference residue, which is why string-matching never saw it. Reconstructed, G437 sits at 0.897 overall (*n *= 1,892): 0.964 in deciduous forest (95% CI 0.939–0.980), 0.941 in coastal savanna (0.900–0.966), 0.875 in Guinea savanna (0.856–0.892) and 0.800 in the transitional zone (0.609–0.911), with chi-square* p = 1.2 x $10^{-6}$. The revertant residue A437 is correspondingly at 0.103. This is the single largest substantive addition to the resistance results, and the previous version of this section reported nothing at this locus at all.

**[Calculated]** *Positive finding two: the* pfdhfr *triple mutant is at 0.771, not near fixation, and the previous figure was not a haplotype frequency. The previous version gave the triple mutant as "0.786–0.908", which is the range of its three marginal codon frequencies and is not a co-occurrence statement. Counted within isolates on the 1,876 called at all three codons, the IRN haplotype (N51I+C59R+S108N) is 0.771 (95% CI 0.751–0.789). As it must, it falls below the smallest of its components on the same complete-case set (51I 0.788; 59R 0.887; 108N 0.908). It is also not a single near-fixed haplotype: NRN sits at 0.116, the fully wild-type NCS at 0.091, ICN at 0.018 and NCN at 0.004. By zone, IRN is 0.867 in coastal savanna (*n *= 203), 0.788 in deciduous forest (*n *= 330), 0.750 in Guinea savanna (*n *= 1,318) and 0.880 in the transitional zone (*n *= 25). The quadruple IRNL is 0.000 (*n *= 1,874; CI 0.000–0.002), with I164L absent from every called isolate. The marginal codon frequencies are 0.786 for Asn51Ile (*n *= 1,891), 0.886 for Cys59Arg (*n *= 1,888) and 0.908 for Ser108Asn (*n *= 1,889); all three differ across the four zones (chi-square p = 2.4 x $10^{-4}$, 4.8 x $10^{-5}$ and 9.4 x $10^{-7}$) and all three are lowest in Guinea savanna. No directional contrast was tested and none should be claimed: those are four-zone heterogeneity tests, and for both Cys59Arg and Ser108Asn the highest frequency is the transitional zone at 1.000 on* n *= 25.* pfdhps *Lys540Glu (0.009) and Ala581Gly (0.014) remain rare, and the* pfdhps *437+540 double GE is 0.010 (*n *= 1,888)* against GK 0.887 and AK 0.103 — so the sulfadoxine-resistance picture is a high-frequency single substitution, not a high-frequency double.

**[Calculated]** *Positive finding three:* pfaat1 *Ser258Leu forms a coastal-to-northern cline. The variant at Pf3D7_06_v3:1215233 is a missense change of moderate predicted impact. Its frequency runs 0.824 in coastal savanna, 0.735 in deciduous forest, 0.640 in the transitional zone and 0.544 in Guinea savanna, against 0.609 overall (*n *= 1,896; chi-square* p *= 1.9 x 10-18). It is independently the second-ranked outlier in the genome-wide scan at XtX 24.20,* q *= 0.0049. The original manuscript carried this locus as an anonymous amino acid transporter in a table while simultaneously reporting that no validated resistance locus had been found. One further reading from the new panel bears directly on how much weight the functional literature can carry here: the second* pfaat1 *codon in the panel, Phe313, is fixed at the reference residue in all 1,900 isolates* — so

F313S, the variant with which S258L shows epistasis in the published experiment, is absent from this sample entirely.

> **One signal, not two.** The second AAT1 outlier, at Pf3D7_06_v3:1214350, is a synonymous variant 883 bp away (XtX 16.54, and the same $q$ = 0.0049). The gene's "two outlier SNPs" are one functional signal plus a linked synonymous site, and must not be presented as two independent lines of evidence.

[Interpreted] **The functional interpretation of S258L in the previous version was wrong, and the error was ours.** It described S258L as "the functionally validated chloroquine-resistance variant" and asserted that the functional work did not test amodiaquine, then built a Ghanaian amodiaquine-pressure argument on that gap. Checked against the source, every step of that is unsafe. Amodiaquine **was** tested: *"Parasite IC50 values for CQ, amodiaquine, lumefantrine, mefloquine and quinine were measured"*, and the shifts were *"well below the threshold associated with clinical resistance"*. The authors explicitly reject the other-quinoline reading — the alternative hypothesis *"is not supported"* — and conclude that chloroquine was *"the primary selective force"*. The effect itself is modest and conditional: a **1.5-fold** rise in chloroquine IC50, in a single chloroquine-resistant Southeast Asian line (NHP4026), with epistasis between S258L and F313S and an explicit *pfcrt*-background caveat. The longitudinal 0% to 97% rise is **Gambian, 1984–2014**; the paper says nothing about Ghana. The defensible description is therefore that S258L is a variant with a demonstrated but **modest, epistasis- and background-dependent** effect that *potentiates* chloroquine resistance, and that it is a **strong candidate** for the Ghanaian cline rather than an explanation of it. On what maintains the cline, three possibilities are offered and none is tested by these data: historical chloroquine pressure persisting after the 2005 switch to artesunate–amodiaquine; the reported fitness cost of S258L, which would predict a cline rather than fixation or loss; and contemporary amodiaquine pressure, which must be labelled an **untested possible** driver, since the one experiment that measured amodiaquine found the shift sub-threshold and the one parasite line tested (NHP4026) does not carry a Ghanaian genetic background, and the effect it showed was conditional on epistasis with F313S — a partner residue absent from every one of the 1,900 isolates here. Nothing is claimed about NHP4026's own F313S genotype: no source available to us states it, and since NHP4026 is itself a Southeast Asian line and the hypothesis the authors reject concerns high F313S frequencies in Southeast Asia, the sources point the other way. Nothing in these data identifies the drug.

[Calculated] **The chloroquine haplotype itself has largely reverted, and it is genuinely a haplotype.** Reconstructed within isolates on the 1,873 called at all five codons, only **two** *pfcrt* 72–76 haplotypes occur: **CVMNK at 0.929 and CVIET at 0.071 (95% CI 0.060–0.084)**, with nothing intermediate. By zone CVIET is 0.054 in coastal savanna ($n$ = 204), 0.104 in deciduous forest ($n$ = 335), 0.064 in Guinea savanna ($n$ = 1,310) and 0.125 in the transitional zone ($n$ = 24). The previous version inferred that the set travelled together from six similar marginal frequencies; it is now demonstrated. The motif's own variant positions carry the haplotype frequency exactly, as they must: Met74Ile, Asn75Glu and Lys76Thr all at 0.071. The **flanking** CVIET-associated residues, which are external to the 72–76 motif and so corroborate it rather than restate it, agree closely: Ala220Ser at 0.070, Gln271Glu at 0.074 and Arg371Ile at 0.076. Codon 75 is **Glu**, as MalariaGEN has it; the previous version's "naming discrepancy" was our annotation-matching failure, not a disagreement between pipelines, and the note in Appendix F has been rewritten accordingly.

[Observed] **Clean negative: no artemisinin-resistance residue anywhere in the panel.** The *pfk13* panel was rebuilt from the WHO compendium of molecular markers, v1.0 (December 2025), after the previous list — written from memory — was found to contain 13 distinct changes rather than the claimed 14, to misclassify **P441L and C469F as validated when both are candidate**, and to omit P574L, G533S and R622I; R622I was requested but never retrieved, so it had become a silent non-result. The panel now covers **all 14 WHO-**

**validated markers (F446I, N458Y, C469Y, M476I, Y493H, G533S, R539T, I543T, P553L, R561H, P574L, C580Y, R622I, A675V), all 6 candidates (E252Q, P441L, C469F, N537I, G538V, V568G)** — 20 variants which, because C469Y and C469F share codon 469, occupy **19 distinct codons** — plus four further potential-tier codons in the same region (449, 481, 515, 527), giving **23 codons in all** (19 + 4). The two WHO tiers are kept distinct throughout: the six candidates are candidates, not validated. **Twenty-two of the 23 codons are fixed at the reference residue, and the twenty-third, codon 458, sits at 0.9995**: of its 1,866 called isolates 1,865 carry Asn and one carries **Asp** — which is not N458Y, the validated residue. No validated, candidate or potential *pfk13* resistance residue was observed in a single isolate. Two caveats belong in the same breath as the result. The denominator is **per codon, 1,845 to 1,876 of the 1,900 isolates called**, not "all 1,900 at every codon" as the previous version said, so between 24 and 55 isolates are uncalled at each codon. And the sampling period is **2009–2021**, so this is a historical negative and not a statement about the present.

> **WHO tiers apply to variants, not to genes.** The compendium classifies individual variants. Three consequences for this panel. First, it contains **no validated or candidate lumefantrine marker** at all — verbatim: *"The review did not identify any validated or candidate molecular markers associated with lumefantrine resistance."* A label such as "*pfmdr1*, WHO-validated, lumefantrine" is a category error twice over and does not appear in this revision. Second, *pfaat1*, *pfap2mu* and *pfmdr2* appear nowhere in the compendium at any tier. *They are retained in the panel because they are scientifically interesting, and S258L is the strongest candidate the scan produced, but none of the three may be described as a WHO-classified marker. Third,* pfcoronin *is in the compendium at the potential tier only; its three panel codons are fixed at the reference residue here. In this sample* pfmdr2 *Ile492Val runs at 0.549 and* pfap2mu Ser160Asn at 0.096, and neither carries any WHO standing.

[Observed] **Per-sample concordance replaces the previous validation, which did not establish what it claimed.** The previous version validated the panel by saying that marginal frequencies agreed with MalariaGEN's published values "within 0.03" at 21 of 23 markers. Agreement of marginals does not show that two pipelines called the same isolates the same way, and it concealed the codon-75 problem rather than exposing it. Compared call by call on the isolates where both pipelines made a call, **47,012 of 47,012 calls agree — 100.0000 per cent across 25 markers** (nine *pfcrt* codons, five *pfmdr1*, five *pfdhfr*, five *pfdhps* and one *pfmdr2*), with per-marker denominators from 1,834 to 1,900. Codon 75 now agrees exactly, because reconstruction yields Glu where string-matching yielded Asp. The limit of this check must be stated with it: **32 of the panel's 57 codons have no independent per-sample corroboration at all.** Twenty-nine of those 32 are accounted for by the four panel genes MalariaGEN publishes no per-codon genotype for — every *pfk13* codon (23), both *pfaat1* codons, all three *pfcoronin* codons and *pfap2mu* 160. The remaining three sit in genes MalariaGEN does publish, but it publishes no column for them: *pfcrt* **73** (concordance covers nine of the panel's ten *pfcrt* codons), *pfdhps* **431** (five of six) and *pfmdr2* **492** (one of two). The log records no MalariaGEN column for these three and gives no reason. *pfmdr2* 492 in particular is therefore an unchecked panel result, which matters because its frequency (0.549) is quoted two paragraphs below.

[Interpreted] *What the artemisinin negative is now worth, and what the* kelch13 *cross-check is worth. The result is a tested negative across the complete WHO-validated set rather than an untested absence, and that is the change that matters. Calling it the most policy-relevant result the analysis produces remains an assessment and not a finding, and three qualifications now attach to it. Surveillance for artemisinin resistance was not among the manuscript's stated objectives. The sampling period is 2009–2021. And the corroboration from MalariaGEN's own kelch13 calls is much weaker than the previous version implied: Pf8 assigns a status over the whole 349–726 domain and classifies any missing call in that range as undetermined, so only 36 isolates have a complete published status* — under 2 per cent of the sample — which is weak corroboration for per-codon calls on roughly 1,860 isolates. The two must not be presented as equivalent evidence. Among those 36, the commonest observed change is Ala578Ser, which is explicitly non-resistance.

[Calculated] **Mixed-infection frequency does not order monotonically along the north–south gradient.** In geographic order from north to south the polyclonal fraction is 0.597 in Guinea savanna (2,153 of 3,605), 0.500 in the transitional zone (25 of 50), 0.539 in deciduous forest (439 of 815) and 0.444 in coastal savanna (193 of 435). The transitional zone is the second-most-northern zone, lies north of the deciduous forest, breaks the trend, and at 25 of 50 carries no information. The original text quoted only the two endpoints. The gradient is also confounded with sequencing study, and the demonstration is in **deciduous forest**, not Guinea savanna: of the five studies contributing to deciduous forest, the four large enough to test (the fifth has 4 isolates) differ in $F_{ws}$ — Kruskal–Wallis H = 14.9, $p$ = 1.9 x $10^{-3}$, $n$ = 811 — and their polyclonal fractions span 0.430 to 0.578. The $p$-value belongs to the $F_{ws}$ distributions; no test of the polyclonal fraction itself exists in the outputs, and the 7.7 x $10^{-88}$ figure printed beside deciduous forest in the corrections document is its call-rate test, not its polyclonality. Guinea savanna is the wrong zone to cite here: its six studies span 0.532 to 0.682 but do not differ significantly (p = 0.103), so it is one of the two testable zones in which this between-study heterogeneity is not detected — coastal savanna is the other (two studies, n = 435, p = 0.378) — and the transitional zone, with a single study, cannot be tested at all. Of the three testable zones the heterogeneity is detected in exactly one and is not detected in two.

[Interpreted] These fractions are a by-product of quality control rather than an analysed result. With one zone non-monotone, one zone uninformative at $n$ = 25, and between-study variation inside a single zone of the same order as the between-zone variation, mixed-infection frequency is not safe to present as the clearest ecological signal in the data.

> **Which isolate set these numbers describe.** The polyclonality fractions, the study-by-zone composition and the within-zone batch tests in this section are computed on the **4,905** isolates that pass sample-level QC — the set on which sequencing-study membership and $F_{ws}$ are defined — and not on the **1,900**-isolate analysis set behind every $F_{ST}$, Mantel, ADMIXTURE, TreeMix, BayPass and codon-panel number above. That is why "the transitional zone, 25 of 50" appears here against "the transitional zone, $n$ = 25" elsewhere. Within the codon panel there is a further and smaller denominator distinction that must be kept: per-codon frequencies are over the isolates called **at that codon** (1,834–1,900 depending on the codon), while a haplotype frequency is over the isolates called at **every** constituent codon (1,873–1,888), which is why 51I is 0.786 as a marginal and 0.788 on the triple-mutant complete-case set. Study nesting within zone is a property of the Pf8 sampling frame and is not created by the later cuts, but the 4,905-set counts are not the analysis-set counts, and the zone-pair study overlap was **not** recomputed on the 1,900 set. The denominator is stated in full at Section 2.6 and cross-referenced from Sections 4.2.2, 6.2, 7.2 and 7.3.

[Interpreted] **What this implies for surveillance design, stated narrowly.** Neutral common-variant differentiation between zones is negligible, so a spatially sparse design is adequate for characterising neutral structure. The withdrawal of the isolation-by-distance result reinforces that narrow claim rather than complicating it: on the present evidence there is no robust distance gradient for a denser neutral design to resolve either. Marker-frequency representativeness is a separate question, and it is the question sentinel surveillance actually answers; on that question this analysis finds significant between-zone variation at *pfdhps* 437 ($p$ = 1.2 x $10^{-6}$), *pfdhfr* ($p$ <= 2.43 x $10^{-4}$ at all three triple-mutant codons; the largest of the three is 2.425 x $10^{-4}$ at Asn51Ile) and *pfaat1* ($p$ = 1.9 x $10^{-18}$). The original Section 7.4 conclusion — that sentinel surveillance at a small number of sites is an adequate design for detecting emergence — is therefore withdrawn in its unqualified form, here and in Section 7.4. This is the clearest case in the manuscript of a conclusion that the resistance panel reverses, and the two sections must not disagree.

> **Robustness, where it stops, and the one conclusion it removed.** The population-genetic conclusions did not change when the SNP panel grew by 16.6 per cent, the BayPass scan became reproducible and false-discovery accounting was added — and **one of them has now been tested against the two thresholds that define the dataset and did change**. What survived: maximum $F_{ST}$ stayed within 0.00204–0.00255 across six threshold scenarios spanning 1,900

> to 4,403 isolates and 3,275 to 10,876 SNPs; the ADMIXTURE cross-validation curve stayed near-flat with a negligible K = 2 improvement; re-running the ordinations on the analysis set left the silhouette slightly more negative rather than less; and the 19 outliers came through a quality check with every significant difference in their favour, though two chromosome-11 sites are downgraded on an ExcessHet and InbreedingCoeff signal the check itself surfaced. What did not survive: isolation by distance, significant in two of six threshold scenarios and four of seven region jackknife replicates, now reported as not robust. Near-panmixia and isolation by distance were graded together in the previous version, both under "robust to the largest correction the pipeline received"; they are not the same kind of result and the same six scenarios separate them cleanly. The resistance results are a third case and should not be described as robust in the same sentence as either: the first correction created the finding and the second changed its content. The triple mutant moved from an implied 0.786–0.908 to a measured 0.771, a locus at 0.897 appeared that had been absent altogether, two *pfk13* variants were reclassified, a marker discrepancy dissolved into an annotation bug, and the central functional interpretation was reversed. Two successive corrections in the same place is itself a result about the process, recorded in Appendix F. The threshold sensitivity was not run on the codon panel, so nothing is claimed about the resistance frequencies' behaviour under other thresholds.

## 10.2 Answers to the research questions

| Question (Section 1.3) | Answer | Confidence |
|---|---|---|
| Is between-zone differentiation distinguishable from zero? | Yes for five of six pairs. The sixth, Guinea savanna against the transitional zone (n = 25), has a confidence interval spanning zero. The maximum is 0.00232 and none of the six is practically significant. The magnitude is insensitive to both thresholds that define the dataset: max FST ranges only 0.00204-0.00255 across six Fws and MAF scenarios spanning 1,900-4,403 isolates and 3,275-10,876 SNPs (Table 10.1), with coastal savanna against Guinea savanna the maximum pair in all six. Relaxing Fws raises FST slightly, so polyclonal isolates were not masking differentiation. | High, and now threshold-tested |
| Does genetic distance increase with geographic distance? | Not robustly, and this answer is a downgrade. At the published thresholds, yes at administrative-region level (r = 0.4269, p = 0.025 with negative pairs floored; r = 0.4138, p = 0.041 without) and not demonstrable at zone level (r = 0.4709, p = 0.12, 0 of 4 jackknife replicates significant, only 24 distinct permutations available). But the region-level test is significant in only two of six Fws and MAF threshold scenarios - the published one and MAF >= 0.02 at p = 0.0467 - and in only four of seven leave-one-region-out replicates, with significance lost on dropping Upper East (n = 1,334), Greater Accra (n = 204) or Eastern (n = 34). Relaxing the Fws cut drops r itself from 0.4269 to 0.2447; relaxing MAF leaves r at about 0.43 and moves only p across the 0.05 line. The result holds at the published thresholds on the full set of seven regions and almost nowhere else, and is reported as a limit on this dataset's resolution rather than as evidence of a distance gradient. It is not deleted; it is regraded. | Not robust (significant in 2 of 6 threshold scenarios and 4 of 7 region jackknife replicates); previously graded Moderate |
| Do isolates assign to zone-associated ancestry components? | Detectably yes, negligibly so. The K = 2 minor component differs across zones (Kruskal-Wallis H = 105.4, p = 1.05 x 10-22) but zone explains 1.9% of its variance (eta-squared = 0.019), and K = 2 improves on K = 1 by 0.54% of cross-validation error. The zone means are not monotone with latitude: coastal 0.085, deciduous forest 0.092, Guinea savanna 0.053, transitional zone 0.042. | High |
| Is there gene flow beyond a bifurcating tree? | Partly. m = 2 is the selected descriptive model: its two edges are significant (p = 7.5 x 10-4 and 3.3 x 10-16), a third is not (weight 0.065, SE 0.147, p = 0.33), and m = 3 is non-identifiable with four populations. That does not identify how many migration events | High that a bifurcating tree is inadequate; m = 2 is a selected descriptive model rather than a |

| Question (Section 1.3) | Answer | Confidence |
|---|---|---|
| | occurred, and the variance-explained criterion across the full sweep still prefers m = 3, so the two criteria disagree. Specific corridors remain unidentifiable. | settled edge count; corridors not identifiable |
| Which loci exceed neutral expectation? | Nineteen SNPs above the calibrated threshold (XtX >= 10.84), 13 of which also pass FDR 10%; 17 intersect 13 annotated genes and two are intergenic, one of those being the fifth-highest XtX genome-wide (20.79). Headed by PTP7 (26.44), with AAT1 second (24.20, q = 0.0049) and the only entry with a direct resistance-literature connection; AAT1's linked synonymous site shares that q, so it ranks the pair without discriminating within it. The callset's quality annotations favour the 19 over the rest of the panel (higher VQSLOD p = 0.008, QD p = 0.016, QUAL p = 0.007; no MQ < 50, no strand or read-position bias, all RegionType Core), and the exported-protein artefact concern is not supported - the eight exported-protein SNPs are the cleanest subset in the panel. Two chromosome-11 sites 369 bp apart are downgraded to lower confidence on ExcessHet and InbreedingCoeff: Pf3D7_11_v3:1054447 (intergenic, q = 0.005, fifth-ranked in the scan) and Pf3D7_11_v3:1054078 (SNRPF, q = 0.031, ExcessHet 69.59, the only nonzero value in the panel). The FDR-passing subset therefore reads as 11 SNPs plus those two, and the mechanism behind the downgrade is unverified. | Moderate for the FDR-passing subset, two of whose 13 are downgraded; low for the remainder; the artefact objection is now tested and cleared |
| Do resistance-marker frequencies differ between agro-ecological zones? | Yes, and the panel behind this answer has now been rebuilt twice. pfdhps G437 is 0.897 overall and zone-structured (0.964 deciduous forest, 0.941 coastal savanna, 0.875 Guinea savanna, 0.800 transitional; p = 1.2 x 10-6) - a locus absent from the previous panel entirely, because the resistance residue is the 3D7 reference residue. pfaat1 S258L runs 0.824 coastal to 0.544 Guinea savanna (p = 1.9 x 10-18, n = 1,896). The pfdhfr triple-mutant haplotype IRN is 0.771 on 1,876 complete cases (CI 0.751-0.789), not the 0.786-0.908 marginal range previously quoted, with NRN 0.116 and wild-type NCS 0.091; its component codons differ across zones at p <= 2.43 x 10-4 (the largest of the three being 2.425 x 10-4 at Asn51Ile). pfcrt CVIET is 0.071 and is genuinely a unit: only CVMNK (0.929) and CVIET occur. All 23 pfk13 codons - all 14 WHO-validated and all 6 candidate variants, which share codon 469 and so occupy 19 codons, plus four potential-tier codons - are fixed at the reference residue (codon 458 at 0.9995, its single alternative call being Asn458Asp and not the validated N458Y) on 1,845-1,876 called isolates per codon, over 2009-2021. This question was originally reported as null; the null was a filter artefact, and the first correction of it was itself incomplete. | High for the frequencies, and 47,012/47,012 per-sample agreement with MalariaGEN at the 25 markers it publishes; moderate for attributing variation to zone rather than to region, year or study; unverified externally at the other 32 codons |

*Table 10.2. Research questions and the answers the current evidence supports. This table carried the number 10.1 in previous versions and is renumbered because the threshold-sensitivity table in Section 10.1 is now Table 10.1; the one internal cross-reference, in Appendix F, Table F.2, is renumbered with it. The distance row is the only answer the robustness round of 10 October 2026 changed, and it changed downwards, from "Moderate" to not robust. The final row replaces the clause "No validated resistance locus among them", which the original table carried, and also replaces the first corrected version of that row, whose triple-mutant range, pfk13 denominator and pfk13 variant classification were all wrong.*

## 10.3 What remains unresolved

- Whether any apparent zone effect survives control for administrative region, collection year and sequencing study. Not resolvable with these data (Section 9.2). Study is a fourth confounder added since the original draft: four of the six zone pairs share no sequencing study at all, including coastal savanna

against Guinea savanna, the pair behind every headline number. That overlap is computed on the 4,905 QC-passing isolates and was not recomputed on the 1,900-isolate analysis set (see the denominator note in Section 10.1).

- Whether the conclusions are stable under alternative $F_{ws}$ and MAF thresholds is **no longer unresolved**. It was tested on 10 October 2026 (`scripts/step13_sensitivity.py`, Table 10.1) and answered in both directions: near-panmixia is stable, with max $F_{ST}$ ranging only 0.00204–0.00255 over six scenarios, and isolation by distance is not, being significant in two of the six. The item is retained in this list only to record that it was the open question whose answer cost the manuscript a finding, and that the finding it cost was the spatial one.
- Whether Ghanaian *Plasmodium falciparum* shows isolation by distance at all. This has moved out of the answered column and into this one. A Mantel correlation that appears at one setting of the two thresholds and vanishes at four others is not evidence either way, and 21 pairs from seven administrative regions is a thin test on which to rest a spatial claim. Settling it needs more populations, finer spatial units, or a method that does not require populations to be defined at all, and none of those is available from Pf8's Ghanaian metadata as it stands (Section 10.4, item 5).
- Whether the outlier positions are alignment or assembly artefacts is now largely settled, and one part of it is not. Mapping quality and the rest of the callset's annotations have been checked at all 19 positions: none has MQ < 50, strand bias or read-position bias, all are RegionType Core, every significant difference from the other 3,897 scanned SNPs runs in the outliers' favour, and the exported-protein worry is specifically not supported — the eight exported-protein SNPs are the cleanest subset in the panel. What remains open is the **mechanism at the two chromosome-11 sites 369 bp apart**, `Pf3D7_11_v3:1 054 078` and `Pf3D7_11_v3:1 054 447`, where ExcessHet = 69.59 at the first and the panel's two lowest InbreedingCoeff values at both point to collapsed paralogy or copy-number variation rather than to mismapping. A read-level depth check would settle it and has not been run; until it is, both sites are lower confidence and one of them is the fifth-ranked SNP in the scan.
- Whether the outlier set is enriched for any functional category beyond chance. A length- and SNP-density-matched permutation test has not been run in this pipeline. Note that two of the 19 outliers are intergenic, so any gene-set test must state how it handles them.
- Whether the pfaat1 S258L, pfdhps 437 and pfdhfr clines reflect spatially varying drug pressure, differing deployment histories, or the design confounders above. The frequencies are measured; their cause is not identified. The amodiaquine argument is now explicitly an untested possibility rather than the preferred explanation: the one experiment that measured amodiaquine found the IC50 shift sub-threshold, its authors conclude chloroquine was the primary selective force, and the F313S variant with which S258L shows epistasis is absent from all 1,900 isolates here.
- Whether the pfk13 negative can be corroborated by anything other than our own pipeline. MalariaGEN publishes no per-codon kelch13 genotype, and its domain-level status (349-726) is complete for only 36 isolates, so 32 of the panel's 57 codons - every pfk13 codon and pfaat1 S258L among them - rest on a single pipeline. The 25 codons that can be checked agree on 47,012 of 47,012 calls, which is reassuring about the method but is not evidence about the other 32.
- Whether the four potential-tier pfk13 codons (449, 481, 515, 527) and the three panel genes with no WHO tier at all (pfaat1, pfap2mu, pfmdr2) belong in a surveillance panel. They are retained here on scientific-interest grounds, which is a judgement of ours and not a WHO recommendation, and the manuscript must not blur the two.

- Whether low differentiation reflects ongoing gene flow or recent common ancestry. Requires demographic inference not attempted here.
- The provenance and licence of the zone polygon layer, which remains a publication blocker (Appendix E). It cannot be closed analytically — no further computation will produce a licence — and the remaining action is correspondence with the likely custodian (Section 10.4, item 2).

## 10.4 Most valuable next steps

**[Proposed]** The resistance-marker work that headed this list two revisions ago is done, and done twice: the first rebuild was itself defective and has been replaced by step12_codon_panel.py, whose outputs are the basis of every resistance number in this revision. Four further items the previous version carried are also now done, and they are struck from the list rather than restated. The $F_{ws}$ and MAF sensitivity analyses were run, and they overturned the isolation-by-distance result rather than defending it (Section 10.1, Table 10.1). Mapping quality and the other callset annotations were checked at all 19 outlier positions, clearing the exported-protein objection and flagging two chromosome-11 sites. Stage 3 was re-run on the 1,900-isolate analysis set (Figure 10.1). And results/baypass/xtx_all_snps.csv, which the previous version's item 6 asked to be restored, was restored and the restoration verified by md5 against three unchanged downstream outputs (Appendix F, Table F.1, row 13). What follows is what is left, ordered by expected value per unit of effort. Only the first two are cheap, and the second of them is not an analysis at all.

38. **Run a read-level depth check at `Pf3D7_11_v3:1054078` and `Pf3D7_11_v3:1054447`.** A per-sample depth ratio against the genome median at two positions, from AD and DP fields the Zarr store already holds: one pass, and the cheapest item on this list. It decides the standing of the fifth-ranked SNP in the whole scan. Either the ExcessHet = 69.59 and the panel's two lowest InbreedingCoeff values are collapsed paralogy or copy-number variation, in which case both sites should be dropped from the outlier set and the FDR-passing count falls from 13 to 11, or they are not, in which case the downgrade can be lifted and the scan's fifth-ranked hit is restored to full standing. Leaving it undone leaves both sites permanently qualified in a list of only 19.
39. **Establish the provenance of `data/ghanaAEZ.geojson`, or retire it.** The layer has been identified with certainty as ArcGIS Online item 0776dae5c5 314 102bd03fdaab8df0356, byte-for-byte identical to the project's copy (md5 96b1a7f4da62d6f0 153 826f41af96 152), but that item carries null licenseInfo and null accessInformation under an anonymous personal account, so it is an unattributed re-host and not a citable source. Two leads remain: whoever supplied the original Colab notebook, whose !wget cells are the most likely surviving record of a fetch URL, and the Centre for Remote Sensing and GIS at the University of Ghana, which the ArcInfo-coverage fingerprint points to as the likely custodian. In parallel, repeat the spatial join against an authoritative six-zone Ghana layer: if zone assignments are stable, the provenance gap becomes a sensitivity result rather than a threat. The practical next action is an email to **cersgis@ug.edu.gh**. This item cannot be closed analytically and no further computation will close it, which is why it has outlasted every other item this list has carried.
40. **Check every externally sourced list against its source document before it is used again.** This is the lesson of the round that produced this revision, and it is cheap. The *pfk13* marker list was written from memory and was wrong in three ways; the reference-allele direction at *pfdhps* 437 was stated in the official Pf8 methodology and was not read; WHO tiers were applied to genes rather than to variants. The lists still carried on trust are the drug-policy dates, the fitness-cost claim for S258L and the gene-to-drug

assignments in the panel definition. Each should be checked against its source and the source quoted in the text.

41. *Corroborate the* pfk13 *and* pfaat1 *calls against something other than this pipeline. The per-sample concordance check covers 25 markers in five genes and is perfect on 47,012 calls, but it cannot reach the 32 codons MalariaGEN does not publish, and the* kelch13 *domain status gives a complete call for only 36 isolates. Options are a direct read of the Pf8 VCF at the 23* pfk13 *codons by an independent route, or comparison against a published Ghanaian* pfk13 survey over the same years. This matters because the artemisinin negative is the result most likely to be quoted.

42. **Test isolation by distance properly, or stop reporting it as a finding.** The region-level Mantel test on 21 pairs is at the limit of what Pf8's Ghanaian metadata supports, and it fails at four of the six threshold settings tried. Re-running it will not help. Two routes would: district- or site-level coordinates wherever the metadata carries them, which raises both the number of populations and the number of pairs; and a continuous-space method that does not require populations to be defined at all. Either is a new analysis rather than a robustness check, and neither is cheap. Until one is done the manuscript should carry near-panmixia as its spatial result and isolation by distance only as a non-robust observation, which is how Section 10.1 and Table 10.2 now report it.

43. **Reframe the thesis objectives in light of the collinearity, and in light of Objective 3 now being answered.** The question "do ecological zones structure the parasite population?" cannot be answered with Pf8 metadata alone. Two questions that can be, and have been, are "is Ghanaian *P. falciparum* spatially structured at all, and what does that imply for surveillance design?" and "do resistance-marker frequencies vary across the country?" The second was originally written off as unanswerable from these data and is in fact the strongest result the analysis holds — with the caveat that it took two attempts to measure correctly. The first of those two questions now has a narrower answer than it did a revision ago: the structure is negligible and the distance gradient is not robust, so what the analysis establishes about space is a magnitude rather than a pattern, and the surveillance-design implication rests on that measured magnitude alone.

44. **Analyse Fws by zone and year as a transmission-intensity phenotype in its own right**, with study as a covariate, since the between-study variation within a single zone is of the same order as the between-zone variation: the polyclonal fraction (Fws < 0.95) spans 0.430 to 0.578 *across the four testable studies in deciduous forest. Those two numbers are fractions of isolates below the monoclonality cut, not* $F_{ws}$ values, and Section 10.1 labels them that way; the two statements must agree.

**[Interpreted] A closing assessment.** The analysis has done what a baseline characterisation should do: established the scale of structure, found it to be small, identified precisely which design features limit what can be inferred, and produced a short candidate list for follow-up. What has changed across the three rounds is the standing of the negative results, the standing of our own corrections, and now the standing of a positive result. The absence of resistance loci from the outlier scan was not a result at all but a filter artefact, and correcting it converted the thesis's third objective from a null finding into its best-supported positive one — *pfdhps* G437 at 0.897 and zone-structured, the *pfdhfr* triple mutant measured as a haplotype at 0.771, and *pfaat1* S258L as a strong candidate whose demonstrated functional effect is modest and background-dependent. The absence of artemisinin resistance is now genuinely tested, across the complete WHO-validated and candidate sets, on 1,845 to 1,876 called isolates per codon over 2009–2021, and genuinely negative. But the first correction of the panel was itself wrong in four ways, and only an external reader caught it. The honest conclusion is therefore two claims rather than one. A negative result is worth nothing

until the thing it denies has been shown to be measurable. And internal consistency is not correctness: every defect in the first corrected panel came from a list, a direction or a label taken on trust, and the verification architecture built to catch stale numbers was never built to ask whether the inputs were the right ones. The third round adds a third claim, and it is the least comfortable of the three. Four checks this list carried as outstanding were run, and the one previously described here as addressing "the two most likely reviewer objections" removed a finding instead of answering them: isolation by distance survives neither a relaxation of the monoclonality cut nor a lower MAF floor, and is significant at two of six threshold settings. The gain and the loss come from the same six runs. The check that destroyed the spatial finding is the check that made near-panmixia the best-evidenced conclusion in the manuscript, because $F_{ST}$ moved by less than 0.0005 while the Mantel *p*-value crossed 0.05 four times. A result that holds only at the thresholds that produced it was never a result; it was a setting. The manuscript is shorter by one finding and better by one method, and the third claim is simply that: a threshold is a hypothesis, and until it has been varied, every conclusion downstream of it is conditional on a number nobody tested.

## 11. References

**[Observed]** Every entry below was checked against a primary source — the publisher's own deposit, the journal's JATS record, the archive record, or the issuing organisation's page — and not against a citation aggregator. The checks corrected eight entries. Three were substantive: Otto et al. (2018) ended on Newbold, C. I., who is author 23 of 24, where APA requires the final author, Berriman, M.; the Pf7 data note is published under the collective name "MalariaGEN" and not the expanded "Malaria Genomic Epidemiology Network (MalariaGEN)" used on Pf8; and Virtanen et al. (2020) omitted the group author "SciPy 1.0 Contributors", which is the final author of that byline. Five were style or completeness: three Wellcome Open Research entries cited an article number as a page number (now "Article 325", "Article 22", "Article 52"); Otto et al. was missing three middle initials; PLINK's journal is "The American Journal of Human Genetics"; Venna and Kaski needed a comma before the series volume; and Su, X.-z. is normalised to Su, X.-Z. Two editorial decisions are recorded rather than hidden. The scikit-allel Zenodo record lists "pyup.io bot" as its second creator; it is omitted here as an automated contributor rather than an author. And Amambua-Ngwa et al. (2019) author 14 is indexed by the publisher, PubMed and Europe PMC as "Kolapo, O.", which may be a given/family inversion of Kolapo M. Oyebola; the published metadata is followed rather than corrected by inference, because the printed byline could not be read. Four sources carry no DOI because none was ever registered — Mantel (1967), the two JMLR papers and the two WHO web publications — and this was verified at Crossref rather than assumed.

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# 12. Appendices

> **Section numbering — resolved.** The previous version had no section 11: the headings ran 1 to 10, then 12, and the gap was flagged here because the reference list had not been written. Section 11 is now the reference list, carrying 42 entries, and the appendices remain section 12. Every entry was resolved against a primary source rather than a citation aggregator, and the eight corrections and two editorial decisions that process produced are itemised at the head of that section and in Appendix D. The numbering defect and the missing-bibliography defect are both closed.

## Appendix A — Software environment

| Component | Version | Role |
|---|---|---|
| Python | 3.10.22 | All analysis scripts |
| NumPy | 1.26.4 | Arrays, and the codon panel's Wilson 95% intervals, which step12 evaluates in closed form with a hard-coded z = 1.96 rather than calling SciPy. Pinned below 2.0 for scikit-allel compatibility |
| pandas | 2.3.3 | Tabular data |
| SciPy | 1.15.2 | Eigendecomposition (PCoA), connected components (clonal pruning), chi-square and Kruskal-Wallis tests; chi-square tests in the codon panel |
| scikit-allel | 1.3.13 | Genotype arrays, Hudson FST, block jackknife |
| Zarr | 2.18.3 | Chunked genotype store. Pinned below 3.0: scikit-allel's writer uses 2.x APIs |
| numcodecs | 0.13.1 | Blosc/LZ4 and Zstd codecs |
| scikit-learn | 1.7.2 | t-SNE, trustworthiness, silhouette |
| umap-learn | 0.5.12 | UMAP |
| scikit-bio | 0.7.3 | Mantel test |
| GeoPandas | 1.1.4 | Spatial join (Stage 2) |
| Matplotlib | 3.10.9 | All figures |
| PLINK | 1.9.0 (64-bit) | Format conversion, LD pruning |
| ADMIXTURE | 1.3 | Ancestry inference |
| TreeMix | 1.13 | Population graph; run with -se for migration-edge standard errors |
| BayPass | 3.1 | Core model and XtX. Run single-threaded (see below) |
| bcftools | 1.24 | VCF validation only; not used in the production path |
| R | 4.5.1 | BayPass POD simulation via baypass_utils.R |
| Operating system | Ubuntu, kernel 5.15.0-194-generic | 40 cores, 629 GB RAM |

*Table A.1. Software versions, re-confirmed from the live environment on 10 October 2026. No version changed between the original run, the corrected run or the codon-panel run; step12_codon_panel.py adds no dependency the pipeline did not already use. Its only import outside the rows above is requests, which step03a_fetch_callset_zarr.py and step10_resistance_panel.py already required and which is not itself listed here. The same holds for step13_sensitivity.py and for the --samples/--suffix re-run of step03_umap.py: both recompute with the packages already listed — scikit-allel for Hudson $F_{ST}$ and the block jackknife, scikit-bio for the Mantel test, scikit-learn for the ordination metrics — and neither adds a dependency.*

The environment lives at /home/gyadu/.conda/envs/spatial and was built with micromamba 2.9.0 against conda-forge and bioconda; the shared conda installation's classic solver was unusable at this dependency scale. Activation:

```
source /home/tools/anaconda3/etc/profile.d/conda.sh && conda activate spatial
```

[Observed] **BayPass is run single-threaded, and this is a correctness requirement rather than a performance choice.** Two 8-thread runs with the same -seed 42 produced different XtX values and therefore different outlier sets; the superseded outputs are retained under results/baypass_8thread_archive/ so the non-reproducibility can be re-demonstrated. NTHREADS=1 is now the default in step08b_baypass.sh and run_all.sh --parallel exports 1 where it previously exported 8.

## Appendix B — Key configuration values

| Constant | Value | Set in | Provenance |
|---|---|---|---|
| RANDOM_SEED | 42 | config.py | Pipeline default. Seeds NumPy wherever a stochastic step is taken (config.py:113, applied at config.py:230); the reproducibility argument in Appendix A rests on it |
| MIN_FWS | 0.95 | config.py | Pre-specified (Manske et al., 2012). config.py:128 records the citation in the constant's own comment: "conventional monoclonality cut (Manske et al. 2012)" |
| MIN_SAMPLE_CALL_RATE | 0.80 | config.py | Pipeline default (config.py:129) |
| CLONAL_THRESHOLD | 0.00005 | config.py | Session decision (Section 4.2.4). No natural break exists in the data; sensitivity quantified over seven thresholds in Table 4.3. It is a pairwise Pf8 genotype distance at or below which two isolates count as clonal, not a similarity cut, and was chosen from the step 3b histogram (median Ghana distance 0.00112) |
| MIN_GENOTYPE_DP | 5 | config.py | Pipeline default (config.py:137). The filter behind the cascade's own first line, "DP < 5: 3,709,090,355 calls" |
| MIN_GENOTYPE_GQ | 0 (GQ filter disabled) | config.py | Session decision (Section 4.2.5). The largest single discretionary choice in the pipeline; the evidence for it is indirect and it is listed for author review |

| Constant | Value | Set in | Provenance |
|---|---|---|---|
| MIN_MAF | 0.05 | config.py | Thesis workflow |
| MAX_MISSING (variant) | 0.10 | config.py | Thesis workflow. Value unchanged; now applied after the sample cut, not before it |
| MAX_SAMPLE_MISSING | 0.20 | config.py | Pipeline default |
| UMAP_N_NEIGHBORS / UMAP_MIN_DIST | 30 / 0.1 | config.py | Thesis workflow |
| TSNE_PERPLEXITY | 30 | config.py | Session addition |
| PCOA_N_AXES | 10 | config.py | Session addition |
| K_MIN, K_MAX | 2, 8 (K = 1 run separately) | step05b_admixture.sh | Thesis workflow. Shell defaults; step05b does not read config.py |
| ADMIXTURE_CV_FOLDS | 10 | step05b_admixture.sh | Pipeline default. Defined in config.py (ADMIXTURE_CV_FOLDS = 10) but not read by step05b, which carries its own shell default CV_FOLDS="${CV_FOLDS:-10}" at line 24; the shell default is what is in force |
| LD_PRUNE_ARGS | 50, 10, 0.2 | step05b_admixture.sh | Pipeline default. Defined in config.py (LD_PRUNE_ARGS = ("50", "10", "0.2")) but not read by step05b, which carries its own shell defaults PRUNE_WIN/PRUNE_STEP/PRUNE_R2 at lines 29-31; the shell defaults are what is in force |
| TREEMIX_BLOCK_SIZE | 50 | config.py | Session decision (Section 5.5). Gives approximately 78 jackknife blocks on 3,916 SNPs |
| TREEMIX_MAX_M / TREEMIX_ROOT | 5 / Guinea_savanna | config.py | Thesis workflow; root name corrected |
| TreeMix SE | 1 (-se enabled) | step07b_treemix.sh | Correction after audit. Produces jackknife means, standard errors and p-values per migration edge, which did not previously exist |
| BayPass NVAL (POD SNPs) | 100,000 | step08b_baypass.sh | Session decision (Section 5.6) |
| BayPass PI_MAF | 0.05 | step08b_baypass.sh | simulate.baypass's own default; retained, not chosen (see note) |
| BayPass NTHREADS | 1 | step08b_baypass.sh; run_all.sh | Correction after audit. Shell default was 4 and --parallel exported 8. Required for reproducibility (Appendix A) |

| Constant | Value | Set in | Provenance |
|---|---|---|---|
| XTX_POD_QUANTILE | 0.999 | config.py | Pipeline default |
| XtX q-value report thresholds | 0.01, 0.05, 0.10, 0.20 | step08c_calibrate_xtx.py | Correction after audit. Benjamini-Hochberg q-values are now written per SNP |
| GENES (codon panel: 9 genes, 57 codons) | pfcrt 10, pfmdr1 5, pfdhfr 5, pfdhps 6, pfk13 23, pfmdr2 2, pfcoronin 3, pfap2mu 1, pfaat1 2 | step12_codon_panel.py | Correction after the external critique. pfk13 codons are taken from the WHO compendium v1.0 (December 2025): all 14 validated and all 6 candidate variants, plus four potential-tier codons in the same region. Codons are requested by number and the reference residue is read from the genome rather than assumed |
| HAPLOTYPES (codon panel) | pfdhfr IRN (51,59,108); pfdhfr IRNL (51,59,108,164); pfcrt CVIET (72-76); pfdhps GE (437,540) | step12_codon_panel.py | Correction after the external critique. Counted on complete cases only; a marginal range is not a haplotype frequency |
| Codon-panel ALT restriction | none (numalt and is_snp both ignored) | step12_codon_panel.py | Correction after the external critique. Deliberately unlike the genome-wide ingest mask, which is what excluded most canonical codons |
| Mixed-call policy (codon panel) | a mixed call makes the codon uncalled | step12_codon_panel.py | Session decision. It is why per-codon denominators run 1,834-1,900 and why pfcrt 75 loses 26 isolates (52 mixed calls across its two variant positions, 403621 and 403623). The mixed-call tally is not the loss at every codon: pfcrt 326 has 24 mixed calls but loses 26 isolates across three variant positions |
| Sensitivity grid ($F_{ws}$, MAF) | $F_{ws}$ >= 0.95 (published), 0.90, 0.80, none; MAF >= 0.05 (published), 0.02, 0.01 | step13_sensitivity.py | Added by the robustness round of 10 October 2026. Not a value in force: it is the grid over which MIN_FWS and MIN_MAF were varied, one threshold at a time from the published pair. The baseline cell reproduces config.py exactly - 1,900 isolates, 3,916 SNPs, max $F_{ST}$ 0.00232, Mantel p = 0.0252 - which is what licenses reading the other five cells against it |

*Table B.1. Configuration values in force for the corrected run and the codon-panel run. Three rows are marked "Correction after audit" and three "Correction after the external critique"; MAX_MISSING kept its value but moved in the cascade; three rows had the wrong file in the "Set in" column and are corrected here. RANDOM_SEED, MIN_SAMPLE_CALL_RATE and MIN_GENOTYPE_DP were dropped from the previous version of this table in error and are restored, as is CLONAL_THRESHOLD's correct value and MIN_FWS's citation. A twenty-eighth row is added in this revision for the grid over which step13_sensitivity.py varied MIN_FWS and MIN_MAF; it records a range rather than a value in force and is marked as such.*

> **PI_MAF is not a MAF filter.** The original text described the POD as "100,000 MAF-matched simulated SNPs". That is not what pi.maf does. In baypass_utils.R it truncates the simulated ancestral frequency to [0.05, 0.95] and binomial sampling then freely produces observed minor-allele frequencies below 0.05, so the POD is not matched to the observed MAF >= 0.05 spectrum. 0.05 is also simulate.baypass's own default, so setting it changed nothing. The "MAF-matched" wording has been removed from Sections 5.6, 6.8 and 8.1 as well as from this table. Section 6.8 had carried it in different words — "with the same MAF filter as the observed data" — which survived two earlier correction rounds precisely because those rounds searched for the phrase rather than for the claim; it has now been removed as well. The pseudo-observed set is not MAF-matched.

[Observed] **Session decisions versus inherited defaults.** Of the 28 rows above, 27 are values in force and the twenty-eighth records the sensitivity grid rather than a setting. Of the 27, five are session decisions taken during this analysis and carry no external justification — CLONAL_THRESHOLD, MIN_GENOTYPE_GQ, TREEMIX_BLOCK_SIZE, BayPass NVAL and the codon panel's mixed-call policy — and two more, TSNE_PERPLEXITY and PCOA_N_AXES, are additions rather than judgements. Three are corrections forced by the internal audit (TreeMix SE, BayPass NTHREADS, the XtX q-value thresholds) and three by the external critique (the GENES list, the HAPLOTYPES definitions and the codon-panel ALT restriction), and all six are marked as such — six in total, not the seven stated in the previous version, whose count did not match the marks actually carried in the table; MAX_MISSING is unchanged in value but changed position in the filtering cascade, which is why the SNP count moved. The remainder are pipeline defaults or thesis-workflow specifications. The rows that most affect reported numbers are MIN_GENOTYPE_GQ, which governs the size of the SNP panel; BayPass NTHREADS, which governs whether the outlier scan is reproducible at all; and the codon panel's GENES list, which governs whether a marker exists in the manuscript or not — a point made the hard way, since *pfdhps* 437 was missing from the previous version because of how markers were identified rather than because of any threshold. Two of those rows have since been varied rather than merely justified: MIN_FWS and MIN_MAF were both relaxed in the sensitivity round, and the outcome is in Table 10.1 — near-panmixia unaffected, isolation by distance not surviving. MIN_GENOTYPE_GQ, which the paragraph above names as the largest discretionary choice in the pipeline, has still not been varied, and it governs the size of the SNP panel that every one of those six scenarios was drawn from.

[Observed] **Dead constants.** config.BAYPASS_THREADS (= 4), config.ADMIXTURE_CV_FOLDS, config.LD_PRUNE_ARGS and config.MANTEL_JSON are referenced by no script. Editing them changes nothing and produces no warning. They should be deleted or the consuming scripts should be made to read them, as step07b_treemix.sh already does for the TreeMix parameters.

## Appendix C — Output inventory

| Path | Contents |
|---|---|
| interim/pf8_ghana.zarr | Ghanaian callset, 20.57 GB, 2,190,026 PASS biallelic SNPs x 6,653 isolates, GT/AD/DP/GQ. Built under FILTER_PASS & is_snp & numalt == 1, which is why most canonical resistance codons are absent from it. The codon panel does not read this store's variant selection; it re-fetches the panel positions from Pf8 with no numalt or is_snp restriction |

| Path | Contents |
|---|---|
| interim/ghana_core_snps.npz | Final genotype matrix, 3,916 x 1,900 (was 3,359 x 1,900 before the filter-order fix) |
| interim/analysis_samples.csv | Canonical sample table; fixes sample order for every stage from 4 onward, and for the codon panel |
| interim/allele_counts_by_zone.npz | Per-zone allele counts, one allele per isolate; feeds stages 7 and 8 |
| interim/baypass_snp_positions.csv | Row-to-position map for the BayPass matrix: baypass_row, chrom, pos for all 3,916 SNPs. The join key for every XtX result |
| results/sample_qc_full.csv | All 4,905 isolates passing sample-level QC, with every QC statistic and flag. This is the denominator for the batch and polyclonality analyses |
| results/fst_pairwise.csv, fst_matrix.csv | Pairwise $F_{ST}$ with jackknife SEs and confidence intervals. One of six intervals spans zero |
| results/fst_prev_run.csv | Pre-correction $F_{ST}$ values, retained for the before/after comparison |
| results/dimred_comparison.csv | Trustworthiness, distance correlation, silhouette for the three ordinations, on the 4,905-isolate set. The analysis-set counterpart is dimred_comparison_qcset.csv below; the two files are both retained and must not be confused, which is what incident 14 of Appendix F nearly caused |
| results/{umap,tsne}_embedding.csv, pcoa_coordinates.csv, pcoa_eigenvalues.csv | Ordination coordinates and PCoA eigenvalues, computed on the 4,905-isolate set. The 1,900-isolate analysis-set re-run writes _qcset counterparts beside these rather than replacing them |
| results/ mantel_result_{zone,region}.json | Mantel r and p both censored and uncensored, the named pairs whose negative $F_{ST}$ was floored, the population list, the permutation count and the seed |
| results/ mantel_jackknife_{zone,region}.csv | Leave-one-out jackknife: r and p with each population dropped in turn (7 region rows, 4 zone rows) |
| results/ geographic_distance_km_{zone,region}.csv | Great-circle distance matrices underlying the Mantel tests |
| results/admixture/ | Q and P matrices for K = 1-8, cross-validation errors. 3,385 LD-pruned SNPs of 3,916 |
| results/treemix/ | Fitted graphs for m = 0-5 with -se output, model_comparison.csv, model_selection.json, run_status.tsv |
| results/baypass/xtx_all_snps.csv | Per-SNP baypass_row, chrom, pos, xtx, pval_pod, qvalue (Benjamini-Hochberg) and is_outlier for all 3,916 SNPs. This file was found truncated to zero bytes on 10 October 2026 and has been regenerated from the raw BayPass summaries; the restoration is verified rather than assumed, with outlier_snps.bed, xtx_calibration.json and xtx_threshold.txt byte-identical by md5 before and after and the restored file carrying the same 3,916 rows and 19 outliers (Appendix F, Table F.1, row 13). The previous version's warning that this file is empty is withdrawn |
| results/baypass/xtx_calibration.json | Threshold and method, outlier count and fraction, expected false positives, the expected-null-exceedance ratio at the threshold, and counts at q <= 0.01/0.05/0.10/0.20 |
| results/baypass/ ghana_core_summary_pi_xtx.out, ghana_pod_summary_pi_xtx.out | Raw BayPass XtX for the 3,916 observed SNPs and for the 100,000 POD SNPs. Every q-value in this revision is reproducible from these two files alone |
| results/baypass/outlier_snps.bed | The 19 SNPs above the calibrated threshold, with XtX. Two carry no gene intersection |
| results/baypass/ | Remaining BayPass outputs: simulated neutral distribution, omega matrices, population order |

| Path | Contents |
|---|---|
| results/baypass_8thread_archive/ | Superseded outputs of the two 8-thread runs, kept so the non-reproducibility can be re-demonstrated (Appendix A) |
| results/outlier_genes.csv, outliers_annotated.txt | Gene-level summary (13 genes) and raw SNP-gene intersections (17 rows). The two intergenic outliers appear in neither |
| results/codon_panel_residues.csv | New, and the source of every resistance frequency in this revision. 57 codons across all nine panel genes, reconstructed from the CDS and the reference genome: gene, gene_id, codon, reference residue, every observed residue with a flag for whether it is the reference residue, called denominator, overall frequency, contig, strand, the three genomic positions, the number of variant records at them, per-zone frequencies with Wilson 95% intervals, and a chi-square test of between-zone heterogeneity. 83 rows, because 24 of the 57 codons are polymorphic and *pfdhps* 436 carries four residues |
| results/codon_panel_haplotypes.csv | New. Four within-isolate haplotypes counted on complete cases only - *pfdhfr* IRN, *pfdhfr* IRNL, *pfcrt* CVIET and *pfdhps* GE - reported by zone only: 16 rows, four haplotypes x four zones, with counts, complete-case denominators and Wilson intervals, and no overall row. The overall figures and their Wilson intervals (IRN 0.771 [0.751-0.789] on n = 1,876; CVIET 0.071 [0.060-0.084] on n = 1,873; GE 0.010 [0.006-0.015] on n = 1,888; IRNL 0.000 [0.000-0.002] on n = 1,874) are in logs/step12_codon_panel.log and must be cited from there; from this CSV the overall denominators are recoverable only by summing n_complete across zones. This file is what replaces the previous version's marginal range 0.786-0.908 |
| results/codon_panel_concordance.csv | New. Per-sample agreement with MalariaGEN's published genotypes at the 25 markers MalariaGEN publishes per codon (*pfcrt* 9, *pfmdr1* 5, *pfdhfr* 5, *pfdhps* 5, *pfmdr2* 1): both-called count, agreeing count, concordance and any disagreements. 47,012 of 47,012 calls agree. It covers 25 of the panel's 57 codons and no *pfk13* codon |
| logs/step12_codon_panel.log | New. The codon-panel run record: the full residue table with per-zone frequencies and chi-square p-values, the haplotype block with each haplotype's components on its own complete-case set, the per-marker concordance table, the list of codons with no variant record in Pf8 (none), and the 24 mixed-call notes that explain the per-codon denominators |
| results/resistance_panel_canonical.csv, resistance_panel_frequencies.csv | SUPERSEDED by the codon panel and must not be cited for any frequency. These are step10's annotation-string output; they omit *pfdhps* A437G entirely, misname *pfcrt* 75 as Asp, carry a *pfk13* list that is incomplete and partly misclassified, and give *pfcrt* N326S as 0.008 against a reconstructed 0.0005. Retained only so the annotation-matching defect can be re-demonstrated (Appendix F, defect 9) |
| results/batch_study_by_zone.csv | Sequencing study x zone crosstab over the 4,905 QC-passing isolates; 11 studies |
| results/batch_zone_pair_overlap.csv | Studies in common for each of the six zone pairs; four pairs have none. Computed on the 4,905 set |
| results/batch_within_zone_tests.csv | Between-study tests of $F_{ws}$, call rate and polyclonal fraction with zone held constant. The transitional zone has a single study and is therefore untestable |
| results/sensitivity_summary.csv | New. One row per threshold scenario from step13_sensitivity.py: the $F_{ws}$ and MAF thresholds, isolate and SNP counts, per-zone isolate counts, max pairwise $F_{ST}$, Mantel r and p both censored and uncensored, and the number of negative $F_{ST}$ pairs floored. Six rows. The source of Table 10.1 and of the isolation-by-distance downgrade |
| results/sensitivity_fst.csv | New. All six pairwise $F_{ST}$ values with jackknife standard errors and per-zone n, for each of the six scenarios: 36 rows. Coastal savanna against Guinea savanna is the maximum pair in all six scenarios, which is what the 0.00204-0.00255 range in Section 10.1 is a range of |
| results/outlier_quality_check.csv | New. Per-SNP callset quality annotations for all 19 outliers with percentiles against |

| Path | Contents |
|---|---|
| | the other 3,897 scanned SNPs: MQ, VQSLOD, QD, FS, SOR, BaseQRankSum, MQRankSum, ReadPosRankSum, DP, AN, QUAL, ExcessHet and InbreedingCoeff, plus culprit, RegionType, the VQSR training flags, scanned-SNP density and mean MQ in 1 kb and 10 kb windows, and a per-SNP verdict. The source of the quality statements in Section 10.1 and of the two chromosome-11 downgrades. ExcessHet is zero in 18 of the 19 rows and 69.59 in the nineteenth |
| results/dimred_comparison_qcset.csv | New. Trustworthiness, distance correlation and silhouette by zone for the three ordinations on the 1,900-isolate analysis set: silhouette by zone on raw distances -0.0196, UMAP trustworthiness 0.499, t-SNE 0.526, PCoA 0.510, PCoA distance correlation 0.287. The _qcset suffix exists so that dimred_comparison.csv is not overwritten |
| figures/ dimred_comparison_zone_qcset.png | New. Figure 10.1, the three ordinations on the 1,900-isolate analysis set. The published figures/dimred_comparison_zone.png (Figure 6.2, 4,905 isolates) is deliberately untouched, and both are retained so the two sample sets can be compared |
| figures/ | All figures reproduced in this report, regenerated by the corrected run. The codon panel produces no figure; its results are tabular |
| logs/ | Per-stage logs including the Zarr fetch verification, the full filtering cascade and the codon-panel log above |

*Table C.1. Principal outputs, relative to /media/biodataA/pf8/gyadu. **Nine** paths are marked "New" in this revision — the three codon_panel files and logs/step12_codon_panel.log from the external-critique round, plus the two sensitivity files, outlier_quality_check.csv, dimred_comparison_qcset.csv and figures/dimred_comparison_zone_qcset.png from the robustness round — and two are marked SUPERSEDED. Six further paths were new at the previous revision and are no longer flagged. One row changed status rather than contents: results/baypass/xtx_all_snps.csv was zero bytes when the previous version went out and has since been regenerated and checked for consistency with the retained downstream outputs.*

## Appendix D — Reference verification status

**[Observed]** Unchanged from the previous version except for the entry counts, which the audit found to be internally inconsistent (26 claimed in the opening sentence, 30 entries listed, subtotals summing to 27). The counts should be recomputed against the final reference list before circulation; no total is asserted here. Three entries change in this revision. *Amambua-Ngwa et al. (2023,* Nature Microbiology*) is cited in Sections 6.8 and 10.1 for the* pfaat1 *S258L functional work, now read correctly: a 1.5-fold chloroquine IC50 rise in a single chloroquine-resistant Southeast Asian line (NHP4026), with epistasis between S258L and F313S and an explicit* pfcrt*-background caveat; amodiaquine, lumefantrine, mefloquine and quinine were all measured and the shifts were sub-threshold; the authors conclude chloroquine was the primary selective force and reject the other-quinoline hypothesis; and the 0% to 97% longitudinal rise is Gambian, 1984–2014, with nothing said about Ghana. The previous version's claim that amodiaquine was untested is withdrawn. The WHO compendium of molecular markers of antimalarial drug resistance, v1.0 (December 2025), is added as the authority for the* pfk13 *panel and for every tier label in it; it classifies variants and not genes, contains no validated or candidate lumefantrine marker, and contains no entry at any tier for* pfaat1, pfap2mu *or* pfmdr2. *The official Pf8 methodology document is added as the authority for the reference-allele direction at* pfdhps 437, quoted in Section 10.1 and in Appendix F.

## Appendix E — Reproducing the analysis

From the project directory as it currently stands on disk, on a machine with the environment of Appendix A — the project is not under version control, so the script state behind the reported run is recoverable only

from those files, the exact commands and script revisions of that run are not archived, and pf8_ghana_pipeline.zip holds only the unmodified originals, which predate this revision's corrections and will not reproduce the reported numbers:

```
source /home/tools/anaconda3/etc/profile.d/conda.sh && conda activate spatial cd /media/biodataA/pf8/gyadu/scripts python check_env.py  # verifies packages and external binaries bash run_all.sh --list  # lists the stages bash run_all.sh # full pipeline, stages 1-9 bash run_all.sh --from 4 --parallel  # stages 5-8 concurrently after stage 4
```

Three analyses are not stages of run_all.sh and are run directly, and one stage is re-runnable on a sample subset. All of them read Pf8 or the Zarr store together with the canonical sample table, and all write only new files under results/ and figures/: none overwrites a published output, and the Stage 3 re-run writes _qcset variants beside the originals rather than replacing them. They can be run at any time after stage 4:

```
python step12_codon_panel.py  # resistance markers by reconstructed codon python step11_batch_confounding.py  # study x zone confounding python step13_sensitivity.py  # Fws and MAF threshold sensitivity, 6 scenarios  # Stage 3 on the 1,900-isolate analysis set; outputs take the suffix, originals are untouched: python step03_umap.py --samples <analysis sample list> --suffix _qcset  # superseded, retained only to re-demonstrate the annotation-matching defect: # python step10_resistance_panel.py
```

**[Observed]** step12_codon_panel.py supersedes step10_resistance_panel.py entirely, and no frequency in the manuscript comes from step10 any longer. It maps each of 57 requested codons to its three genomic positions through the CDS (171 positions in total), fetches every variant record at those positions with no numalt or is_snp restriction (125 records; every codon has at least one), applies each isolate's alleles to the reference codon read from the genome, translates, and reports the full residue spectrum including the reference residue. Reference residues are read rather than assumed, so a wrong expectation in the panel definition cannot bias the output; a mixed call makes a codon uncalled. On the 1,900-isolate analysis set the run fetches about 2.53 GB of GT data in 9 x 76 chunks and completes in minutes. Note that step11_batch_confounding.py operates on the 4,905 isolates that pass sample-level QC, not on the 1,900-isolate analysis set, because sequencing-study membership and $F_{ws}$ are defined on that larger set; step12_codon_panel.py operates on the 1,900. Any comparison of their outputs must respect the difference (see the denominator note in Section 10.1).

**[Observed]** step13_sensitivity.py loads the 2.19 M × 4,682 genotype matrix once and then rebuilds the analysis matrix six times, varying one threshold at a time from the published pair, and recomputes pairwise Hudson $F_{ST}$ with block-jackknife standard errors and the region-level Mantel test under each. It writes results/sensitivity_summary.csv and results/sensitivity_fst.csv and touches nothing else; the baseline scenario reproduces config.py exactly, which is the check that makes the other five comparable. step03_umap.py --samples --suffix re-slices the distance submatrix for the requested isolates instead of reusing the cached full-set matrix — the bug that would otherwise make a subset run silently return the full-set answer — and suffixes every output it writes. Both are re-runnable at any time and neither has a skip-if-output-exists guard.

Stage 3a is resumable: completed variant chunks are recorded and skipped, and the store is built under a .partial name and renamed only on completion, so an interrupted run cannot be mistaken for a finished one. It also self-verifies:

```
python step03a_fetch_callset_zarr.py --validate-only \ --validate-vcf ../data/Pf8_Ghana_Pf3D7_10_v3.vcf.gz \ --validate-region Pf3D7_10_v3:400 001-440 000
```

**[Observed]** In --parallel mode the thread budget is divided as: ADMIXTURE 7 concurrent K values x 4 threads; TreeMix 6 concurrent m values, single-threaded; **BayPass single-threaded** (NTHREADS=1, where --parallel previously exported 8 — see Appendix A); NumPy/BLAS and Numba capped at 8 threads each. Stages 5-8 write disjoint outputs and depend only on stage 4, so concurrency does not affect results. Fixed seeds make each stage deterministic given its inputs, and the Mantel permutation test is now among them: it previously set no seed, so both reported p-values were irreproducible Monte Carlo estimates.

> **Cache invalidation — read this before re-running.** The "skip if the output exists" guards in step05b_admixture.sh, step07b_treemix.sh and step08b_baypass.sh originally tested only for the presence of an output file, not for whether its input had changed. When the SNP panel grew from 3,359 to 3,916, every stage regenerated its inputs, and then ADMIXTURE and TreeMix found their previous outputs in place, skipped the work entirely, and reported "stage finished". Only stage 8 caught the problem, because step08c asserts that its position table and the BayPass output have matching row counts and so failed loudly with the right diagnosis. All three guards now require the output to be **newer than its input** ([ output -nt input ]). A successor re-running after any change to stage 4 should confirm this behaviour before trusting a "skipping" message. step12_codon_panel.py has no such guard and always recomputes.

[Observed] **Reproducibility caveat.** The zone polygon layer (data/ghanaAEZ.geojson) still has no citable provenance, so Stage 2 cannot be reproduced from public sources alone. The file has been identified with certainty — it is byte-for-byte identical to ArcGIS Online item 0776dae5c5 314 102bd03fdaab8df0356, md5 96b1a7f4da62d6f0 153 826f41af96 152 on both sides — but that item's licenseInfo and accessInformation are both null and its owner is an anonymous personal account, so it is an unattributed re-host rather than an origin. It must not be attributed to any named institution. Cite it as accessed, record the checksum for identity, cite the underlying classification separately (the scheme is a hybrid: MoFA/FAO savanna and coastal names with a Hall & Swaine-lineage evergreen split), and state the provenance gap as a limitation. This remains a publication blocker. Three further gaps are unchanged or newly noted: the project is not under version control, so the exact script state at run time is recoverable only from the files on disk; t-SNE and the ADMIXTURE runs have not been verified to be bitwise reproducible across thread counts, although BayPass now is; and the results/baypass/xtx_all_snps.csv gap is narrowed — the file was found truncated to zero bytes, has been regenerated from the raw BayPass summaries, and the regeneration is consistent with three unchanged downstream outputs by md5, which does not establish row-for-row identity with the lost table (Appendix F, Table F.1, row 13) — but the incident itself belongs in this list, because nothing under results/ is write-protected and the project is not under version control, so a stray redirect into an output file is undetectable except by looking for it. A fifth, added in this revision: the codon panel's *pfk13* codon list is transcribed from the WHO compendium into step12_codon_panel.py by hand, which is exactly the kind of step that produced defect 10 below, so a successor should re-check the list against the compendium rather than against the previous script.

## Appendix F — Audit and correction record

[Observed] This appendix exists for traceability, and it now records three rounds. **Round one, 9–10 October 2026: an adversarial multi-agent audit.** Eight reviewers working across separate dimensions raised 110 findings, each of which was then put to three independent verifiers prompted to refute it and defaulting to refuted where they could not substantiate the claim. A finding counts as confirmed only where at least two of three verifiers failed to refute it after checking the code, data or outputs themselves. **Sixty-one findings were confirmed** — 14 high severity, 31 medium, 16 low — with 19 disputed and 21 refuted. Thirty verifications in the reproducibility dimension were lost to a rate limit and are recorded as unverified rather than confirmed. Seven confirmed defects required code changes; stages 4 to 9 were re-run on 10 October 2026 with all seven in place. **Round two, 10 October 2026: an external critique of the corrected manuscript.** It raised five points. **All five were upheld and four were defects in the analysis code, not in the prose** — in the resistance panel that round one had introduced to fix its own largest finding. The panel was rebuilt as scripts/step12_codon_panel.py and every resistance number in this revision was read from that script's outputs. This is the fourth consecutive round in which a correction contained its own defect, and the pattern is

consistent: each layer was checked for internal consistency against our own outputs, and what escaped was external correctness — whether the marker list matched an authoritative source, whether the annotation direction matched the reference genome, whether a co-occurrence claim had co-occurrence data behind it. The verification architecture was built to catch stale numbers, and it did. It was not built to ask whether the inputs were the right ones. **Round three, 10 October 2026: the four outstanding robustness checks.** This round was not an audit of the manuscript but the execution of four items that Section 10.4 carried as outstanding, and it is recorded here because one of them overturned a conclusion instead of confirming it. The $F_{ws}$ and MAF sensitivity analysis (scripts/step13_sensitivity.py) withdrew isolation by distance as a supported result — significant in two of six threshold scenarios — and, from the same six runs, made near-panmixia the best-evidenced conclusion in the manuscript. The outlier quality check cleared the exported-protein artefact concern and downgraded two chromosome-11 SNPs on an ExcessHet and InbreedingCoeff signal. Stage 3 was re-run on the analysis set. The GeoJSON provenance cannot be closed analytically and remains open. Round three found no defect in the prose and none in the analysis code; what it found was that a published result depended on the thresholds that produced it. Two integrity incidents occurred during the round and are registered as rows 13 and 14 of Table F.1 rather than left in the logs.

| # | Confirmed defect | Fix | Effect on reported results |
|---|---|---|---|
| 1 | Most canonical resistance codon-alleles were excluded at ingest. The callset is built under FILTER_PASS & is_snp & numalt == 1, and in Pf8 these records are frequently multiallelic or carry spanning-deletion ALTs; the mask dropped 28 of the 44 annotation records the first panel assembled. Across the nine panel genes the filter kept 3,330 of 13,367 PASS records (figure taken from the corrections document, not re-derived). The resulting absence was interpreted biologically. | scripts/step10_resistance_panel.py re-fetched the panel genes with no numalt restriction. That script has since been superseded by step12 (defect 9), but the diagnosis of the ingest mask stands and is unaffected. | Objective 3 changed from a null result to an answered question. Of the rebuilt panel's 171 codon positions four ever reached the MAF-filtered scan matrix, not three - *pfdhfr* C59R, *pfdhfr* N51I, *pfaat1* S258L and Pf3D7_07_v3:403623, the second of *pfcrt* codon 75's two variant positions (BayPass MRK 1406, M_XtX 4.18, XtXst 6.21 against the 10.840 threshold; M_P 0.914, so MAF 0.086 cleared the 0.05 cut). The count of three was carried over from step10's annotation-matched panel, whose resistance_panel_canonical.csv holds 403621 but never 403623 - a residual instance of the two-SNP-codon failure mode of defect 9. So the scan's silence at *pfdhfr* 51 and 59 and at *pfcrt* 75 is a tested negative, while its silence on *pfmdr1*, *pfdhps* and *pfk13* is an ascertainment artefact: those three genes contribute no scanned codon position at all. *pfcrt* is struck from the artefact list. Note also that "the antifolate codons" cannot stand as a blanket description of the tested negative - only two of the panel's five *pfdhfr* codons reached the matrix and none of its six *pfdhps* codons did, and *pfdhps* is itself an antifolate target. |
| 2 | Variant missingness was measured over 2,017 isolates, 117 of which were then dropped, so every SNP was charged for isolates that did not survive. | step04a_prepare_callset.py is now two-pass and blockwise: the sample cut is decided on the same provisional variant set as before, then variant missingness is recomputed on the | 3,916 SNPs instead of 3,359 at identical thresholds and with an identical sample set, a 16.6% gain. No population-genetic conclusion changed, which is reported in Section |

| # | Confirmed defect | Fix | Effect on reported results |
|---|---|---|---|
| | | isolates that remain. | 10.1 as a robustness result. |
| 3 | BayPass was run with -nthreads 8 and is not reproducible at that setting despite -seed 42. | NTHREADS default changed from 4 to 1 in step08b_baypass.sh, with the reason in a comment; run_all.sh --parallel exports NTHREADS=1 where it previously exported 8. | Bitwise reproducible. XtX threshold 10.840 and 19 outliers, replacing a threshold and outlier count that were a single random draw from the first of two mutually inconsistent 8-thread runs. |
| 4 | TreeMix ran without -se, so every migration line read "weight NA NA NA" and no edge could be tested. The manuscript left the m = 2 versus m = 3 choice open in the abstract and Section 6.6. | SE=1 is now the default in step07b_treemix.sh. | Edge p-values exist for the first time. m = 2 is supported (p = 7.5 x 10-4 and 3.3 x 10-16), m = 3's additional edge is not (weight 0.065, SE 0.147, p = 0.328) and m = 3 is non-identifiable with four populations. Report m = 2 as a selected descriptive model, and keep the disagreement with the full-sweep variance-explained criterion, which still prefers m = 3. |
| 5 | Negative pairwise $F_{ST}$ values were silently floored to zero in the Mantel test that carries the isolation-by-distance claim, and the permutation test set no seed. | step06_mantel.py seeds the permutation, names the floored pairs, reports r and p both censored and uncensored, adds a leave-one-out jackknife, and records all of it in the JSON. | Three of 21 region pairs floored on the final matrix. Censored r = 0.4269, p = 0.0252; uncensored r = 0.4138, p = 0.0413; significant in 4 of 7 jackknife replicates, with significance lost on dropping Upper East, Greater Accra or Eastern. The censoring no longer determines significance, but both versions are reported and the disclosure stands. |
| 6 | No false-discovery accounting for the XtX outliers. A 0.999 quantile is not a significance test. | step08c_calibrate_xtx.py now reports expected false positives, the implied expected-null-exceedance ratio, per-SNP p-values against the POD null, Benjamini-Hochberg q-values written into xtx_all_snps.csv, and counts at q <= 0.01/0.05/0.10/0.20. | 19 SNPs above the quantile threshold, 13 of which also pass q <= 0.10; 3.9 expected false positives, expected-null-exceedance ratio 20.6%. Both figures are now reported. The q-values are reproducible from the raw BayPass summaries, which is what made the file recoverable when it was later found truncated to zero bytes (row 13); it has been regenerated and verified, so the previous version's statement that it is empty on disk no longer holds. |
| 7 | Zone is confounded with sequencing study, a fourth design confounder that the manuscript did not list. | New scripts/step11_batch_confounding.py quantifies it rather than asserting it, on the 4,905 QC-passing isolates. | Four of six zone pairs share no sequencing study, including coastal savanna against Guinea savanna. Within deciduous forest the four testable studies of five differ in $F_{ws}$ (H = 14.9, p = 1.9 x 10-3, n = 811) with polyclonal fractions spanning 0.430-0.578; Guinea savanna's six studies do not differ significantly (p = 0.103). Added to the limitations in Section 9 and to Section 10.3, with the denominator disclosed. |
| 8 | Found by the re-run itself. The "skip if output exists" guards in step05b, step07b and step08b did | All three guards now require the output to be newer than its input. | No incorrect result reached the manuscript, but only because stage 8 asserts matching row counts and failed |

| # | Confirmed defect | Fix | Effect on reported results |
|---|---|---|---|
| | not check whether the input had changed, so ADMIXTURE and TreeMix silently reported the previous day's results against the new SNP matrix and logged "stage finished". | | loudly. ADMIXTURE and TreeMix had no such assertion. Documented in Appendix E. |
| 9 | Markers were identified by matching annotation strings. step10 matched SnpEff ANN_HGVS_p protein-change strings against a hand-written list. Two failure modes, both of which occurred: (i) a resistance allele that IS the 3D7 reference residue can never appear as an ALT change, so at *pfdhps* 437 the only annotation is the revertant Gly437Ala and a search for "Ala437Gly" matched nothing; (ii) a residue reachable only by two changes in one codon is never annotated, so *pfcrt* N75E - the CVIET residue - is named by neither 403621 (A>G, Asn75Asp) nor 403623 (T>A, Asn75Lys). | New scripts/step12_codon_panel.py reconstructs the codon instead of reading the annotation: each requested codon is mapped to its three genomic positions through the CDS, every variant record there is collected regardless of numalt or is_snp, each isolate's alleles are applied to the reference codon read from the genome, and the result is translated. The output is the full residue spectrum including the reference residue. step10 is superseded and its outputs are marked so in Appendix C. | 57 codons across all nine panel genes, 24 polymorphic and 33 fixed at the reference residue. *pfdhps* A437G, absent from the previous panel entirely, is recovered at 0.897 (n = 1,892) and is zone-structured at p = 1.2 x 10-6 - the largest single addition to the resistance results. *pfcrt* codon 75 resolves to Glu, which dissolves the "naming discrepancy" reported in the previous version. *pfcrt* N326S corrects from 0.008 to 0.0005. *pfcoronin*, previously said to contribute no canonical codon, contributes three. |
| 10 | *The* pfk13 *marker list was written from memory*. It contained 13 distinct protein changes while claiming 14 (a duplicated M476I record inflated the count), classified P441L and C469F as validated when both are candidate, and omitted P574L, G533S and R622I. R622I was requested but never retrieved, so it became a silent non-result. | The list is rebuilt from the WHO compendium v1.0 (December 2025) in step12_codon_panel.py: all 14 validated variants (F446I N458Y C469Y M476I Y493H G533S R539T I543T P553L R561H P574L C580Y R622I A675V), all 6 candidates (E252Q P441L C469F N537I G538V V568G) and four potential-tier codons in the same region (449, 481, 515, 527) - 20 variants at 19 distinct codons, because C469Y and C469F share codon 469, so 19 + 4 = 23 codons in all. Script comments record the tier of each. | Twenty-two of the 23 codons are fixed at the reference residue and the twenty-third, codon 458, is at 0.9995 (1,865 of 1,866 called isolates carry Asn); the only non-reference *pfk13* call in the dataset is the one isolate carrying Asp at 458, which is not the validated Tyr. "Non-reference anywhere" must be read as anywhere in *pfk13*: *pfcrt* N326S is likewise a single non-reference call, at 0.0005. The artemisinin negative is now complete across the validated and candidate sets rather than resting on a partial list. Two corrected caveats go with it: the denominator is per codon, 1,845-1,876 of 1,900 called, not "all 1,900"; and the sampling period is 2009-2021, so it is a historical negative. |
| 11 | A haplotype frequency was reported as the range of its marginals. The *pfdhfr* "triple mutant" was given as 0.786-0.908, which is the spread of three independent per-codon frequencies and says nothing about co-occurrence. The *pfcrt* CVIET set was likewise inferred to travel together from six similar marginals. | step12_codon_panel.py reconstructs haplotypes within isolates on complete cases only, and prints each haplotype's components on the same complete-case set so that the arithmetic constraint is visible. | *pfdhfr* IRN = 0.771 (n = 1,876, CI 0.751-0.789), below the smallest component on the same set (51I 0.788) as it must be, with NRN 0.116 and wild-type NCS 0.091 - so not a near-fixed single haplotype. IRNL = 0.000. *pfcrt* CVIET = 0.071 and is genuinely a unit: only CVMNK (0.929) and CVIET occur, with nothing intermediate. *pfdhps* GE = 0.010 against GK 0.887, so sulfadoxine resistance here is a high-frequency single substitution and not a high- |

| # | Confirmed defect | Fix | Effect on reported results |
|---|---|---|---|
| | | | frequency double. |
| 12 | The panel was validated by agreement of marginal frequencies, reported as "21 of 23 markers agree within 0.03". That does not show that two pipelines called the same isolates the same way, and it concealed the codon-75 error rather than exposing it. | step12_codon_panel.py compares call by call against MalariaGEN's published genotypes on the isolates where both pipelines called. | 47,012 of 47,012 calls agree - 100.0000% across 25 markers (*pfcrt* 9, *pfmdr1* 5, *pfdhfr* 5, *pfdhps* 5, *pfmdr2* 1), per-marker denominators 1,834-1,900. Codon 75 agrees exactly. The limit is now stated too: MalariaGEN publishes no per-codon genotype for *pfk13*, *pfaat1*, *pfcoronin* or *pfap2mu*, which accounts for 29 codons, and none for *pfcrt* 73, *pfdhps* 431 or *pfmdr2* 492 either, so 32 of the 57 codons have no independent per-sample check. The "within 0.03" claim is withdrawn wherever it appeared. |
| 13 | A results file was destroyed and restored. results/baypass/xtx_all_snps.csv was found at 0 bytes, truncated at 08:29:53 — hours after the 01:42 run that produced it, so not a failed write but a later overwrite, almost certainly a shell redirect where a read was intended during a verification pass. It was the only zero-byte file anywhere under results/ or interim/. Found during the robustness round rather than by either audit. | Regenerated from the intact raw BayPass outputs, ghana_core_summary_pi_xtx.out and ghana_pod_summary_pi_xtx.out, which were never touched. The regeneration was verified rather than assumed: outlier_snps.bed, xtx_calibration.json and xtx_threshold.txt are byte-identical by md5 before and after, and the restored file carries the same 3,916 rows and the same 19 outliers. | No reported number changes, and the file the text cites now exists and can be checked, so the previous version's next-steps item 6 and the matching warnings in Appendices C and E are withdrawn. What the incident does establish belongs in the record: nothing under results/ is write-protected, the project is not under version control, and this file was recoverable only because its own inputs happened to survive. A file whose inputs had also been overwritten would have been unrecoverable and, on this evidence, unnoticed until someone opened it. |
| 14 | A log misreported its own output path. The --suffix patch written for the Stage 3 re-run suffixed step03_umap.py's file writes but not its print statements, so the run announced that it was writing the published results/dimred_comparison.csv when it had in fact correctly written results/dimred_comparison_qcset.csv. | The print statements now carry the suffix. The published outputs were checked by mtime and by contents and were never modified; the data were never at risk. | No reported number changes. It is registered because a log that misreports its own output path is precisely the failure mode that produces a later misattribution between the 4,905-isolate and 1,900-isolate figures — the very distinction the re-run existed to remove — and because it was caught by reading the log against the directory rather than by any assertion in the code. Defect 8 was caught by an assertion; this one was not, because step03_umap.py has none. |

*Table F.1. Confirmed defects that required code changes: eight from the internal audit (the eighth found by the re-run itself), four from the external critique, and two integrity incidents from the robustness round of 10 October 2026. All fourteen are fixed. Defects 1-8 were re-run on 10 October 2026; defects 9-12 are fixed in step12_codon_panel.py, whose outputs supersede step10's. Rows 13 and 14 are operational rather than analytical: neither changed a reported number, and they are registered here rather than left in the logs because a defect register that records only the defects that changed a number is not a defect register.*

| Confirmed defect | Correction |
|---|---|
| "No validated antimalarial resistance locus appears among the outliers", in the abstract | Removed. Replaced by the measured panel frequencies, by the ascertainment cascade stated once in Section 10.1, and by the distinction |

| Confirmed defect | Correction |
|---|---|
| and Section 6.8, read as biology when only four of the panel's 171 codon positions - two in *pfdhfr*, one in *pfaat1* and one of the two *pfcrt* codon-75 positions - ever reached the MAF-filtered scan matrix. | between the tested negative at *pfdhfr* Cys59Arg (q = 0.43), *pfdhfr* Asn51Ile (q = 0.67) and *pfcrt* codon 75 (XtXst 6.21 against a threshold of 10.840) and the untested *pfmdr1*, *pfdhps* and *pfk13* markers. *pfcrt* is not on the untested side: codon 75 was in the scan matrix. |
| *pfaat1* S258L was called "the functionally validated chloroquine-resistance variant", and the manuscript stated that the functional work did not test amodiaquine, then built a Ghanaian amodiaquine-pressure argument on that gap. *Found by the external critique; the error was ours. Amodiaquine was tested ("Parasite IC50 values for CQ, amodiaquine, lumefantrine, mefloquine and quinine were measured"), the shifts were "well below the threshold associated with clinical resistance", the authors state the other-quinoline hypothesis "is not supported" and conclude CQ was "the primary selective force". The effect is 1.5-fold, in a single chloroquine-resistant Southeast Asian line (NHP4026), with S258L-F313S epistasis and an explicit* pfcrt-background caveat; the 0%-to-97% rise is Gambian, 1984-2014, and the paper says nothing about Ghana. | Rewritten in Sections 6.8 and 10.1. S258L is now described as a variant with a demonstrated but modest, epistasis- and background-dependent effect that potentiates chloroquine resistance, and as a strong candidate for the cline rather than an explanation of it. Three possible drivers are offered and none is claimed: historical chloroquine pressure, the reported fitness cost, and amodiaquine as an explicitly untested possibility. The codon panel adds a further caution now stated in the text - F313, the epistatic partner, is fixed at the reference residue in all 1,900 isolates. |
| WHO tiers were applied to genes rather than to variants, including the label "*pfmdr1*, WHO-validated, lumefantrine". | Removed. The compendium classifies variants, and it contains no validated or candidate lumefantrine marker at all ("The review did not identify any validated or candidate molecular markers associated with lumefantrine resistance"). *pfaat1*, *pfap2mu* and *pfmdr2* appear nowhere in it at any tier and are never described as WHO-classified, though they stay in the panel; *pfcoronin* is in it at the potential tier only. Stated as a note in Section 10.1. |
| *The* pfcrt *codon 75 disagreement with MalariaGEN was reported as a "marker-naming discrepancy" and MalariaGEN was said to be "correct on the biology".* It was neither a naming question nor a disagreement between datasets. | Replaced. N75E requires two changes in one codon, so no single-variant annotation names it; reconstruction yields Glu at 0.071 and agrees with MalariaGEN on all 1,874 jointly called isolates. The old note is rewritten below and the phrase "naming discrepancy" is removed from Section 10.1, Appendix F and the marker tables. |
| *The* kelch13 *cross-check was presented as corroboration of the per-codon* pfk13 *calls.* Pf8 assigns status over the whole 349-726 domain and classifies any missing call in that range as undetermined. | Weakened explicitly. Only 36 isolates have a complete published status - under 2% of the sample - which is weak corroboration for per-codon calls on roughly 1,860 isolates, and the two are no longer presented as equivalent evidence. Stated in Sections 6.8, 10.1 and 10.3. |
| "No ancestry structure corresponds to geography" was asserted and never tested. | Tested. Kruskal-Wallis H = 105.4, p = 1.05 x 10-22; zone explains 1.9% of the K = 2 minor component's variance (eta-squared = 0.019, computed from the final Q matrix). The magnitude claim survives, the wording does not, and the zone means are reported as non-monotone with latitude. |
| Section 7.4 concluded that sentinel surveillance at a few sites is adequate for detecting emergence, which the project's own literature review argues against. | Restricted to what the evidence supports: negligible neutral differentiation licenses a claim about neutral structure only, and marker-frequency representativeness is a separate question on which this analysis finds significant between-zone variation at *pfdhps* 437 (p = 1.2 x 10-6), *pfdhfr* (p <= 2.43 x 10-4) and *pfaat1* (p = 1.9 x 10-18). The qualified form now appears in Section 7.4, in the abstract's conclusion and in Section 10.1; the unqualified form is removed. |
| "100,000 MAF-matched POD SNPs" is false: | Wording withdrawn in Sections 5.6, 6.8 and 8.1 and in Appendix B, with a |

| Confirmed defect | Correction |
|---|---|
| pi.maf truncates the simulated ancestral frequency and does not filter on MAF. The claim appeared three times plus the configuration table. | note on what pi.maf actually does. Section 6.8's variant phrasing was missed by two earlier rounds and is corrected here. |
| The abstract said two of six $F_{ST}$ confidence intervals span zero where Section 6.4 said three. Raised independently three times. | On the final matrix it is one of six - Guinea savanna against the transitional zone. Both remaining transitional pairs have strictly positive lower bounds, so the small-n caveat for that zone now rests on n = 25 and the wide standard error rather than on the intervals. Abstract, Section 6.4, Table 6.4's caption and Table 10.2 row 1 now agree and cite the same table. |
| Section 10.1 said the outlier list is "headed by AAT1"; the table it summarises is headed by PTP7. | Corrected. PTP7 heads at max XtX 26.44, AAT1 is second at 24.20 and is described as the strongest candidate on literature grounds rather than by rank. The gene list is also qualified: 17 of the 19 outliers intersect a gene, and one of the two intergenic SNPs is fifth-highest genome-wide at XtX 20.79. |
| Polyclonality was said to "order cleanly along the north-south gradient (0.597 to 0.444)" by quoting only the endpoints; the transitional zone breaks the trend and its 25 of 50 is uninformative. | All four fractions reported in geographic order in Section 10.1, with the non-monotonicity, the transitional zone's position as the second-most-northern zone, the 4,905-isolate denominator, and the between-study confounding demonstrated inside deciduous forest - not inside Guinea savanna, where it is not significant. |
| The between-study polyclonality range 0.430-0.578 was attributed to Guinea savanna, and a p-value was attached to the polyclonal fraction rather than to the $F_{ws}$ distributions. | The range belongs to deciduous forest (four testable studies of five, n = 811); Guinea savanna's six studies span 0.532-0.682 at p = 0.103 and are not significantly different. The Kruskal-Wallis p = 1.9 x 10-3 is a test of $F_{ws}$, not of the fraction, and no test of the fraction exists in the outputs. Corrected in Sections 4.2.2, 6.2, 7.2, 7.3 and 10.1. |
| Appendix B attributed K_MIN/K_MAX, ADMIXTURE_CV_FOLDS and LD_PRUNE_ARGS to config.py, which does not control them. | "Set in" column corrected to step05b_admixture.sh, and the four constants that no script reads are listed as dead. |
| ghanaAEZ.geojson had no recorded provenance. | Traced with certainty to an ArcGIS Online item, byte-for-byte identical, md5 96b1a7f4da62d6f0153826f41af96152, with null licence and accession information under an anonymous account. Documented as an unattributed re-host, cited as accessed, and retained as a publication blocker. No institution is named as author. |

*Table F.2. Confirmed defects corrected in the text rather than in code, from both rounds. No re-run was needed for any of them with one exception, which is noted rather than hidden: the pfcrt codon 75 entry's correction comes from the panel rebuild recorded as defect 9 of Table F.1 and did require a re-run; it is repeated here because the wording it replaces was prose. The four entries shown in bold were raised by the external critique, whose five points are otherwise recorded as defects 9-12 of Table F.1; on the critique's own four-code/one-prose split only the pfaat1 interpretation and the gene-level WHO tier labels are text-only, so the bold marks here should not be read as four further text-only defects.*

**[Observed]** **Confirmed and not yet acted on.** Three items have come off this list since the previous version, all in the robustness round of 10 October 2026: the $F_{ws}$ and MAF sensitivity analyses, now reported in Section 10.1 and Table 10.1; the mapping-quality check at the outlier positions, now in Section 10.1 and results/outlier_quality_check.csv; and the re-run of Stage 3 on the analysis set, now Figure 10.1. What remains: the functional-enrichment permutation test for the outlier gene set; the temporal control at a single collection year; the robustness join against an authoritative six-zone Ghana layer; the recomputation of the zone-pair study overlap on the 1,900-isolate set; independent corroboration of the 32 panel codons MalariaGEN does not publish per sample; and, added by the robustness round, a read-level depth check at Pf3D7_11_v3:1 054 078 and Pf3D7_11_v3:1 054 447, without which two of the 19 outliers stay qualified. These are carried in Sections 8.2, 10.3 and 10.4 rather than here. Four items that the corrections document listed as outstanding — the filter order, the TreeMix run without -se, the study/batch confounding and the absence of

FDR accounting — were subsequently fixed and are reported as results in Sections 4.2.6, 6.1, 6.7, 2.6, 6.2, 7.2 and 6.8; any defect register that still lists them as open is out of date. Two high-severity findings — that the canonical markers were excluded at ingest, and that the top AAT1 outlier is S258L — were confirmed by the audit but not independently re-verified before the server went off-LAN; both have since been re-derived directly from Pf8 by the codon panel. The previous version's cross-validation claim for that panel, "21 of 23 shared markers agree within 0.03", is withdrawn and replaced: per-sample comparison against MalariaGEN's published genotypes now gives **47,012 of 47,012 calls in agreement across 25 markers**, which is a stronger result obtained by a stricter test, and which also exposed the codon-75 error that the weaker test had hidden.

> **On *pfcrt* codon 75 — not a naming discrepancy.** The previous version reported that Pf8's SnpEff annotation labels the ALT at *pfcrt* codon 75 as Asp while MalariaGEN calls it Glu, described this as the one marker-naming discrepancy among 23, and credited MalariaGEN with being right on the biology. The diagnosis was wrong. Codon 75 carries two variants — 403621 (A>G, annotated Asn75Asp) and 403623 (T>A, annotated Asn75Lys) — and Asn→Glu requires both, so no single-variant annotation can name N75E at all. The appearance of a discrepancy was created entirely by identifying markers from annotation strings. Reconstructed from the codon, residue 75 is *Glu at 0.071 (*n = *1,874)*, exactly matching the other CVIET component residues, and it agrees with MalariaGEN on every one of the 1,874 jointly called isolates. There is no discrepancy to follow a convention about: there is one residue, Glu, and it is ours and MalariaGEN's alike.

— End of manuscript —